Volume III.

First Section.

The Elliptic Integrals.

§ 1. Definition of the elliptic integrals.

When the systematic presentation of the integral calculus has reached the point where algebraic integrals containing the square root of a function of the first or second degree are reduced to integrals of rational functions, the learner encounters there a barrier which he is not able to pass with the aids available up to that point. The effort to enlarge these aids, so as to make even the next class of integrals accessible to investigation, is, as it was historically the occasion for a more thorough study of the elliptic integrals and for the introduction of the elliptic functions, also the most natural and intelligible point of departure for one who is first to be introduced into the theory of these functions. Hence our next task is to become acquainted with the elliptic integrals and with their most important properties.

The definition of an elliptic integral from which we shall start is the following. Let $f(x)$ denote an integral rational function of the third or fourth degree with four distinct roots,

$$f(x) = a_0x^4 + 4a_1x^3 + 6a_2x^2 + 4a_3x + a_4, \tag{1}$$

where a_0 and a_1 do not vanish simultaneously; and let $\Phi(x, y)$ be an arbitrary integral or fractional rational function of the two arguments x, y. Then

$$\int \Phi(x, \sqrt{f(x)})\, dx$$

is the general elliptic integral, and

$$\Phi(x, \sqrt{f(x)})\, dx$$

is the general elliptic differential. But this general integral and differential can now be reduced to substantially simpler ones. First Φ can be put in the form

$$\frac{A + B\sqrt{f(x)}}{C + D\sqrt{f(x)}},$$

where A, B, C, D are integral rational functions of x; and, by multiplying this fraction by $C - D\sqrt{f(x)}$, into the form

$$\Psi(x) + \frac{\Phi(x)}{\sqrt{f(x)}},$$

where $\Psi(x)$ and $\Phi(x)$ are integral or fractional rational functions of x, namely

$$\Psi(x) = \frac{AC - BDf(x)}{C^2 - D^2f(x)},$$

$$\Phi(x) = \frac{(BC - AD)f(x)}{C^2 - D^2f(x)}.$$

We now leave aside the rational integral

$$\int \Psi(x)\, dx,$$

which leads to logarithms and algebraic functions, and concern ourselves only with the elliptic integral

$$\int \frac{\Phi(x)\, dx}{\sqrt{f(x)}} \tag{2}$$

or the corresponding elliptic differential

$$\Phi(x)\frac{dx}{\sqrt{f(x)}}. \tag{3}$$

It is sometimes useful to consider this differential in homogeneous form. To this end we put for x the ratio of two variables $x : y$, and accordingly for dx

$$\frac{y\,dx - x\,dy}{y^2}.$$

If then

$$f(x, y) = a_0x^4 + 4a_1x^3y + 6a_2x^2y^2 + 4a_3xy^3 + a_4y^4,$$

and $\Phi(x, y)$ is a homogeneous function of order 0, our elliptic differential takes the form

$$\Phi(x, y)\frac{y\,dx - x\,dy}{\sqrt{f(x, y)}}. \tag{4}$$

The problem that will occupy us first is twofold. First, by the introduction of new variables, the differential

$$\frac{dx}{\sqrt{f(x)}} \quad \text{or} \quad \frac{y\,dx - x\,dy}{\sqrt{f(x, y)}} \tag{5}$$

is to be transformed into another of the same form, but in which the function under the square-root sign is made as simple as possible, and in particular dependent on as small a number of parameters as possible (§§ 3, 4, 5). Second, the general differential (3) or (4) is to be decomposed into other similar differentials in which the function $\Phi(x)$ or $\Phi(x, y)$ is as simple as possible (§ 12).

The function $f(x)$ can be decomposed into linear factors (Vol. I, § 33). Accordingly we also put

$$f(x) = a_0(x - x_1)(x - x_2)(x - x_3)(x - x_4).$$

If we think of the quantities $x, x_1, x_2, \ldots$ as abscissae of points laid off on a straight line from an arbitrary origin, positive on one side and negative on the other side (for example positive to the right, negative to the left), then the differences $x - x_1$, $x - x_2$, $x - x_3$, $x - x_4$ represent the distances of the point x from x_1, x_2, x_3, x_4, counted positively when x lies on the positive side of $x_1, x_2, \ldots$. In the same sense $x_2 - x_1$ is the distance of the point x_2 from x_1, and so forth.

We shall denote the differences more briefly by (xx_1), $(xx_2), \ldots, (x_2x_1), \ldots$. We use this notation only for easier survey, and we do not exclude imaginary points of the line.

§ 2. Cross-ratios.

The simplest means of simplifying the elliptic differential rests on the linear transformation of the function $f(x)$ (Vol. I, § 67). Let $\alpha, \beta, \gamma, \delta$ denote four arbitrary constants whose determinant

$$\alpha\delta - \beta\gamma = r \tag{1}$$

is different from zero. We put

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}, \tag{2}$$

and call this a linear substitution. If x and x' are interpreted, as in § 1, as points on two straight lines L and L', then (2) determines a one-to-one correspondence of the points of L and L', a mapping. To the values $x = \infty$ and $x' = \infty$ correspond the points at infinity of the lines, which we likewise regard as definite points. Thus the point $x' = \infty$ corresponds to the point $x = \alpha : \gamma$, and the point $x = \infty$ to the point $x' = -\delta : \gamma$; and only when $\gamma = 0$, so that the substitution (2) is integral, do the points at infinity on L and L' correspond to one another.

The substitution (2) is not changed when the four transformation numbers are multiplied by the same factor. The determinant r is multiplied by the square of this factor, and therefore this factor can be so chosen that the determinant takes any prescribed value, for example the value 1.

If the substitution (2) is applied to any two points x_1, x_2 and the corresponding x'_1, x'_2, then one obtains

$$(x_1 x_2) = \frac{r(x'_1 x'_2)}{(\gamma x'_1 + \delta)(\gamma x'_2 + \delta)},$$

and, if one adds a second pair of points x_3, x_4 and the corresponding x'_3, x'_4,

$$(x_1 x_2)(x_3 x_4) = \frac{r^2 (x'_1 x'_2)(x'_3 x'_4)}{(\gamma x'_1 + \delta)(\gamma x'_2 + \delta)(\gamma x'_3 + \delta)(\gamma x'_4 + \delta)}.$$

Interchanging here x_2 with x_3 and forming the quotient, one obtains

$$\frac{(x_1 x_2)(x_3 x_4)}{(x_1 x_3)(x_2 x_4)} = \frac{(x'_1 x'_2)(x'_3 x'_4)}{(x'_1 x'_3)(x'_2 x'_4)}. \tag{3}$$

The expression on the left side is called the cross-ratio of the pair of points $x_1 x_4$ to the pair of points $x_2 x_3$, and thus this formula contains the theorem:

The cross-ratio of two pairs of points is invariant under simultaneous linear transformation of the four points.

Four points can be separated into two pairs in six ways. But if we put

$$a = (x_2 x_3)(x_1 x_4), \qquad b = (x_3 x_1)(x_2 x_4), \qquad c = (x_1 x_2)(x_3 x_4),$$

then the identity holds:

$$a + b + c = 0 \tag{4}$$

and we obtain the six cross-ratios

$$-\frac{c}{b}, \quad -\frac{a}{c}, \quad -\frac{b}{a}, \quad -\frac{b}{c}, \quad -\frac{c}{a}, \quad -\frac{a}{b}; \tag{5}$$

if we put the first of them, $-c : b$, equal to $\varkappa^2$, then from (4) we obtain the six:

$$\varkappa^2, \quad 1 - \frac{1}{\varkappa^2}, \quad \frac{1}{1 - \varkappa^2}, \quad \frac{1}{\varkappa^2}, \quad \frac{\varkappa^2}{\varkappa^2 - 1}, \quad 1 - \varkappa^2. \tag{6}$$

Thus the six cross-ratios of the four points are expressed linearly through any one among them.

If x_1, x_2, x_3, x_4 are permuted among themselves, then a, b, c are permuted among themselves and change their signs, but in such a way that either all three signs are changed simultaneously or none is changed. For example the transpositions give

$$\begin{array}{ll}
(14) \text{ and } (23) & \begin{pmatrix} a & b & c \\ -a & -c & -b \end{pmatrix} \\
(24) \text{ and } (31) & \begin{pmatrix} a & b & c \\ -c & -b & -a \end{pmatrix} \\
(34) \text{ and } (12) & \begin{pmatrix} a & b & c \\ -b & -a & -c \end{pmatrix}.
\end{array}$$

If x_1, x_2, x_3, x_4 are real, then $\varkappa^2$ is a positive proper fraction if and only if $\varkappa^2$ and $1 - \varkappa^2$ are positive; that is, if either a and c are positive and b negative, or a and c negative and b positive. This occurs when x_1, x_2, x_3, x_4 follow one another in this order increasingly in magnitude, and for all arrangements that arise from it by the permutations which leave b unchanged. These are the following eight:

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3
1	4	3	2
4	3	2	1
3	2	1	4
2	1	4	3

Thus, if one imagines x_1, x_2, x_3, x_4 placed in this order on the circumference of a circle, then one obtains a positive proper fraction $\varkappa^2$ precisely when the x_i follow one another in magnitude in such a way that, beginning with an arbitrary one as the least, one then continues around the circle either to the right or to the left. We express this as follows:

In order that $\varkappa^2$ be a positive proper fraction, the quantities x_1, x_2, x_3, x_4 must follow one another cyclically in order of magnitude.

§ 3. Linear transformation of the elliptic differential.

When the two points x_3 and x_4 approach one and the same point x, then x_3' and x_4' approach the corresponding point x'. We then put $(x_3x_4) : (x_3'x_4') = dx : dx'$ and obtain from (3), § 2, the relation between the differentials dx, dx':

$$\frac{(x_1x_2)\,dx}{(x_1x)(x_2x)} = \frac{(x_1'x_2')\,dx'}{(x_1'x')(x_2'x')}. \tag{1}$$

Similarly, if in place of x_1, x_2 one puts two other points x_3, x_4, one obtains

$$\frac{(x_3x_4)\,dx}{(x_3x)(x_4x)} = \frac{(x_3'x_4')\,dx'}{(x_3'x')(x_4'x')}, \tag{2}$$

and, multiplying and taking the square root,

$$\frac{\sqrt{(x_1x_2)(x_3x_4)}\,dx}{\sqrt{(x_1x)(x_2x)(x_3x)(x_4x)}} = \frac{\sqrt{(x_1'x_2')(x_3'x_4')}\,dx'}{\sqrt{(x_1'x')(x_2'x')(x_3'x')(x_4'x')}}, \tag{3}$$

and therefore a transformation of the elliptic differential by a linear substitution:

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}.$$

This substitution is completely determined if to three arbitrary points x_1, x_2, x_3 the corresponding values x_1', x_2', x_3' are arbitrarily assigned; it is obtained from the equality of the cross-ratios

$$\frac{(x_1x_2)(x_3x)}{(x_1x_3)(x_2x)} = \frac{(x_1'x_2')(x_3'x')}{(x_1'x_3')(x_2'x')}, \tag{4}$$

by solving for x, and the fourth value x_4' belonging to x_4 is obtained from

$$\frac{(x_1x_2)(x_3x_4)}{(x_1x_3)(x_2x_4)} = \frac{(x_1'x_2')(x_3'x_4')}{(x_1'x_3')(x_2'x_4')}. \tag{5}$$

If the function $f(x)$ is given in the elliptic differential

$$du = \frac{dx}{\sqrt{f(x)}},$$

then x_1, x_2, x_3, x_4 are thereby also given, and by means of the linear transformation (3) one can bring the differential into another form in which three of the quantities x_1', x_2', x_3', x_4' have arbitrarily assigned values. One obtains the normal form if one assumes

$$x' = z, \qquad x_1' = 0, \qquad x_2' = 1, \qquad x_3' = \frac{1}{\varkappa^2}, \qquad x_4' = \infty.$$

The quantity $\varkappa$ is called by Legendre the modulus of the elliptic differential, and $\sqrt{1-\varkappa^2} = \varkappa'$ the complementary modulus. Thus, if the points are made to correspond in the following way,

$$\begin{array}{ccccc} z, & 0, & 1, & \frac{1}{\varkappa^2}, & \infty, \\ x, & x_1, & x_2, & x_3, & x_4, \end{array}$$

then from the equality of cross-ratios one easily obtains the relations

$$\begin{aligned} z &= \frac{(xx_1)(x_2x_4)}{(xx_4)(x_2x_1)}, \\ 1 - z &= \frac{(xx_2)(x_1x_4)}{(xx_4)(x_1x_2)}, \\ 1 - \varkappa^2 z &= \frac{(xx_3)(x_1x_4)}{(xx_4)(x_1x_3)}, \end{aligned} \tag{6}$$

$$dz = \frac{(x_4x_1)(x_4x_2)}{(x_2x_1)} \frac{dx}{(xx_4)^2}, \tag{7}$$

$$\varkappa^2 = \frac{(x_3x_4)(x_1x_2)}{(x_1x_3)(x_2x_4)}, \qquad \varkappa'^2 = \frac{(x_2x_3)(x_1x_4)}{(x_2x_4)(x_1x_3)}, \tag{8}$$

and from (3)

$$\frac{\sqrt{(x_3x_1)(x_4x_2)}\,dx}{\sqrt{-(x_1x)(x_2x)(x_3x)(x_4x)}} = \frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}. \tag{9}$$

By § 2, $\varkappa^2$ is a positive proper fraction when x_1, x_2, x_3, x_4 are real and follow one another cyclically in order of magnitude. This gives eight different transformations of this kind, and among them one can choose two for each case in which the interval $z = 0$ to $z = 1$ corresponds to one of the prescribed intervals $(x_1x_2), (x_2x_3), (x_3x_4), (x_4x_1)$; for one of these transformations increasing z corresponds to increasing x, and for the other to decreasing x (with passage from $+\infty$ to $-\infty$). In the first case the square roots in (9) have the same sign on both sides, in the second the opposite sign.

If one does not insist that $\varkappa^2$ be a positive proper fraction, then one may permute the points x_1, x_2, x_3, x_4 in all possible ways and obtains 24 distinct linear transformations into the normal form, of which any four give the same $\varkappa^2$; in all there are six different moduli, derived from one another according to § 2, (6).

The totality of these transformations is most easily surveyed if one assumes that the integral to be transformed already has the normal form

$$\frac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}};$$

then in the formulae (6) to (9) one has only to replace x_1, x_2, x_3, x_4 in all possible ways by $0, 1, 1:\lambda^2, \infty$. As examples we take the following correspondences:

	x_1	x_2	x_3	x_4
1)	0	∞	$\frac{1}{\lambda^2}$	1
2)	0	$\frac{1}{\lambda^2}$	1	∞
3)	1	0	∞	$\frac{1}{\lambda^2}$.

From (6) and (8) one then obtains

$$\begin{array}{lll} 1) & z = -\dfrac{x}{1-x}, & \varkappa^2 = 1-\lambda^2 = \lambda'^2, \\ 2) & z = \lambda^2 x, & \varkappa^2 = \dfrac{1}{\lambda^2}, \\ 3) & z = \dfrac{1-x}{1-\lambda^2 x}, & \varkappa^2 = \lambda^2, \end{array}$$

and for the differential

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

one obtains, in the three cases,

$$\begin{array}{ll} 1) & \dfrac{dx}{\sqrt{-x(1-x)(1-\lambda^2 x)}}, \\ 2) & \dfrac{\lambda\, dx}{\sqrt{x(1-x)(1-\lambda^2 x)}}, \\ 3) & \dfrac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}}. \end{array}$$

§ 4. Legendre's normal form.

If the function $f(x)$ which stands under the root sign in the elliptic differential has real coefficients, three cases are possible:

1. $f(x)$ has four real roots x_1, x_2, x_3, x_4;
2. $f(x)$ has two real roots x_1, x_2 and a pair of conjugate imaginary roots x_3, x_4;
3. $f(x)$ has two pairs of conjugate imaginary roots x_1, x_2 and x_3, x_4.

In all three cases the elliptic differential can be brought by a real linear transformation to the form

$$\frac{dz}{\sqrt{(z^2-\alpha)(z^2-\beta)}}, \tag{1}$$

where α and β are real, positive or negative, constants.

We determine the linear dependence between the variables x and z so that the following values correspond:

$$\begin{array}{ccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \sqrt{\alpha}, & -\sqrt{\alpha}, & \sqrt{\beta}, & -\sqrt{\beta}, \end{array} \tag{2}$$

and obtain a substitution of the form

$$h\frac{x-x_1}{x-x_2} = \frac{z-\sqrt{\alpha}}{z+\sqrt{\alpha}}. \tag{3}$$

To determine h, we put $x = x_3$, $x = x_4$, and correspondingly $z = \sqrt{\beta}$, $z = -\sqrt{\beta}$; this gives

$$h\frac{(x_3x_1)}{(x_3x_2)} = \frac{\sqrt{\beta}-\sqrt{\alpha}}{\sqrt{\beta}+\sqrt{\alpha}}, \qquad h\frac{(x_4x_1)}{(x_4x_2)} = \frac{\sqrt{\beta}+\sqrt{\alpha}}{\sqrt{\beta}-\sqrt{\alpha}}.$$

By multiplication and division it follows that

$$h = \sqrt{\frac{(x_3x_2)(x_4x_2)}{(x_3x_1)(x_4x_1)}}, \qquad \frac{\sqrt{\alpha}-\sqrt{\beta}}{\sqrt{\alpha}+\sqrt{\beta}} = \sqrt{\frac{(x_3x_1)(x_4x_2)}{(x_3x_2)(x_4x_1)}}. \tag{4}$$

If, as in § 2, we put

$$
\begin{aligned}
a &= (x_2x_3)(x_1x_4),\\
b &= (x_3x_1)(x_2x_4),\\
c &= (x_1x_2)(x_3x_4),\\
a+b+c &= 0,
\end{aligned}
\tag{5}
$$

then, since only the ratio of α to β matters, we can satisfy the last of equations (4) by putting

$$\sqrt{\alpha} = \sqrt{\pm a} + \sqrt{\mp b}, \qquad \sqrt{\beta} = \sqrt{\pm a} - \sqrt{\mp b}, \tag{6}$$

and from (3) one obtains for z the expression

$$z = \sqrt{\alpha}\,\frac{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} + (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} - (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}. \tag{7}$$

Putting beside (3) the equation following from it,

$$h'\frac{x-x_3}{x-x_4} = \frac{z-\sqrt{\beta}}{z+\sqrt{\beta}},$$

one obtains by logarithmic differentiation

$$\frac{(x_1x_2)dx}{(xx_1)(xx_2)} = \frac{2\sqrt{\alpha}\,dz}{z^2-\alpha}, \qquad \frac{(x_3x_4)dx}{(xx_3)(xx_4)} = \frac{2\sqrt{\beta}\,dz}{z^2-\beta},$$

and hence, by multiplication and with regard to (5) and (6),

$$\frac{dx}{\sqrt{\pm(xx_1)(xx_2)(xx_3)(xx_4)}} = \frac{2dz}{\sqrt{(z^2-\alpha)(z^2-\beta)}}. \tag{8}$$

If now the four roots of $f(x)$ are real, then, as we saw in § 2, there are eight ways to assign these roots to the signs x_1, x_2, x_3, x_4 so that a and b have opposite signs; and if therefore $\pm a, \mp b$ are positive, $\sqrt{\alpha}, \sqrt{\beta}$ become real, hence α, β positive, and then by (7) z also becomes real.

Second, if x_1, x_2 are real and x_3, x_4 conjugate imaginary, then a and $-b$ are conjugate imaginary; hence, if the signs of the square roots $\sqrt{\pm a}, \sqrt{\mp b}$ are suitably chosen, $\sqrt{\alpha}$ is real and $\sqrt{\beta}$ purely imaginary, so that α is positive, β negative, and z real, since $(x_3x_1)(x_4x_1)$ and $(x_3x_2)(x_4x_2)$ are positive as products of conjugate imaginary quantities.

Finally, if x_1, x_2 and x_3, x_4 are two conjugate imaginary pairs, then a and $-b$ are both positive. If therefore in (6) we take the lower signs, then $\sqrt{\alpha}, \sqrt{\beta}$ are purely imaginary, hence α and β are negative, and from (7) z is a real expression.

Putting next

$$z^2 = y, \tag{9}$$

one obtains

$$\frac{2dz}{\sqrt{(z^2-\alpha)(z^2-\beta)}} = \frac{dy}{\sqrt{y(y-\alpha)(y-\beta)}} \tag{10}$$

and hence, by the quadratic substitution (9), an elliptic differential in which the expression under the root sign is a function of the third degree with real roots. By § 3 this can be brought by a linear substitution to the normal form

$$\frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}},$$

and in it ξ and $\varkappa$ can be taken as positive proper fractions.

Legendre's normal form follows from this by putting

$$\xi = \sin^2\varphi, \qquad d\xi = 2\sin\varphi\cos\varphi\,d\varphi;$$

then

$$\frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}} = \frac{2d\varphi}{\sqrt{1-\varkappa^2\sin^2\varphi}}. \tag{11}$$

§ 5. Weierstrass's normal form.

Another normal form of the elliptic differential, the one on which Weierstrass based his investigations, is

$$du = \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}, \tag{1}$$

where g_2, g_3 are constants, called the first and second invariants of the differential. The general elliptic differential

$$\frac{dx}{\sqrt{f(x)}}, \tag{2}$$

where

$$f(x) = a_0 x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4, \tag{3}$$

can be brought by a linear transformation to Weierstrass's normal form if the roots x_1, x_2, x_3, x_4 of $f(x)$ are known. This is most simply done as follows.

Let the roots of the cubic function

$$\varphi(z) = 4z^3 - g_2 z - g_3$$

be denoted by e_1, e_2, e_3. Then

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3)$$

and $e_1 + e_2 + e_3 = 0$.

We let the values of x and z correspond to one another as follows:

$$\begin{array}{ccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \infty, & e_1, & e_2, & e_3, \end{array}$$

and it follows, if m denotes a constant factor that may be chosen arbitrarily, that

$$\begin{aligned} z - e_1 &= \frac{m(xx_2)}{(x_1x_2)(xx_1)} = m\left(\frac{1}{(x_1x_2)} - \frac{1}{(x_1x)}\right), \\ z - e_2 &= \frac{m(xx_3)}{(x_1x_3)(xx_1)} = m\left(\frac{1}{(x_1x_3)} - \frac{1}{(x_1x)}\right), \\ z - e_3 &= \frac{m(xx_4)}{(x_1x_4)(xx_1)} = m\left(\frac{1}{(x_1x_4)} - \frac{1}{(x_1x)}\right). \end{aligned} \tag{4}$$

The factor m must be the same in all three equations in order that the differences $(z-e_3)-(z-e_2) = e_2 - e_3$, and so on, be independent of x.

Forming these differences and putting, as in Algebra, Vol. I, § 70,

$$(x_1x_2)(x_3x_4) = U, \qquad (x_1x_3)(x_4x_2) = V, \qquad (x_1x_4)(x_2x_3) = W,$$

one obtains

$$e_2 - e_3 = \frac{-mU}{(x_1x_2)(x_1x_3)(x_1x_4)}.$$

Introducing a new arbitrary factor μ by putting

$$m = 3\mu a_0 (x_1x_2)(x_1x_3)(x_1x_4), \tag{5}$$

one obtains

$$e_2 - e_3 = -3\mu a_0 U, \qquad e_3 - e_1 = -3\mu a_0 V, \qquad e_1 - e_2 = -3\mu a_0 W,$$

and, since $e_1 + e_2 + e_3 = 0$,

$$e_1 = \mu a_0(V - W), \qquad e_2 = \mu a_0(W - U), \qquad e_3 = \mu a_0(U - V).$$

The quantities y_1, y_2, y_3 introduced in Algebra I, § 70, are therefore equal to $e_1 : \mu$, $e_2 : \mu$, $e_3 : \mu$, and by formula (11) there one obtains for e_1, e_2, e_3 the cubic equation

$$z^3 - 3A\mu^2 z + \mu^3 B = 0.$$

Thus, if $\varphi(z) = z^3 - g_2 z - g_3$ and if

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3) = 4z^3 - g_2 z - g_3$$

is to hold, then

$$g_2 = 12\mu^2 A, \qquad g_3 = -4\mu^3 B. \tag{6}$$

Here

$$\begin{aligned} A &= a_2^2 - 3a_1a_3 + 12a_0a_4, \\ B &= 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3 \end{aligned} \tag{7}$$

are the first and second invariants of the biquadratic form $f(x)$.

Logarithmic differentiation of the first two equations (4) gives

$$\frac{dz^2}{(z - e_1)(z - e_2)} = \frac{dx^2(x_1x_2)(x_1x_3)}{(xx_1)^2(xx_2)(xx_3)},$$

and with the help of the last equation (4), by (5),

$$\frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} = \frac{dx}{\sqrt{12\mu f(x)}}. \tag{8}$$

If we wish to apply these results to the case where $f(x)$ has the normal form

$$f(x) = x(1 - x)(1 - \varkappa^2 x) = x - x^2(1 + \varkappa^2) + \varkappa^2 x^3,$$

and the point $z = \infty$ corresponds to the point $x = 0$, then we let a_0 tend to zero and x_4 to infinity, but let the product a_0x_4 tend to a finite value, which we may take to be 1.

Then

$$\begin{gathered} a_0x_4 = 1, \qquad a_0 = 0, \qquad a_1 = \varkappa^2, \\ a_2 = -(1 + \varkappa^2), \qquad a_3 = 1, \qquad a_4 = 0, \end{gathered}$$

and consequently

$$\begin{aligned} A &= (1 + \varkappa^2)^2 - 3\varkappa^2 = 1 - \varkappa^2 - \varkappa^4 = 1 - \varkappa^2\varkappa'^2, \\ B &= -(1 + \varkappa^2)\{2(1 + \varkappa^2)^2 - 9\varkappa^2\} \\ &= -(1 + \varkappa^2)(2 - \varkappa^2)(1 - 2\varkappa^2) \\ &= (1 + \varkappa^2)(1 + \varkappa'^2)(\varkappa^2 - \varkappa'^2) \\ &= (2 + \varkappa^2\varkappa'^2)(\varkappa^2 - \varkappa'^2), \end{aligned}$$

where $\varkappa'^2 = 1 - \varkappa^2$.

Thus, if we also add the discriminant

$$\Delta = 16 \cdot 27\, \mu^6(4A^3 - B^2),$$

then

$$\begin{aligned} g_2 &= 12\mu^2(1 - \varkappa^2\varkappa'^2), \\ g_3 &= -4\mu^3(2 + \varkappa^2\varkappa'^2)(\varkappa^2 - \varkappa'^2), \\ \Delta &= g_2^3 - 27g_3^2 = 27^2 \cdot 16\, \mu^6\varkappa^4\varkappa'^4, \end{aligned} \tag{9}$$

and from (6) one obtains the transformation

$$\frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} = \frac{dx}{\sqrt{12\mu x(1-x)(1-\varkappa^2 x)}}. \tag{10}$$

To find the substitution, we put $x_1 = 0$, $x_2 = 1$, $x_3 = 1 : \varkappa^2$, $x_4 = \infty$, $a_0 x_4 = 1$, and obtain

$$a_0 U = 1, \qquad a_0 V = -\frac{1}{\varkappa^2}, \qquad a_0 W = \frac{\varkappa'^2}{\varkappa^2},$$

so that

$$e_1 = \mu\frac{\varkappa^2 - 2}{\varkappa^2}, \qquad e_2 = \mu\frac{1 - 2\varkappa^2}{\varkappa^2}, \qquad e_3 = \mu\frac{1 + \varkappa^2}{\varkappa^2},$$

$$z = \frac{3\mu}{\varkappa^2}\left(\frac{\varkappa^2 + 1}{3} - \frac{1}{x}\right). \tag{11}$$

In this transformation of the elliptic differential into Weierstrass's normal form, the decomposition of the function $f(x)$ into its linear factors, and hence the solution of the biquadratic equation $f(x) = 0$, is presupposed. Another transformation, not linear, to be sure, which reaches the same goal without this presupposition, was given by Hermite (Crelle's Journal, vol. 52). To present it, we use the homogeneous form of the elliptic differential

$$\frac{y\,dx - x\,dy}{\sqrt{f(x,y)}}, \tag{12}$$

where

$$f(x,y) = a_0 x^4 + a_1 x^3 y + a_2 x^2 y^2 + a_3 x y^3 + a_4 y^4. \tag{13}$$

Besides the two invariants A, B, this biquadratic form has two further covariants:

$$H = \frac{1}{3}\left[\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2\right], \qquad T = \frac{1}{12}\left(\frac{\partial f}{\partial x}\frac{\partial H}{\partial y} - \frac{\partial f}{\partial y}\frac{\partial H}{\partial x}\right), \tag{14}$$

where H and T are forms of the fourth and sixth degrees whose coefficients are composed rationally, and with integral numerical coefficients, from the coefficients of f. Between these forms there is the further relation

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2 \tag{15}$$

(Vol. I, §§ 70, 72).

But by Euler's theorem on homogeneous functions,

$$4f = \frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y, \qquad df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy,$$

$$4H = \frac{\partial H}{\partial x}x + \frac{\partial H}{\partial y}y, \qquad dH = \frac{\partial H}{\partial x}dx + \frac{\partial H}{\partial y}dy,$$

whence

$$f\,dH - H\,df = -3T(y\,dx - x\,dy),$$

and if, according to (15), one puts

$$3\sqrt{3}\,T = \sqrt{-H^3 + 48Af^2H + 64Bf^3}, \tag{16}$$

then

$$\frac{y\,dx - x\,dy}{\sqrt{f}} = \frac{-\sqrt{3}(f\,dH - H\,df)}{\sqrt{-(H^3 - 48Af^2H - 64Bf^3)f}}.$$

If one now makes the substitution

$$z = -\frac{\mu H}{4f} \tag{17}$$

with an undetermined factor μ, then one obtains

$$\frac{y\,dx - x\,dy}{\sqrt{3\mu f}} = \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}, \tag{18}$$

where, as before,

$$g_2 = 12\mu^2 A, \qquad g_3 = -4\mu^3 B. \tag{19}$$

This transformation can be applied to a differential which already has the normal form. If one puts

$$f(x,y) = 4x^3 y - g_2 x y^3 - g_3 y^4,$$

then one has to replace

$$\begin{array}{ccccc} a_0 & a_1 & a_2 & a_3 & a_4 \\ 0 & 4 & 0 & -g_2 & -g_3 \end{array}$$

and from (7) one obtains

$$A = 12 g_2, \qquad B = -27 \cdot 16\, g_3.$$

If therefore one puts $\mu = \frac{1}{12}$, then equations (19) pass into the identities $g_2 = g_2$, $g_3 = g_3$.

If one then puts $y = 1$, transformation (18) gives

$$\frac{2dx}{\sqrt{4x^3 - g_2 x - g_3}} = \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}. \tag{20}$$

By (16), this transformation is mediated by

$$z = -\frac{H}{48 f}.$$

But from (14) one obtains

$$H = -48\left[\left(x^2 + \frac{1}{4} g_2\right)^2 + 2 g_3 x\right],$$

and hence

$$z = \frac{\left(x^2 + \frac{1}{4} g_2\right)^2 + 2 g_3 x}{4x^3 - g_2 x - g_3}. \tag{21}$$

Furthermore, from (14) one obtains

$$T = 64x^6 - 80 g_2 x^4 - 320 g_3 x^3 - 20 g_2^2 x^2 - 16 g_2 g_3 x + g_2^3 - 32 g_3^2,$$

and then, by (16),

$$4\left(\sqrt{4x^3 - g_2 x - g_3}\right)^3 \sqrt{4z^3 - g_2 z - g_3} = T. \tag{22}$$

Equations (20), (21), (22) carry out the multiplication of the elliptic differential by 2. In this way not only is z represented as a single-valued rational function of x, but one square root is also represented uniquely by the other, that is, to a point x there is assigned a point z uniquely.

§ 6. Elliptic curves.

Every algebraic dependence between two variables x, y is expressed by an equation of the form

$$F(x, y) = 0, \tag{1}$$

where $F(x, y)$ denotes an integral function of the two variables x, y. If one regards x and y as Cartesian coordinates of a point in the plane, then (1) is the equation of a curve of the nth degree when F is of the nth dimension with respect to x and y together. This curve is then the geometric image of the algebraic dependence, although imaginary points too must be taken into account.

If $\Phi(x, y)$ is an integral or fractional rational function of x and y, and y depends on x through equation (1), then

$$\Phi(x, y)\, dx, \qquad \int \Phi(x, y)\, dx \tag{2}$$

are the algebraic differentials and integrals belonging to the curve F.

For example, if $f(x)$ is a function of the third or fourth degree in x, then

$$y^2 - f(x) = 0$$

is the equation of a curve of the third or fourth degree to which belong the elliptic differentials and integrals affected with the irrationality $\sqrt{f(x)}$.

We shall here call all curves whose differentials and integrals are reducible to elliptic ones elliptic curves. As the example shows, curves of the third and fourth degrees occur among them.

Conic sections do not belong to the elliptic curves, because, if an equation of the second degree holds between x and y, then x and y can be represented as rational functions of a parameter t, whereby the algebraic differential can be reduced to a rational one in t. Such curves are called rational curves.

We shall see that all curves of the third degree without a double point or cusp belong to the elliptic ones. Curves of higher degree can belong here only when they have a certain number of singular points. This is a fundamental chapter in the general theory of algebraic functions and does not belong to the plan of this work.[1]

For the investigation of algebraic differentials from this point of view, the introduction of homogeneous variables is expedient. We put $x = x_1 : x_3$, $y = x_2 : x_3$, $x_3^n F(x, y) = f(x_1, x_2, x_3)$; then $f(x_1, x_2, x_3)$ is a homogeneous function of order n of the three variables x_1, x_2, x_3, and

$$f(x_1, x_2, x_3) = 0 \tag{3}$$

is the equation of a curve of order n in homogeneous coordinates. We have

$$dx = \frac{x_3\, dx_1 - x_1\, dx_3}{x_3^2}$$

and

$$d\Omega = \Phi(x, y) dx = \frac{1}{x_3^2} \Phi\left(\frac{x_1}{x_3}, \frac{x_2}{x_3}\right)(x_3\, dx_1 - x_1\, dx_3). \tag{4}$$

But by Euler's theorem on homogeneous functions, if f_1, f_2, f_3 denote the partial derivatives of f with respect to x_1, x_2, x_3, then

$$f_1 x_1 + f_2 x_2 + f_3 x_3 = 0, \qquad f_1 dx_1 + f_2 dx_2 + f_3 dx_3 = 0.$$

[1]The most important works concerning this subject are: Riemann, Theorie der Abelschen Funktionen [Crelle's Journal, vol. 54 (1857)]. Mathematische Werke, 2nd ed., pp. 88, 467. Nachträge, edited by Noether and Wirtinger (1902). Aronhold, Monatsberichte der Berliner Akademie of 25 April 1861. Clebsch, Über die Anwendung der Abelschen Funktionen in der Geometrie (Crelle's Journal, vol. 63). Clebsch, Über diejenigen ebenen Kurven, deren Koordinaten rationale Funktionen eines Parameters sind (ibid., vol. 64). Clebsch, Über diejenigen Kurven, deren Koordinaten sich als elliptische Funktionen eines Parameters darstellen lassen (ibid., vol. 64). Brill, Über diejenigen Kurven, deren Koordinaten sich als hyperelliptische Funktionen eines Parameters darstellen lassen (ibid., vol. 65). Clebsch and Gordan, Theorie der Abelschen Funktionen (Teubner, Leipzig 1866).

Therefore, if ρ denotes a factor of proportionality,

$$f_1 = \rho(x_2\,dx_3 - x_3\,dx_2), \qquad f_2 = \rho(x_3\,dx_1 - x_1\,dx_3), \qquad f_3 = \rho(x_1\,dx_2 - x_2\,dx_1),$$

and if c_1, c_2, c_3 denote wholly arbitrary quantities, for example constants,

$$c_1 f_1 + c_2 f_2 + c_3 f_3 = \rho \sum \pm c_1 x_2\,dx_3,$$

where $\sum \pm c_1 x_2\,dx_3$ denotes, in the usual way, the determinant

$$\begin{vmatrix} c_1 & c_2 & c_3 \\ x_1 & x_2 & x_3 \\ dx_1 & dx_2 & dx_3 \end{vmatrix}.$$

It follows that

$$x_3\,dx_1 - x_1\,dx_3 = \frac{f_2 \sum \pm c_1 x_2\,dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3},$$

and, if one puts

$$\frac{f_2}{x_3^2}\Phi\left(\frac{x_1}{x_3}, \frac{x_2}{x_3}\right) = \Psi,$$

there results from (4) the expression for the most general algebraic differential belonging to the curve f:

$$d\Omega = \frac{\Psi \sum \pm c_1 x_2\,dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}, \tag{5}$$

where Ψ is an integral or fractional homogeneous function of order $n-3$. The arbitrary quantities c_1, c_2, c_3 occur in this expression only in appearance. In reality it is wholly independent of them.

We again consider the case $n = 3$. Then the simplest case is the one in which Ψ is a constant, and (5) passes into the elliptic differential of the first kind:

$$du = \frac{\sum \pm c_1 x_2\,dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}. \tag{6}$$

We shall now transform this, with the aid of the covariants of the ternary form of the third degree, to Weierstrass's form.

In Vol. II, § 107, we became acquainted with the fundamental covariants of the cubic form. Apart from f itself, these were

$$\Delta = \frac{1}{6^3}\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}, \qquad J = \frac{1}{36}\begin{vmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{vmatrix},$$

$$K = \frac{1}{9}\begin{vmatrix} f_1 & \Delta_1 & J_1 \\ f_2 & \Delta_2 & J_2 \\ f_3 & \Delta_3 & J_3 \end{vmatrix},$$

where the indices $1, 2, 3$ denote differentiation with respect to the variables x_1, x_2, x_3.

From the multiplication theorem for determinants it follows, since f and df are zero, that

$$\begin{vmatrix} c_1 & c_2 & c_3 \\ x_1 & x_2 & x_3 \\ dx_1 & dx_2 & dx_3 \end{vmatrix}\begin{vmatrix} f_1 & f_2 & f_3 \\ \Delta_1 & \Delta_2 & \Delta_3 \\ J_1 & J_2 & J_3 \end{vmatrix} = 3\sum c_i f_i(\Delta\,dJ - 2J\,d\Delta),$$

and hence

$$du = 3\frac{\Delta\,dJ - 2J\,d\Delta}{K}. \tag{7}$$

With the help of the covariants we have reduced the form $f(x)$ to the canonical form

$$\varphi(y) = y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3.$$

If r denotes the substitution determinant, we put

$$P = \frac{(2+m^3)m^3f^2 + (2-5m^3)mr^2f\Delta + 3m^2r^4\Delta^2 - r^6J}{(1+8m^3)^2},$$

$$Q = \frac{(1+2m^3)f - 6mr^2\Delta}{1+8m^3}, \qquad R = \frac{m^2f + r^2\Delta}{1+8m^3},$$

and obtained y_1^3, y_2^3, y_3^3 as the roots of the cubic equation

$$u^3 - Qu^2 + Pu - R^3 = 0,$$

and the discriminant of this cubic equation is

$$D_1 = \frac{r^{18}K^2}{(1+8m^3)^6}.$$

If one expresses this discriminant through P, Q, R, then K^2 is represented rationally through f, Δ, J. At the place mentioned in the Algebra we did not give this expression explicitly. Now, however, we must form it, even though only under the presupposition $f = 0$, that is, only for the points of the curve. But [by Vol. I, § 50, (10)]

$$D_1 = P^2Q^2 + 18PQR^3 - 4P^3 - 4Q^3R^3 - 27R^6.$$

In this one must set

$$P = \frac{3m^2r^4\Delta^2 - r^6J}{(1+8m^3)^2}, \qquad Q = \frac{-6mr^2\Delta}{1+8m^3}, \qquad R = \frac{r^2\Delta}{1+8m^3},$$

and by a not difficult calculation one obtains, denoting by S and T the two invariants of the curve of the third order (Vol. II, § 108),

$$(1+8m^3)^6D_1 = r^{18}(4J^3 + 108SJ\Delta^4 - 27T\Delta^6),$$

and consequently

$$K^2 = 4J^3 + 108SJ\Delta^4 - 27T\Delta^6. \tag{8}$$

This equation, however, is not identical, but is satisfied only under the presupposition $f = 0$. The complete expression of K^2 through J, Δ, f becomes much more complicated.

If one puts

$$\frac{J}{\Delta^2} = 3z, \qquad dz = \frac{1}{3}\frac{\Delta\, dJ - 2J\, d\Delta}{\Delta^3}, \tag{9}$$

then from (8)

$$K^2 = 27\Delta^6(4z^3 + 12Sz - T),$$

and from (7)

$$du = \frac{1}{\sqrt{3}}\frac{dz}{\sqrt{4z^3 + 12Sz - T}}, \tag{10}$$

and this is Weierstrass's normal form for $g_2 = -12S$, $g_3 = T$.

§ 7. Elliptic space curves of fourth order.

One can also represent algebraic functions of one variable by equations between several variables, if the number of equations is increased correspondingly. For example, take three variables x, y, z and let two equations

$$\varphi(x, y, z) = 0, \qquad \psi(x, y, z) = 0 \tag{1}$$

hold between them. Then from these two equations one can eliminate, for example, x and obtains an equation between y and z by which y is defined as an algebraic function of z. Then, if certain exceptional cases are excluded, x and every rational function of x, y, z can be represented rationally by y and z. The integrals of the form

$$\int F(x, y, z)\, dz, \tag{2}$$

in which F denotes a rational function, then belong to the algebraic configuration represented by (1). If x, y, z are taken as Cartesian coordinates in space, then (1) represents a space curve as the intersection of two surfaces $\varphi = 0$, $\psi = 0$, and the integrals (2) belong to this space curve. The curve is again called elliptic when these integrals are elliptic.

We shall apply these considerations to the space curves of fourth order and first species, that is, to the curves of fourth order which can be represented as the complete intersection of two surfaces of the second degree.

We again introduce homogeneous coordinates x_1, x_2, x_3, x_4 and assume the equations of two given surfaces of the second degree in the form

$$\varphi(x_1, x_2, x_3, x_4) = \sum a_{ik} x_i x_k, \qquad \psi(x_1, x_2, x_3, x_4) = \sum b_{ik} x_i x_k. \tag{3}$$

The two surfaces determine a pencil of surfaces of the second order $\xi\varphi = \eta\psi$. All surfaces of the pencil cut one another in a space curve of fourth order, which we call the base curve. One obtains the same curve and the same pencil if the functions φ and ψ are replaced by φ', ψ', which are derived from φ, ψ by means of the linear substitution $\begin{pmatrix} m & n \\ p & q \end{pmatrix}$, that is, by

$$\varphi' = m\varphi + n\psi, \qquad \psi' = p\varphi + q\psi, \tag{4}$$

whose determinant $r = mq - np$ is different from zero. This gives the identity

$$\xi\varphi + \eta\psi = \xi'\varphi' + \eta'\psi', \tag{5}$$

where

$$\xi = m\xi' + p\eta', \qquad \eta = n\xi' + q\eta' \tag{6}$$

represents a linear transformation of the variables ξ, η. Besides this binary substitution there also comes into consideration a linear quaternary substitution of the variables x_1, x_2, x_3, x_4 (coordinate transformation), whose determinant we may assume to be equal to 1.

Every quadratic form has, with respect to these latter transformations, an invariant, namely the Hessian determinant (Vol. I, §§ 62, 63, 66), and we therefore obtain as invariant of the form (5)

$$f(\xi, \eta) = \sum \pm (\xi a_{11} + \eta b_{11})(\xi a_{22} + \eta b_{22})(\xi a_{33} + \eta b_{33})(\xi a_{44} + \eta b_{44}). \tag{7}$$

This is a binary biquadratic form in the variables ξ, η, which, when developed, we write as

$$f(\xi, \eta) = a_0\xi^4 + a_1\xi^3\eta + a_2\xi^2\eta^2 + a_3\xi\eta^3 + a_4\eta^4. \tag{8}$$

The coefficients of this biquadratic form, a_0, a_1, a_2, a_3, a_4, do not change under a coordinate transformation. They are therefore called simultaneous invariants of the pair of forms φ, ψ (a_0 and a_4 are the determinants of φ and of ψ).

The a_i change when ξ, η are subjected to a substitution (6). But if one forms the invariants of the form (8) (as in Algebra, Vol. I, § 70), one obtains functions of the coefficients which are also invariant under these substitutions, and which therefore do not belong to the individual forms φ, ψ, but to the whole pencil and consequently also to their intersection, the base curve. We call them invariants of the base curve. With respect to the system of forms φ, ψ they are also called combinants. Thus the base curve has two invariants:

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \qquad B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3, \tag{9}$$

from which one derives the discriminant D by the formula

$$27D = 4A^3 - B^2. \tag{10}$$

With respect to the coefficients of φ and ψ, the invariants A, B, D are of degrees $8, 12, 24$.

The system of forms φ, ψ has two simultaneous quadratic covariants, which are derived in the same way as the covariant C (in Algebra I, § 65). The first of them is, if we put $\psi_i = \frac{1}{2}\frac{\partial \psi}{\partial x_i}$,

$$\Phi = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & \psi_1 \\ a_{21} & a_{22} & a_{23} & a_{24} & \psi_2 \\ a_{31} & a_{32} & a_{33} & a_{34} & \psi_3 \\ a_{41} & a_{42} & a_{43} & a_{44} & \psi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 & 0 \end{vmatrix}, \tag{11}$$

and the second, Ψ, is obtained from this by interchanging a_{ik} with b_{ik} and ψ with φ.

The equation $\Phi = 0$ expresses a surface of the second degree (not belonging to the pencil), which is the geometric locus of the points x whose polar planes, with respect to ψ, touch the surface φ.

We now assume that the discriminant D is different from zero. Then the two functions φ, ψ can be transformed simultaneously into the sum of four squares:

$$\varphi = \alpha_1 y_1^2 + \alpha_2 y_2^2 + \alpha_3 y_3^2 + \alpha_4 y_4^2, \qquad \psi = \beta_1 y_1^2 + \beta_2 y_2^2 + \beta_3 y_3^2 + \beta_4 y_4^2, \tag{12}$$

where the y_i are linear functions of the x_i. This is the canonical form of the pair of functions. Then

$$f(\xi, \eta) = (\xi\alpha_1 + \eta\beta_1)(\xi\alpha_2 + \eta\beta_2)(\xi\alpha_3 + \eta\beta_3)(\xi\alpha_4 + \eta\beta_4), \tag{13}$$

and the functions Φ, Ψ receive the canonical form

$$\Phi = \sum_{i=1}^{4} \beta_i^2 \prod_{j\neq i} \alpha_j \, y_i^2, \qquad \Psi = \sum_{i=1}^{4} \alpha_i^2 \prod_{j\neq i} \beta_j \, y_i^2. \tag{14}$$

The determinant of the system (12), (14), considered as linear equations for $y_1^2, y_2^2, y_3^2, y_4^2$, is

$$\begin{vmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \beta_1 & \beta_2 & \beta_3 & \beta_4 \\ \beta_1^2\alpha_2\alpha_3\alpha_4 & \beta_2^2\alpha_1\alpha_3\alpha_4 & \beta_3^2\alpha_1\alpha_2\alpha_4 & \beta_4^2\alpha_1\alpha_2\alpha_3 \\ \alpha_1^2\beta_2\beta_3\beta_4 & \alpha_2^2\beta_1\beta_3\beta_4 & \alpha_3^2\beta_1\beta_2\beta_4 & \alpha_4^2\beta_1\beta_2\beta_3 \end{vmatrix}. \tag{15}$$

It vanishes when $\alpha_i : \alpha_k = \beta_i : \beta_k$, because then the first two columns become proportional to one another, and likewise when $\alpha_i : \alpha_k = \beta_i : \beta_k$. Therefore (15), considered as an integral function of the α, β, is divisible by

$$\alpha_i\beta_k - \alpha_k\beta_i = (\alpha_i\beta_k), \tag{16}$$

and hence also by the product of all these factors

$$(\alpha_1\beta_2)(\alpha_1\beta_3)(\alpha_1\beta_4)(\alpha_2\beta_3)(\alpha_2\beta_4)(\alpha_3\beta_4) = \Delta. \tag{17}$$

Comparison of the degrees shows that the two functions (15), (17) can differ only by a numerical factor, and this factor is found from the assumption $\beta_1 = 0$, $\alpha_2 = 0$, $\alpha_4 = 0$ to be equal to 1. The determinant (15) is therefore equal to the product Δ, and the square of Δ is the discriminant D of the function $f(\xi, \eta)$, hence is equal to the function D, which we have assumed to be different from zero. Thus the $y_1^2, y_2^2, y_3^2, y_4^2$ can be represented linearly through $\varphi, \psi, \Phi, \Psi$.

The covariants Φ, Ψ change when φ and ψ are replaced by φ', ψ' according to (4). They therefore do not belong to the base curve, but to the pair of forms φ, ψ. But under this substitution Φ and Ψ, as linear functions of $y_1^2, y_2^2, y_3^2, y_4^2$, pass into linear functions of $\varphi, \psi, \Phi, \Psi$. These expressions are rather complicated. We need them here, however, only for the points of the base curve, that is, for $\varphi = 0$, $\psi = 0$, and under this assumption they become very simple. Namely

$$\begin{aligned} \Phi' &= \sum (p\alpha_1 + q\beta_1)^2 (m\alpha_2 + n\beta_2)(m\alpha_3 + n\beta_3)(m\alpha_4 + n\beta_4) y_1^2, \\ \Psi' &= \sum (m\alpha_1 + n\beta_1)^2 (p\alpha_2 + q\beta_2)(p\alpha_3 + q\beta_3)(p\alpha_4 + q\beta_4) y_1^2. \end{aligned} \tag{18}$$

and from this, by (12) and (14), one has to derive

$$\Phi' = M\Phi + N\Psi, \qquad \Psi' = P\Phi + Q\Psi,$$

where the coefficients M, N, P, Q still have to be determined. We can carry this out by computation. Without computation the result is obtained as follows.

The M, N are integral functions of degree two in p, q and of degree three in m, n. If one now takes $p : q = m : n$, so that $r = mq - np = 0$, then Φ' and $\partial\Phi'/\partial p$ pass by (18) into linear combinations of φ and ψ, and thus vanish. It follows that M and N are divisible by r^2, and the quotients are linear functions of m and n. But since in the two cases

$$\begin{pmatrix} m & n \\ p & q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Φ' must pass into Φ and into Ψ, and since the same consideration can be applied to Ψ', it follows that

$$\Phi' = r^2(m\Phi + n\Psi), \qquad \Psi' = r^2(p\Phi + q\Psi). \tag{19}$$

Thus, apart from the factor r^2, the functions Φ, Ψ are transformed cogrediently with φ, ψ. (For the strict justification of these conclusions see Vol. I, § 20.)

From these results we can in a simple way, always under the assumption $\varphi = 0$, $\psi = 0$, express the y_i^2 by Φ and Ψ, that is, solve the linear equations (12), (14). If we apply the formulae (19) to the substitution

$$\begin{pmatrix} m & n \\ p & q \end{pmatrix} = \begin{pmatrix} \beta_1 & -\alpha_1 \\ 0 & 1 \end{pmatrix},$$

then $r = \beta_1$, the function ψ' remains unchanged $= \psi$, and φ' is obtained from φ if

$$\alpha_1, \alpha_2, \alpha_3, \alpha_4$$

are replaced by

$$0, \quad (\alpha_2\beta_1), \quad (\alpha_3\beta_1), \quad (\alpha_4\beta_1).$$

Thus by (14)

$$\Phi' = \beta_1^2 (\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1) y_1^2,$$

and consequently by (19)

$$\begin{aligned} (\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1) y_1^2 &= \alpha_1\Phi - \beta_1\Psi, \\ (\alpha_1\beta_2)(\alpha_3\beta_2)(\alpha_4\beta_2) y_2^2 &= \alpha_2\Phi - \beta_2\Psi, \\ (\alpha_1\beta_3)(\alpha_2\beta_3)(\alpha_4\beta_3) y_3^2 &= \alpha_3\Phi - \beta_3\Psi, \\ (\alpha_1\beta_4)(\alpha_2\beta_4)(\alpha_3\beta_4) y_4^2 &= \alpha_4\Phi - \beta_4\Psi. \end{aligned} \tag{20}$$

The product of the constant factors on the left is the discriminant D, and therefore by (13) one obtains

$$Dy_1^2y_2^2y_3^2y_4^2 = f(\Phi, -\Psi) = a_0\Phi^4 - a_1\Phi^3\Psi + a_2\Phi^2\Psi^2 - a_3\Phi\Psi^3 + a_4\Psi^4, \tag{21}$$

and the y_i^2 are therefore the linear factors of this biquadratic form.

There is still another simultaneous covariant of the form φ, ψ, namely the Jacobi functional determinant of the four functions $\varphi, \psi, \Phi, \Psi$. We denote it by

$$K = \begin{vmatrix} \varphi_1 & \varphi_2 & \varphi_3 & \varphi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 \\ \Phi_1 & \Phi_2 & \Phi_3 & \Phi_4 \\ \Psi_1 & \Psi_2 & \Psi_3 & \Psi_4 \end{vmatrix}. \tag{22}$$

If it is formed for the canonical form, then by (12), (14), (15) one obtains

$$K = -\Delta y_1 y_2 y_3 y_4,$$

and since $\Delta^2 = D$,

$$K^2 = Dy_1^2y_2^2y_3^2y_4^2.$$

Therefore by (21) we have

$$K^2 = f(\Phi, -\Psi). \tag{23}$$

The biquadratic form $f(\Phi, -\Psi)$ has two covariants, which we denote as in Vol. I, § 70:

$$H = \frac{1}{3}\begin{vmatrix} \dfrac{\partial^2 f}{\partial \Phi^2} & -\dfrac{\partial^2 f}{\partial \Phi\, \partial \Psi} \\ \dfrac{\partial^2 f}{\partial \Psi\, \partial \Phi} & -\dfrac{\partial^2 f}{\partial \Psi^2} \end{vmatrix}, \qquad T = \frac{1}{12}\begin{vmatrix} f'(\Phi) & f'(-\Psi) \\ H'(\Phi) & H'(-\Psi) \end{vmatrix}. \tag{24}$$

They are of the fourth and sixth degrees in Φ, Ψ, hence of the eighth and twelfth degrees in the x_i. They are covariants of the base curve (combinants of φ and ψ), and between them there is the relation

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2. \tag{25}$$

The most general integral (2) belonging to the base curve passes, by introduction of homogeneous variables $z = x_3 : x_4$, into the following form:

$$\int F'(x, y, z) \frac{x_3\, dx_4 - x_4\, dx_3}{x_4^2},$$

or, if one puts

$$F(x_1, x_2, x_3, x_4) = x_4^2\, \frac{F'(x, y, z)}{\varphi_1\psi_2 - \varphi_2\psi_1},$$

into

$$\int F(x_1, x_2, x_3, x_4)\, \frac{x_3\, dx_4 - x_4\, dx_3}{\varphi_1\psi_2 - \varphi_2\psi_1},$$

where F is a homogeneous function of degree zero. The simplest case is $F = 1$, and we therefore consider the integral of the first kind

$$u = \int \frac{x_3\, dx_4 - x_4\, dx_3}{\varphi_1\psi_2 - \varphi_2\psi_1}. \tag{26}$$

Now, by the multiplication theorem for determinants, we form the product $K(x_3 dx_4 - x_4 dx_3)$, hence, since $\varphi = 0, \psi = 0$,

$$\begin{vmatrix} \varphi_1 & \varphi_2 & \varphi_3 & \varphi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 \\ \Phi_1 & \Phi_2 & \Phi_3 & \Phi_4 \\ \Psi_1 & \Psi_2 & \Psi_3 & \Psi_4 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ x_1 & x_2 & x_3 & x_4 \\ dx_1 & dx_2 & dx_3 & dx_4 \end{vmatrix} = \begin{vmatrix} \varphi_1 & \varphi_2 & 0 & 0 \\ \psi_1 & \psi_2 & 0 & 0 \\ \Phi_1 & \Phi_2 & d\Phi & \Phi \\ \Psi_1 & \Psi_2 & d\Psi & \Psi \end{vmatrix}$$

$$= (\varphi_1\psi_2 - \varphi_2\psi_1)(\Phi\, d\Psi - \Psi\, d\Phi),$$

and consequently from (26) and (23) one obtains

$$u = \int \frac{\Phi\, d\Psi - \Psi\, d\Phi}{K} = \int \frac{\Phi\, d\Psi - \Psi\, d\Phi}{\sqrt{f(\Phi, -\Psi)}}. \tag{27}$$

If one now applies the transformation of § 5 to this, replacing x, y by $\Phi, -\Psi$, one finds

$$u = \sqrt{3} \int \frac{f\, dH - H\, df}{\sqrt{-(H^3 - 48Af^2H - 64Bf^3)\, f}},$$

and if in § 5, (17), one puts $\mu = 3$, that is,

$$z = -\frac{3H}{4f},$$

then

$$u = 3 \int \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}}, \qquad g_2 = 12 \cdot 9A, \quad g_3 = -4 \cdot 27B.$$

§ 8. Jacobi's transformation principle.

Jacobi's theory of transformation poses, in general, the problem of reducing an elliptic differential to another elliptic differential by an algebraic substitution. In setting out here the foundations of this theory, we use the representation of the elliptic differential in homogeneous variables [§ 1, (4)]:

$$\frac{x\, dy - y\, dx}{\sqrt{f(x, y)}}, \tag{1}$$

and substitute for x, y two relatively prime integral homogeneous functions of equal, but arbitrary, degree n in two new variables ξ, η:

$$x = U(\xi, \eta), \qquad y = V(\xi, \eta). \tag{2}$$

From the equations

$$nU = \xi \frac{\partial U}{\partial \xi} + \eta \frac{\partial U}{\partial \eta}, \qquad dU = d\xi \frac{\partial U}{\partial \xi} + d\eta \frac{\partial U}{\partial \eta},$$
$$nV = \xi \frac{\partial V}{\partial \xi} + \eta \frac{\partial V}{\partial \eta}, \qquad dV = d\xi \frac{\partial V}{\partial \xi} + d\eta \frac{\partial V}{\partial \eta},$$

there follows

$$U\, dV - V\, dU = H(\xi\, d\eta - \eta\, d\xi), \tag{3}$$

where H denotes the functional determinant

$$H = \frac{1}{n} \left(\frac{\partial U}{\partial \xi} \frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi} \frac{\partial U}{\partial \eta} \right). \tag{4}$$

Accordingly the elliptic differential (1) passes into

$$H \frac{\xi\, d\eta - \eta\, d\xi}{\sqrt{f(U, V)}}. \tag{5}$$

In order that this differential should again acquire the form of an elliptic differential, it is necessary that from the function $f(U, V)$, whose degree is $4n$, a square factor T^2 of degree $4n - 4$ be separated, hence, if $\varphi(\xi, \eta)$ denotes a function of the fourth degree, that

$$f(U, V) = T^2 \varphi(\xi, \eta) \tag{6}$$

become true, whereby, if

$$\frac{T}{H} = M \tag{7}$$

is put, the differential (5) passes into

$$\frac{1}{M}\,\frac{\xi\,d\eta - \eta\,d\xi}{\sqrt{\varphi(\xi,\eta)}}. \tag{8}$$

It can now be proved that as soon as the condition (6) is fulfilled, H is divisible by T, and therefore, since the degree of both functions is the same, M becomes a constant. This constant is called the multiplier of the transformation.

Indeed, observe that, since $f(x,y)$ contains no quadratic factor, the two functions

$$\frac{\partial f}{\partial x}, \qquad \frac{\partial f}{\partial y}$$

have no common divisor, and that consequently, because U and V are relatively prime, also

$$\frac{\partial f}{\partial U}, \qquad \frac{\partial f}{\partial V}$$

have no common divisor (with respect to ξ, η). Thus two functions α, β of ξ, η can be chosen in such a way that

$$\alpha\frac{\partial f}{\partial U} + \beta\frac{\partial f}{\partial V}$$

is relatively prime to an arbitrarily given function T. For example, one can assume α relatively prime to T, and β divisible by the divisors of T which do not occur in $\frac{\partial f}{\partial U}$, but not divisible by the common divisors of T and $\frac{\partial f}{\partial U}$. If we now differentiate the equation (6), which we suppose fulfilled, with respect to ξ, η, we get

$$\frac{\partial f}{\partial U}\frac{\partial U}{\partial \xi} + \frac{\partial f}{\partial V}\frac{\partial V}{\partial \xi} = TX,$$

$$\frac{\partial f}{\partial U}\frac{\partial U}{\partial \eta} + \frac{\partial f}{\partial V}\frac{\partial V}{\partial \eta} = TY,$$

and from this, by solving with respect to $\frac{\partial f}{\partial U}$, $\frac{\partial f}{\partial V}$:

$$H\left(\alpha\frac{\partial f}{\partial U} + \beta\frac{\partial f}{\partial V}\right) = TZ,$$

where X, Y, Z are integral homogeneous functions of ξ, η. From the last equation, however, one concludes that H must be divisible by T.

Accordingly our whole problem is contained in the equation (6), which says nothing other than that the function $f(U,V)$ of degree $4n$ is to contain $2n-2$ quadratic factors. To satisfy the $2n-2$ conditions resulting from this, one has at one's disposal the $2n+2$ coefficients contained in U, V, so that four of them remain undetermined. This was to be foreseen, since arbitrary homogeneous linear functions of ξ, η can be introduced for ξ, η. These four remaining constants can be used in order to bring the differential (8) into a normal form. But since the equations of condition for the coefficients of U, V are not linear, there are several transformations for a given degree of transformation.

We shall later encounter this transformation problem again from an entirely different side and shall have to enter much more deeply into it. Therefore the details of the algebraic problem are not pursued further here; instead we shall illustrate what has been said by two examples.

§ 9. The transformation of the second degree.

We put, in order to obtain the elliptic differentials in normal form,

$$f(x,y) = xy(x-y)(x-\lambda^2 y), \qquad \varphi(\xi,\eta) = \xi\eta(\xi-\eta)(\xi-\varkappa^2\eta),$$

and let U, V be of the second degree.

The equation (6), § 8, now becomes, since T is of the second degree,

$$UV(U-V)(U-\lambda^2 V) = (a\xi+b\eta)^2(a'\xi+b'\eta)^2\xi\eta(\xi-\eta)(\xi-\varkappa^2\eta), \tag{1}$$

and two of the four factors of the second degree on the left side of this equation must be squares. We shall further restrict the large number of possibilities lying here by supposing that η and y are to vanish simultaneously, so that V is divisible by η. This gives (apart from a constant factor) the following three possibilities for V:

$$1.\ V = \xi\eta, \qquad 2.\ V = \eta(\xi-\eta), \qquad 3.\ V = \eta(\xi-\varkappa^2\eta).$$

Each of these three cases again comprises three subcases, according as any two of the three remaining factors U, $U-V$, $U-\lambda^2 V$ are assumed to be squares. Of these latter three cases, two pass into one another by interchanging V, λ^2 with $V:\lambda^2, 1:\lambda^2$.

We shall carry out completely two transformations among these which are especially important for what follows.

1. The Gaussian transformation.

$$V = \xi\eta, \qquad U = (a\xi+b\eta)^2.$$

If now $U-\lambda^2 V$ is also to be a square, one must put

$$\lambda^2 = 4ab,$$

and then

$$U - \lambda^2 V = (a\xi - b\eta)^2.$$

Since $U-V$ must then be divisible by $\xi-\eta$ and by $\xi-\varkappa^2\eta$, one obtains, by putting $\xi=\eta$, $\xi=\varkappa^2\eta$ and using $U=V$, the conditions

$$a+b=\pm 1, \qquad a\varkappa^2+b=\pm\varkappa.$$

If the upper signs are taken, it follows that

$$a = \frac{1}{1+\varkappa}, \qquad b = \frac{\varkappa}{1+\varkappa}, \qquad \lambda = \frac{2\sqrt{\varkappa}}{1+\varkappa},$$

$$V = \xi\eta, \qquad U = \left(\frac{\xi+\varkappa\eta}{1+\varkappa}\right)^2, \qquad U-\lambda^2 V = \left(\frac{\xi-\varkappa\eta}{1+\varkappa}\right)^2,$$

$$U - V = \frac{(\xi-\eta)(\xi-\varkappa^2\eta)}{(1+\varkappa)^2},$$

$$T = \frac{\xi^2-\varkappa^2\eta^2}{(1+\varkappa)^3}, \qquad H = \frac{\xi^2-\varkappa^2\eta^2}{(1+\varkappa)^2}, \qquad M = \frac{1}{1+\varkappa}.$$

Putting again

$$\frac{y}{x} = z, \qquad \frac{\eta}{\xi} = \zeta,$$

we therefore have the result that the substitution

$$z = \frac{(1+\varkappa)^2\zeta}{(1+\varkappa\zeta)^2}, \qquad \lambda^2 = \frac{4\varkappa}{(1+\varkappa)^2} \tag{2}$$

effects the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-\lambda^2 z)}} = \frac{(1+\varkappa)d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}. \tag{3}$$

2. The Landen transformation.

$$V = a\eta(\xi - \eta), \qquad U = \xi(\xi - \varkappa^2\eta).$$

The conditions that $U - V$, $U - \lambda^2 V$ be squares are

$$4a = (\varkappa^2 + a)^2, \qquad 4a\lambda^2 = (\varkappa^2 + a\lambda^2)^2,$$

whence

$$\lambda(\varkappa^2 + a) = \varkappa^2 + a\lambda^2,$$

or, after division by $\lambda - 1$,

$$a\lambda = \varkappa^2.$$

Substitution of this into one of the above equations gives

$$4\lambda = \varkappa^2(1 + \lambda)^2,$$

and by solving this quadratic equation, if $\varkappa' = \sqrt{1 - \varkappa^2}$ is put,

$$\lambda = \frac{1 - \varkappa'}{1 + \varkappa'}, \qquad a = (1 + \varkappa')^2,$$

$$U - V = \{\xi - (1 + \varkappa')\eta\}^2, \qquad U - \lambda^2 V = \{\xi - (1 - \varkappa')\eta\}^2,$$
$$H = (1 + \varkappa')^2\{\xi - (1 + \varkappa')\eta\}\{\xi - (1 - \varkappa')\eta\},$$
$$T = (1 + \varkappa')\{(\xi - \eta)^2 - \varkappa'^2\eta^2\}, \qquad M = \frac{1}{1 + \varkappa'}.$$

Thus by the substitution

$$z = \frac{(1 + \varkappa')^2\zeta(1 - \zeta)}{1 - \varkappa^2\zeta}, \qquad \lambda = \frac{1 - \varkappa'}{1 + \varkappa'}, \tag{4}$$

one obtains the transformation

$$\frac{dz}{\sqrt{z(1 - z)(1 - \lambda^2 z)}} = \frac{(1 + \varkappa')\, d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}}. \tag{5}$$

If one again applies the Gaussian transformation to the left side of this equation, replacing in formulae (2), (3) ξ, λ by z, λ and replacing z by a variable η, then, as follows from (2) and (4), λ in (3) is to be replaced by $\varkappa$, and by combining (5) and (3) one finds

$$\frac{d\eta}{\sqrt{\eta(1 - \eta)(1 - \varkappa^2\eta)}} = \frac{2\, d\zeta}{\sqrt{\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}}, \tag{6}$$

and the connection of (2) and (4) gives the following relation between the variables η and ζ:

$$\eta = \frac{4\zeta(1 - \zeta)(1 - \varkappa^2\zeta)}{(1 - \varkappa^2\zeta^2)^2}. \tag{7}$$

The combination of the two transformations of the second degree therefore gives multiplication by 2.

§ 10. The transformation of the third degree.

As a second example we consider the transformation of the third degree. The equation

$$UV(U-V)(U-\lambda^2 V) = T^2 \xi\eta(\xi-\eta)(\xi-\varkappa^2\eta) \tag{1}$$

requires that each of the four factors of the third degree, $U, V, U-V, U-\lambda^2 V$, be divisible by one of the linear factors $\xi, \eta, \xi-\eta, \xi-\varkappa^2\eta$, and that the quotient be a square. We therefore put

$$U = \xi(a\xi+b\eta)^2, \qquad V = \eta(a'\xi+b'\eta)^2, \tag{2}$$

and further require that

$$\frac{U-V}{\xi-\eta}, \qquad \frac{U-\lambda^2 V}{\xi-\varkappa^2\eta} \tag{3}$$

become the squares of linear functions.

Of the two latter requirements, one follows from the other if U, V are arranged so that, apart from constant factors, U and V pass into one another by the interchange of ξ, η with $\varkappa^2\eta, \xi$. Under this interchange, however,

$$\xi(a\xi+b\eta)^2, \qquad \eta(a'\xi+b'\eta)^2$$

passes into

$$\varkappa^2\eta(b\xi+a\varkappa^2\eta)^2, \qquad \xi(b'\xi+a'\varkappa^2\eta)^2,$$

and our requirement is fulfilled if

$$\varkappa^2 = \frac{bb'}{aa'} \tag{4}$$

holds. If then

$$\lambda^2 = \varkappa^2\left(\frac{ab}{a'b'}\right)^2 \tag{5}$$

is true, this interchange carries

$$\frac{U-V}{\xi-\eta} \quad \text{into} \quad \frac{b'^2}{a^2}\frac{U-\lambda^2 V}{\xi-\varkappa^2\eta}. \tag{6}$$

After this has been fixed, there remains only the condition that the first of the two quantities (3) be the square of a linear function, which we denote by $(\alpha^2\xi-\beta^2\eta)$, that is,

$$U-V = (\xi-\eta)(\alpha^2\xi-\beta^2\eta)^2 \tag{7}$$

or

$$U-\xi(\alpha^2\xi-\beta^2\eta)^2 = V-\eta(\alpha^2\xi-\beta^2\eta)^2.$$

Since U is divisible by ξ and V by η, this function must be divisible by $\xi\eta$; if we put it equal to $\xi\eta(m\xi+n\eta)$, then from (2) we obtain

$$(a\xi+b\eta)^2 = (\alpha^2\xi-\beta^2\eta)^2 + \eta(m\xi+n\eta),$$

$$(a'\xi+b'\eta)^2 = (\alpha^2\xi-\beta^2\eta)^2 + \xi(m\xi+n\eta).$$

Comparison of coefficients gives

$$a = \alpha^2, \qquad b = -\beta^2 + \frac{m}{2\alpha^2}, \qquad b^2 = \beta^4 + n,$$

$$b' = \beta^2, \qquad a' = -\alpha^2 + \frac{n}{2\beta^2}, \qquad a'^2 = \alpha^4 + m,$$

and from the two last equations of each row,

$$m^2 = 4\alpha^2(n\alpha^2+m\beta^2), \qquad n^2 = 4\beta^2(n\alpha^2+m\beta^2).$$

If therefore one puts

$$m = h\alpha, \qquad n = h\beta,$$

then

$$h = 4\alpha\beta(\alpha+\beta).$$

Accordingly a, b, a', b' can be expressed by the two quantities α, β in the form

$$a = \alpha^2, \qquad b = \beta^2 + 2\alpha\beta, \qquad a' = \alpha^2 + 2\alpha\beta, \qquad b' = \beta^2, \tag{8}$$

and then from (4), (5)

$$\varkappa^2 = \frac{\beta^3}{\alpha^3}\frac{\beta+2\alpha}{\alpha+2\beta}, \qquad \lambda^2 = \frac{\alpha}{\beta}\frac{\beta+2\alpha}{\alpha+2\beta}, \tag{9}$$

or, by multiplying and dividing these two equations,

$$\sqrt{\lambda\varkappa} = \frac{\beta}{\alpha}\frac{\beta+2\alpha}{\alpha+2\beta}, \qquad \frac{\varkappa^3}{\lambda} = \frac{\beta^4}{\alpha^4}. \tag{10}$$

By eliminating $\beta : \alpha$ one obtains an equation between $\varkappa, \lambda$, which is called the modular equation, and whose degree gives the number of distinct transformations of the third degree. It assumes the simplest form if one puts

$$\sqrt[4]{\varkappa} = u, \qquad \sqrt[4]{\lambda} = v. \tag{11}$$

Then equations (10) become

$$u^2v^2 = \frac{\beta}{\alpha}\frac{\beta+2\alpha}{\alpha+2\beta}, \qquad \frac{u^3}{v} = \frac{\beta}{\alpha},$$

and therefore, by substituting the value of $\beta : \alpha$ into the first equation and removing the factor u^2,

$$v^4 - u^4 + 2u^3v^3 - 2uv = 0, \tag{12}$$

an equation of the fourth degree.

Furthermore one obtains [without lengthy calculation, by comparing the coefficients of the highest power of ξ in (1)]

$$T = \frac{a}{b'}(a\xi + b\eta)(a'\xi + b'\eta)(\alpha^2\xi - \beta^2\eta)(\beta^2\xi - \varkappa^2\alpha^2\eta),$$

$$H = \frac{1}{3}\left(\frac{\partial U}{\partial \xi}\frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi}\frac{\partial U}{\partial \eta}\right) = \frac{a'}{a}T,$$

whence one easily finds by (8)

$$M = \frac{a}{a'} = \frac{\alpha}{\alpha+2\beta} = \frac{v}{v+2u^3}. \tag{13}$$

If the expression derived from this,

$$v = \frac{2u^3M}{1-M},$$

is substituted into equation (12), one obtains for M an equation of the fourth degree, the multiplier equation, which can replace the modular equation. But one obtains this equation more simply in the following way. By (13)

$$\frac{1}{M} - 1 = 2\frac{\beta}{\alpha}, \qquad \frac{1}{M} + 3 = \frac{2(\beta+2\alpha)}{\alpha},$$

and hence by (9)

$$\left(\frac{1}{M} - 1\right)^3\left(\frac{1}{M} + 3\right) = 16\varkappa^2\frac{\alpha+2\beta}{\alpha},$$

and consequently by (13)

$$\frac{16\varkappa^2}{M} = \left(\frac{1}{M} - 1\right)^3 \left(\frac{1}{M} + 3\right),$$

or, ordered,

$$\frac{1}{M^4} - \frac{6}{M^2} + \frac{8(1-2\varkappa^2)}{M} - 3 = 0. \tag{14}$$

If we therefore express our formulae by u, v, the result of this consideration is the following. The substitution

$$z = \frac{\zeta\left[(v^2 + 2vu^3) + u^6\zeta\right]^2}{\left[v^2 + (u^6 + 2vu^3)\zeta\right]^2}$$

effects the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-v^8 z)}} = \frac{v + 2u^3}{v} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-u^8\zeta)}},$$

provided the modular equation (12) holds between u, v.

For a given u, the modular equation gives four values of v, hence four distinct transformations of the third degree.

§ 11. The three kinds of elliptic integrals.

Let now

$$f(x) = a_0 x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4 \tag{1}$$

be an integral function of the third or fourth degree, and

$$\Omega(x) = \int \frac{\Phi(x)\, dx}{\sqrt{f(x)}} = \int R(x)\, dx, \tag{2}$$

where, for abbreviation,

$$R(x) = \frac{\Phi(x)}{\sqrt{f(x)}} \tag{3}$$

has been put, an elliptic integral. Here x denotes an unrestrictedly variable (also complex) quantity, and $\sqrt{f(x)}$ has, for each value of x for which $f(x)$ does not vanish, two opposite values. By a point we understand not a value of x alone, but an associated pair of values of x and $\sqrt{f(x)}$, hence a value of x with a definite sign of $\sqrt{f(x)}$. Thus if x_0 is any value of x, there are two points x_0 if $f(x_0)$ is different from zero, but only one if $f(x_0) = 0$. For $x = \infty$ we determine the points according to the sign of $\sqrt{f(x)} : x^2$, and therefore we have two points $x = \infty$ if $f(x)$ is of the fourth degree, and only one if $f(x)$ is of the third degree.

By known theorems of function theory, if $f(x_0)$ is different from zero, there is in a certain region a convergent expansion according to Taylor's theorem:

$$R(x) = A_m(x - x_0)^m + A_{m+1}(x - x_0)^{m+1} + A_{m+2}(x - x_0)^{m+2} + \cdots, \tag{4}$$

where m is a positive or negative integer or also zero, while $A_m, A_{m+1}, A_{m+2}, \ldots$ denote constants, of which A_m does not vanish. This is an expansion in ascending powers. If $f(x)$ is of the fourth (not of the third) degree, there is added the expansion in descending powers

$$R(x) = C_m x^{-m-2} + C_{m+1} x^{-m-3} + C_{m+2} x^{-m-4} + \cdots, \tag{5}$$

of which we say that it holds in a neighborhood of an infinitely distant point.

But if $f(x_0)$ vanishes, then the expansion of $R(x)$ receives the form

$$R(x) = A_m(x - x_0)^{m-\frac{1}{2}} + A_{m+1}(x - x_0)^{m-\frac{1}{2}+1} + A_{m+2}(x - x_0)^{m-\frac{1}{2}+2} + \cdots, \tag{6}$$

and if $f(x)$ is of the third degree, then in a neighborhood of the infinitely distant point the expansion

$$R(x) = C_m x^{-m-\frac{3}{2}} + C_{m+1} x^{-m-\frac{5}{2}} + C_{m+2} x^{-m-\frac{7}{2}} + \cdots \tag{7}$$

holds.

From the developments (4) to (7) we can derive the developments of Ω, an additive constant remaining undetermined.

The development of Ω is then likewise a power series, to which, if m is negative, in the cases (4) or (5) there is still added a term $A_{-1}\log(x - x_0)$, $C_{-1}\log x$.

We now call x_0 a point of the first kind of Ω if m in these developments is not negative. Then Ω is finite at the point x_0 (also in the case $x_0 = \infty$).

The point x_0 is called a point of the second kind of Ω if in the developments (4), (5) m is negative and $A_{-1} = 0$, $C_{-1} = 0$, and in the case of the developments (6), (7), with negative m.

Finally x_0 is called a point of the third kind if, in the developments (4) or (5), A_{-1} or C_{-1} is different from zero, that is, if a logarithmic term occurs in the development of Ω.

The zeros of $f(x)$ cannot be of the third kind for the integral (2); but they can become of the third kind if we consider the elliptic integral in the general form

$$\int \Phi(x, \sqrt{f(x)})\, dx.$$

The integral Ω is called an integral of the first kind if it has only points of the first kind. Such an integral then has a finite value for all finite and infinite values of x.

Ω is called an integral of the second kind if it has only points of the first and second kinds, and Ω is called an integral of the third kind if, besides points of the first and second kinds, it also has points of the third kind.

§ 12. Representation of the elliptic integrals by the simplest fundamental integrals.

In order to represent the integral $\Omega(x)$ by integrals of the simplest possible type, we decompose the rational function $\Phi(x)$ into an integral function and a sum of partial fractions; that is, we represent $\Phi(x)$ as a sum of expressions of the form

$$x^n, \qquad \frac{1}{(x-\alpha)^n}$$

with constant coefficients, where n denotes a positive integer (also zero) and α denotes a constant value for which $\Phi(x)$ becomes infinite.

If we then put, for positive and for negative or vanishing n,

$$S_n(\alpha) = \int \frac{(x-\alpha)^n\, dx}{\sqrt{f(x)}}, \tag{1}$$

then $\Omega(x)$ becomes a sum of the form

$$\Omega(x) = \sum M S_n(\alpha),$$

where the sum may extend over several different values of n and of α, and the M are constants.

The functions S_n can be reduced, through the mediation of algebraic functions, to a small number. To this end we form

$$d\Big((x-\alpha)^n \sqrt{f(x)}\Big) = \frac{n(x-\alpha)^{n-1} f(x) + (x-\alpha)^n \frac{1}{2} f'(x)}{\sqrt{f(x)}}\, dx,$$

and if in this we put

$$f(x) = f(\alpha) + (x-\alpha)f'(\alpha) + \frac{(x-\alpha)^2}{2}f''(\alpha) + \frac{(x-\alpha)^3}{6}f'''(\alpha) + \frac{(x-\alpha)^4}{24}f''''(\alpha),$$

$$f'(x) = f'(\alpha) + (x-\alpha)f''(\alpha) + \frac{(x-\alpha)^2}{2}f'''(\alpha) + \frac{(x-\alpha)^3}{6}f''''(\alpha),$$

then it follows that

$$\begin{aligned}(x-\alpha)^n\sqrt{f(x)} &= nf(\alpha)S_{n-1} + \left(n+\frac{1}{2}\right)f'(\alpha)S_n \\ &\quad + \frac{n+1}{2}f''(\alpha)S_{n+1} + \left(\frac{n}{6}+\frac{1}{4}\right)f'''(\alpha)S_{n+2} \\ &\quad + \left(\frac{n}{24}+\frac{1}{12}\right)f''''(\alpha)S_{n+3}.\end{aligned} \tag{2}$$

We distinguish four cases.

1. $f'''(\alpha)$ and $f(\alpha)$ are not $=0$ [that is, $f(x)$ is of the fourth degree and α is no root of $f(x)$]. In this case, by formula (2) for $n=0$, S_3 can be expressed through S_0, S_1, S_2, and therefore S_n for every positive n through S_0, S_1, S_2. But if one puts $n=-1,-2,\ldots$, then one obtains $S_{-2}, S_{-3}, \ldots$ expressed through S_{-1}, S_0, S_1, S_2. Hence in this case there remain the four fundamental integrals

$$S_{-1}, \quad S_0, \quad S_1, \quad S_2.$$

2. $f'''(\alpha)$ is not $=0$, $f(\alpha)=0$. In this case S_{-1} can still be expressed through S_0, S_1, S_2, and the fundamental integrals are

$$S_0, \quad S_1, \quad S_2.$$

3. $f'''(\alpha)=0$, $f(\alpha)$ not $=0$. Here the fundamental integrals are

$$S_{-1}, \quad S_0, \quad S_1.$$

4. $f'''(\alpha)=0$, $f(\alpha)=0$. Here the fundamental integrals are

$$S_0, \quad S_1.$$

The result is therefore:

I. If $f(x)$ is of the fourth degree ($a_0=1$), then all integrals $\Omega(x)$ can be expressed by integrals of the form

$$S_0 = \int\frac{dx}{\sqrt{f(x)}}, \qquad S_1 = \int\frac{x\,dx}{\sqrt{f(x)}}, \qquad S_2 = \int\frac{x^2\,dx}{\sqrt{f(x)}},$$
$$S_{-1} = \int\frac{dx}{(x-\alpha)\sqrt{f(x)}},$$

where S_{-1} still depends on the parameter α.

II. If, however, $f(x)$ is of the third degree, the three integrals

$$S_0 = \int\frac{dx}{\sqrt{f(x)}}, \qquad S_1 = \int\frac{x\,dx}{\sqrt{f(x)}}, \qquad S_{-1} = \int\frac{dx}{(x-\alpha)\sqrt{f(x)}}$$

suffice.

In both cases S_0 is of the first kind. If $f(x)$ is of the third degree, then S_1 is of the second kind and S_{-1} of the third kind. If, on the other hand, $f(x)$ is of the fourth degree, then S_1, S_2 and S_{-1} are of the third kind. But from S_1 and S_2 one can form an integral of the second kind. For

$$\frac{1}{\sqrt{f(x)}} = (x^4 + a_1x^3 + a_2x^2 + a_3x + a_4)^{-1/2} = x^{-2} - \frac{a_1}{2}x^{-3} + \cdots,$$

and in the development of

$$\left(x^2 + \frac{a_1}{2}x\right) : \sqrt{f(x)}$$

in descending powers no term with x^{-1} occurs. Consequently

$$S_2 + \frac{a_1}{2}S_1$$

is an integral of the second kind.

By the mediation of a logarithmic function one can also reduce the four integrals of case I to three. To this end bring $f(x)$ into the form

$$x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = P^2 + ax + b,$$

where

$$P = x^2 + px + q$$

is a quadratic function and a and b are constants. It follows that

$$p = \frac{a_1}{2}, \qquad q = \frac{a_2}{2} - \frac{a_1^2}{8},$$

$$a = a_3 - \frac{a_1a_2}{2} + \frac{a_1^3}{8}, \qquad b = a_4 - \left(\frac{a_2}{2} - \frac{a_1^2}{8}\right)^2.$$

Now

$$\begin{gathered} d\log\frac{\sqrt{f} - P}{\sqrt{f} + P} = \frac{f'(x)P - 2f(x)P'(x)}{f - P^2}\frac{dx}{\sqrt{f(x)}}, \\ f - P^2 = ax + b, \qquad f'(x) = 2PP' + a. \end{gathered} \tag{3}$$

and consequently

$$f'(x)P - 2f(x)P'(x) = a(P - 2xP') - 2bP' = Q,$$

a quadratic function of x, and therefore from (3)

$$\log\frac{\sqrt{f} - P}{\sqrt{f} + P} = \int \frac{Q}{ax + b}\frac{dx}{\sqrt{f(x)}}. \tag{4}$$

The right hand side, however, is a linear function of S_0, S_1 and S_{-1} (for $\alpha = -b/a$). In the special case where $a = 0$, Q is of the first degree, and one can express a linear combination of S_0 and S_{-1} by a logarithmic function.

If we assume a_1 and a_3 to be 0, to which the general case can be reduced by a linear transformation according to § 4, then we obtain as fundamental integrals

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \qquad S_2 = \int \frac{x^2\,dx}{\sqrt{f(x)}}, \qquad S_{-1} = \int \frac{dx}{(x - \alpha)\sqrt{f(x)}},$$

whereas $S_1 = \int \frac{x\,dx}{\sqrt{f(x)}}$ is reduced by the substitution $x^2 = y$ to the non-elliptic integral

$$\frac{1}{2}\int \frac{dy}{\sqrt{y^2 + a_2y + a_4}}.$$

Here S_0 is of the first kind, S_2 of the second, S_{-1} of the third kind. For Legendre's normal form we put $f(x) = (1 - x^2)(1 - \varkappa^2x^2)$ and take as the normal integral of the second kind not S_2 itself, but $S_0 - \varkappa^2S_2$. In this way the normal integrals of the three kinds are

$$\int \frac{dx}{\sqrt{(1 - x^2)(1 - \varkappa^2x^2)}}, \qquad \int \sqrt{\frac{1 - \varkappa^2x^2}{1 - x^2}}\,dx, \qquad \int \frac{dx}{(x - \alpha)\sqrt{(1 - x^2)(1 - \varkappa^2x^2)}},$$

and, if one puts $x = \sin\varphi$,

$$\int \frac{d\varphi}{\sqrt{1-\varkappa^2\sin^2\varphi}}, \qquad \int \sqrt{1-\varkappa^2\sin^2\varphi}\, d\varphi, \qquad \int \frac{d\varphi}{(\sin\varphi-\alpha)\sqrt{1-\varkappa^2\sin^2\varphi}}.$$

The same result is obtained if one assumes the elliptic differential in the normal form

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

and proceeds according to case II.

§ 13. The addition theorem.

The addition theorem of elliptic integrals, discovered by Euler, consists in the theorem, fundamental for the whole further theory, that if, of the three pairs of values

$$x_1, \sqrt{f(x_1)}; \qquad x_2, \sqrt{f(x_2)}; \qquad x_3, \sqrt{f(x_3)}$$

two are assumed given, one can determine the third algebraically in such a way that the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0 \tag{1}$$

is satisfied. Here $f(x)$ is a given function of the third or fourth degree. This is a special case of Abel's great theorem, and is also to be understood and derived here in that way[2]. We precede the proof by the following elementary algebraic theorem.

Let $F(x)$ be an integral rational function of degree n without multiple factors, $F'(x)$ its derivative, let $x_1, x_2, \ldots, x_n$ be the roots of the equation $F(x) = 0$, and let $\varphi(x)$ be an integral function of x whose degree is at most $n-2$. Then

$$\frac{\varphi(x_1)}{F'(x_1)} + \frac{\varphi(x_2)}{F'(x_2)} + \cdots + \frac{\varphi(x_n)}{F'(x_n)} = 0. \tag{2}$$

The proof of this theorem follows immediately from decomposing the rational fraction

$$\frac{x\varphi(x)}{F(x)}$$

into partial fractions

$$\frac{x_1\varphi(x_1)}{F'(x_1)(x-x_1)} + \frac{x_2\varphi(x_2)}{F'(x_2)(x-x_2)} + \cdots + \frac{x_n\varphi(x_n)}{F'(x_n)(x-x_n)},$$

and then putting $x = 0$ [Vol. I, § 15 (9)].

Let now P, Q denote integral functions of x of degrees m and $m-2$, and let us ask for the values of x, $\sqrt{f(x)}$ for which the function

$$P + Q\sqrt{f(x)} \tag{3}$$

vanishes. There are $2m$ such pairs of values (points), and the values of x are found as roots of the equation of degree $2m$:

$$F(x) = P^2 - Q^2 f(x) = 0, \tag{4}$$

[2] This very comprehensive theorem is derived with magnificent simplicity in a scarcely two-page paper in the fourth volume of Crelle's Journal: "Démonstration d'une propriété générale d'une certaine classe de fonctions transcendantes"; Oeuvres complètes de N. H. Abel. Nouvelle édition, T. I, p. 515. Detailed presentation and applications are found in the posthumous large memoir: "Mémoire sur une propriété générale d'une classe très étendue de fonctions transcendantes".

and the corresponding values of $\sqrt{f(x)}$ from

$$P + Q\sqrt{f(x)} = 0. \tag{5}$$

We now assume that the coefficients in the functions P, Q are variable, either independent variables or in some way dependent on other variables. Then the values x and $\sqrt{f(x)}$ determined by (4), (5) are also functions of these variables, and (5) can be differentiated.

Let δ denote differentiation with respect to the variables occurring in the coefficients of P, Q, where x is regarded as constant; and let dx denote the corresponding differential of x when x is determined by (4) or (5) as a function of these coefficients. Differentiating (4) gives

$$2P\delta P - 2Q\delta Q f(x) + F'(x)\,dx = 0,$$

whence, using (5),

$$\frac{Q\delta P - P\delta Q}{F'(x)} = \frac{dx}{2\sqrt{f(x)}}. \tag{6}$$

Since the numerator on the left has degree $2m-2$, while $F(x)$ has degree $2m$, formula (2) can be applied, and it follows that

$$\sum \frac{dx}{\sqrt{f(x)}} = 0, \tag{7}$$

where the sum is extended over all roots of the equation (5).

We apply this theorem to the simplest case, namely $m = 2$, and obtain the following theorem.

If

$$x_1, \sqrt{f(x_1)}; \quad x_2, \sqrt{f(x_2)}; \quad x_3, \sqrt{f(x_3)}; \quad x_4, \sqrt{f(x_4)} \tag{8}$$

are four pairs of values for which some function of the form

$$a + bx + cx^2 + \sqrt{f(x)} \tag{9}$$

vanishes, then these pairs of values are not wholly independent of one another, but between them there exists the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} + \frac{dx_4}{\sqrt{f(x_4)}} = 0. \tag{10}$$

The dependence of these four pairs of values, that is, an integration of the differential equation (10), can also be expressed algebraically by putting the function (9) equal to zero for $x = x_1, x_2, x_3, x_4$ and then eliminating a, b, c. One obtains the determinant equation

$$\begin{vmatrix} 1 & x_1 & x_1^2 & \sqrt{f(x_1)} \\ 1 & x_2 & x_2^2 & \sqrt{f(x_2)} \\ 1 & x_3 & x_3^2 & \sqrt{f(x_3)} \\ 1 & x_4 & x_4^2 & \sqrt{f(x_4)} \end{vmatrix} = 0. \tag{11}$$

From this equation x_1 and $\sqrt{f(x_1)}$ can be expressed rationally through $x_2, \sqrt{f(x_2)};\ x_3, \sqrt{f(x_3)};\ x_4, \sqrt{f(x_4)}$; for x_1 is the root of a biquadratic equation whose three other roots are x_2, x_3, x_4, and whose coefficients depend rationally on $x_2, \sqrt{f(x_2)};\ x_3, \sqrt{f(x_3)};\ x_4, \sqrt{f(x_4)}$, while $\sqrt{f(x_1)}$ is representable rationally through x_1 by means of (11). If x_4 is set equal to an arbitrary constant, then the differential equation (10) passes into (1), and (11) gives its integral.

We first apply this theorem to the normal form and put

$$f(x) = x(1-x)(1-\varkappa^2 x).$$

If in (11) we assume $x_4 = 0$, then this equation reduces, after division by $\sqrt{x_1 x_2 x_3}$, to

$$\begin{vmatrix} \sqrt{x_1} & x_1\sqrt{x_1} & \sqrt{(1-x_1)(1-\varkappa^2 x_1)} \\ \sqrt{x_2} & x_2\sqrt{x_2} & \sqrt{(1-x_2)(1-\varkappa^2 x_2)} \\ \sqrt{x_3} & x_3\sqrt{x_3} & \sqrt{(1-x_3)(1-\varkappa^2 x_3)} \end{vmatrix} = 0, \tag{12}$$

or, expanding the determinant by the elements of the first row,

$$\sqrt{x_1}(a + bx_1) + c\sqrt{(1-x_1)(1-\varkappa^2 x_1)} = 0, \tag{13}$$

where, for abbreviation,

$$\begin{aligned} a &= x_2\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2 x_3)} - x_3\sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2 x_2)}, \\ b &= -\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2 x_3)} + \sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2 x_2)}, \\ c &= -(x_2 - x_3)\sqrt{x_2}\sqrt{x_3}. \end{aligned} \tag{14}$$

If we rationalize (13), we obtain the cubic equation

$$x(a+bx)^2 - c^2(1-x)(1-\varkappa^2 x) = 0,$$

whose three roots are x_1, x_2, x_3. It is therefore identically

$$b^2(x-x_1)(x-x_2)(x-x_3) = x(a+bx)^2 - c^2(1-x)(1-\varkappa^2 x). \tag{15}$$

Putting $x = 0, 1, 1 : \varkappa^2$ in this identity, one obtains

$$\begin{aligned} \sqrt{x_1 x_2 x_3} &= -\frac{c}{b}, \\ \sqrt{(1-x_1)(1-x_2)(1-x_3)} &= \frac{b+a}{b}, \\ \sqrt{(1-\varkappa^2 x_1)(1-\varkappa^2 x_2)(1-\varkappa^2 x_3)} &= \frac{b+\varkappa^2 a}{b}. \end{aligned} \tag{16}$$

To determine the signs in these equations we further obtain from (13), by replacing x_1 by x_2 and x_3 and multiplying the results,

$$x_1 x_2 x_3 (a+bx_1)(a+bx_2)(a+bx_3) = -c^3\sqrt{f(x_1)}\sqrt{f(x_2)}\sqrt{f(x_3)},$$

and (15) gives for $x = -a : b$

$$(a+bx_1)(a+bx_2)(a+bx_3) = \frac{c^2}{b^2}(b+a)(b+\varkappa^2 a).$$

If one also uses the square of the first equation (16), it follows that

$$\sqrt{f(x_1)}\sqrt{f(x_2)}\sqrt{f(x_3)} = -\frac{c}{b^3}(b+a)(b+\varkappa^2 a).$$

The same result, however, is obtained by multiplying the three equations (16), and thus the signs in these formulae are chosen so that the product of the left hand sides gives the value of $\sqrt{f(x_1)}$ required by the differential equation (1). A further determination of the signs in (16) is not possible by the nature of the matter, and these equations serve for the unambiguous definition of $\sqrt{x_1}$, $\sqrt{1-x_1}$, $\sqrt{1-\varkappa^2 x_1}$ when $\sqrt{x_2}$, $\sqrt{1-x_2}$, $\sqrt{1-\varkappa^2 x_2}$, $\sqrt{x_3}$, $\sqrt{1-x_3}$, $\sqrt{1-\varkappa^2 x_3}$ are assumed given.

To bring equations (16) into the customary form, we first rationalize the numerator of b in (14) and obtain

$$b = -\frac{(x_2 - x_3)(1-\varkappa^2 x_2 x_3)}{\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2 x_3)} + \sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2 x_2)}},$$

and further, by (14),

$$\frac{b+a}{b} = \frac{\sqrt{1-x_2}\sqrt{1-x_3} - \sqrt{x_2 x_3}\sqrt{(1-\varkappa^2 x_2)(1-\varkappa^2 x_3)}}{1-\varkappa^2 x_2 x_3},$$

$$\frac{b+\varkappa^2 a}{b} = \frac{\sqrt{1-\varkappa^2 x_2}\sqrt{1-\varkappa^2 x_3} - \varkappa^2\sqrt{x_2 x_3}\sqrt{(1-x_2)(1-x_3)}}{1-\varkappa^2 x_2 x_3}.$$

Thus equations (16) are easily brought into the form

$$\begin{aligned} \sqrt{x_1} &= -\frac{\sqrt{x_2}\sqrt{(1-x_3)(1-\varkappa^2 x_3)} + \sqrt{x_3}\sqrt{(1-x_2)(1-\varkappa^2 x_2)}}{1-\varkappa^2 x_2 x_3}, \\ \sqrt{1-x_1} &= \frac{\sqrt{1-x_2}\sqrt{1-x_3} - \sqrt{x_2 x_3}\sqrt{(1-\varkappa^2 x_2)(1-\varkappa^2 x_3)}}{1-\varkappa^2 x_2 x_3}, \\ \sqrt{1-\varkappa^2 x_1} &= \frac{\sqrt{1-\varkappa^2 x_2}\sqrt{1-\varkappa^2 x_3} - \varkappa^2\sqrt{x_2 x_3}\sqrt{(1-x_2)(1-x_3)}}{1-\varkappa^2 x_2 x_3}. \end{aligned} \tag{17}$$

and our consideration shows that, when x_1 and the roots $\sqrt{x_1}, \sqrt{1-x_1}, \sqrt{1-\varkappa^2 x_1}$ are determined by these equations as functions of x_2, x_3, the differential equation (1) is identically satisfied.

In order also to obtain the addition theorem for Weierstrass's normal form in the simplest possible shape, put

$$f(x) = 4x^3 - g_2 x - g_3,$$

and in equation (11), after division by x_4^2, let x_4 become infinitely large. Then the integral of the differential equation (1)

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0$$

is obtained in the form

$$\begin{vmatrix} 1 & x_1 & \sqrt{f(x_1)} \\ 1 & x_2 & \sqrt{f(x_2)} \\ 1 & x_3 & \sqrt{f(x_3)} \end{vmatrix} = 0, \tag{18}$$

or

$$a + bx_1 + c\sqrt{f(x_1)} = 0, \tag{19}$$

where

$$a = x_2\sqrt{f(x_3)} - x_3\sqrt{f(x_2)}, \qquad b = \sqrt{f(x_2)} - \sqrt{f(x_3)}, \qquad c = x_3 - x_2. \tag{20}$$

Then x_1, x_2, x_3 become the roots of the cubic equation

$$(a+bx)^2 - c^2 f(x) = 0,$$

and, if the coefficient of x^2 divided by that of x^3 is set equal to the negative sum of the roots, one obtains

$$x_1 + x_2 + x_3 = \frac{1}{4}\left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3}\right)^2, \tag{21}$$

and from (19)

$$\sqrt{f(x_1)} = \frac{1}{4}\left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3}\right)^3 - (x_2 + x_3)\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3} - \frac{x_3\sqrt{f(x_2)} - x_2\sqrt{f(x_3)}}{x_2 - x_3}. \tag{22}$$

§ 14. Origin of the elliptic functions.

If in the elliptic differential of the first kind in Legendre's normal form

$$\frac{1}{2}\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

the substitution

$$z = \sin^2\varphi \tag{1}$$

is made, and the integration is then carried out from $\varphi = 0$, then there arises the elliptic integral of the first kind

$$\int_0^\varphi \frac{d\varphi}{\sqrt{1-\varkappa^2\sin^2\varphi}} = u. \tag{2}$$

Jacobi calls the upper limit φ of this integral its amplitude and writes

$$\varphi = \operatorname{am} u. \tag{3}$$

The trigonometric functions of this arc,

$$\sin\varphi, \quad \cos\varphi, \quad \sqrt{1-\varkappa^2\sin^2\varphi} = \Delta\varphi, \tag{4}$$

are now precisely those which, considered as functions of u, are called elliptic functions and denoted by Jacobi by

$$\sin\operatorname{am} u, \qquad \cos\operatorname{am} u, \qquad \Delta\operatorname{am} u,$$

or, after Gudermann, more briefly by

$$\operatorname{sn} u, \qquad \operatorname{cn} u, \qquad \operatorname{dn} u. \tag{5}$$

These functions depend on the two arguments u and $\varkappa^2$, and if a more exact notation is necessary one also writes

$$\operatorname{sn}(u,\varkappa^2), \qquad \operatorname{cn}(u,\varkappa^2), \qquad \operatorname{dn}(u,\varkappa^2).$$

If in formula (1) φ is transformed into $-\varphi$, then u passes into $-u$; hence $\operatorname{am}(-u) = -\operatorname{am} u$, and consequently

$$\operatorname{sn}(-u) = -\operatorname{sn} u, \qquad \operatorname{cn}(-u) = \operatorname{cn} u, \qquad \operatorname{dn}(-u) = \operatorname{dn} u,$$

that is, $\operatorname{sn} u$ is an odd function, while $\operatorname{cn} u$ and $\operatorname{dn} u$ are even functions.

If therefore one puts

$$\frac{d\varphi_2}{\Delta\varphi_2} = du, \qquad \frac{d\varphi_3}{\Delta\varphi_3} = dv, \qquad \frac{d\varphi_1}{\Delta\varphi_1} = -d(u+v),$$

then the formulae (17), § 13, can be applied, and one obtains the addition theorems:

$$\begin{aligned}
\operatorname{sn}(u+v) &= \frac{\operatorname{sn} u\ \operatorname{cn} v\ \operatorname{dn} v + \operatorname{sn} v\ \operatorname{cn} u\ \operatorname{dn} u}{1-\varkappa^2\operatorname{sn}^2 u\ \operatorname{sn}^2 v},\\
\operatorname{cn}(u+v) &= \frac{\operatorname{cn} u\ \operatorname{cn} v - \operatorname{sn} u\ \operatorname{sn} v\ \operatorname{dn} u\ \operatorname{dn} v}{1-\varkappa^2\operatorname{sn}^2 u\ \operatorname{sn}^2 v},\\
\operatorname{dn}(u+v) &= \frac{\operatorname{dn} u\ \operatorname{dn} v - \varkappa^2\operatorname{sn} u\ \operatorname{sn} v\ \operatorname{cn} u\ \operatorname{cn} v}{1-\varkappa^2\operatorname{sn}^2 u\ \operatorname{sn}^2 v}.
\end{aligned} \tag{6}$$

From these one can compute the three elliptic functions unambiguously for arbitrary values of the argument, once they are determined for any, however small, region around the origin. Weierstrass takes this theorem as the point of departure for the theory of elliptic functions.

If one puts

$$\int_0^{\pi/2}\frac{d\varphi}{\Delta\varphi} = K,$$

then $\operatorname{sn} K = 1$, $\operatorname{cn} K = 0$, $\operatorname{dn} K = \sqrt{1-\varkappa^2} = \varkappa'$, and formulae (6), with $v = K$, give

$$\operatorname{sn}(u+K) = \frac{\operatorname{cn} u}{\operatorname{dn} u}, \qquad \operatorname{cn}(u+K) = \frac{-\varkappa' \operatorname{sn} u}{\operatorname{dn} u}, \qquad \operatorname{dn}(u+K) = \frac{\varkappa'}{\operatorname{dn} u}, \tag{7}$$

and if this formula is applied once more:

$$\operatorname{sn}(u+2K) = -\operatorname{sn} u, \qquad \operatorname{cn}(u+2K) = -\operatorname{cn} u, \qquad \operatorname{dn}(u+2K) = \operatorname{dn} u, \tag{8}$$

and applied again:

$$\operatorname{sn}(u+4K) = \operatorname{sn} u, \qquad \operatorname{cn}(u+4K) = \operatorname{cn} u, \qquad \operatorname{dn}(u+4K) = \operatorname{dn} u. \tag{9}$$

Thus the elliptic functions have the property of periodicity in common with the trigonometric functions. They have the period $4K$ ($\operatorname{dn} u$ also has the period $2K$).

In § 3, as an example, we derived the transformation, formula (1):

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}} = \frac{dx}{i\sqrt{x(1-x)(1-\varkappa'^2 x)}}, \tag{10}$$

when

$$z = \frac{-x}{1-x} \tag{11}$$

held. Therefore, if one puts

$$\frac{dx}{\sqrt{x(1-x)(1-\varkappa'^2 x)}} = i\,du,$$

then, if x and u vanish simultaneously, $x = \operatorname{sn}^2(iu,\varkappa')$, $1 - x = \operatorname{cn}^2(iu,\varkappa')$, and from (10) it follows that

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}} = du,$$

hence

$$z = \operatorname{sn}^2(u,\varkappa).$$

Accordingly (11) gives

$$\operatorname{sn}(u,\varkappa) = \frac{-i\,\operatorname{sn}(iu,\varkappa')}{\operatorname{cn}(iu,\varkappa')} \tag{12}$$

(where the sign is to be determined from the special value $u = 0$).

If now

$$\int_0^{\pi/2} \frac{d\varphi}{\sqrt{1-\varkappa'^2\sin^2\varphi}} = K'$$

is put, then from (8) one obtains

$$\operatorname{sn}(iu+2K',\varkappa') = -\operatorname{sn}(iu,\varkappa'), \qquad \operatorname{cn}(iu+2K',\varkappa') = -\operatorname{cn}(iu,\varkappa'),$$

and consequently from (12)

$$\operatorname{sn}(u-2iK',\varkappa) = \operatorname{sn}(u,\varkappa),$$

or also, by replacing u by $u + 2iK'$ and omitting the notation $\varkappa$ again:

$$\operatorname{sn}(u+2iK') = \operatorname{sn} u. \tag{13}$$

Thus the function $\operatorname{sn} u$ has a double periodicity. One of these periods, $4K$, is real, the other, $2iK'$, purely imaginary (at least when the modulus $\varkappa$ is a positive proper fraction). This double periodicity is a property which occurs nowhere in the realm of the elementary functions, and thus showed most clearly that one was dealing here with a new kind of function. At the same time, the consequences derived from double periodicity are those which, as experience has shown, lead by far the most quickly and with satisfactory rigor into the interior of the theory, so that here the most suitable point of departure for a methodical development is to be sought. Our next considerations will therefore be devoted to the doubly periodic functions in general.

Second Section.

Theta Functions.

§ 15. Presuppositions from function theory.

The concept of doubly periodic functions can be rightly grasped only when they are considered as functions of a complex argument

$$u = v + iw,$$

where i has, as usual, the meaning of $\sqrt{-1}$. The variable u is to us an independent and unboundedly variable magnitude, which, following Gauss, is represented geometrically by the points of a plane in such a way that the point whose coordinates in a rectangular system are v, w is regarded as the carrier of the value $u = v + iw$, and is briefly called the point u.

We restrict our consideration here to single-valued analytic functions $\varphi(u)$ which in the finite part have no essentially singular places. What happens at infinity we leave undecided. Thus we assume the following about the functions considered.

1. In every finite piece of surface there lies a finite number of points at which the function $\varphi(u)$ becomes infinite; these points are called points of discontinuity.

The number of the points of discontinuity in all, that is, in the whole infinite plane, can of course also be infinitely large.

2. If u_0 is a point not belonging to the points of discontinuity, then the function is developable into a series proceeding according to whole ascending positive powers of $u - u_0$, which is convergent in a circle having u_0 as center and reaching to the nearest point of discontinuity. One says that the function has, in the neighborhood of the point u_0, the character of an entire function.

3. If u_0 is a point of discontinuity, then there is a whole positive number m such that $(u-u_0)^m\varphi(u)$ remains finite, continuous, and different from zero at the point u_0, and hence is developable according to whole positive powers of $u - u_0$. In this case the point of discontinuity is called one of the mth order. From this there results an expansion of $\varphi(u)$ according to rising powers of $u - u_0$, which begins with $(u - u_0)^{-m}$ and converges in a circle described about u_0 and reaching to the nearest point of discontinuity. The coefficient of $(u - u_0)^{-1}$ in this expansion is called the residue of this function for the point u_0.

4. Cauchy's theorem. The integral

$$\frac{1}{2\pi i} \int \varphi(u)\, du,$$

extended in the positive sense over the boundary of a finite piece of surface which contains no point of discontinuity on its boundary line, is equal to the sum of the residues for the points of discontinuity lying in the interior of the piece of surface. As positive direction of integration one is here to regard the direction which lies with respect to the interior of the piece of surface as the v-axis does to the w-axis, so that, with the usual way of determining these axes, when the boundary is traversed positively the interior of the surface remains on the left.

5. The function

$$\psi(u) = \frac{d \log \varphi(u)}{du} = \frac{1}{\varphi(u)} \frac{d\varphi(u)}{du}$$

has only points of discontinuity of the first order, and if at a point u_0 the product $(u - u_0)^{-m}\varphi(u)$ is finite and different from zero, then m is the residue of $\psi(u)$ for this point. If m is positive, then u_0 is called a zero of mth order of $\varphi(u)$.

Accordingly the following is an immediate consequence of Cauchy's theorem.

6. The integral

$$\frac{1}{2\pi i} \int d \log \varphi(u),$$

extended in the positive direction over the boundary of a piece of surface, is equal to the number of zeros, diminished by the number of points of discontinuity, which lie in the interior of this piece of surface, zeros and points of discontinuity of mth order being counted as m such points of first order. Hence it follows that in the interior of a finite piece of surface there also lies only a finite number of zeros.

In the same sense one obtains:

7. The integral

$$\frac{1}{2\pi i}\int u\,d\log\varphi(u)$$

is equal to the sum of the values of u for which $\varphi(u)$ vanishes, diminished by the sum of the values of u for which $\varphi(u)$ becomes infinite.

8. A function which has no points of discontinuity at all in the finite part is called an entire function. An entire function which also remains finite at infinity is a constant. An entire function which also at infinity has no essentially singular place, so that there is therefore an exponent m such that $u^{-m}\varphi(u)$ does not become infinite for $u=\infty$, is a rational function.

9. An entire function whose reciprocal is likewise entire is called a unit function. Such a function becomes neither infinite nor zero in the finite part. Its logarithm is therefore likewise an entire function, and it follows that every unit function can be represented in the form $e^{g(u)}$, where $g(u)$ is an entire function. Conversely, every expression of this form is a unit function.

10. According to the fundamental investigations of Weierstrass on single-valued analytic functions, there is always an entire function $G(u)$ which vanishes at the points of discontinuity of a function $\varphi(u)$ and only at these, and this function $G(u)$ is determined by the points of discontinuity of $\varphi(u)$ up to a unit function as factor. Then $\varphi(u)G(u)=G_1(u)$ is likewise an entire function, and thus every function $\varphi(u)$ can be represented as the quotient of two entire functions

$$\varphi(u)=\frac{G_1(u)}{G(u)},$$

where $G(u)$ and $G_1(u)$ have no common zeros.

11. Numerator and denominator of this representation are uniquely determined by $\varphi(u)$ itself, apart from unit factors.

The functions $G(u)$, $G_1(u)$ can be decomposed into prime factors, that is, into factors which become zero at only one point. If one pursues this path for the doubly periodic functions, one arrives at the Weierstrass σ-functions. We shall, however, take another path, on which the ϑ-functions first meet us at a later place.

§ 16. Periodicity.

A function $\varphi(u)$ of u which has the property that, for a constant ω and for every value of u,

$$\varphi(u+\omega)=\varphi(u) \tag{1}$$

is called periodic, and ω is called the period.

The object of our investigation is functions $\varphi(u)$ with two periods ω_1,ω_2, of which we shall assume once for all that their ratio $\omega_2:\omega_1$ is not real and that the imaginary part of this ratio is positive. In the latter assumption there lies only a convention concerning the designation ω_1,ω_2. For if $\omega_2:\omega_1$ has a negative imaginary part, then that of $\omega_1:\omega_2$ is positive.

Every combination $m_1\omega_1+m_2\omega_2$ is likewise a period, when m_1,m_2 are integers.

In the u-plane we take an arbitrary point u_0 and denote the four points

$$u_0,\qquad u_0+\omega_1,\qquad u_0+\omega_1+\omega_2,\qquad u_0+\omega_2.$$

If we connect these four points in order by straight lines, returning from the last to the first, we obtain a parallelogram, which is called the period parallelogram. The straight sides of the period parallelogram may also be replaced by curved arcs, provided that opposite sides come to coincide by parallel displacement and the individual arcs form no loops.

If $\omega_1 = \alpha_1 + \beta_1 i$, $\omega_2 = \alpha_2 + \beta_2 i$ and $\alpha_1, \alpha_2, \beta_1, \beta_2$ are real, then by the assumption $\alpha_1\beta_2 - \alpha_2\beta_1$ is positive, and the geometrical meaning of this magnitude is the area of the period parallelogram.

By adjoining congruent period parallelograms the whole u-plane can be covered simply and without gaps. Corresponding points of two different such parallelograms are representatives of u-values which differ by integral multiples of the periods, and which are called congruent modulo (ω_1, ω_2), or, if no doubt is possible, simply congruent. The sign of congruence is

$$u \equiv u' \pmod{\omega_1, \omega_2},$$

and means that u' has the form $u + m_1\omega_1 + m_2\omega_2$, where m_1, m_2 are integers.

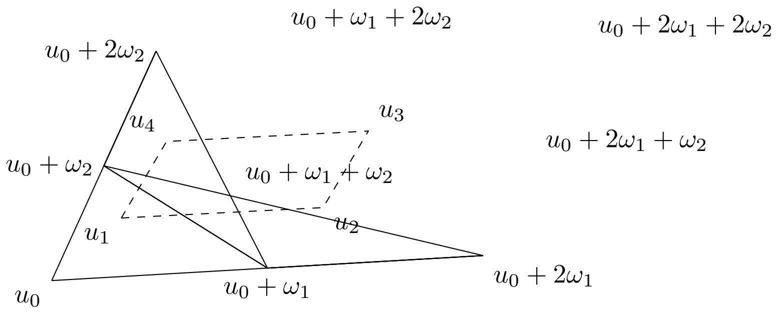

Figure 1: Fig. 1.

Thus Fig. 1 represents four period parallelograms laid alongside one another; the points u_1, u_2, u_3, u_4 are congruent:

$$u_2 = u_1 + \omega_1, \qquad u_3 = u_1 + \omega_1 + \omega_2, \qquad u_4 = u_1 + \omega_2,$$

and u_1, u_2, u_3, u_4 likewise form the corners of a period parallelogram.

The doubly periodic function $\varphi(u)$ has the same value at all congruent points, and the entire stock of values of this function is therefore exhausted in one period parallelogram, if of two opposite sides only one is counted with the parallelogram.

From our general assumptions we draw the conclusion:

1. There exists, apart from the constant, no doubly periodic function which is free of points of discontinuity in the period parallelogram; for such a function would be finite in the whole u-plane and hence by § 15, 8 would have to be a constant.

If $\varphi(u)$ is a doubly periodic function with periods ω_1, ω_2, then the integral taken over the boundary of the period parallelogram is, because of (1),

$$\int d\log\varphi(u) = \int_{u_0}^{u_0+\omega_1} [d\log\varphi(u) - d\log\varphi(u+\omega_2)] - \int_{u_0}^{u_0+\omega_2} [d\log\varphi(u) - d\log\varphi(u+\omega_1)] = 0.$$

From this follows, by § 15, 6, the theorem:

2. A doubly periodic function becomes zero at just as many points in a period parallelogram as it becomes infinite. If m is this number, then m is called the order of the doubly periodic function. By 1. there are no doubly periodic functions of order zero.

If in the proof one replaces the function $\varphi(u)$ by $\varphi(u) - c$, it follows that a doubly periodic function of mth order assumes every arbitrary value c at equally many points.

If we take the integral of the formula in 7., § 15 over the boundary of the period parallelogram, we obtain

$$\begin{aligned}\int u\, d\log\varphi(u) &= \int_{u_0}^{u_0+\omega_1} [u\, d\log\varphi(u) - (u+\omega_2) d\log\varphi(u+\omega_2)] \\ &\quad - \int_{u_0}^{u_0+\omega_2} [u\, d\log\varphi(u) - (u+\omega_1) d\log\varphi(u+\omega_1)] \\ &= -\omega_2 \int d\log\varphi(u) + \omega_1 \int d\log\varphi(u).\end{aligned}$$

Now, because of the many-valuedness of the logarithm,

$$\int d\log\varphi(u) = \log\varphi(u+\omega_1) - \log\varphi(u) = 2\pi i m_1,$$

$$\int d\log\varphi(u) = \log\varphi(u+\omega_2) - \log\varphi(u) = -2\pi i m_2,$$

where m_1 and m_2 are undetermined integers.

If therefore the function $\varphi(u)$ becomes infinite at the points $\alpha_1, \alpha_2, \dots, \alpha_m$ and zero at the points $\beta_1, \beta_2, \dots, \beta_m$, then, by § 15, 7,

$$\sum \alpha \equiv \sum \beta \pmod{\omega_1, \omega_2}.$$

If one replaces $\varphi(u)$ by $\varphi(u) - c$, where c denotes an arbitrary value, the important theorem follows:

3. The sum of the argument values at which a doubly periodic function assumes one and the same value c in the interior of a period parallelogram is independent of c, or changes under continuous variation of c at most by jumps of a period.

4. From this it follows also that there is no doubly periodic function of the first order.

For such a function would have to assume every value c at one point u. If it were therefore equal to c' at a point u' and equal to c' at a different point u'', then by 3. $u' = u''$ would follow, contrary to the assumption.

By a doubly periodic function of the second kind one understands, following Hermite, a function $\psi(u)$ which, with respect to the periods ω_1, ω_2, behaves according to the equations

$$\psi(u+\omega_1) = A_1\psi(u), \qquad \psi(u+\omega_2) = A_2\psi(u),$$

where A_1, A_2 are constants.

5. Also for a doubly periodic function of the second kind the theorem holds that it becomes zero and infinite at equally many points of the period parallelogram. If the number of these points is m, then here too the function is called of the mth order.

The proof is the same as that of theorem 2. But here the extension no longer holds that the function $\psi(u)$ assumes every value equally often, because $\psi(u) - c$ is no longer periodic.

6. A doubly periodic function $\psi(u)$ of the second kind and of order zero is necessarily a constant or an exponential function. For if $\psi(u)$ does not become zero, then

$$\varphi(u) = \frac{\psi'(u)}{\psi(u)}$$

is a doubly periodic function of the first kind which does not become infinite, and hence by 1. is a constant a. It follows that

$$\psi(u) = Ae^{au},$$

where A is a new constant.

§ 17. The functions T.

If doubly periodic functions $\varphi(u)$ of the mth order exist, then they must be fractional functions. We therefore set them, according to § 15, 10, in the form

$$\varphi(u) = \frac{G_1(u)}{G(u)}. \tag{1}$$

If $G_1(u)$ and $G(u)$ are entire functions without a common zero, then

$$\frac{G(u+\omega_1)}{G(u)} = \frac{G_1(u+\omega_1)}{G_1(u)}, \qquad \frac{G(u+\omega_2)}{G(u)} = \frac{G_1(u+\omega_2)}{G_1(u)}.$$

In these equations the three functions $G(u)$, $G(u+\omega_1)$, $G(u+\omega_2)$ have the same m zeros, namely the points of discontinuity of $\varphi(u)$. Their quotients are therefore unit functions, and by § 15, 9, if $l_1(u), l_2(u)$ are entire functions,

$$G(u+\omega_1) = e^{l_1(u)} G(u), \qquad G(u+\omega_2) = e^{l_2(u)} G(u), \tag{2}$$

and the function $G_1(u)$ satisfies the same equations.

If

$$\varphi(u) = \frac{T_1(u)}{T(u)} \tag{3}$$

is a second representation of $\varphi(u)$ of the form (1), then

$$T(u) = e^{g(u)} G(u), \tag{4}$$

where $g(u)$ is again an entire function, and if therefore, by (2),

$$T(u+\omega_1) = e^{l_1'(u)} T(u), \qquad T(u+\omega_2) = e^{l_2'(u)} T(u), \tag{5}$$

then

$$l_1'(u) = l_1(u) + g(u) - g(u+\omega_1), \qquad l_2'(u) = l_2(u) + g(u) - g(u+\omega_2).$$

Everything therefore depends on being able, under some suitable assumption about the entire functions $l_1(u), l_2(u)$, to form entire functions $T(u)$ which satisfy the conditions (5) and vanish at m arbitrarily given points of the period parallelogram. By the quotients of two such functions with the same $l_1(u), l_2(u)$ we can then represent all doubly periodic functions $\varphi(u)$, and the more general functions $G(u)$ which accomplish the same thing are obtained by (4) through adjoining an arbitrary unit factor.

If we wanted to assume $l_1(u), l_2(u)$ to be constants, then $T(u)$ would be an entire doubly periodic function of the second kind and consequently by § 16, 6 an exponential function, from which no doubly periodic functions can be formed. The simplest assumption that remains to us after this is that $l_1(u), l_2(u)$ are linear functions of u, including as a special case also the case that one of the two functions l_1, l_2 is constant.

Thus we set up the following definition:

1. A T-function, $T(u)$, is an entire function of u which satisfies the following two equations:

$$\begin{aligned} T(u+\omega_1) &= e^{-\pi i[a_1(2u+\omega_1)+b_1]} T(u), \\ T(u+\omega_2) &= e^{-\pi i[a_2(2u+\omega_2)+b_2]} T(u). \end{aligned} \tag{6}$$

Here a_1, b_1, a_2, b_2 are constants, that is, magnitudes independent of u. The periods ω_1, ω_2 are taken here throughout as constants given once for all, whose ratio $\omega_2 : \omega_1$ has a positive imaginary part.

The factors

$$e_1 = e_1(u) = e^{-\pi i[a_1(2u+\omega_1)+b_1]}, \qquad e_2 = e_2(u) = e^{-\pi i[a_2(2u+\omega_2)+b_2]}$$

are called the two periodicity factors of the function T.

The periodicity factors do not change if b_1, b_2 are changed by even integers. That functions of this kind really exist will first appear in the further course of the investigation.

2. If the function $T(u)$ vanishes at any point, then it also vanishes at all congruent points. If it vanishes at m points of the period parallelogram, then it is called of the mth order.

From this definition it follows at once:

3. By multiplying two T-functions T, T' of orders m, m' one obtains a T-function of order $m+m'$. The periodicity factors of the product TT' are the products of the periodicity factors of T and T'.

We first ask about the possibility of T-functions of zeroth order. If $L(u)$ is such a function and $L'(u)$ its derivative, then by twice differentiating (6) it follows that

$$\frac{dL(u+\omega_1)}{du\,L(u+\omega_1)} = \frac{dL(u)}{du\,L(u)},$$

and correspondingly from the second equation (6). Hence $d^2 \log L(u)/du^2$, as an entire doubly periodic function, is a constant, and consequently $\log L(u)$ is an entire function of the second degree in u. We therefore obtain:

4. A T-function of zeroth order is of the form

$$L(u) = Ce^{-\pi i(\lambda u^2+\mu u)}, \tag{7}$$

where λ, μ, C are constants.

That every function of this form is a T-function of zeroth order is seen immediately.

The product

$$T'(u) = L(u)T(u) \tag{8}$$

is, like $T(u)$, a T-function of mth order and vanishes at the same points.

The periodicity factors of T' are obtained from those of T when the exponents a_1, a_2, b_1, b_2 are replaced by

$$a_1' = a_1 + \lambda\omega_1, \quad b_1' = b_1 + \mu\omega_1, \qquad a_2' = a_2 + \lambda\omega_2, \quad b_2' = b_2 + \mu\omega_2, \tag{9}$$

where however the b_1', b_2' are determined only up to additive even integers.

5. By (9) one can determine μ always, and in only one way, so that b_1', b_2' become real.

For if one sets the imaginary parts of b_1', b_2' equal to zero, one obtains for the imaginary and real part of μ two linear equations whose determinant, by § 16, is the area of the period parallelogram and therefore different from zero.

The periodicity factors and the zeros of a T-function of mth order stand in a certain dependence on one another, which we obtain easily from the theorems of § 15. First the order m is obtained when one extends the integral

$$\int d\log T(u) = 2\pi i\, m$$

over the boundary of the period parallelogram. For simplicity let the corner u_0 lie at the origin of coordinates; then this integral can be decomposed as

$$2\pi i m = \int_0^{\omega_1} d[\log T(u) - \log T(u+\omega_2)] - \int_0^{\omega_2} d[\log T(u) - \log T(u+\omega_1)],$$

and by (6)

$$d[\log T(u) - \log T(u+\omega_2)] = 2\pi i a_2, \qquad d[\log T(u) - \log T(u+\omega_1)] = 2\pi i a_1.$$

It follows that

$$a_2\omega_1 - a_1\omega_2 = m. \tag{10}$$

If we further denote by $\alpha_1, \alpha_2, \ldots, \alpha_m$ the zeros of a T-function in the interior of a period parallelogram, then from § 15, 7 we obtain

$$2\pi i \sum \alpha = \int u\, d\log T(u),$$

when the integral is again extended over the boundary of the parallelogram. But

$$\begin{aligned}\int u\,d\log T(u) = &\int_0^{\omega_1} [u\,d\log T(u) - (u+\omega_2)d\log T(u+\omega_2)] \\ &- \int_0^{\omega_2} [u\,d\log T(u) - (u+\omega_1)d\log T(u+\omega_1)]\,.\end{aligned}$$

The first of these two integrals is, by (6),

$$\begin{aligned}
&-\omega_2\int_0^{\omega_1} d\log T(u) + 2\pi i a_2\int_0^{\omega_1}(u+\omega_2)\,du\\
&\quad= -\omega_2[\log T(\omega_1)-\log T(0)] + \pi i a_2\omega_1(\omega_1+2\omega_2)\\
&\quad= \pi i\omega_2(a_1\omega_1+b_1) + \pi i a_2(\omega_1^2+2\omega_1\omega_2) + 2\pi i N_2\omega_2,
\end{aligned}$$

where N_2, because of the many-valuedness of the logarithm, is an integer not more closely determined. Likewise, for the second of the integrals one obtains

$$-\pi i\omega_1(a_2\omega_2+b_2) - \pi i a_1(\omega_2^2+2\omega_1\omega_2) + 2\pi i N_1\omega_1,$$

and consequently, written as a congruence,

$$\sum\alpha \equiv \frac{1}{2}(b_1\omega_2 - b_2\omega_1) + \frac{m}{2}(\omega_1+\omega_2). \tag{11}$$

This sum, or rather the totality of the numbers congruent with it modulo (ω_1, ω_2), is called the character of the T-function.

The character of the function T is not changed if one replaces T by one of the functions LT. Thus by 5. one can also represent the character in the form

$$\sum\alpha \equiv \frac{1}{2}(g_1\omega_2 - g_2\omega_1) + \frac{m}{2}(\omega_1+\omega_2), \tag{12}$$

where g_1, g_2 are real. These real numbers are determined by the character up to multiples of 2 and can take every value between 0 and 2. The symbol (g_1, g_2) is called the characteristic of the T-function. From the characteristic the character of the T-function is determined by (12).

If γ is the character of a function $T(u)$ of order m and v is an arbitrary constant, then $T(u+v)$ has the character

$$\gamma' = \gamma - vm.$$

Accordingly, when $m > 0$, the different characters can be reduced to one another.

With the help of equation (10) one can combine the two equations (6) into a general one. If in the first equation (6) one repeatedly changes u into $u+\omega_1$, one obtains the system

$$\begin{aligned}
T(u+\omega_1) &= e^{-\pi i[a_1(2u+\omega_1)+b_1]}T(u),\\
T(u+2\omega_1) &= e^{-\pi i[a_1(2u+3\omega_1)+b_1]}T(u+\omega_1),\\
T(u+n_1\omega_1) &= e^{-\pi i[a_1(2u+(2n_1-1)\omega_1)+b_1]}T(u+(n_1-1)\omega_1),
\end{aligned}$$

and by multiplication of all these formulae:

$$T(u+n_1\omega_1) = e^{-\pi i[a_1(2n_1u+n_1^2\omega_1)+n_1b_1]}T(u). \tag{13}$$

This formula is first derived for a positive integral n_1. But if in it one replaces u by $u - n_1\omega_1$, its correctness for negative n_1 also results.

Likewise one obtains

$$T(u+n_2\omega_2) = e^{-\pi i[a_2(2n_2u+n_2^2\omega_2)+n_2b_2]}T(u), \tag{14}$$

and if in this formula one changes u into $u+n_1\omega_1$ and again applies (13),

$$T(u+n_1\omega_1+n_2\omega_2) = e^{-\pi i[2(a_1n_1+a_2n_2)u+2a_2n_1n_2\omega_1+a_1n_1^2\omega_1+a_2n_2^2\omega_2+n_1b_1+n_2b_2]}T(u). \tag{15}$$

But by (10)

$$2a_2n_1n_2\omega_1 = a_1n_1n_2\omega_1 + a_2n_1n_2\omega_2 + mn_1n_2,$$

and therefore this last formula becomes

$$T(u+n_1\omega_1+n_2\omega_2) = e^{-\pi i[(n_1a_1+n_2a_2)(2u+n_1\omega_1+n_2\omega_2)+b_1n_1+b_2n_2+mn_1n_2]}T(u). \tag{16}$$

Here n_1 and n_2 are arbitrary positive or negative integers.

If one multiplies two T-functions of order m and m' and of characters γ, γ', there arises a new T-function whose order is $m+m'$ and whose character is $\gamma+\gamma'$. If (g_1, g_2) and (g_1', g_2') are the characteristics of the two factors, then (g_1+g_1', g_2+g_2') is the characteristic of the product.

§ 18. Relations between related T-functions.

Two T-functions with the same periods, the same order, and the same character shall be called related. If $T'(u)$ is a T-function related to $T(u)$, and if a_1', a_2', b_1', b_2' have the same meaning for T' as a_1, a_2, b_1, b_2 for T, then, in consequence of the assumed relatedness,

$$a_1'\omega_2 - a_2'\omega_1 = a_1\omega_2 - a_2\omega_1, \tag{1}$$
$$b_1'\omega_2 - b_2'\omega_1 = b_1\omega_2 - b_2\omega_1 + 2n_1\omega_2 - 2n_2\omega_1, \tag{2}$$

where n_1, n_2 are integers.

Therefore λ and μ can be determined according to § 17, (9) so that the periodicity factors of the function

$$e^{-\pi i(\lambda u^2+\mu u)}T(u)$$

become the same as those of the function $T'(u)$, and hence we have the theorem:

1. Related T-functions can, by adjoining a T-function of zeroth order

$$L = Ce^{-\pi i(\lambda u^2+\mu u)}$$

as factor, be given the same periodicity factors.

Furthermore, from the equality of the characters it follows immediately:

2. If related T-functions of mth order have $m-1$ common zeros in the period parallelogram, then they also have the mth in common; and from this:

3. Between at most $m+1$ related T-functions of mth order $T, T_1, \ldots, T_m$ there exists an identical equation of the form

$$LT + L_1T_1 + L_2T_2 + \cdots + L_mT_m = 0.$$

For if this relation does not already hold for $L = 0$, then one can first determine the constants available in $L_1, L_2, \ldots, L_m$ so that all the products $L_1T_1, L_2T_2, \ldots, L_mT_m$ receive the same periodicity factors, and then the remaining constants so that $m-1$, and consequently all, zeros of the function $L_1T_1 + L_2T_2 + \cdots + L_mT_m$ coincide with the zeros of the function T. From this, however, it follows that the ratio of $L_1T_1 + L_2T_2 + \cdots + L_mT_m$ to T is equal to a T-function of zeroth order, thus equal to $-L$, as was to be proved.

4. The quotients of related T-functions can be transformed into doubly periodic functions by adjoining a factor L.

5. If we assume the existence of a T-function of the first order $t(u)$, then every T-function of mth order can be derived from it.

Namely, let γ be the character of $t(u)$ and let $\alpha_1, \alpha_2, \ldots, \alpha_m$ be arbitrarily given values. Then the function

$$t(u - \alpha_i + \gamma) = t_i(u)$$

vanishes at the point α_i and all points congruent with α_i, but at no other.

The product

$$T(u) = t_1(u)t_2(u)\cdots t_m(u)$$

is therefore a T-function of mth order with the arbitrarily given zeros $\alpha_1, \alpha_2, \ldots, \alpha_m$.

6. From this it is easy to prove that every doubly periodic function which has only a finite number of points of discontinuity in a period parallelogram can be represented as the quotient of two T-functions.

Let $\varphi(u)$ be a doubly periodic function with periods ω_1, ω_2 and with points of discontinuity $\alpha_1, \alpha_2, \ldots, \alpha_m$, which may also partly coincide to points of discontinuity of higher order.

If according to 5. one determines a function $T(u)$ with zeros $\alpha_1, \alpha_2, \ldots, \alpha_m$, then

$$T(u)\varphi(u) = T_1(u)$$

is a T-function of mth order, and

$$\varphi(u) = \frac{T_1(u)}{T(u)}.$$

The functions $T(u), T_1(u)$ are related, since they have the same periodicity factors and hence also the same characteristic.

The zeros of $T_1(u)$ are at the same time the zeros of $\varphi(u)$, and the sum of these zero values is therefore congruent with the sum of the discontinuity values.

§ 19. T-functions of first order.

We now pass to the closer determination of the T-functions of first order, which we denote by $t(u)$, and from which, as we have seen, the remaining T-functions can be derived.

From 3. of the preceding paragraph it follows that two t-functions of the same characteristic differ only by a factor of the form

$$Ce^{-\pi i(\lambda u^2 + \mu u)},$$

and we shall now, by an extension of the definition, determine this factor still more closely.

This shall be done, as is possible without restriction of generality by § 17, (9), so that in the periodicity factors

$$a_1 = 0, \qquad b_1 = g_1, \qquad b_2 = g_2$$

when (g_1, g_2) is the characteristic; because of the relation

$$a_2\omega_1 - a_1\omega_2 = 1$$

one then has

$$a_2 = \frac{1}{\omega_1},$$

and the conditions for this t-function read:

$$t(u + \omega_1) = e^{-\pi i g_1} t(u), \qquad t(u + \omega_2) = e^{-\pi i \frac{2u+\omega_2}{\omega_1}} e^{-\pi i g_2} t(u). \tag{1}$$

By this the function $t(u)$ is determined up to a factor independent of u. In order to determine this factor still more closely, we consider also the dependence of the function t on ω_1, ω_2, and denote it, regarding g_1, g_2 as given numbers independent of ω_1, ω_2, by $t(u, \omega_1, \omega_2)$. It then appears that if h is an arbitrary factor,

$$t(hu, h\omega_1, h\omega_2)$$

likewise satisfies the conditions (1), and hence that

$$\frac{t(hu, h\omega_1, h\omega_2)}{t(u, \omega_1, \omega_2)} = f(\omega_1, \omega_2) \tag{2}$$

is independent of u.

But the function $f(\omega_1, \omega_2)$ can always be put into the form

$$f(\omega_1, \omega_2) = \frac{\varphi(h\omega_1, h\omega_2)}{\varphi(\omega_1, \omega_2)}.$$

One has only, if $t(u)$ does not vanish for $u = 0$, to put

$$\varphi(\omega_1, \omega_2) = t(0, \omega_1, \omega_2),$$

and if $t(0) = 0$, for instance

$$\varphi(\omega_1, \omega_2) = \omega_1 t'(0, \omega_1, \omega_2),$$

where t' denotes the derivative of t with respect to u. If therefore we now again denote

$$\frac{t(u, \omega_1, \omega_2)}{\varphi(\omega_1, \omega_2)} \tag{3}$$

by $t(u, \omega_1, \omega_2)$, then one can add to the conditions (1) the condition that for an arbitrary h

$$t(u, \omega_1, \omega_2) = t(hu, h\omega_1, h\omega_2). \tag{4}$$

The function t now depends only on two variables, namely the ratios $u : \omega_1 : \omega_2$. We put in (4)

$$h = \frac{1}{\omega_1}, \qquad \frac{u}{\omega_1} = v, \qquad \frac{\omega_2}{\omega_1} = \omega,$$

and obtain

$$t(u, \omega_1, \omega_2) = \vartheta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right) = \vartheta(v, \omega), \tag{5}$$

where ϑ is a new function sign.

Thus a new function $\vartheta(v)$ is defined, of only two variables v, ω, the first of which is unrestrictedly variable, while ω has, by the assumption made in § 16, a positive imaginary constituent; v is called the argument, ω the modulus of the ϑ-function.

§ 20. The ϑ-function.

The function $\vartheta(v)$ is an entire function of v which satisfies the conditions

$$\vartheta(v+1) = e^{-\pi i g_1}\vartheta(v), \qquad \vartheta(v+\omega) = e^{-\pi i(2v+\omega)}e^{-\pi i g_2}\vartheta(v), \tag{1}$$

and is thereby determined up to a factor independent of v. It is therefore included among the t-functions as a special case, while conversely the general t-function can be expressed by the ϑ-function in the way

$$t(u) = Ce^{-\pi i(\lambda u^2 + \mu u)}\vartheta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right),$$

where C, λ, μ are arbitrary magnitudes, constant or depending on ω_1, ω_2.

The function ϑ still depends on the numbers g_1, g_2 occurring in the characteristic, and if a notation for this dependence is necessary, one shall write for $\vartheta(v, \omega)$

$$\vartheta_{g_1, g_2}(v, \omega);$$

here g_1, g_2 may be arbitrary numbers. By what was said earlier it is enough if we assume them real and between 0 and 2.

Corresponding to the T-functions of higher order we shall also introduce Θ-functions of mth order, and understand by this an entire function of v which satisfies the conditions

$$\Theta(v+1) = e^{-\pi i g_1}\Theta(v), \qquad \Theta(v+\omega) = e^{-m\pi i(2v+\omega)}e^{-\pi i g_2}\Theta(v). \tag{2}$$

Here also the characteristic g_1, g_2 can be included in the notation:

$$\Theta_{g_1, g_2}(v, \omega).$$

These functions are contained as special cases among the T-functions, and one can form general T-functions in the way

$$Ce^{-\pi i(\lambda u^2 + \mu u)}\Theta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right). \tag{3}$$

Thus from § 18, 3 there results the fundamental theorem for the Θ-functions, which we state as follows.

If we call Θ-functions with the same periods, equal order, and equal characteristic related, and if furthermore we call a system of functions linearly dependent or independent according as a homogeneous linear relation with constant coefficients, not all vanishing, exists between these functions or not, then the theorem holds:

$m+1$ related Θ-functions of mth order are always linearly dependent;

or:

from m linearly independent related Θ-functions of mth order

every other related Θ-function can be linearly composed, with constant coefficients.

This theorem is of the most fundamental importance for our following considerations; from it, by § 18, 3, it follows that every T-function of mth order can, by use of formula (3), be composed out of m linearly independent Θ-functions.

By § 17, (11), the sum of the m argument values for which the function $\Theta_{g_1,g_2}(v)$ vanishes in the period parallelogram is congruent modulo $(1,\omega)$ to

$$m\frac{1+\omega}{2}+\frac{g_1\omega-g_2}{2}.$$

The definition of the function ϑ given up to now leaves undetermined a factor which may be a function of ω. But by an addition to the definition we can determine this factor more closely.

If the defining equations (1) are differentiated twice with respect to v and once with respect to ω, regarding g_1, g_2 as constants, then by a simple calculation it follows that the function

$$\frac{\partial^2\vartheta(v,\omega)}{\partial v^2}-4\pi i\,\frac{\partial\vartheta(v,\omega)}{\partial\omega}$$

itself satisfies the conditions (1), and therefore has the form

$$\varphi(\omega)\vartheta(v,\omega),$$

where $\varphi(\omega)$ is constant with respect to v, hence a function of ω alone. If one now sets

$$e^{-\frac{1}{4\pi i}\int\varphi(\omega)\,d\omega}\vartheta(v,\omega)$$

equal to a new ϑ, then for this one obtains the partial differential equation

$$\frac{\partial^2\vartheta(v,\omega)}{\partial v^2}-4\pi i\,\frac{\partial\vartheta(v,\omega)}{\partial\omega}=0, \tag{4}$$

and by (1) and (4) the function ϑ is now defined up to a factor independent of v and ω, thus depending only on g_1, g_2.

§ 21. The theta functions of different characteristics. Main characteristics.

Before we go on to the determination of the remaining factor in the ϑ-functions that is independent of v and ω, we shall now reduce the different characteristics to one another, which by § 17 is always possible. If $\vartheta(v)$ denotes the ϑ-function belonging to the characteristic $(0,0)$, then

$$\Phi(v,\omega)=e^{-\pi i g_1 v}\,\vartheta\left(v-\frac{g_1\omega-g_2}{2}\right)$$

is a function satisfying the conditions (1) of § 20. But, with regard to the differential equation (4), one has

$$\frac{\partial^2\Phi}{\partial v^2}-4\pi i\,\frac{\partial\Phi}{\partial\omega}=-\pi^2 g_1^2\Phi,$$

and if therefore we put

$$\vartheta_{g_1,g_2}(v) = e^{\frac{\pi i\omega g_1^2}{4} - \pi i g_1 v + \frac{g_1 g_2 \pi i}{2}}\,\vartheta\left(v - \frac{g_1\omega - g_2}{2}\right), \tag{1}$$

then the differential equation (4) of § 20 is satisfied by all these functions. Thus the functions ϑ_{g_1,g_2} are determined up to one numerical factor common to all.

From formula (1) a more general formula can be derived by replacing v by

$$v - \frac{g_1'\omega - g_2'}{2}.$$

One obtains

$$\vartheta_{g_1,g_2}\left(v - \frac{g_1'\omega - g_2'}{2}\right) = e^{-\frac{\pi i\omega}{4}g_1'^2 + \pi i g_1' v - \pi i g_1 g_2' - \frac{\pi i}{2}g_1'(g_2+g_2')}\vartheta_{g_1+g_1',g_2+g_2'}(v). \tag{2}$$

From (2) there also follows, using (1) of § 20, if one sets $g_1' = 2, g_2' = 0$, or $g_1' = 0, g_2' = 2$:

$$\vartheta_{g_1+2,g_2}(v) = e^{2\pi i g_2}\vartheta_{g_1,g_2}(v), \qquad \vartheta_{g_1,g_2+2}(v) = e^{\pi i g_1}\vartheta_{g_1,g_2}(v), \tag{3}$$

and by (2) at the same time the periodicity of the functions $\vartheta_{g_1,g_2}(v)$ is completely expressed. For example, if μ and ν are integers, one obtains

$$\vartheta_{00}\left(v - \frac{(2\nu-1)\omega + 2\mu - 1}{2}\right) = (-1)^{\nu+1} i\, e^{-\frac{\pi i\omega}{4}(2\nu-1)^2} e^{\pi i(2\nu-1)v}\vartheta_{11}(v), \tag{4}$$

$$\vartheta_{11}(v - \nu\omega - \mu) = (-1)^{\mu+\nu} e^{-\pi i\omega\nu^2} e^{2\pi i\nu v}\vartheta_{11}(v). \tag{5}$$

Formula (2) is valid for quite arbitrary, even complex, values of g_1, g_2.

If in the defining formulae of the ϑ-functions § 20, (1) one replaces v by $-v-1$ and by $-v-\omega$, and also replaces g_1, g_2 by $-g_1, -g_2$, then it is seen that $\vartheta_{-g_1,-g_2}(-v)$ satisfies the same conditions as $\vartheta_{g_1,g_2}(v)$, and hence that

$$\vartheta_{g_1,g_2}(v), \qquad \vartheta_{-g_1,-g_2}(-v)$$

are identical up to a constant factor; in particular, as follows from $v = 0$,

$$\vartheta_{0,0}(v) = \vartheta_{0,0}(-v),$$

and then formula (1), when v, g_1, g_2 are replaced by $-v, -g_1, -g_2$, gives

$$\vartheta_{g_1,g_2}(v) = \vartheta_{-g_1,-g_2}(-v). \tag{6}$$

If the elements g_1, g_2 are integers, then (g_1, g_2) is called a main characteristic. There are four essentially different ones, namely

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1),$$

and therefore also four essentially different main ϑ-functions:

$$\vartheta_{00}(v), \qquad \vartheta_{01}(v), \qquad \vartheta_{10}(v), \qquad \vartheta_{11}(v), \tag{7}$$

of which the first is also denoted by $\vartheta(v)$.

In what follows, by characteristics and ϑ-functions only main characteristics and main functions will be meant.

Formula (2), applied to this system of functions, gives the following frequently used table:

$$
\begin{aligned}
\vartheta_{00}(v) &= \vartheta_{01}\left(v+\frac{1}{2}\right) = \varepsilon\,\vartheta_{10}\left(v+\frac{\omega}{2}\right) = \varepsilon\,\vartheta_{11}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{01}(v) &= \vartheta_{00}\left(v+\frac{1}{2}\right) = -i\varepsilon\,\vartheta_{11}\left(v+\frac{\omega}{2}\right) = i\varepsilon\,\vartheta_{10}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{10}(v) &= \vartheta_{11}\left(v+\frac{1}{2}\right) = \varepsilon\,\vartheta_{00}\left(v+\frac{\omega}{2}\right) = \varepsilon\,\vartheta_{01}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{11}(v) &= -\vartheta_{10}\left(v+\frac{1}{2}\right) = -i\varepsilon\,\vartheta_{01}\left(v+\frac{\omega}{2}\right) = -i\varepsilon\,\vartheta_{00}\left(v+\frac{1+\omega}{2}\right),
\end{aligned}
\tag{8}
$$

where, as an abbreviation,

$$\varepsilon = e^{\frac{\pi i\omega}{4}+\pi i v}$$

is put.

By (3), (4),

$$\vartheta_{00}(-v) = \vartheta_{00}(v), \quad \vartheta_{01}(-v) = \vartheta_{01}(v), \quad \vartheta_{10}(-v) = \vartheta_{10}(v), \quad \vartheta_{11}(-v) = -\vartheta_{11}(v), \tag{9}$$

that is, $\vartheta_{00}(v), \vartheta_{01}(v), \vartheta_{10}(v)$ are even functions, while $\vartheta_{11}(v)$ is an odd function.

By § 17, (12), the four functions (7) vanish respectively for the following values of the argument

$$\frac{1+\omega}{2}, \qquad \frac{\omega}{2}, \qquad \frac{1}{2}, \qquad 0,$$

thus for certain half-periods. Of importance are the values which all the functions (7) assume for these argument values, and these can be reduced by means of table (8) to the three

$$\vartheta_{00}(0) = \vartheta_{00}, \qquad \vartheta_{01}(0) = \vartheta_{01}, \qquad \vartheta_{10}(0) = \vartheta_{10}.$$

Thus from (8) one obtains the following system of formulae:

$$
\begin{aligned}
\vartheta_{00} &= \vartheta_{01}\left(\frac{1}{2}\right) = \varepsilon_0\vartheta_{10}\left(\frac{\omega}{2}\right) = \varepsilon_0\vartheta_{11}\left(\frac{1+\omega}{2}\right),\\
\vartheta_{01} &= \vartheta_{00}\left(\frac{1}{2}\right) = -i\varepsilon_0\vartheta_{11}\left(\frac{\omega}{2}\right) = i\varepsilon_0\vartheta_{10}\left(\frac{1+\omega}{2}\right),\\
\vartheta_{10} &= \vartheta_{11}\left(\frac{1}{2}\right) = \varepsilon_0\vartheta_{00}\left(\frac{\omega}{2}\right) = \varepsilon_0\vartheta_{01}\left(\frac{1+\omega}{2}\right),
\end{aligned}
\tag{10}
$$

where

$$\varepsilon_0 = e^{\frac{\pi i\omega}{4}},$$

and

$$\vartheta_{11}(0) = 0, \quad \vartheta_{10}\left(\frac{1}{2}\right) = 0, \quad \vartheta_{01}\left(\frac{\omega}{2}\right) = 0, \quad \vartheta_{00}\left(\frac{1+\omega}{2}\right) = 0. \tag{11}$$

The squares of the functions (7),

$$\vartheta_{00}^2(v), \quad \vartheta_{01}^2(v), \quad \vartheta_{10}^2(v), \quad \vartheta_{11}^2(v), \tag{12}$$

are Θ_{00}-functions of the second order, and consequently there are two linear relations between them. If we assume these relations in the form

$$\vartheta_{10}^2(v) = A\vartheta_{01}^2(v) + B\vartheta_{11}^2(v), \qquad \vartheta_{00}^2(v) = A'\vartheta_{01}^2(v) + B'\vartheta_{11}^2(v),$$

then the coefficients can be easily determined on the basis of formulae (10), (11), if one sets $v = 0$ and $v = \frac{\omega}{2}$. Thus one obtains

$$\vartheta_{01}^2\vartheta_{10}^2(v) = \vartheta_{10}^2\vartheta_{01}^2(v) - \vartheta_{00}^2\vartheta_{11}^2(v), \qquad \vartheta_{01}^2\vartheta_{00}^2(v) = \vartheta_{00}^2\vartheta_{01}^2(v) - \vartheta_{10}^2\vartheta_{11}^2(v), \tag{13}$$

and from this also, putting $v = \frac{1}{2}$,

$$\vartheta_{00}^4 = \vartheta_{01}^4 + \vartheta_{10}^4. \tag{14}$$

By any two of the squares (12) all functions Θ_{00} of the second order can be represented linearly; but the remaining Θ-functions of the second order can also be composed from the functions ϑ_{g_1,g_2}, for for each characteristic one obtains two linearly independent products, of which one is an even and the other an odd function, namely:

$$\begin{array}{lll} \vartheta_{00}(v)\vartheta_{01}(v), \quad \vartheta_{10}(v)\vartheta_{11}(v) & & \text{characteristic } (0,1), \\ \vartheta_{00}(v)\vartheta_{10}(v), \quad \vartheta_{01}(v)\vartheta_{11}(v) & & (1,0), \\ \vartheta_{10}(v)\vartheta_{01}(v), \quad \vartheta_{00}(v)\vartheta_{11}(v) & & (1,1). \end{array} \tag{15}$$

By the same principle all Θ-functions of arbitrary order and arbitrary main characteristic can now be formed from the functions ϑ. To prove this, we denote by Θ_0, Θ_1 any two of the ϑ-squares (12), and by $F^{(n)}(\Theta_0, \Theta_1)$ an entire rational homogeneous function of nth order of the two arguments Θ_0, Θ_1. We obtain hereafter for the Θ-functions of mth order $\Theta^{(m)}(v)$ the following expressions.

For even m:

even functions:

$$\Theta_{00}^{(m)}(v) = F^{(m/2)}(\Theta_0, \Theta_1), \qquad \Theta_{01}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$

$$\Theta_{10}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)F^{(m/2-1)}(\Theta_0, \Theta_1), \qquad \Theta_{11}^{(m)}(v) = \vartheta_{10}(v)\vartheta_{01}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$

odd functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)\vartheta_{01}(v)\vartheta_{11}(v)F^{((m-4)/2)}(\Theta_0, \Theta_1),$$

$$\Theta_{01}^{(m)}(v) = \vartheta_{10}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1), \qquad \Theta_{10}^{(m)}(v) = \vartheta_{01}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$

$$\Theta_{11}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1). \tag{16}$$

For odd m:

even functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{00}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1), \qquad \Theta_{01}^{(m)}(v) = \vartheta_{01}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1),$$

$$\Theta_{10}^{(m)}(v) = \vartheta_{10}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1), \qquad \Theta_{11}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)\vartheta_{10}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$

odd functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{01}(v)\vartheta_{10}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$

$$\Theta_{01}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$

$$\Theta_{10}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1), \quad \Theta_{11}^{(m)}(v) = \vartheta_{11}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1). \tag{17}$$

That all Θ-functions are representable in this form follows, on the basis of § 20, from the following three considerations.

1. Between even and odd functions no linear dependence can exist unless a linear dependence already exists between the even functions alone or between the odd functions alone.

2. Since two of the ϑ-squares (12) do not stand in a constant ratio, there also exists no equation of the form

$$F^{(n)}(\Theta_0, \Theta_1) = 0.$$

3. The total number of constants which occur, according to (16), (17), in the even and odd functions belonging to one and the same characteristic is exactly equal to the order m.

§ 22. The addition theorem.

If u, v are two variables, then the products

$$\vartheta_{g_1,g_2}(u+v)\vartheta_{g'_1,g'_2}(u-v)$$

belong to the Θ-functions of the second order, with characteristic

$$(g_1+g'_1,\, g_2+g'_2),$$

and indeed for each of the two variables u, v; regarded as functions of v they are therefore linearly representable by the functions (12) and (15) of § 21, so that the variable u occurs in the coefficients. When the characteristic is $(0,0)$ this representation is possible in several ways, since one can choose two arbitrary functions from (12); in the other cases it is possible in only one way. One obtains in this manner sixteen formulae, of which, however, six can be derived from the others by mere interchange of v with $-v$, so that only ten essentially different formulae remain. These formulae are very easily derived by first writing them with undetermined coefficients and then determining these coefficients by putting for v such special values for which one of the functions $\vartheta(v)$ vanishes. Thus, for example,

$$\vartheta_{00}(u+v)\vartheta_{00}(u-v) = A\vartheta_{00}^2(v) + B\vartheta_{11}^2(v),$$

and if one puts $v = 0$ and $v = (1+\omega)/2$, one obtains, using the formulae of § 21,

$$A\vartheta_{00}^2 = \vartheta_{00}^2(u), \qquad B\vartheta_{00}^2 = \vartheta_{11}^2(u),$$

and hence

$$\vartheta_{00}^2\vartheta_{00}(u+v)\vartheta_{00}(u-v) = \vartheta_{00}^2(u)\vartheta_{00}^2(v) + \vartheta_{11}^2(u)\vartheta_{11}^2(v). \tag{1}$$

If one replaces u in this by

$$u+\frac{1}{2}, \qquad u+\frac{\omega}{2}, \qquad u+\frac{1+\omega}{2},$$

one obtains three further relations, which could also have been derived directly in the same way as (1):

$$\vartheta_{11}^2\vartheta_{01}(u+v)\vartheta_{01}(u-v) = \vartheta_{01}^2(u)\vartheta_{01}^2(v) - \vartheta_{11}^2(u)\vartheta_{11}^2(v), \tag{2}$$
$$\vartheta_{11}^2\vartheta_{10}(u+v)\vartheta_{10}(u-v) = \vartheta_{10}^2(u)\vartheta_{10}^2(v) - \vartheta_{11}^2(u)\vartheta_{11}^2(v), \tag{3}$$
$$\vartheta_{01}^2\vartheta_{11}(u+v)\vartheta_{11}(u-v) = \vartheta_{11}^2(u)\vartheta_{01}^2(v) - \vartheta_{01}^2(u)\vartheta_{11}^2(v). \tag{4}$$

These four equations can, as already mentioned, be transformed in many ways by applying the formulae (13) of the preceding paragraph. The following six formulae have only one form. It is, to begin again with an arbitrary example,

$$\vartheta_{00}(u+v)\vartheta_{11}(u-v) = A\vartheta_{01}(v)\vartheta_{10}(v) + B\vartheta_{00}(v)\vartheta_{11}(v),$$

and if one puts $v = 0$,

$$A\vartheta_{01}\vartheta_{10} = \vartheta_{00}(u)\vartheta_{11}(u);$$

from this one obtains B by interchanging u with v. Thus the three following formulae are derived:

$$\vartheta_{01}\vartheta_{10}\vartheta_{00}(u+v)\vartheta_{11}(u-v) = \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v) - \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v), \tag{5}$$
$$\vartheta_{00}\vartheta_{01}\vartheta_{10}(u+v)\vartheta_{11}(u-v) = \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(v)\vartheta_{11}(v), \tag{6}$$
$$\vartheta_{00}\vartheta_{10}\vartheta_{01}(u+v)\vartheta_{11}(u-v) = \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{10}(v) - \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{11}(v), \tag{7}$$

which are verified immediately by putting first u, then v, equal to zero. From these the three others follow by increasing u by

$$\frac{1}{2}, \qquad \frac{\omega}{2}, \qquad \frac{1}{2}:$$

$$\vartheta_{01}\vartheta_{10}\vartheta_{01}(u+v)\vartheta_{10}(u-v) = \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{10}(v) + \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{11}(v), \tag{8}$$
$$\vartheta_{00}\vartheta_{01}\vartheta_{00}(u+v)\vartheta_{01}(u-v) = \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{10}(v)\vartheta_{11}(v), \tag{9}$$
$$\vartheta_{00}\vartheta_{10}\vartheta_{00}(u+v)\vartheta_{10}(u-v) = \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{10}(v) + \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{11}(v). \tag{10}$$

The formulae whose totality is designated by the name of the addition theorem can be generalized in many ways; the following may serve as an example.

If u, v are two variables, then the four products

$$\vartheta_{00}(u)\vartheta_{00}(u+v), \quad \vartheta_{01}(u)\vartheta_{01}(u+v), \quad \vartheta_{10}(u)\vartheta_{10}(u+v), \quad \vartheta_{11}(u)\vartheta_{11}(u+v),$$

regarded as functions of u, are related T-functions of the second order with the same periodicity factors, and hence any three of them are linearly dependent. Thus

$$A\vartheta_{00}(u)\vartheta_{00}(u+v) + B\vartheta_{10}(u)\vartheta_{10}(u+v) + C\vartheta_{11}(u)\vartheta_{11}(u+v) = 0,$$

from which, for $u = 0$, $u = \frac{1}{2}$,

$$A\vartheta_{00}\vartheta_{00}(v) + B\vartheta_{10}\vartheta_{10}(v) = 0, \qquad A\vartheta_{01}\vartheta_{01}(v) + C\vartheta_{10}\vartheta_{10}(v) = 0,$$

and hence

$$\vartheta_{10}\vartheta_{10}(v)\vartheta_{00}(u)\vartheta_{00}(u+v) - \vartheta_{00}\vartheta_{00}(v)\vartheta_{10}(u)\vartheta_{10}(u+v) - \vartheta_{01}\vartheta_{01}(v)\vartheta_{11}(u)\vartheta_{11}(u+v) = 0. \tag{11}$$

Likewise, if w is a third variable, the products contained in the form

$$\vartheta_{g_1,g_2}(u+v)\vartheta_{g_1,g_2}(u+w), \qquad \vartheta_{g_1,g_2}(u)\vartheta_{g_1,g_2}(u+v+w),$$

regarded as functions of u, are related T-functions of the second order with the same periodicity factors, so that a linear dependence exists between any three among them. The coefficients are determined, as above, by special values of u. Thus follows the system of formulae:

$$\begin{aligned}
\vartheta_{00}\vartheta_{00}(v+w)\vartheta_{00}(u+v)\vartheta_{00}(u+w) &= \vartheta_{00}(u)\vartheta_{00}(v)\vartheta_{00}(w)\vartheta_{00}(u+v+w) \\
&\quad - \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w), && (12)\\
\vartheta_{01}\vartheta_{01}(v+w)\vartheta_{01}(u+v)\vartheta_{01}(u+w) &= \vartheta_{01}(u)\vartheta_{01}(v)\vartheta_{01}(w)\vartheta_{01}(u+v+w) \\
&\quad + \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w), && (13)\\
\vartheta_{10}\vartheta_{10}(v+w)\vartheta_{10}(u+v)\vartheta_{10}(u+w) &= \vartheta_{10}(u)\vartheta_{10}(v)\vartheta_{10}(w)\vartheta_{10}(u+v+w) \\
&\quad + \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w). && (14)
\end{aligned}$$

All these formulae can be regarded as special cases of a general formula which Jacobi derived from the series expansions and used for the foundation of the theory of elliptic functions.[3] This formula can also be obtained from the concept of theta functions as follows.

The four functions

$$\vartheta_{00}(2v), \quad \vartheta_{01}(2v), \quad \vartheta_{10}(2v), \quad \vartheta_{11}(2v) \tag{15}$$

are Θ_{00}-functions of the fourth order in v, and since they are linearly independent, all Θ_{00}-functions of the fourth order can be expressed linearly by them. But the product

$$\vartheta(v+a_1)\vartheta(v+a_2)\vartheta(v+a_3)\vartheta(v+a_4) \tag{16}$$

is also such a function, provided that the quantities a satisfy the condition

$$a_1 + a_2 + a_3 + a_4 = 0. \tag{17}$$

[3] Theory of elliptic functions, derived from the properties of theta series. Jacobi's collected works, vol. I, p. 497.

Thus, denoting by A_1, A_2, A_3, A_4 constants with respect to v, we obtain

$$\vartheta(v+a_1)\vartheta(v+a_2)\vartheta(v+a_3)\vartheta(v+a_4) = A_1\vartheta_{00}(2v) + A_2\vartheta_{01}(2v) + A_3\vartheta_{10}(2v) + A_4\vartheta_{11}(2v).$$

From this, by increasing v by $\frac{1}{2}(g_1\omega + g_2)$, one obtains four formulae:

$$\begin{gathered}\vartheta_{g_1,g_2}(v+a_1)\vartheta_{g_1,g_2}(v+a_2)\vartheta_{g_1,g_2}(v+a_3)\vartheta_{g_1,g_2}(v+a_4)\\ = A_1\vartheta_{00}(2v) + (-1)^{g_2}A_2\vartheta_{01}(2v) + (-1)^{g_1}A_3\vartheta_{10}(2v) + (-1)^{g_1+g_2}A_4\vartheta_{11}(2v).\end{gathered}$$

If one adds these four formulae, then

$$4A_1\vartheta_{00}(2v) = \sum_{g_1,g_2} \vartheta_{g_1,g_2}(v+a_1)\vartheta_{g_1,g_2}(v+a_2)\vartheta_{g_1,g_2}(v+a_3)\vartheta_{g_1,g_2}(v+a_4), \tag{18}$$

where in the sum all combinations of 0 and 1 are to be put for g_1 and g_2.

We now introduce a slightly different notation by putting

$$v+a_1 = v_1', \qquad v+a_2 = v_2', \qquad v+a_3 = v_3', \qquad v+a_4 = v_4',$$

so that, because of (17),

$$4v = v_1' + v_2' + v_3' + v_4',$$

and now define the four quantities v_1, v_2, v_3, v_4 by the equations

$$\begin{aligned} v_1 &= \frac{1}{2}(v_1'+v_2'+v_3'+v_4'), \quad v_2 = \frac{1}{2}(v_1'+v_2'-v_3'-v_4'),\\ v_3 &= \frac{1}{2}(v_1'-v_2'+v_3'-v_4'), \quad v_4 = \frac{1}{2}(v_1'-v_2'-v_3'+v_4'). \end{aligned} \tag{19}$$

From this one obtains conversely

$$\begin{aligned} v_1' &= \tfrac{1}{2}(v_1+v_2+v_3+v_4), \quad a_1 = \tfrac{1}{2}(v_2+v_3+v_4),\\ v_2' &= \tfrac{1}{2}(v_1+v_2-v_3-v_4), \quad a_2 = \tfrac{1}{2}(v_2-v_3-v_4),\\ v_3' &= \tfrac{1}{2}(v_1-v_2+v_3-v_4), \quad a_3 = \tfrac{1}{2}(-v_2+v_3-v_4),\\ v_4' &= \tfrac{1}{2}(v_1-v_2-v_3+v_4), \quad a_4 = \tfrac{1}{2}(-v_2-v_3+v_4), \end{aligned} \tag{20}$$

and thereafter formula (18) gives

$$4A_1\vartheta_{00}(v_1) = \sum_{g_1,g_2} \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4'),$$

where now either the v_i or the v_i' can be regarded as independent variables.

If the v_i are regarded as independent variables, then in this formula A_1 is independent of v_1, but still depends on v_2, v_3, v_4. Thus put

$$4A_1 = c\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4).$$

Then

$$c\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4) = \sum \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4'),$$

and here c is in any case independent of v_1. But since the right hand side of this formula remains unchanged under arbitrary interchanges of v_1, v_2, v_3, v_4, c is independent of all v_i; and its value is obtained by putting all $v_i = 0$:

$$c\vartheta_{00}^4 = \vartheta_{00}^4 + \vartheta_{01}^4 + \vartheta_{10}^4,$$

whence by § 21, (14), $c = 2$ follows. Thus we have the formula

$$2\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4) = \sum_{g_1,g_2} \vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4').$$

If in it each of the variables v_1, v_2, v_3, v_4 is increased by the half-period $-\frac{1}{2}(g_1'\omega - g_2')$, then v_1' is increased by a whole period $-(g_1'\omega - g_2')$, while v_2', v_3', v_4' remain unchanged. Accordingly, with the aid of § 21, (2) and (3), one obtains a system of four formulae:

$$2\vartheta_{g_1',g_2'}(v_1)\vartheta_{g_1',g_2'}(v_2)\vartheta_{g_1',g_2'}(v_3)\vartheta_{g_1',g_2'}(v_4) = \sum_{g_1,g_2}(-1)^{g_1g_2'+g_2g_1'+g_1'g_2'}\vartheta_{g_1,g_2}(v_1')\vartheta_{g_1,g_2}(v_2')\vartheta_{g_1,g_2}(v_3')\vartheta_{g_1,g_2}(v_4'). \tag{21}$$

These are Jacobi's formulae.

If, for example, one puts

$$v_1 = 0, \quad v_2 = v + w, \quad v_3 = u + v, \quad v_4 = u + w,$$

$$v_1' = u + v + w, \quad v_2' = -u, \quad v_3' = -w, \quad v_4' = -v,$$

then from (21), for $g_1', g_2' = 1, 1$,

$$0 = \sum_{g_1,g_2}(-1)^{g_1+g_2}\vartheta_{g_1,g_2}(u)\vartheta_{g_1,g_2}(v)\vartheta_{g_1,g_2}(w)\vartheta_{g_1,g_2}(u+v+w),$$

and from this one obtains, by putting in (21) the three other characteristics for g_1', g_2', the formulae (12), (13), (14).

§ 23. The derivatives of the ϑ-functions.

In what follows we denote by $\vartheta_{g_1,g_2}'(v)$, $\vartheta_{g_1,g_2}''(v)$ the derivatives, taken with respect to v, of the function $\vartheta_{g_1,g_2}(v)$, and by ϑ_{g_1,g_2}', ϑ_{g_1,g_2}'' the values of this function for $v = 0$. Then $\vartheta_{01}', \vartheta_{10}', \vartheta_{00}'$ vanish, because $\vartheta_{01}(v), \vartheta_{10}(v), \vartheta_{00}(v)$ are even functions of v, and likewise ϑ_{11} vanishes.

The six functions

$$\vartheta_{g_1,g_2}(v)\vartheta_{g_1',g_2'}'(v) - \vartheta_{g_1',g_2'}(v)\vartheta_{g_1,g_2}'(v)$$

are, as follows from the fundamental equations § 20, (1), ϑ-functions of the second order with characteristic

$$(g_1 + g_1',\ g_2 + g_2'),$$

and at the same time either even or odd functions; accordingly they can be represented by ϑ-functions according to § 21. A constant factor is determined by a special value of v, namely $v = 0$. Thus

$$\vartheta_{11}'(v)\vartheta_{01}(v) - \vartheta_{01}'(v)\vartheta_{11}(v) = A\vartheta_{10}(v)\vartheta_{00}(v), \tag{1}$$

$$A = \frac{\vartheta_{11}'\vartheta_{01}}{\vartheta_{10}\vartheta_{00}}. \tag{2}$$

But this expression for A can still be simplified. If we differentiate (1) twice with respect to v and then put $v = 0$, we get

$$\vartheta_{11}'''\vartheta_{01} - \vartheta_{01}''\vartheta_{11}' = A(\vartheta_{10}''\vartheta_{00} + \vartheta_{10}\vartheta_{00}''),$$

and if for A we insert the value (2),

$$\frac{\vartheta_{11}'''}{\vartheta_{11}'} = \frac{\vartheta_{01}''}{\vartheta_{01}} + \frac{\vartheta_{10}''}{\vartheta_{10}} + \frac{\vartheta_{00}''}{\vartheta_{00}}. \tag{3}$$

According to § 20, (4), however, the four functions $\vartheta_{11}, \vartheta_{01}, \vartheta_{10}, \vartheta_{00}$ satisfy the differential equation

$$\vartheta'' = 4\pi i\frac{\partial\vartheta}{\partial\omega}, \tag{4}$$

and accordingly (3) can be written

$$\frac{d\log\vartheta_{11}'}{d\omega} = \frac{d\log\vartheta_{00}\vartheta_{10}\vartheta_{01}}{d\omega},$$

or, by integration,

$$\vartheta'_{11} = c\vartheta_{00}\vartheta_{10}\vartheta_{01},$$

where c is independent of ω. By § 20, 21 the ϑ-functions were determined up to a factor independent of v, ω and common to all of them. This factor is now to be chosen so that the constant c obtains the value π; thus

$$\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{10}\vartheta_{01}, \tag{5}$$

and thereby the ϑ-functions are now completely defined, up to the common sign $\pm$. By application of (5), formula (1) assumes the form

$$\vartheta'_{11}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{11}(v) = \pi\vartheta_{01}^2\vartheta_{10}(v)\vartheta_{00}(v), \tag{6}$$

and similarly, using § 21, (8), one obtains

$$\vartheta'_{11}(v)\vartheta_{00}(v) - \vartheta'_{00}(v)\vartheta_{11}(v) = \pi\vartheta_{00}^2\vartheta_{10}(v)\vartheta_{01}(v), \tag{7}$$

$$\vartheta'_{11}(v)\vartheta_{10}(v) - \vartheta'_{10}(v)\vartheta_{11}(v) = \pi\vartheta_{10}^2\vartheta_{01}(v)\vartheta_{00}(v), \tag{8}$$

$$\vartheta'_{10}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{10}(v) = -\pi\vartheta_{00}^2\vartheta_{00}(v)\vartheta_{11}(v), \tag{9}$$

$$\vartheta'_{00}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{00}(v) = -\pi\vartheta_{10}^2\vartheta_{10}(v)\vartheta_{11}(v), \tag{10}$$

$$\vartheta'_{00}(v)\vartheta_{10}(v) - \vartheta'_{10}(v)\vartheta_{00}(v) = \pi\vartheta_{01}^2\vartheta_{01}(v)\vartheta_{11}(v). \tag{11}$$

If the equations (9), (10), (11) are differentiated with respect to v and then $v = 0$ is put, we obtain by means of (4) and (5) the relations

$$4\frac{d}{d\omega}\log\frac{\vartheta_{10}}{\vartheta_{01}} = i\pi\vartheta_{00}^4, \tag{12}$$

$$4\frac{d}{d\omega}\log\frac{\vartheta_{00}}{\vartheta_{01}} = i\pi\vartheta_{10}^4, \tag{13}$$

$$4\frac{d}{d\omega}\log\frac{\vartheta_{10}}{\vartheta_{00}} = i\pi\vartheta_{01}^4. \tag{14}$$

If the fundamental formulae for the ϑ-functions are differentiated logarithmically twice, one sees that the functions

$$\vartheta^2(v)\frac{d^2\log\vartheta(v)}{dv^2} = \vartheta''(v)\vartheta(v) - \vartheta'(v)\vartheta'(v)$$

are Θ-functions of the second order with characteristic $(0, 0)$, and can therefore be expressed linearly by the functions $\vartheta^2(v)$. In this way one obtains

$$\vartheta_{00}^2\vartheta_{00}^2(v)\frac{d^2\log\vartheta_{00}(v)}{dv^2} = \vartheta_{00}\vartheta''_{00}\vartheta_{00}^2(v) + \vartheta'^2_{11}\vartheta_{11}^2(v), \tag{15}$$

$$\vartheta_{01}^2\vartheta_{01}^2(v)\frac{d^2\log\vartheta_{01}(v)}{dv^2} = \vartheta_{01}\vartheta''_{01}\vartheta_{01}^2(v) - \vartheta'^2_{11}\vartheta_{11}^2(v), \tag{16}$$

$$\vartheta_{10}^2\vartheta_{10}^2(v)\frac{d^2\log\vartheta_{10}(v)}{dv^2} = \vartheta_{10}\vartheta''_{10}\vartheta_{10}^2(v) - \vartheta'^2_{11}\vartheta_{11}^2(v), \tag{17}$$

$$\vartheta_{00}^2\vartheta_{11}^2(v)\frac{d^2\log\vartheta_{11}(v)}{dv^2} = \vartheta_{00}\vartheta''_{00}\vartheta_{11}^2(v) - \vartheta'^2_{11}\vartheta_{00}^2(v). \tag{18}$$

From this further relations can be derived by continued differentiation. We mention one more of these formulae, which results when one differentiates (15) twice more with respect to v and then puts $v = 0$. Expressing the differentiations with respect to v by such with respect to ω, by means of the partial differential equation (4), one obtains

$$\frac{d^2\log\vartheta_{00}}{d\omega^2} - 2\left(\frac{d\log\vartheta_{00}}{d\omega}\right)^2 = -\frac{\pi^2}{8}\vartheta_{10}^4\vartheta_{01}^4,$$

or

$$\frac{d}{d\omega}\left(\frac{1}{\vartheta_{00}^4}\frac{d\vartheta_{00}^2}{d\omega}\right) = -\frac{\pi^2}{4}\frac{\vartheta_{10}^4\vartheta_{01}^4}{\vartheta_{00}^2}. \tag{19}$$

§ 24. Representation of the ϑ-functions by infinite products.

The theory of the ϑ-functions has now advanced far enough that their representation is obtained very easily. Thereby the existence of these functions is proved, and the previous considerations obtain their secure ground. Two paths to this goal stand open to us. The first starts from the zeros of the ϑ-functions already known to us and composes infinite products from them.

By § 21, the function $\vartheta_{00}(v)$ vanishes when

$$2v = (2\nu - 1)\omega + (2\mu - 1)$$

where ν, μ are integers, or when

$$e^{\pm 2\pi i v} = -e^{\pi i \omega(2\nu-1)},$$

where we may now restrict ν to positive numbers. Thus, for abbreviation, put

$$e^{\pi i \omega} = q;$$

then q is a quantity whose absolute value is a proper fraction. The convergent infinite product

$$P(v) = \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1} e^{2\pi i v})(1 + q^{2\nu-1} e^{-2\pi i v})$$

therefore vanishes in all, and only in, those points in which $\vartheta_{00}(v)$ vanishes. Moreover,

$$P(v+1) = P(v),$$

and

$$P(v+\omega) = \frac{1 + q^{-1} e^{-2\pi i v}}{1 + q e^{2\pi i v}} P(v) = q^{-1} e^{-2\pi i v} P(v).$$

But this is, by § 20, (1), the periodicity property of the function $\vartheta_{00}(v)$, and therefore

$$\vartheta_{00}(v) = Q \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1} e^{2\pi i v})(1 + q^{2\nu-1} e^{-2\pi i v}), \tag{1}$$

where Q is a factor independent of v. From this, by the formulae (8) of § 21, replacing v by

$$v + \frac{1}{2}, \qquad v + \frac{\omega}{2}, \qquad v + \frac{1+\omega}{2},$$

one obtains

$$\vartheta_{01}(v) = Q \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1} e^{2\pi i v})(1 - q^{2\nu-1} e^{-2\pi i v}), \tag{2}$$

$$\vartheta_{10}(v) = Q q^{1/4} (e^{\pi i v} + e^{-\pi i v}) \prod_{\nu=1}^{\infty} (1 + q^{2\nu} e^{2\pi i v})(1 + q^{2\nu} e^{-2\pi i v}), \tag{3}$$

$$\vartheta_{11}(v) = -i Q q^{1/4} (e^{\pi i v} - e^{-\pi i v}) \prod_{\nu=1}^{\infty} (1 - q^{2\nu} e^{2\pi i v})(1 - q^{2\nu} e^{-2\pi i v}). \tag{4}$$

If in these formulae one puts $v = 0$, in the last after differentiating once, then

$$\left.\begin{aligned}
\vartheta_{00} &= Q \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1})^2, \\
\vartheta_{01} &= Q \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1})^2, \\
\vartheta_{10} &= 2Q q^{1/4} \prod_{\nu=1}^{\infty} (1 + q^{2\nu})^2, \\
\vartheta'_{11} &= 2\pi Q q^{1/4} \prod_{\nu=1}^{\infty} (1 - q^{2\nu})^2.
\end{aligned}\right\} \tag{5}$$

By this, using § 23, (5),

$$\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{01}\vartheta_{10},$$

the factor Q can be determined. First one obtains

$$1 = Q^2 \prod_{\nu=1}^{\infty} \frac{(1-q^{2\nu-1})^2(1+q^{2\nu-1})^2(1+q^{2\nu})^2}{(1-q^{2\nu})^2},$$

or, disposing of the still undetermined sign of Q, and thereby of the signs of the ϑ-functions,

$$Q = \prod_{\nu=1}^{\infty} \frac{1-q^{2\nu}}{(1+q^{2\nu})(1+q^{2\nu-1})(1-q^{2\nu-1})}.$$

In the denominator one can put, in place of $\prod(1+q^{2\nu})(1+q^{2\nu-1})$, the product $\prod(1+q^{\nu})$, and the numerator $\prod(1-q^{2\nu})$ can be decomposed as

$$\prod(1-q^{\nu})(1+q^{\nu}) = \prod(1-q^{2\nu})(1-q^{2\nu-1})(1+q^{\nu}).$$

Thus finally

$$Q = \prod_{\nu=1}^{\infty}(1-q^{2\nu}). \tag{6}$$

The expressions (1), (2), (3), (4) can be represented in real form when the multiplication of the conjugate imaginary factors is carried out. One obtains

$$\begin{aligned}
\vartheta_{00}(v) &= \prod_{\nu=1}^{\infty}(1-q^{2\nu})(1+2q^{2\nu-1}\cos 2\pi v + q^{4\nu-2}),\\
\vartheta_{01}(v) &= \prod_{\nu=1}^{\infty}(1-q^{2\nu})(1-2q^{2\nu-1}\cos 2\pi v + q^{4\nu-2}),\\
\vartheta_{10}(v) &= 2q^{1/4}\cos\pi v \prod_{\nu=1}^{\infty}(1-q^{2\nu})(1+2q^{2\nu}\cos 2\pi v + q^{4\nu}),\\
\vartheta_{11}(v) &= 2q^{1/4}\sin\pi v \prod_{\nu=1}^{\infty}(1-q^{2\nu})(1-2q^{2\nu}\cos 2\pi v + q^{4\nu}).
\end{aligned} \tag{7}$$

If one introduces the expression (6) into the last equation (5), then

$$\vartheta'_{11} = 2\pi q^{1/4} \prod_{\nu=1}^{\infty}(1-q^{2\nu})^3,$$

and if therefore

$$\eta(\omega) = q^{1/12} \prod_{\nu=1}^{\infty}(1-q^{2\nu}), \tag{8}$$

then

$$\vartheta'_{11} = 2\pi\eta(\omega)^3. \tag{9}$$

Eliminating Q from (5) with the aid of the relation

$$Q = q^{-1/12}\eta(\omega),$$

we put

$$\vartheta_{00} = \eta(\omega)f(\omega)^2, \qquad \vartheta_{01} = \eta(\omega)f_1(\omega)^2, \qquad \vartheta_{10} = \eta(\omega)f_2(\omega)^2, \tag{10}$$

where the functions $f(\omega), f_1(\omega), f_2(\omega)$ are defined as follows:

$$\begin{aligned} f(\omega) &= q^{-1/24} \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1}), \\ f_1(\omega) &= q^{-1/24} \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1}), \\ f_2(\omega) &= \sqrt{2}\, q^{1/12} \prod_{\nu=1}^{\infty} (1 + q^{2\nu}). \end{aligned} \tag{11}$$

The functions $\eta(\omega), f(\omega), f_1(\omega), f_2(\omega)$ will play an important role in our later considerations. From § 21, (14), by (10), one gets the relation

$$f(\omega)^8 = f_1(\omega)^8 + f_2(\omega)^8, \tag{12}$$

and from § 23, (5), by (9),

$$f(\omega) f_1(\omega) f_2(\omega) = \sqrt{2}. \tag{13}$$

The last formula also follows from (11) by use of the identical relation

$$\prod (1 + q^\nu)(1 - q^{2\nu-1}) = \prod \frac{(1 + q^\nu)(1 - q^\nu)}{1 - q^{2\nu}} = 1.$$

§ 25. Representation of the ϑ-functions by infinite series.

The second way of arriving at the representation of the ϑ-functions consists in seeking to form a convergent infinite series satisfying the fundamental equations.

First observe that, because of the condition

$$\vartheta_{00}(v + 1) = \vartheta_{00}(v),$$

the function $\vartheta_{00}(v)$ can be regarded as a single-valued function of $e^{2\pi i v}$, and put accordingly

$$\vartheta_{00}(v) = \sum_{\nu=-\infty}^{\infty} A_\nu e^{2\pi i \nu v}.$$

Then the differential equation § 20, (4) gives for A_ν the condition

$$\frac{dA_\nu}{d\omega} = \pi i \nu^2 A_\nu, \qquad A_\nu = c_\nu e^{\pi i \omega \nu^2} = c_\nu q^{\nu^2},$$

where c_ν is constant with respect to ω. Thus

$$\vartheta_{00}(v) = \sum c_\nu q^{\nu^2} e^{2\pi i \nu v},$$

and hence

$$\vartheta_{00}(v + \omega) = \sum c_\nu q^{\nu^2 + 2\nu} e^{2\pi i \nu v} = q^{-1} e^{-2\pi i v} \sum c_\nu q^{(\nu+1)^2} e^{2\pi i (\nu+1) v}.$$

Since ν runs from $-\infty$ to $+\infty$, it is permissible in this formula to put $\nu - 1$ in place of ν; doing this gives

$$\vartheta_{00}(v + \omega) = q^{-1} e^{-2\pi i v} \sum c_{\nu-1} q^{\nu^2} e^{2\pi i \nu v}.$$

According to the second of the fundamental equations § 20, (1), the two series

$$\sum c_\nu q^{\nu^2} e^{2\pi i \nu v} \qquad \text{and} \qquad \sum c_{\nu-1} q^{\nu^2} e^{2\pi i \nu v}$$

must therefore agree; that is,

$$c_{\nu-1} = c_\nu.$$

All the c_ν are consequently equal to one another; that they have the value 1 follows by comparing the two developments

$$\sum_{\nu=-\infty}^{\infty} q^{\nu^2} e^{2\pi i\nu v}, \qquad \prod_{\nu=1}^{\infty}(1-q^{2\nu})(1+q^{2\nu-1}e^{2\pi iv})(1+q^{2\nu-1}e^{-2\pi iv})$$

for the value $q=0$.

The development thus found for $\vartheta_{00}(v)$ can also be written in both forms:

$$\begin{aligned}\vartheta_{00}(v) &= \sum_{\nu=-\infty}^{\infty} q^{\nu^2} e^{2\pi i\nu v} \\ &= 1 + 2q\cos 2\pi v + 2q^4\cos 4\pi v + 2q^9 \cos 6\pi v + \cdots, \end{aligned} \tag{1}$$

and from it, by § 21, (8), through increasing v by

$$\frac{1}{2}, \qquad \frac{\omega}{2}, \qquad \frac{1+\omega}{2},$$

one obtains the developments for the three remaining ϑ-functions:

$$\begin{aligned}\vartheta_{01}(v) &= \sum_{\nu=-\infty}^{\infty} (-1)^\nu q^{\nu^2} e^{2\pi i\nu v} \\ &= 1 - 2q\cos 2\pi v + 2q^4\cos 4\pi v - 2q^9 \cos 6\pi v + \cdots, \end{aligned} \tag{2}$$

$$\begin{aligned}\vartheta_{10}(v) &= \sum_{\nu=-\infty}^{\infty} q^{(2\nu+1)^2/4} e^{(2\nu+1)\pi iv} \\ &= 2q^{1/4}\cos \pi v + 2q^{9/4}\cos 3\pi v + 2q^{25/4} \cos 5\pi v + \cdots, \end{aligned} \tag{3}$$

$$\begin{aligned}\vartheta_{11}(v) &= -i\sum_{\nu=-\infty}^{\infty} (-1)^\nu q^{(2\nu+1)^2/4} e^{(2\nu+1)\pi iv} \\ &= 2q^{1/4}\sin \pi v - 2q^{9/4}\sin 3\pi v + 2q^{25/4} \sin 5\pi v + \cdots. \end{aligned} \tag{4}$$

For later applications we draw the following conclusions from these developments:

If the imaginary part of ω grows without bound, so that the absolute value of q vanishes, then

$$\vartheta_{00}(v) = 1, \qquad \vartheta_{01}(v) = 1, \qquad q^{-1/4}\vartheta_{10}(v) = 2\cos \pi v,$$
$$q^{-1/4}\vartheta_{11}(v) = 2\sin\pi v.$$

Assume ω purely imaginary, hence q real, positive, and a proper fraction. Then for real v:

1. $\vartheta_{00}(v), \vartheta_{01}(v), \vartheta_{10}(v), \vartheta_{11}(v)$ are real, and if v lies between 0 and 1/2, positive by § 24, (7); consequently also $\vartheta_{00}, \vartheta_{01}, \vartheta_{10}, \vartheta'_{11}$ are positive.

2. $\vartheta_{00}(iv)$, $\vartheta_{01}(iv)$, $\vartheta_{10}(iv)$, $-i\vartheta_{11}(iv)$ are real, and so long as v lies between 0 and $-i\omega/2$, positive.

3. $\vartheta_{00}(\frac{1}{2}+iv)$, $\vartheta_{01}(\frac{1}{2}+iv)$, $i\vartheta_{10}(\frac{1}{2}+iv)$, $\vartheta_{11}(\frac{1}{2}+iv)$ are real, and so long as v lies between 0 and $-i\omega/2$, positive.

4. The functions

$$q^{1/4}e^{\pi iv}\vartheta_{00}\left(\frac{\omega}{2}+v\right), \quad -iq^{1/4}e^{\pi iv}\vartheta_{01}\left(\frac{\omega}{2}+v\right),$$
$$q^{1/4}e^{\pi iv}\vartheta_{10}\left(\frac{\omega}{2}+v\right), \quad -iq^{1/4}e^{\pi iv}\vartheta_{11}\left(\frac{\omega}{2}+v\right)$$

are real, and so long as v lies between 0 and 1/2, positive.

§ 26. Development of ϑ-quotients.

If v denotes a variable and a an arbitrary constant, then the quotients

$$\frac{\vartheta_{g_1,g_2}(v+a)}{\vartheta_{g_1',g_2'}(v)}$$

are doubly periodic functions of the second kind and first order with periodicity factors $(-1)^{g_1+g_1'}$, $(-1)^{g_2+g_2'}e^{-2\pi ia}$. If one appends an exponential factor $e^{\lambda v}$, then λ and a can be so determined that the periodicity factors become arbitrary prescribed quantities. Taking the two main characteristics $(g_1, g_2), (g_1', g_2')$ in all possible ways, one obtains sixteen such functions. We start from one among them, for which, appending a constant factor, we choose

$$F = \frac{\vartheta_{11}'\vartheta_{11}(v+a)}{2\pi i\,\vartheta_{11}(v)\vartheta_{11}(a)}. \tag{1}$$

This is a single-valued function $F(z)$ of the variable

$$z = e^{2\pi iv},$$

and, if as before $q = e^{\pi i\omega}$, it has the periodicity property

$$F(q^2z) = e^{-2\pi ia}F(z), \tag{2}$$

from which, for every integral ν,

$$F(q^{2\nu}z) = e^{-2\pi i\nu a}F(z).$$

The function $F(z)$ becomes infinite for a finite z only when one of the factors $q^{2\nu}z - 1$ vanishes, and in particular

$$(z-1)F(z) = 1 \qquad (z = 1). \tag{3}$$

But by (2)

$$(q^{2\nu}z-1)F(q^{2\nu}z) = e^{-2\pi i\nu a}F(z)(q^{2\nu}z-1),$$

and consequently, by (3),

$$F(z)(q^{2\nu}z-1) = e^{2\pi i\nu a} \qquad (z = q^{-2\nu}). \tag{4}$$

Put, under the supposition of convergence still to be examined more closely,

$$S(z) = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu}z-1}. \tag{5}$$

Then, replacing z by q^2z and ν by $\nu-1$, one obtains

$$S(q^2z) = e^{-2\pi ia}S(z), \tag{6}$$

and the difference $F(z) - S(z)$, considered as a function of v, would therefore by (3) and (4) be an entire doubly periodic function of the second kind. Such a function must, however, by § 16, 6, be an exponential function; hence

$$F(z) = S(z) + Ce^{-2\pi imv},$$

where m is an integer and C a constant. From (2) and (6), however, if $C \neq 0$, it follows that

$$e^{-2\pi im\omega} = e^{-2\pi ia};$$

thus $a = m\omega + n$, where m, n are integers. But if a is a period, $F(z)$ reduces to a constant or an exponential function. Otherwise $C = 0$, and the development follows:

$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{2\pi i\,\vartheta_{11}(v)\vartheta_{11}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi iv}-1}. \tag{7}$$

We have presupposed the convergence of this series for all values of v. To judge it, observe that the general term of this series becomes, for $\nu = +\infty$, equal to $e^{2\pi i\nu a}$, and for $\nu = -\infty$, equal to $e^{-2\pi i\nu a}e^{-2\pi i\nu(\omega - a)}$. Thus convergence will take place when the imaginary parts of a and of $\omega - a$ are positive. Hence, if

$$\omega = \omega' + i\omega'', \qquad a = a' + ia'',$$

the condition of convergence is

$$0 < a'' < \omega''. \tag{8}$$

If in (7) one replaces a by $a + \frac{1}{2}$, then one obtains the development for a second of the sixteen functions:

$$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{2\pi i\,\vartheta_{11}(v)\vartheta_{10}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{(-1)^\nu e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi i v} - 1}. \tag{9}$$

If in (7) and (9) one changes a into $a + \frac{1}{2}\omega$, then

$$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{2\pi i\,\vartheta_{11}(v)\vartheta_{01}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a}q^\nu e^{\pi i v}}{q^{2\nu}e^{2\pi i v} - 1}, \tag{10}$$

$$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{2\pi i\,\vartheta_{11}(v)\vartheta_{00}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{(-1)^\nu e^{2\pi i\nu a}q^\nu e^{\pi i v}}{q^{2\nu}e^{2\pi i v} - 1}, \tag{11}$$

where it is to be noted, however, that in the last two formulae the condition of convergence is changed, namely to

$$-\frac{\omega''}{2} < a'' < \frac{\omega''}{2}.$$

From the four formulae the remaining twelve result if v is replaced by $v + \frac{1}{2}$, $v + \frac{1}{2}\omega$, $v + \frac{1}{2}(1+\omega)$; the regions of convergence are not further changed. In this way the formulae of Table I at the end of the volume are derived.

The series thus obtained do not all converge for real values of a; for example, the series (7), in which the part running toward the side of positive ν,

$$\sum_{\nu=1}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi i v} - 1}, \tag{12}$$

ceases to converge when the imaginary part of a becomes equal to zero, while the other part remains convergent even then. But one obtains from (12) an expression which still converges for a real a if one adds the sum

$$\sum_{\nu=1}^{\infty} e^{-2\pi i\nu a} = -\frac{e^{-2\pi i a}}{e^{-2\pi i a} - 1}. \tag{13}$$

It thereby goes over into

$$\sum_{\nu=1}^{\infty} \frac{e^{2\pi i\nu a}q^{2\nu}e^{2\pi i v}}{q^{2\nu}e^{2\pi i v} - 1},$$

and this series remains convergent so long as the imaginary part a'' of a lies between $-\omega''$ and $+\omega''$.

Accordingly, from (7), if we separate off the term corresponding to the value $\nu = 0$ and replace the negative ν by $-\nu$, one obtains

$$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \frac{2i}{e^{2\pi i v} - 1} + \frac{2ie^{2\pi i a}}{e^{2\pi i a} - 1} + 2i\sum_{\nu=1}^{\infty} \frac{e^{-2\pi i\nu a}}{q^{-2\nu}e^{2\pi i v} - 1} + 2i\sum_{\nu=1}^{\infty} \frac{q^{2\nu}e^{2\pi i\nu a}e^{2\pi i v}}{q^{2\nu}e^{2\pi i v} - 1}.$$

But

$$\frac{2i}{e^{2\pi i v} - 1} = \frac{e^{-\pi i v}}{\sin\pi v} = \operatorname{cotg}\pi v - i, \qquad \frac{2ie^{2\pi i a}}{e^{2\pi i a} - 1} = \frac{e^{\pi i a}}{\sin\pi a} = \operatorname{cotg}\pi a + i,$$

and consequently this development can also be written

$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \operatorname{cotg}\pi v + \operatorname{cotg}\pi a - 2i \sum_{m=2,4,6,\ldots} \left(\frac{q^m e^{m\pi ia}}{e^{-2\pi iv} - q^m} - \frac{q^m e^{-m\pi ia}}{e^{2\pi iv} - q^m} \right), \tag{14}$$

where m runs through the series of even numbers

$$m = 2, 4, 6, 8, \ldots.$$

The domain of validity of this development is

$$-\omega'' < a'' < \omega''.$$

In Table II these sixteen developments are assembled.

From these formulae three others are derivable by increasing v, or a, or both by $1/2$. But three others, which correspond to increasing v and a by $\omega/2$, must be derived directly. They show real form when q, a, v are real.

A third kind of development, in which the symmetry of the functions with respect to the two variables v, a comes to expression, is obtained by expanding, according to powers of q, the fractions that occurred in the preceding developments. Thus one obtains

$$\frac{q^m}{e^{-2\pi iv} - q^m} = \sum_{\nu=1}^{\infty} q^{m\nu} e^{2\pi i\nu v}, \qquad \frac{q^m}{e^{2\pi iv} - q^m} = \sum_{\nu=1}^{\infty} q^{m\nu} e^{-2\pi i\nu v},$$

and these developments are valid as long as the absolute value of $qe^{2\pi iv}$ is a proper fraction, that is, if $v = v' + iv''$, so long as

$$-\omega'' < v'' < \omega''.$$

Thus from (14) there results

$$\frac{\vartheta_{11}'\vartheta_{11}(v+a)}{\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \operatorname{cotg}\pi v + \operatorname{cotg}\pi a + 4\sum q^{mm'/2}\sin(ma + m'v)\pi,$$

where m, m' independently run through the even numbers

$$2, 4, 6, 8, \ldots.$$

This formula is valid for real a and v and, beyond that, as long as the imaginary part both of v and of a is in absolute value smaller than ω''.

In Table III the sixteen formulae of this kind are assembled.[4]

Third Section.

Transformation of the Theta-Functions.

§ 27. The transformation principle.

We return to the defining equations, given in § 17, for the T-functions of m-th order, and, for a reason which will presently be evident, provide the letters ω, a, b, m occurring there with primes, so that these equations read

$$\begin{aligned} T(u+\omega_1') &= e^{-\pi i[a_1'(2u+\omega_1')+b_1']}T(u), \\ T(u+\omega_2') &= e^{-\pi i[a_2'(2u+\omega_2')+b_2']}T(u), \end{aligned} \tag{1}$$

$$a_2'\omega_1' - a_1'\omega_2' = m'. \tag{2}$$

[4] Such developments were considered by Jacobi, "sur la rotation d'un corps" (collected works, vol. II), then by Hermite, Annales de l'ecole normale, 1885, and by Kronecker, Berlin Academy, 1885. Compare also the Strasbourg dissertation of L. Vockerodt, 1905.

If a, c are arbitrary integral (positive or negative) numbers, then, by replacing n_1, n_2 in § 17, (16), by $-c, a$, one obtains

$$\begin{aligned} &T(u - c\omega_1' + a\omega_2') \\ &\quad = e^{-\pi i(-ca_1'+aa_2')(2u-c\omega_1'+a\omega_2')-\pi i(-cb_1'+ab_2'-m'ac)}T(u), \end{aligned} \tag{3}$$

an equation which also follows from one of equations (1), if in it a', b', ω' are replaced by

$$-ca_1' + aa_2', \qquad -cb_1' + ab_2' - m'ac, \qquad -c\omega_1' + a\omega_2'.$$

Herein is contained the principle of transformation of the T-functions.

Let b, ∂ be two other integral numbers for which the determinant

$$n = a\partial - bc \tag{4}$$

has a positive value. We put

$$\omega_1 = \partial\omega_1' - b\omega_2', \qquad\qquad \omega_2 = -c\omega_1' + a\omega_2', \tag{5}$$

and consequently

$$n\omega_1' = a\omega_1 + b\omega_2, \qquad\qquad n\omega_2' = c\omega_1 + \partial\omega_2. \tag{6}$$

It then has, as one sees from

$$\frac{\omega_2'}{\omega_1'} = \frac{c\omega_1 + \partial\omega_2}{a\omega_1 + b\omega_2}$$

by separating the real and imaginary parts, the imaginary part of $\omega_2' : \omega_1'$ the same sign as that of $\omega_2 : \omega_1$, namely the positive sign.

We set

$$a_1 = \partial a_1' - ba_2', \qquad\qquad a_2 = -ca_1' + aa_2', \tag{7}$$

$$b_1 = \partial b_1' - bb_2' - m'b\partial, \qquad\qquad b_2 = -cb_1' + ab_2' - m'ac. \tag{8}$$

From (3) one then concludes that the function $T(u)$ satisfies not only conditions (1), but also the equations obtained from (1) by interchanging $\omega_1', \omega_2', a_1', a_2', b_1', b_2'$ with $\omega_1, \omega_2, a_1, a_2, b_1, b_2$, that is, the equations (1) of § 17. Thus it is simultaneously a T-function of the periods ω_1', ω_2' and of the periods ω_1, ω_2, which we indicate by the equation

$$T'(u, \omega_1', \omega_2') = T(u, \omega_1, \omega_2). \tag{9}$$

But from (5) and (7) one has

$$a_2\omega_1 - a_1\omega_2 = (a_2'\omega_1' - a_1'\omega_2')(a\partial - bc),$$

and hence, if m is the order of T,

$$m = m'n. \tag{10}$$

According to (8), the characteristic (g_1, g_2) of T, if (g_1', g_2') is that of T' (§ 17), is

$$(g_1, g_2) = (\partial g_1' - bg_2' - m'b\partial,\ -cg_1' + ag_2' - m'ac). \tag{11}$$

By the transformation of the T-functions one understands the representation of the functions T' with the periods ω_1', ω_2' by T-functions with the periods ω_1, ω_2.

The numbers a, b, c, ∂ are called the transformation numbers, and $n = a\partial - bc$ the degree of transformation.

In order to survey the form of this representation more clearly, we seek the conditions under which $T'(u, \omega_1, \omega_2)$ becomes a Θ-function of the m-th order $\Theta(u, \omega)$ (§ 20). We take b'_1, b'_2, and consequently also b_1, b_2, as integers, so that b_1, b_2, b'_1, b'_2 may be replaced by g_1, g_2, g'_1, g'_2. Then one must put

$$\omega_1 = 1, \qquad \omega_2 = \omega, \qquad a_1 = 0, \qquad a_2 = m,$$

and therefore, by (6), (7), (10),

$$\omega'_1 = \frac{a + b\omega}{n}, \qquad \omega'_2 = \frac{c + \partial\omega}{n}, \qquad a'_1 = m'b, \qquad a'_2 = m'\partial.$$

The function $T'(u, \omega'_1, \omega'_2)$ therefore satisfies the conditions (1):

$$\begin{aligned} T'(u + \omega'_1) &= (-1)^{g'_1} e^{-\pi i m' b(2u + \omega'_1)} T'(u), \\ T'(u + \omega'_2) &= (-1)^{g'_2} e^{-\pi i m' \partial(2u + \omega'_2)} T'(u), \end{aligned}$$

and from this it follows that the product

$$e^{\pi i m' b u^2 / \omega'_1} T'(u, \omega'_1, \omega'_2)$$

is a Θ-function of order m', with the arguments

$$\frac{u}{\omega'_1}, \qquad \frac{\omega'_2}{\omega'_1}$$

and the characteristic (g'_1, g'_2). We can express this in the equation

$$e^{-\frac{\pi i m' n b u^2}{a + b\omega}} \Theta^{(m')}_{g'_1, g'_2} \left(\frac{nu}{a + b\omega}, \frac{c + \partial\omega}{a + b\omega} \right) = \Theta^{(m'n)}_{g_1, g_2}(u, \omega), \tag{12}$$

where the characteristics are determined by (11). The means for representing these functions are contained in § 21.

We designate the transformation of T and T' by a single letter S, or by (T', T), thus

$$S = (T', T).$$

If S' denotes a second transformation, by which T' passes into T'', thus

$$S' = (T'', T'),$$

then from this we can derive a new transformation S'', by which T passes into T''. This is called the transformation compounded from S and S' and is denoted by

$$S'' = S'S$$

or

$$(T'', T) = (T'', T')(T', T).$$

With this composition the commutative law does not hold in general; thus SS' is different from $S'S$. But the associative law holds, and is expressed by the formula

$$(T''', T'')[(T'', T')(T', T)] = [(T''', T'')(T'', T')](T', T) = (T''', T).$$

§ 28. Composition of the transformations.

A transformation [§ 27, (9)] is completely determined by the transformation numbers a, b, c, ∂, and these four integers may be arbitrarily given, provided only that their determinant $n = a\partial - bc$ is positive. If one gives these four numbers the opposite sign, then ω_1', ω_2', by § 27, (6), pass into $-\omega_1', -\omega_2'$; and if one replaces a, b, c, ∂ by $ma, mb, mc, m\partial$, where m is an arbitrary natural number, then ω_1', ω_2' pass into $\omega_1'/m, \omega_2'/m$, and n into $m^2 n$. The period ratio $\omega' = \omega_2'/\omega_1'$ remains unchanged in both cases. For the present, however, we shall always regard two transformations as different when their transformation numbers are different. According to this convention we can denote a transformation unambiguously by a matrix

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}. \tag{1}$$

The determinant

$$n = a\partial - bc \tag{2}$$

is the degree of transformation.

In this notation we also write the relations (6) of § 27 as

$$n(\omega_1', \omega_2') = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} (\omega_1, \omega_2). \tag{3}$$

If

$$S' = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix}, \qquad a'\partial' - b'c' = n',$$

then

$$n'(\omega_1'', \omega_2'') = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix} (\omega_1', \omega_2'), \tag{4}$$

and if in (4) one expresses ω_1', ω_2' by (3) in terms of ω_1, ω_2, one obtains

$$nn'(\omega_1'', \omega_2'') = \begin{pmatrix} a'a + b'c & a'b + b'\partial \\ c'a + \partial'c & c'b + \partial'\partial \end{pmatrix} (\omega_1, \omega_2). \tag{5}$$

If we put, therefore,

$$S'' = S'S, \tag{6}$$

then

$$S'' = \begin{pmatrix} a'a + b'c & a'b + b'\partial \\ c'a + \partial'c & c'b + \partial'\partial \end{pmatrix} = \begin{pmatrix} a'' & b'' \\ c'' & \partial'' \end{pmatrix},$$

and

$$a''\partial'' - b''c'' = n'' = nn'.$$

The transformations S therefore compose according to the same rule as the linear substitutions and matrices which we considered in the sixth section of the second volume. The degree of a compound transformation is equal to the product of the degrees of its components. Here these matrices are tied to the assumption that their elements are integers and their determinant is positive.

These properties are preserved under the composition of transformations. Nevertheless, the totality $\mathfrak{S}$ of transformations S does not form a group, any more than the totality of natural numbers, under composition by multiplication, forms a group; for for given S', S'' one cannot always determine an S satisfying condition (6), as would be required for a group.

By the special transformation of degree m^2,

$$M = \begin{pmatrix} \pm m & 0 \\ 0 & \pm m \end{pmatrix},$$

the periods ω_1, ω_2 pass into $\omega_1' = \omega_1/m$, $\omega_2' = \omega_2/m$, and the period ratio $\omega = \omega_2/\omega_1$ remains unchanged. These transformations are called multiplications (similarity transformations). Among them is contained the identical substitution

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which leaves everything unchanged and plays the role of the unit in composition.

The multiplications are interchangeable, under composition, with every transformation S:

$$SM = MS. \tag{7}$$

If in SM or MS one holds the transformation S fixed and lets M run through the totality $\mathfrak{M}$ of multiplications, then one obtains a system $\mathfrak{M}S$ which one would, according to Vol. II, § 46, have to call a collineation. If S_1 and S_2 belong to a collineation C, and S_1' and S_2' to a collineation C', then $S_1'S_1$ and $S_2'S_2$ also belong to the same collineation C''. Thus, putting $C'' = C'C$, one can compose the collineations. With this composition the collineation $\mathfrak{M}$ plays the role of the unit. If $S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$ is an arbitrary transformation, then

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}$$

is a multiplication. The two collineations

$$C = \mathfrak{M} \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}, \qquad C^{-1} = \mathfrak{M} \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix}$$

give, therefore, under composition

$$CC^{-1} = C^{-1}C = \mathfrak{M},$$

and are accordingly reciprocal to one another.

Consequently the totality of the collineations forms a group.

The transformations of degree 1 are called linear transformations. For these we denote the transformation numbers, to distinguish them, by the Greek letters $\alpha, \beta, \gamma, \delta$, so that

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1,$$

denotes a linear transformation. The system $\mathfrak{L}$ of linear transformations is a group contained in $\mathfrak{S}$; for if

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad L' = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix},$$

then

$$L'' = L'L = \begin{pmatrix} \alpha'\alpha + \beta'\gamma & \alpha'\beta + \beta'\delta \\ \gamma'\alpha + \delta'\gamma & \gamma'\beta + \delta'\delta \end{pmatrix} = \begin{pmatrix} \alpha'' & \beta'' \\ \gamma'' & \delta'' \end{pmatrix}$$

is also linear, and, for given L', L'', one can determine L uniquely from the equations

$$\alpha'\alpha + \beta'\gamma = \alpha'', \quad \alpha'\beta + \beta'\delta = \beta'',$$
$$\gamma'\alpha + \delta'\gamma = \gamma'', \quad \gamma'\beta + \delta'\delta = \delta''.$$

The unit of the group $\mathfrak{L}$ is the identical substitution $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and every substitution L has its reciprocal L^{-1}, as follows from the composition

$$LL^{-1} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The group $\mathfrak{L}$ is infinite and is not commutative.

From the equation

$$\alpha\delta - \beta\gamma = 1 \tag{8}$$

it follows that neither α, β, nor α, γ, nor δ, β, nor δ, γ can have a common factor. Conversely, if α, β have been assumed arbitrarily without a common divisor, then γ, δ can still be determined from (8) in infinitely many ways. If γ, δ is one of these determinations, then all are contained in the form

$$\gamma + \lambda\alpha, \qquad \delta + \lambda\beta, \tag{9}$$

where λ is an arbitrary integer.

§ 29. Composition of the transformations from simpler ones.

In the system $\mathfrak{S}$ of all transformations S there is contained a system $\mathfrak{S}_0$, consisting of all transformations S_0 whose second transformation number $b = 0$, while a and ∂ are positive:

$$S_0 = \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}. \tag{1}$$

On composing two S_0 there again arises an S_0, but nevertheless $\mathfrak{S}_0$ is no more a group than is $\mathfrak{S}$.

One can reduce any arbitrary transformation

$$S = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

by a composition LS to an S_0. For if

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \tag{2}$$

is to hold, then α, β must satisfy

$$\alpha q + \beta s = 0,$$

and hence, if ∂ is the greatest common divisor of q and s, one puts

$$\partial\alpha = s, \qquad \partial\beta = -q,$$

and determines, after α and β have thereby been found as relative primes, γ and δ from formula (8) of § 28. Then (2) is fulfilled if

$$a\partial = n, \qquad c = \gamma p + \delta r,$$

is put; a and ∂ are thereby determined uniquely, but with another choice of γ and δ, c can be replaced by $c + \lambda a$. One can therefore choose λ so that c is contained in the series of numbers

$$0, 1, 2, \ldots, a - 1,$$

and thereby the substitution S_0 is completely determined.

If the four transformation numbers have a common factor, then this can be separated off by means of the formula

$$\begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} = \begin{pmatrix} ma & mb \\ mc & m\partial \end{pmatrix}$$

by composition with the multiplication; hence we now assume that a, b, c, ∂ have no common divisor. One can then always determine two integers ξ, η such that

$$a\eta - c\xi = \alpha, \qquad b\eta - \partial\xi = \beta \tag{3}$$

have no common divisor.

To see this, we first assume ξ, η relatively prime. Then every common divisor of α, β is necessarily a divisor of n, as is seen from the solutions of (3),

$$n\xi = b\alpha - a\beta, \qquad n\eta = \partial\alpha - c\beta. \tag{4}$$

Thus one takes ξ not divisible by all the prime numbers which occur simultaneously in a and b, but ξ divisible, η indivisible, by all the other prime numbers occurring in n, and moreover ξ, η relatively prime, which is always possible. Then α and β have no common divisor. Thereupon one determines γ, δ so that

$$\alpha\delta - \beta\gamma = 1.$$

Then, by (3), also

$$(a\delta - b\gamma)\eta - (c\delta - \partial\gamma)\xi = 1,$$

and the following composition results, as is easily recognized with the aid of (4):

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} = \begin{pmatrix} a\delta - b\gamma & \xi \\ c\delta - \partial\gamma & \eta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{5}$$

If we call

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \tag{6}$$

the principal transformation of degree n, then it is proved that all transformations of degree n can be composed from a multiplication, a principal transformation, and linear transformations.

From the composition

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & m \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & mn \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}, \tag{7}$$

we can still further conclude that every transformation of degree n can be composed from such transformations whose degree is a prime number. If $n = pq$ is decomposed into two factors p and q, which are relatively prime to one another, then, by determining the numbers β, δ from

$$p\delta - q\beta = 1,$$

the composition

$$\begin{pmatrix} p & \beta \\ q & \delta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} p\delta & -q\beta \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

is obtained, from which it is seen that, instead of the principal transformation, one may also use any of these transformations $\begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$ for the derivation of all the others.

§ 30. The linear fundamental transformations.

The whole group $\mathfrak{L}$ of linear transformations can be derived by repetition of two among them, which we call the linear fundamental transformations.

If

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1,$$

is an arbitrary linear transformation, then

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{1}$$

and, since the identical transformation is the unit in the group L,

$$L^{-1} = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

is the transformation reciprocal to L.

We denote by the power $L^{\pm m}$ what arises by m-fold repetition of L or L^{-1}, and will now prove that, by the powers of the fundamental transformations

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{2}$$

every substitution L of the group $\mathfrak{L}$ can be composed. To begin with,

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad A^{\lambda} = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \tag{3}$$

(for every integral positive or negative λ), and

$$B^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad B^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B^3 = B^{-1}. \tag{4}$$

We also put

$$C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = B^{-1}AB, \qquad C^{\lambda} = \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix}, \tag{5}$$

so that C is derivable from A and B.

Now let

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

be an arbitrary linear transformation. We derive from it the series

$$L' = LA^{\lambda}, \qquad L'' = L'C^{\lambda'}, \qquad L''' = L''A^{\lambda''}, \qquad L'''' = L'''C^{\lambda'''},$$

whose first and second elements are formed as follows:

$$\alpha' = \alpha + \lambda\beta, \qquad \beta'' = \beta' - \lambda'\alpha', \qquad \alpha''' = \alpha'' + \lambda''\beta'', \qquad \beta'''' = \beta''' - \lambda'''\alpha''', \ldots,$$

and one can dispose of $\lambda, \lambda', \lambda'', \lambda''', \ldots$ so that, in absolute value,

$$\alpha' \leq \frac{1}{2}\beta, \qquad \beta'' \leq \frac{1}{2}\alpha', \qquad \alpha''' \leq \frac{1}{2}\beta'', \qquad \beta'''' \leq \frac{1}{2}\alpha''', \ldots,$$

as long as none of these numbers vanishes. The numbers

$$\beta, \alpha', \beta'', \alpha''', \beta'''', \ldots$$

therefore form a sequence decreasing in absolute value, and after a finite number of compositions of this kind one number of this series must vanish.

If $\beta^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} \pm 1 & 0 \\ \gamma^{(\nu)} & \pm 1 \end{pmatrix} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} A^{\pm\gamma^{(\nu)}},$$

and if $\alpha^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} 0 & \pm 1 \\ \mp 1 & \delta^{(\nu)} \end{pmatrix} = A^{\pm\delta^{(\nu)}} B^{\pm 1}.$$

And since

$$L = L' A^{-\lambda} = L'' C^{-\lambda'} A^{-\lambda} = L''' A^{-\lambda''} C^{-\lambda'} A^{-\lambda} = \cdots,$$

the theorem is proved.

§ 31. The linear fundamental transformations of the ϑ-functions.

In the application to the transformation of the Θ-functions [§ 27, (12)], the transformed modulus

$$\omega' = \frac{c + \partial\omega}{a + b\omega}$$

first comes under consideration. It does not change when the four transformation numbers receive a common factor m. The transformation is called a proper one if a, b, c, ∂ have no common factor.

The transformed argument

$$u' = \frac{nu}{a + b\omega}$$

passes into mu', when a, b, c, ∂ are replaced by $ma, mb, mc, m\partial$, and hence n by $m^2 n$. The multiplication $\begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}$ leaves the modulus ω unchanged and changes u into mu.

According to the results of the preceding two paragraphs, the whole system of proper transformations can be derived by repeated application of the three transformations

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix},$$

and we therefore first consider the linear fundamental transformations of the ϑ-functions.

I. $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ or $(\omega, \omega + 1)$.

According to § 27, (11), (12), one has

$$\vartheta_{11}(u, \omega + 1) = A\vartheta_{11}(u, \omega), \tag{1}$$

where A is independent of u. If one replaces u by

$$u + \frac{1}{2}, \qquad u + \frac{\omega}{2}, \qquad u + \frac{1 + \omega}{2},$$

then the formulae § 21, (8), give

$$\vartheta_{10}(u, \omega + 1) = A\vartheta_{10}(u, \omega), \tag{2}$$

$$e^{\pi i/4}\vartheta_{00}(u, \omega + 1) = A\vartheta_{01}(u, \omega), \tag{3}$$

$$e^{\pi i/4}\vartheta_{01}(u, \omega + 1) = A\vartheta_{00}(u, \omega). \tag{4}$$

To determine the constant A we use, as will often happen in what follows, the method of putting $u = 0$ in (1) after differentiation, and then making use on the right and left of formula (5), § 23,

$$\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{10}\vartheta_{01}; \tag{5}$$

one obtains

$$A^2 = e^{\pi i/2}, \qquad A = e^{\pi i/4}.$$

That the positive sign stands with A follows from one of formulae (3), (4), according to the concluding remark of § 25, when one lets ω become infinite. Consequently one obtains

$$\begin{aligned}
\vartheta_{11}(u, \omega + 1) &= e^{\pi i/4}\vartheta_{11}(u),\\
\vartheta_{10}(u, \omega + 1) &= e^{\pi i/4}\vartheta_{10}(u),\\
\vartheta_{01}(u, \omega + 1) &= \vartheta_{00}(u),\\
\vartheta_{00}(u, \omega + 1) &= \vartheta_{01}(u).
\end{aligned} \tag{6}$$

II. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ or $\left(\omega, -\frac{1}{\omega}\right)$.

Again by § 27, (12),

$$e^{-\pi i u^2/\omega}\vartheta_{11}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = A\vartheta_{11}(u, \omega), \tag{7}$$

and by increasing u by $1/2$, $\omega/2$, $(1 + \omega)/2$,

$$e^{-\pi i u^2/\omega}\vartheta_{01}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{10}(u, \omega), \tag{8}$$

$$e^{-\pi i u^2/\omega}\vartheta_{10}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{01}(u, \omega), \tag{9}$$

$$e^{-\pi i u^2/\omega}\vartheta_{00}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) = iA\vartheta_{00}(u, \omega). \tag{10}$$

From this, as above, one obtains

$$A = \pm i\sqrt{-i\omega}.$$

From $u = 0$ and a purely imaginary ω one concludes that, if $\sqrt{-i\omega}$ is taken so that its real part is positive, the lower sign must stand, and hence

$$\begin{aligned}
e^{-\pi i u^2/\omega}\vartheta_{11}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= -i\sqrt{-i\omega}\,\vartheta_{11}(u),\\
e^{-\pi i u^2/\omega}\vartheta_{01}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{10}(u),\\
e^{-\pi i u^2/\omega}\vartheta_{10}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{01}(u),\\
e^{-\pi i u^2/\omega}\vartheta_{00}\left(\frac{u}{\omega}, -\frac{1}{\omega}\right) &= \sqrt{-i\omega}\,\vartheta_{00}(u).
\end{aligned} \tag{11}$$

§ 32. The principal transformations of second order of the ϑ-functions.

The two principal transformations of n-th order

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \qquad \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix}$$

transform, according to § 27, (11), the characteristic (g_1, g_2) into

$$(ng_1, g_2), \qquad (g_1, ng_2),$$

that is, for odd n the characteristic remains unaltered, while for even n it passes into

$$(0, g_2) \quad \text{or} \quad (g_1, 0). \tag{1}$$

Since, according to this, the transformation of even degree behaves essentially otherwise than that of odd degree, we first consider the case $n = 2$. The first and the second principal transformation of second degree are called the Landen and the Gauss transformation.

According to § 27, (12),

$$\begin{aligned} \vartheta_{g_1,g_2}\left(u, \frac{\omega}{2}\right) &= \Theta_{g_1,0}(u, \omega), \\ \vartheta_{g_1,g_2}(2u, 2\omega) &= \Theta_{0,g_2}(u, \omega) \end{aligned} \tag{2}$$

are Θ-functions of second order in u, ω, which can be represented according to § 21.

We first obtain the following two pairs of formulae, in which A, B are independent of u:

$$\begin{aligned} A\vartheta_{11}\left(u, \frac{\omega}{2}\right) &= \vartheta_{01}(u, \omega)\vartheta_{11}(u, \omega), \\ A\vartheta_{10}\left(u, \frac{\omega}{2}\right) &= \vartheta_{00}(u, \omega)\vartheta_{10}(u, \omega), \end{aligned} \tag{3}$$

$$\begin{aligned} B\vartheta_{11}(2u, 2\omega) &= \vartheta_{10}(u, \omega)\vartheta_{11}(u, \omega), \\ B\vartheta_{01}(2u, 2\omega) &= \vartheta_{00}(u, \omega)\vartheta_{01}(u, \omega), \end{aligned} \tag{4}$$

and in each case the second formula can be derived from the first by increasing the argument by a half-period.

Putting $u = 0$ in these equations gives

$$\begin{aligned} A\vartheta'_{11}\left(0, \frac{\omega}{2}\right) &= \vartheta_{01}\vartheta'_{11}, & 2B\vartheta'_{11}(0, 2\omega) &= \vartheta_{10}\vartheta'_{11}, \\ A\vartheta_{10}\left(0, \frac{\omega}{2}\right) &= \vartheta_{00}\vartheta_{10}, & B\vartheta_{01}(0, 2\omega) &= \vartheta_{00}\vartheta_{01}, \end{aligned} \tag{5}$$

whence, by division and with use of the relation

$$\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{11}\vartheta_{01},$$

one obtains

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)\vartheta_{01}\left(0, \frac{\omega}{2}\right) = \vartheta_{01}^2, \tag{6}$$

$$2\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega) = \vartheta_{10}^2, \tag{7}$$

and, replacing ω in (6) by 2ω and ω in (7) by $\omega/2$,

$$\vartheta_{01}(0, 2\omega)^2 = \vartheta_{00}\vartheta_{01}, \tag{8}$$

$$\vartheta_{10}\left(0, \frac{\omega}{2}\right)^2 = 2\vartheta_{00}\vartheta_{10}. \tag{9}$$

From (8) and (9), the second equations (5) give

$$2A = \vartheta_{10}\left(0, \frac{\omega}{2}\right), \qquad B = \vartheta_{01}(0, 2\omega),$$

and hence for the Gauss transformation one obtains

$$\begin{aligned}\vartheta_{10}\left(0,\frac{\omega}{2}\right)\vartheta_{11}\left(u,\frac{\omega}{2}\right) &= 2\vartheta_{01}(u,\omega)\vartheta_{11}(u,\omega),\\ \vartheta_{10}\left(0,\frac{\omega}{2}\right)\vartheta_{10}\left(u,\frac{\omega}{2}\right) &= 2\vartheta_{00}(u,\omega)\vartheta_{10}(u,\omega),\end{aligned}\tag{10}$$

and for the Landen transformation

$$\begin{aligned}\vartheta_{01}(0,2\omega)\vartheta_{11}(2u,2\omega) &= \vartheta_{10}(u,\omega)\vartheta_{11}(u,\omega),\\ \vartheta_{01}(0,2\omega)\vartheta_{01}(2u,2\omega) &= \vartheta_{00}(u,\omega)\vartheta_{01}(u,\omega).\end{aligned}\tag{11}$$

For each of the two transformations two ϑ-functions still remain to be expressed. These expressions may be derived from (10), (11) according to § 21, (13), but one also obtains them directly in the following way. The functions

$$\vartheta_{01}\left(u,\frac{\omega}{2}\right),\qquad \vartheta_{10}(2u,2\omega)\tag{12}$$

vanish for

$$u=\frac{\omega}{4},\qquad u=\frac{1}{4}.$$

On the other hand, from the formulae (8) of § 21, putting there $v=-\omega/4$ and $-1/4$, one obtains

$$\vartheta_{11}\left(\frac{\omega}{4}\right)=-i\vartheta_{01}\left(\frac{\omega}{4}\right),\qquad \vartheta_{11}\left(\frac{1}{4}\right)=\vartheta_{10}\left(\frac{1}{4}\right),$$

and consequently the two functions (12), which are expressible linearly by two ϑ-squares, agree, apart from constant factors, with

$$\vartheta_{01}^2(u)+\vartheta_{11}^2(u),\qquad \vartheta_{10}^2(u)-\vartheta_{11}^2(u).$$

The constant factors are obtained immediately by setting $u=0$ and using the relations (6), (7):

$$\vartheta_{00}\left(0,\frac{\omega}{2}\right)\vartheta_{01}\left(u,\frac{\omega}{2}\right)=\vartheta_{01}^2(u)+\vartheta_{11}^2(u),\tag{13}$$

$$2\vartheta_{00}(0,2\omega)\vartheta_{10}(2u,2\omega)=\vartheta_{10}^2(u)-\vartheta_{11}^2(u).\tag{14}$$

The two final formulae are obtained from this by transforming u into $u+\frac{1}{2}$ and $u+\frac{\omega}{2}$, or also by the same method as for (13), (14):

$$\vartheta_{00}\left(0,\frac{\omega}{2}\right)\vartheta_{00}\left(u,\frac{\omega}{2}\right)=\vartheta_{00}^2(u)+\vartheta_{10}^2(u),\tag{15}$$

$$2\vartheta_{00}(0,2\omega)\vartheta_{00}(2u,2\omega)=\vartheta_{00}^2(u)+\vartheta_{01}^2(u).\tag{16}$$

From these one may derive manifold relations between the zero-values of the ϑ-functions. We record only the following three, of which the first two flow from (8), (9), while the last follows from the first equation (11), after replacing ω by $\omega/2$ and taking $u=1/4$, with the fact that $\vartheta_{11}(1/4)=\vartheta_{10}(1/4)$:

$$\begin{aligned}\sqrt{\vartheta_{00}\vartheta_{01}} &= \vartheta_{01}(0,2\omega),\\ \sqrt{\vartheta_{00}\vartheta_{10}} &= \frac{1}{\sqrt{2}}\vartheta_{10}\left(0,\frac{\omega}{2}\right),\\ \sqrt{\vartheta_{01}\vartheta_{10}} &= \vartheta_{10}\left(\frac{1}{4},\frac{\omega}{2}\right).\end{aligned}\tag{17}$$

These formulae are of interest because they represent the square roots as single-valued functions of ω.

We make one further application of the transformation of second order to the proof of a formula needed for the transformation of odd order. Replacing ω by 2ω in the second equation (10), we have

$$2\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega) = \vartheta_{10}\vartheta_{10}(u).$$

Multiplying by the second equation (11) gives

$$2\vartheta_{01}(0, 2\omega)\vartheta_{01}(2u, 2\omega)\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega) = \vartheta_{10}\vartheta_{10}(u)\vartheta_{00}(u)\vartheta_{01}(u).$$

Dividing by the product of the two equations (7), (8),

$$2\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega)\vartheta_{01}(0, 2\omega)^2 = \vartheta_{10}^2\vartheta_{00}\vartheta_{01},$$

one obtains

$$\frac{\vartheta_{00}(u, 2\omega)\vartheta_{10}(u, 2\omega)\vartheta_{01}(2u, 2\omega)}{\vartheta_{00}(0, 2\omega)\vartheta_{10}(0, 2\omega)\vartheta_{01}(0, 2\omega)} = \frac{\vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(u)}{\vartheta_{00}\vartheta_{10}\vartheta_{01}}. \tag{18}$$

If now n denotes any odd integer, the function

$$\vartheta_{01}\left(\frac{v}{n}\right)$$

remains unaltered when v is increased by a multiple of n, and hence the product

$$\prod \vartheta_{01}\left(\frac{v}{n}\right),$$

when taken over a complete system of residues modulo n, is independent of the special choice of this residue system. Therefore, since $2v$ runs through such a residue system together with v,

$$\prod_{v=1}^{n-1} \vartheta_{01}\left(\frac{v}{n}\right) = \prod_{v=1}^{n-1} \vartheta_{01}\left(\frac{2v}{n}\right). \tag{19}$$

Putting $u = v/n$ in (18), forming the product, and applying (19) to the last factor on the left, one obtains that the product

$$\frac{\prod_{v=1}^{n-1} \vartheta_{00}\left(\frac{v}{n}\right)\vartheta_{10}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}\right)}{\vartheta_{00}^{n-1}\vartheta_{10}^{n-1}\vartheta_{01}^{n-1}} \tag{20}$$

remains unaltered when ω is replaced by 2ω.

The values of v occurring in (20) can be arranged in pairs

$$v,\ n - v, \qquad v = 1, 2, \ldots, \frac{n-1}{2},$$

and, since

$$\vartheta_{00}\left(\frac{v}{n}\right) = \vartheta_{00}\left(\frac{n-v}{n}\right), \quad \vartheta_{10}\left(\frac{v}{n}\right) = -\vartheta_{10}\left(\frac{n-v}{n}\right), \quad \vartheta_{01}\left(\frac{v}{n}\right) = \vartheta_{01}\left(\frac{n-v}{n}\right),$$

(20), except for the sign, agrees with the square of

$$\frac{\prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right)\vartheta_{10}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}\right)}{\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}}. \tag{21}$$

The latter quotient, which is a continuous non-vanishing function of ω as long as the imaginary part of ω is positive, therefore also remains unchanged when ω is replaced by 2ω, and hence by $4\omega, 8\omega, \ldots$. Its value can be determined by allowing the imaginary part of ω to become infinite, that is by taking $q = 0$. But for $q = 0$ one has, by § 25,

$$\vartheta_{00}(v) = 1, \qquad \vartheta_{01}(v) = 1, \qquad \frac{\vartheta_{10}(v)}{\vartheta_{10}} = \cos \pi v,$$

and hence the value of (21) is

$$\prod_{v=1}^{(n-1)/2} \cos \frac{v\pi}{n}. \tag{22}$$

Since here $v\pi/n$ is less than $\frac{1}{2}\pi$, this product has a positive value. But

$$\cos \frac{v\pi}{n} = -\cos \frac{(n-v)\pi}{n},$$

and by Vol. I, § 144,

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \cos \frac{v\pi}{n} = 1.$$

Thus the formula is proved:

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right) \vartheta_{10}\left(\frac{v}{n}\right) \vartheta_{01}\left(\frac{v}{n}\right) = \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}. \tag{23}$$

With regard to the formula Vol. I, § 145, (3), one can give this relation also the form

$$2^{(n-1)/2} \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{2v}{n}\right) \vartheta_{10}\left(\frac{2v}{n}\right) \vartheta_{01}\left(\frac{2v}{n}\right) = (-1)^{(n^2-1)/8} \vartheta_{00}^{(n-1)/2} \vartheta_{10}^{(n-1)/2} \vartheta_{01}^{(n-1)/2}. \tag{24}$$

§ 33. The principal transformations of odd order.

The formula proved last is useful to us in carrying out the transformation of odd order n. We first consider the first principal transformation. According to § 27, (12),

$$\vartheta_{11}(nu, n\omega) \tag{1}$$

is a Θ_{11}-function of n-th order in u and ω, and it can readily be formed from its zeros, of which there are n in the period-parallelogram. The zeros of (1) are the values

$$u = \frac{v + \mu n\omega}{n} = \frac{v}{n} + \mu\omega,$$

where v and μ are integers, and all incongruent values among these are obtained when μ is fixed and v runs through a complete system of residues modulo n. We choose the system of residues

$$0,\ \pm 1,\ \pm 2, \ldots, \pm \frac{n-1}{2},$$

and hence, if C denotes a factor independent of u, obtain

$$C\vartheta_{11}(nu, n\omega) = \vartheta_{11}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n} + u\right) \vartheta_{11}\left(\frac{v}{n} - u\right), \tag{2}$$

a formula which, by § 22, (4), can also be expressed in the following way through the functions $\vartheta_{11}(u), \vartheta_{01}(u)$:

$$C\vartheta_{01}^{n-1}\vartheta_{11}(nu, n\omega) = \vartheta_{11}(u) \prod_{v=1}^{(n-1)/2} \left[\vartheta_{11}^2\left(\frac{v}{n}\right) \vartheta_{01}^2(u) - \vartheta_{01}^2\left(\frac{v}{n}\right) \vartheta_{11}^2(u)\right].$$

We replace u by

$$u + \frac{1}{2}, \qquad u + \frac{\omega}{2}, \qquad u + \frac{1+\omega}{2},$$

and obtain from (8) of § 21:

$$
\begin{aligned}
C\vartheta_{10}(nu, n\omega) &= \vartheta_{10}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{10}\left(\frac{v}{n}+u\right)\vartheta_{10}\left(\frac{v}{n}-u\right),\\
C\vartheta_{01}(nu, n\omega) &= \vartheta_{01}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{01}\left(\frac{v}{n}+u\right)\vartheta_{01}\left(\frac{v}{n}-u\right),\\
C\vartheta_{00}(nu, n\omega) &= \vartheta_{00}(u) \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}+u\right)\vartheta_{00}\left(\frac{v}{n}-u\right).
\end{aligned} \tag{3}
$$

Putting $u = 0$ and using $\vartheta'_{11} = \pi\vartheta_{00}\vartheta_{01}\vartheta_{10}$ gives

$$
C\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) = \sqrt{n}\prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}\right)\vartheta_{10}\left(\frac{v}{n}\right), \tag{4}
$$

or, with use of (23) of the preceding paragraph,

$$
C\,2^{(n-1)/2}\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right) = \sqrt{n}\,\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}. \tag{5}
$$

The sign is obtained from the fact, following from one of the formulae (3), that $C = 1$ for $q = 0$.

After this determination of C, the formulae (2), (3) may be written as

$$
\begin{aligned}
&\sqrt{n}\,\vartheta_{11}(nu, n\omega)\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}\\
&\quad= 2^{(n-1)/2}\vartheta_{11}(u)\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right)\vartheta_{11}\left(\frac{v}{n}+u\right)\vartheta_{11}\left(\frac{v}{n}-u\right),
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
&\sqrt{n}\,\vartheta_{10}(nu, n\omega)\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}\\
&\quad= 2^{(n-1)/2}\vartheta_{10}(u)\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}+u\right)\vartheta_{01}\left(\frac{v}{n}-u\right),
\end{aligned} \tag{7}
$$

$$
\begin{aligned}
&\sqrt{n}\,\vartheta_{01}(nu, n\omega)\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}\\
&\quad= 2^{(n-1)/2}\vartheta_{01}(u)\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right)\vartheta_{01}\left(\frac{v}{n}+u\right)\vartheta_{01}\left(\frac{v}{n}-u\right),
\end{aligned} \tag{8}
$$

$$
\begin{aligned}
&\sqrt{n}\,\vartheta_{00}(nu, n\omega)\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}\\
&\quad= 2^{(n-1)/2}\vartheta_{00}(u)\prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right)\vartheta_{00}\left(\frac{v}{n}+u\right)\vartheta_{00}\left(\frac{v}{n}-u\right).
\end{aligned} \tag{9}
$$

§ 34. The functions $\eta(\omega), f(\omega), f_1(\omega), f_2(\omega)$.

The functions $\eta(\omega), f(\omega), f_1(\omega), f_2(\omega)$ were already mentioned in § 24. There they arose almost of themselves as single-valued functions of ω in the representation of the ϑ-functions by infinite products, whereas they did not arise in the second representation by infinite series. Connected with this is the remarkable fact that many of our formulae, for example the formulae (17) of § 32 belonging to the transformation of second order, or the formula (23) of § 32, may easily be verified by the infinite products, but only with difficulty, or not at all, by the infinite series. It was therefore of interest to us to derive these results, without using either of those expressions, from the theory of transformation; and from this same source we shall now obtain the functions $\eta(\omega)$, $f(\omega)$, $f_1(\omega)$, $f_2(\omega)$ and their fundamental properties.

Formula (6) of the preceding paragraph gives, for $n = 3$,

$$\sqrt{3}\,\vartheta_{11}(3u, 3\omega)\vartheta_{00}\vartheta_{10}\vartheta_{01} = 2\vartheta_{11}\left(\frac{1}{3}\right)\vartheta_{11}(u)\vartheta_{11}\left(\frac{1}{3}+u\right)\vartheta_{11}\left(\frac{1}{3}-u\right),$$

and if one differentiates and then sets $u = 0$, using (4),

$$3\sqrt{3}\,\vartheta_{11}'(0, 3\omega) = 2\pi\vartheta_{11}\left(\frac{1}{3}\right)^3,$$

or, replacing ω by $\omega/3$,

$$3\sqrt{3}\,\vartheta_{11}' = 2\pi\left[\vartheta_{11}\left(\frac{1}{3}, \frac{\omega}{3}\right)\right]^3.$$

Thus, putting

$$\eta(\omega) = \frac{1}{\sqrt{3}}\vartheta_{11}\left(\frac{1}{3}, \frac{\omega}{3}\right), \tag{1}$$

one obtains, in agreement with the notation of § 24, (9),

$$\vartheta_{11}' = \pi\vartheta_{00}\vartheta_{01}\vartheta_{10} = 2\pi\eta(\omega)^3. \tag{2}$$

From § 25, (4), one obtains for $\eta(\omega)$ the series expansion

$$\eta(\omega) = q^{1/12}\sum_{\nu=-\infty}^{\infty}(-1)^\nu q^{3\nu^2+\nu}. \tag{3}$$

The function $\eta(\omega)$ is real for purely imaginary ω (real q), and for infinitely large ω (that is, for vanishing q),

$$q^{-1/12}\eta(\omega) = 1.$$

Hence, applying to (2) the formulae § 31, (6), (11), one finds for the linear fundamental transformations of $\eta(\omega)$

$$\eta(\omega+1) = e^{\pi i/12}\eta(\omega), \tag{4}$$

$$\eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\,\eta(\omega), \tag{5}$$

of which the first also follows immediately from the series representation (3), while the second is not so easily proved by a direct transformation of the series.

We now pass to the consideration of the two principal transformations of second order of the η-function. From § 32, (11), by differentiating and then setting $u = 0$, one obtains

$$2\vartheta_{01}(0, 2\omega)\eta(2\omega)^3 = \vartheta_{10}\eta(\omega)^3,$$

and therefore, after squaring and using § 32, (8),

$$4\vartheta_{00}\vartheta_{10}\vartheta_{01}\eta(2\omega)^6 = \vartheta_{10}^3\eta(\omega)^6.$$

With use of (2) and extraction of the cube root this gives

$$2\eta(2\omega)^2 = \vartheta_{10}\eta(\omega). \tag{6}$$

Replacing ω in (6) by $\omega/2$ gives

$$2\eta(\omega)^2 = \vartheta_{10}\left(0, \frac{\omega}{2}\right)\eta\left(\frac{\omega}{2}\right),$$

and, after squaring and applying § 32, (9),

$$2\eta(\omega)^4 = \eta\left(\frac{\omega}{2}\right)^2 \vartheta_{00}\vartheta_{10}.$$

Multiplication on both sides by ϑ_{01} gives, according to (2),

$$\eta\left(\frac{\omega}{2}\right)^2 = \vartheta_{01}\eta(\omega). \tag{7}$$

By changing ω into $\omega + 1$, using (4) and § 31, (6), one obtains

$$e^{-\pi i/12}\eta\left(\frac{\omega+1}{2}\right)^2 = \vartheta_{00}\eta(\omega). \tag{8}$$

Accordingly we introduce the three functions

$$f(\omega) = \frac{e^{-\pi i/24}\eta\left(\frac{\omega+1}{2}\right)}{\eta(\omega)}, \qquad f_1(\omega) = \frac{\eta\left(\frac{\omega}{2}\right)}{\eta(\omega)}, \qquad f_2(\omega) = \frac{\sqrt{2}\,\eta(2\omega)}{\eta(\omega)}. \tag{9}$$

Then, by (6), (7), (8),

$$\begin{aligned} \vartheta_{00} &= \eta(\omega)f(\omega)^2, \\ \vartheta_{01} &= \eta(\omega)f_1(\omega)^2, \\ \vartheta_{10} &= \eta(\omega)f_2(\omega)^2. \end{aligned} \tag{10}$$

It follows from (8) and (9) that $f(\omega), f_1(\omega), f_2(\omega)$ are real and positive for purely imaginary ω. Equation (10) agrees with § 24, (10), and the functions f, f_1, f_2 introduced there are the same as these. From (10) the relations [(2) and § 21, (14)] follow easily:

$$f(\omega)^8 = f_1(\omega)^8 + f_2(\omega)^8, \qquad \sqrt{2} = f(\omega)f_1(\omega)f_2(\omega). \tag{11}$$

With the aid of these formulae one can, by square roots, express any one of the three functions $f(\omega)^8, f_1(\omega)^8, f_2(\omega)^8$ in terms of any other. This is furnished by the following formulae derived from (11):

$$\begin{aligned} [f_1(\omega)^8 - f_2(\omega)^8]^2 &= f(\omega)^{16} - \frac{64}{f(\omega)^8}, \\ [f(\omega)^8 + f_2(\omega)^8]^2 &= f_1(\omega)^{16} + \frac{64}{f_1(\omega)^8}, \\ [f(\omega)^8 + f_1(\omega)^8]^2 &= f_2(\omega)^{16} + \frac{64}{f_2(\omega)^8}. \end{aligned} \tag{12}$$

For the linear fundamental transformations of the functions f, one obtains first from (4) and (9):

$$\begin{aligned} f(\omega+1) &= e^{-\pi i/24} f_1(\omega), \\ f_1(\omega+1) &= e^{-\pi i/24} f(\omega), \\ f_2(\omega+1) &= e^{\pi i/12} f_2(\omega). \end{aligned} \tag{13}$$

Furthermore, from (5) and the two last equations (9),

$$f_1\left(-\frac{1}{\omega}\right) = f_2(\omega), \qquad f_2\left(-\frac{1}{\omega}\right) = f_1(\omega), \tag{14}$$

and using these in the second equation (11),

$$f\left(-\frac{1}{\omega}\right) = f(\omega). \tag{15}$$

For the transformation of second order, one obtains immediately from the two last equations (9)

$$f_1(2\omega)f_2(\omega) = \sqrt{2}, \tag{16}$$

from which, by means of (12), it follows also that

$$f_2(\omega)^4[f(2\omega)^8 + f_2(2\omega)^8] = 2[f(\omega)^8 + f_1(\omega)^8]. \tag{17}$$

Putting in (16), according to (13), (14),

$$f_1(2\omega) = e^{\pi i/24} f(2\omega - 1), \qquad f_2(\omega) = e^{\pi i/24} f\left(1 - \frac{1}{\omega}\right),$$

one obtains

$$f\left(1 - \frac{1}{\omega}\right) f(2\omega - 1) = \sqrt{2},$$

and, putting $2\omega - 1$ equal to a new ω,

$$f(\omega) f\left(\frac{\omega - 1}{\omega + 1}\right) = \sqrt{2}. \tag{18}$$

Replacing ω in (16) by $\omega/2$ and $(\omega+1)/2$ gives

$$\begin{aligned} f_1(\omega) f_2\left(\frac{\omega}{2}\right) &= \sqrt{2}, \\ f(\omega) f_2\left(\frac{\omega+1}{2}\right) &= e^{\pi i/24}\sqrt{2}. \end{aligned} \tag{19}$$

Finally, we set down the formulae for the principal transformation of odd order n of the functions η, f. For $\eta(n\omega)$ one obtains easily, if in formulae (6) of § 33 one differentiates and sets $u = 0$, and then extracts the cube root,

$$\sqrt{n}\, \eta(n\omega)\eta(\omega)^{(n-3)/2} = \prod_{v=1}^{(n-1)/2} \vartheta_{11}\left(\frac{v}{n}\right). \tag{20}$$

Putting $u = 0$ in (7), (8), (9) of § 33, using the relations (10) and (20), and extracting the square root, one finds

$$\begin{aligned} f(n\omega)\eta(\omega)^{(n-1)/2} &= f(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{00}\left(\frac{v}{n}\right), \\ f_1(n\omega)\eta(\omega)^{(n-1)/2} &= f_1(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{01}\left(\frac{v}{n}\right), \\ f_2(n\omega)\eta(\omega)^{(n-1)/2} &= f_2(\omega) \prod_{v=1}^{(n-1)/2} \vartheta_{10}\left(\frac{v}{n}\right). \end{aligned} \tag{21}$$

The signs here are obtained from the assumption of infinitely small q.

§ 35. The Weierstrass σ-function.

By the general linear substitution

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1,$$

applied to the variables ω_1, ω_2, every t-function passes again into a t-function, while the characteristic is changed. If (g_1, g_2) is the characteristic of the original function and (g_1', g_2') that of the transformed function, then, by § 27, (11),

$$(g_1, g_2) = (\delta g_1' - \beta g_2' - \beta\delta, -\gamma g_1' + \alpha g_2' - \alpha\gamma).$$

Solving the two linear equations contained in this for g_1', g_2', and observing that $\alpha\beta(\gamma+\delta+1)$ and $\gamma\delta(\alpha+\beta+1)$ are necessarily even numbers, and that a characteristic is not changed if its elements are changed by multiples of 2, one may put equivalently

$$(g_1', g_2') = (\alpha g_1 + \beta g_2 + \alpha\beta, \gamma g_1 + \delta g_2 + \gamma\delta). \tag{1}$$

Thus, apart from exponential factors, the ϑ-functions pass into one another. It is, however, important to form a function which is absolutely invariant under linear transformations, and such a function is the σ-function introduced into the theory by Weierstrass; we now pass to its definition.

If formula (1) is applied to the four principal characteristics $(0,0), (0,1), (1,0), (1,1)$, these pass, in order, into $(\alpha\beta, \gamma\delta)$, $((\alpha+1)\beta, (\gamma+1)\delta)$, $(\alpha(\beta+1), \gamma(\delta+1))$, $((\alpha+1)(\beta+1)-1, (\gamma+1)(\delta+1)-1) = (1,1)$. Thus only the last characteristic, $(1,1)$, remains unchanged under all linear transformations, since neither α and β nor γ and δ can both be even. Let, therefore, $t(u, \omega_1, \omega_2)$ be a t-function with characteristic $(1,1)$, which by § 17, (12), must vanish for $u = 0$.

If

$$(\omega_1', \omega_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2),$$

then, by our transformation principle,

$$t(u, \omega_1, \omega_2), \qquad t(u, \omega_1', \omega_2')$$

are related T-functions of first order, and hence, if C, λ, μ are quantities independent of u,

$$t(u, \omega_1', \omega_2') = C e^{\lambda u^2 + \mu u} t(u, \omega_1, \omega_2). \tag{2}$$

It follows by logarithmic differentiation that

$$2\lambda u + \mu = \frac{d \log t(u, \omega_1', \omega_2')}{du} - \frac{d \log t(u, \omega_1, \omega_2)}{du}. \tag{3}$$

Expanding in powers of u by Taylor's theorem, and denoting by t', t'', t''' the first, second, and third derivatives of t with respect to u for $u = 0$, one has

$$\frac{d \log t(u, \omega_1, \omega_2)}{du} = \frac{1}{u} + \frac{t''}{2t'} + u\left(\frac{t'''}{3t'} - \frac{t''^2}{4t'^2}\right) + \cdots,$$

from which, comparing with (3), it follows that λ and μ can be written in the form

$$\lambda = \varphi(\omega_1', \omega_2') - \varphi(\omega_1, \omega_2), \qquad \mu = \psi(\omega_1', \omega_2') - \psi(\omega_1, \omega_2),$$

where

$$\varphi(\omega_1, \omega_2) = \frac{t'''}{6t'} - \frac{t''^2}{8t'^2}, \qquad \psi(\omega_1, \omega_2) = \frac{t''}{2t'}. \tag{4}$$

If one further determines the factor C in (2) by the special value $u = 0$, then

$$C = \frac{t'(\omega_1', \omega_2')}{t'(\omega_1, \omega_2)}. \tag{5}$$

Consequently formula (2) can be expressed in the following theorem: the function

$$\sigma(u, \omega_1, \omega_2) = e^{-u^2\left(\frac{t'''}{6t'} - \frac{t''^2}{8t'^2}\right) - u\frac{t''}{2t'}} \frac{t(u, \omega_1, \omega_2)}{t'} \tag{6}$$

remains unchanged when ω_1, ω_2 are replaced by ω_1', ω_2', that is, when any linear transformation is applied:

$$\sigma(u, \omega_1', \omega_2') = \sigma(u, \omega_1, \omega_2). \tag{7}$$

The function defined by (6) remains, as a simple calculation shows, unchanged if t is replaced by any related t-function.

The function $t(u)$ has, by assumption, the characteristic $(1, 1)$, and its zeros are therefore congruent to 0. Hence $t(-u)$ has the same zeros and also the same character as $t(u)$. The two functions differ only by an exponential factor, and one easily obtains, determining the constant factor from $u = 0$,

$$t(-u) = -e^{2\pi i\mu u} t(u),$$

where μ is a constant. Differentiating this equation twice with respect to u and then putting $u = 0$ gives

$$2\pi i\mu = -\frac{t''}{t'},$$

and hence by (6) one obtains that the function $\sigma(u)$ satisfies

$$\sigma(-u) = -\sigma(u), \tag{8}$$

that is, is an odd function of u.

Since $\sigma(u)$ is a t-function of first order, it suffices to satisfy the two conditions

$$\sigma(u + \omega_1) = c_1 e^{\eta_1(2u+\omega_1)} \sigma(u), \qquad \sigma(u + \omega_2) = c_2 e^{\eta_2(2u+\omega_2)} \sigma(u),$$

where c_1, c_2, η_1, η_2 are constants. The function $\sigma(u)$ vanishes for $u = 0$, but not for $u = \omega_1/2$ or $u = \omega_2/2$; if, therefore, in the preceding formulae one puts $u = -\omega_1/2$ and $u = -\omega_2/2$, respectively, then by (8) one obtains $c_1 = -1, c_2 = -1$, and hence the formulae

$$\begin{aligned} \sigma(u + \omega_1) &= -e^{\eta_1(2u+\omega_1)} \sigma(u), \\ \sigma(u + \omega_2) &= -e^{\eta_2(2u+\omega_2)} \sigma(u). \end{aligned} \tag{9}$$

The quantities η_1, η_2 here introduced are functions of ω_1, ω_2, satisfying the equation

$$\eta_1\omega_2 - \eta_2\omega_1 = \pi i, \tag{10}$$

according to the relation § 17, (10), in which $\pi i a_1, \pi i a_2, m$ are to be replaced by $-\eta_1, -\eta_2, 1$.

By repeated application of (9), if a, b are integers, one obtains [cf. § 17, (16)]

$$\sigma(u + a\omega_1 + b\omega_2) = (-1)^{a+b+ab} e^{(a\eta_1+b\eta_2)(2u+a\omega_1+b\omega_2)} \sigma(u). \tag{11}$$

When the function $\sigma(u)$, as given by formula (6), is expanded in powers of u, one finds that not only the second, but also the third power of u does not occur in the expansion; hence

$$\sigma(0) = 0, \qquad \sigma'(0) = 1, \qquad \sigma''(0) = 0, \qquad \sigma'''(0) = 0. \tag{12}$$

If the linear substitution

$$(\omega_1', \omega_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2) \tag{13}$$

is applied to ω_1, ω_2, then the quantities η_1, η_2 undergo the corresponding substitution

$$(\eta_1', \eta_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\eta_1, \eta_2), \tag{14}$$

as follows immediately from the relations (7) and (11).

Logarithmically differentiating the formulae (9) with respect to u gives

$$\begin{aligned}\frac{\sigma'(u+\omega_1)}{\sigma(u+\omega_1)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_1, \\ \frac{\sigma'(u+\omega_2)}{\sigma(u+\omega_2)} &= \frac{\sigma'(u)}{\sigma(u)} + 2\eta_2, \end{aligned} \tag{15}$$

and by putting in the first equation $u = -\omega_1/2$ and in the second $u = -\omega_2/2$, and observing that $\sigma(u)$ is odd and hence $\sigma'(u)$ is even,

$$\eta_1 = \frac{\sigma'(\omega_1/2)}{\sigma(\omega_1/2)}, \qquad \eta_2 = \frac{\sigma'(\omega_2/2)}{\sigma(\omega_2/2)}. \tag{16}$$

§ 36. The functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$.

It remains to obtain the analogous results for the remaining characteristics. We proceed from the following observation. If $x_1, x_2; y_1, y_2$ are any two pairs of quantities which are transformed by the same linear substitution S into $x_1', x_2'; y_1', y_2'$, so that

$$(x_1', x_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (x_1, x_2), \qquad (y_1', y_2') = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (y_1, y_2),$$

then also

$$x_1 y_2 - x_2 y_1 = x_1' y_2' - x_2' y_1',$$

as follows from the multiplication of determinants. Hence, whatever x_1, x_2 may be, by (7) of § 35,

$$\sigma(u + x_2\omega_1 - x_1\omega_2, \omega_1, \omega_2) = \sigma(u + x_2'\omega_1' - x_1'\omega_2', \omega_1', \omega_2'). \tag{1}$$

This function, by formula (11) of the preceding paragraph, changes when x_1 and x_2 are changed by integers. This change can be avoided if one considers instead the function

$$e^{-2(\eta_1 x_2 - \eta_2 x_1)u} \frac{\sigma(u + x_2\omega_1 - x_1\omega_2)}{\sigma(x_2\omega_1 - x_1\omega_2)},$$

which we shall for the moment denote by $\sigma(u, x_1, x_2, \omega_1, \omega_2)$. This function satisfies the conditions

$$\sigma(u, x_1', x_2', \omega_1', \omega_2') = \sigma(u, x_1, x_2, \omega_1, \omega_2), \tag{2}$$

$$\begin{aligned}\sigma(u, x_1 + 1, x_2, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \\ \sigma(u, x_1, x_2 + 1, \omega_1, \omega_2) &= \sigma(u, x_1, x_2, \omega_1, \omega_2), \end{aligned} \tag{3}$$

$$\begin{aligned}\sigma(u + \omega_1) &= -e^{-2\pi i x_2} e^{\eta_1(2u+\omega_1)} \sigma(u), \\ \sigma(u + \omega_2) &= -e^{-2\pi i x_1} e^{\eta_2(2u+\omega_2)} \sigma(u), \end{aligned} \tag{4}$$

and thus remains unchanged when x_1 and x_2 are changed by integers.

Restricting ourselves again to the principal characteristics, we put

$$(x_1, x_2) = \left(-\frac{1}{2}, \frac{1}{2}\right), \qquad \left(0, \frac{1}{2}\right), \qquad \left(-\frac{1}{2}, 0\right),$$

and so obtain the following three functions:

$$\begin{aligned}\sigma_{00}(u) &= \sigma\left(u, -\frac{1}{2}, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-(\eta_1+\eta_2)u} \frac{\sigma\left(u + \frac{\omega_1+\omega_2}{2}\right)}{\sigma\left(\frac{\omega_1+\omega_2}{2}\right)}, \\ \sigma_{10}(u) &= \sigma\left(u, 0, \frac{1}{2}, \omega_1, \omega_2\right) = e^{-\eta_1 u} \frac{\sigma\left(u + \frac{\omega_1}{2}\right)}{\sigma\left(\frac{\omega_1}{2}\right)}, \\ \sigma_{01}(u) &= \sigma\left(u, -\frac{1}{2}, 0, \omega_1, \omega_2\right) = e^{-\eta_2 u} \frac{\sigma\left(u + \frac{\omega_2}{2}\right)}{\sigma\left(\frac{\omega_2}{2}\right)}. \end{aligned} \tag{5}$$

The function $\sigma(u)$ itself can correspondingly be denoted by $\sigma_{11}(u)$.

For these functions, (4) gives the characteristic period equations

$$\begin{aligned}\sigma_{g_1,g_2}(u+\omega_1) &= (-1)^{g_1} e^{\eta_1(2u+\omega_1)} \sigma_{g_1,g_2}(u),\\ \sigma_{g_1,g_2}(u+\omega_2) &= (-1)^{g_2} e^{\eta_2(2u+\omega_2)} \sigma_{g_1,g_2}(u),\\ \sigma_{g_1,g_2}(u+a\omega_1+b\omega_2) &= (-1)^{g_1a+g_2b+ab} e^{(a\eta_1+b\eta_2)(2u+a\omega_1+b\omega_2)} \sigma_{g_1,g_2}(u).\end{aligned} \tag{6}$$

By application of a linear transformation the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ are permuted among one another, as formula (2) shows, or § 35, (1). According to this permutation the linear transformations decompose into six classes, the first of which comprises all transformations leaving the functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$ unchanged. These are characterized by α, δ being odd and β, γ even, which we write briefly as

$$\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} \equiv \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \pmod 2.$$

Accordingly the six classes of linear transformations are characterized as follows:

$$\begin{aligned}
&\text{I.}\ \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} \equiv \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \pmod 2 \quad (00, 01, 10),\\
&\text{II.}\ \equiv \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} \pmod 2 \quad (00, 10, 01),\\
&\text{III.}\ \equiv \begin{pmatrix}1 & 1\\ -1 & 0\end{pmatrix} \pmod 2 \quad (10, 00, 01),\\
&\text{IV.}\ \equiv \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix} \pmod 2 \quad (10, 01, 00),\\
&\text{V.}\ \equiv \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix} \pmod 2 \quad (01, 00, 10),\\
&\text{VI.}\ \equiv \begin{pmatrix}0 & 1\\ -1 & 1\end{pmatrix} \pmod 2 \quad (01, 10, 00),
\end{aligned} \tag{7}$$

where in the last column the corresponding permutation of the characteristics 00, 01, 10 is indicated.

The transformations of the first class form a group $\mathfrak{A}$ contained in the group $\mathfrak{L}$ of all linear substitutions, that is, a divisor of the group $\mathfrak{L}$ (§ 28). If

$$\alpha_1 = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \quad \alpha_2 = \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}, \quad \alpha_3 = \begin{pmatrix}1 & 1\\ -1 & 0\end{pmatrix},$$

$$\alpha_4 = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}, \quad \alpha_5 = \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}, \quad \alpha_6 = \begin{pmatrix}0 & 1\\ -1 & 1\end{pmatrix},$$

then $\alpha_1\mathfrak{A}, \alpha_2\mathfrak{A}, \alpha_3\mathfrak{A}, \alpha_4\mathfrak{A}, \alpha_5\mathfrak{A}, \alpha_6\mathfrak{A}$ are the cosets relative to $\mathfrak{A}$, and

$$\mathfrak{L} = \alpha_1\mathfrak{A} + \alpha_2\mathfrak{A} + \alpha_3\mathfrak{A} + \alpha_4\mathfrak{A} + \alpha_5\mathfrak{A} + \alpha_6\mathfrak{A},$$

and $\mathfrak{A}$ is a divisor of $\mathfrak{L}$ of finite index 6:

$$(\mathfrak{L}, \mathfrak{A}) = 6 \qquad (\text{Vol. II, § 2}).$$

The totality $\alpha_1\mathfrak{A} + \alpha_2\mathfrak{A}$ is likewise a group $\mathfrak{A}'$, and

$$\mathfrak{L} = \alpha_1\mathfrak{A}' + \alpha_3\mathfrak{A}' + \alpha_4\mathfrak{A}', \qquad (\mathfrak{L}, \mathfrak{A}') = 3.$$

Just as all linear transformations can be derived from the two fundamental transformations, so the transformations of the first class can be derived by repeated application of

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$

which may be proved in the manner of § 30.

§ 37. Representation of the σ-functions by ϑ-functions.

In order to represent the function $\sigma(u)$ as a ϑ-function, one can simply apply formula (6) of § 35 to the function

$$t(u) = \vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right),$$

that is, put

$$\sigma(u, \omega_1, \omega_2) = \omega_1 e^{-\frac{u^2}{6\omega_1^2}\frac{\vartheta_{11}'''}{\vartheta_{11}'}} \frac{\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{11}'}. \tag{1}$$

In consequence of the differential equation § 20, (4), however,

$$\vartheta_{11}''' = 4\pi i \frac{d\vartheta_{11}'}{d\omega},$$

and hence, by the definition of the function $\eta(\omega)$ in § 34, (2),

$$\frac{\vartheta_{11}'''}{\vartheta_{11}'} = 4\pi i \frac{d\log\vartheta_{11}'}{d\omega} = 12\pi i \frac{d\log\eta(\omega)}{d\omega}. \tag{2}$$

Logarithmic differentiation of (1), using (2), gives

$$\frac{d\log\sigma(u)}{du} = -\frac{4\pi i u}{\omega_1^2}\frac{d\log\eta(\omega)}{d\omega} + \frac{d\log\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{du},$$

and, putting herein $u = \omega_1/2, \omega_2/2$, and applying the formulae (8) of § 21 together with § 35, (16),

$$\begin{aligned} \eta_1 &= -\frac{2\pi i}{\omega_1}\frac{d\log\eta(\omega)}{d\omega}, \\ \eta_2 &= -\frac{2\pi i\omega_2}{\omega_1^2}\frac{d\log\eta(\omega)}{d\omega} - \frac{\pi i}{\omega_1}. \end{aligned} \tag{3}$$

Consequently one can put for σ:

$$\sigma(u) = \omega_1 e^{\eta_1 u^2/\omega_1} \frac{\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{11}'}, \tag{4}$$

and for the three functions $\sigma_{00}, \sigma_{01}, \sigma_{10}$, by § 36, (5), when one determines a constant factor from

$$\sigma_{00}(0) = 1, \qquad \sigma_{01}(0) = 1, \qquad \sigma_{10}(0) = 1, \tag{5}$$

one obtains

$$\begin{aligned} \sigma_{00}(u) &= e^{\eta_1 u^2/\omega_1} \frac{\vartheta_{00}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}}, \\ \sigma_{01}(u) &= e^{\eta_1 u^2/\omega_1} \frac{\vartheta_{01}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{01}}, \\ \sigma_{10}(u) &= e^{\eta_1 u^2/\omega_1} \frac{\vartheta_{10}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{10}}. \end{aligned} \tag{6}$$

The functions $\sigma_{00}(u), \sigma_{01}(u), \sigma_{10}(u)$ are even functions of u, and by twice differentiating the logarithms of (6) one further obtains:

$$\sigma''_{00}(0) + \sigma''_{01}(0) + \sigma''_{10}(0) = 0. \tag{7}$$

The expressions (4), (6) show at first glance an important property of the σ-functions, namely that $\sigma_{00}, \sigma_{01}, \sigma_{10}$ depend only on the ratios $u : \omega_1 : \omega_2$, while the same is true of σ, apart from the factor ω_1. Thus $\sigma, \sigma_{00}, \sigma_{01}, \sigma_{10}$ are homogeneous functions of the three variables u, ω_1, ω_2, the first of order one and the other three of order zero, or, in symbols, if λ denotes an arbitrary factor,

$$\begin{aligned} \sigma(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \lambda\sigma(u, \omega_1, \omega_2), \\ \sigma_{00}(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \sigma_{00}(u, \omega_1, \omega_2), \\ \sigma_{01}(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \sigma_{01}(u, \omega_1, \omega_2), \\ \sigma_{10}(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \sigma_{10}(u, \omega_1, \omega_2). \end{aligned} \tag{8}$$

§ 38. Linear transformations of the function $\eta(\omega)$.

The formulae (3), § 37 lead to the linear transformation of the function $\eta(\omega)$. If

$$(\omega'_1, \omega'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\omega_1, \omega_2) \tag{1}$$

is put, then (η_1, η_2) passes into

$$(\eta'_1, \eta'_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\eta_1, \eta_2) \tag{2}$$

[§ 35, (14)], and $\omega = \omega_2 : \omega_1$ passes into

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{3}$$

It now follows from (2), with regard to (3) of the preceding paragraph, that

$$\begin{aligned} \eta'_1 &= -\frac{2\pi i}{\omega'_1} \frac{d \log \eta(\omega')}{d\omega'} = -\frac{2\pi i \omega'_1}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} - \frac{\pi i \beta}{\omega_1}, \\ \eta'_2 &= -\frac{2\pi i \omega'_2}{\omega_1'^2} \frac{d \log \eta(\omega')}{d\omega'} - \frac{\pi i}{\omega'_1} = -\frac{2\pi i \omega'_2}{\omega_1^2} \frac{d \log \eta(\omega)}{d\omega} - \frac{\pi i \delta}{\omega_1}. \end{aligned}$$

These two relations give concordantly

$$\frac{d \log \eta(\omega')}{\omega_1'^2 \, d\omega'} = \frac{d \log \eta(\omega)}{\omega_1^2 \, d\omega} + \frac{\beta}{2\omega_1 \omega'_1},$$

or finally, since by (3)

$$d\omega' = \frac{d\omega}{(\alpha + \beta\omega)^2}, \qquad \omega_1'^2 \, d\omega' = \omega_1^2 \, d\omega,$$

$$\frac{d \log \eta(\omega')}{d\omega} = \frac{d \log \eta(\omega)}{d\omega} + \frac{\beta}{2(\alpha + \beta\omega)}.$$

By integration this gives

$$\eta\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \varepsilon\sqrt{\alpha + \beta\omega}\, \eta(\omega), \tag{4}$$

where ε is independent of ω, and hence depends only on the numbers $\alpha, \beta, \gamma, \delta$.

The exact determination of this constant ε, particularly with regard to the sign, is a well-known important problem, which presents peculiar difficulties, but whose solution is indispensable for us.

Solutions have been given, by various routes, by Hermite,[5] Dedekind,[6] and recently also by Mertens and Scheibner.[7] We shall here take a route which has the advantage of great simplicity, though admittedly it contains not a derivation, but only a proof of the completed formula.

For two special cases we have already carried out this determination earlier [§ 34, (4), (5)], and we shall base ourselves here on that result. The formulae are

$$\eta(\omega \pm 1) = e^{\pm\pi i/12}\eta(\omega) \tag{5}$$

and

$$\eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\,\eta(\omega), \tag{6}$$

where $\sqrt{-i\omega}$ is to be taken with positive real part.

We now put

$$\frac{\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right)}{\eta(\omega)} = E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega\Bigg), \tag{7}$$

and have to determine the value of this symbol. By (5), (6), (7), one has

$$E\left(\begin{matrix}-\alpha & -\beta \\ -\gamma & -\delta\end{matrix}; \omega\right) = E\left(\begin{matrix}\alpha & \beta \\ \gamma & \delta\end{matrix}; \omega\right), \tag{8}$$

$$E\left(\begin{matrix}1 & 0 \\ 0 & 1\end{matrix}; \omega\right) = 1, \tag{9}$$

$$E\left(\begin{matrix}1 & 0 \\ \pm 1 & 1\end{matrix}; \omega\right) = e^{\pm\pi i/12}, \tag{10}$$

$$E\left(\begin{matrix}0 & 1 \\ -1 & 0\end{matrix}; \omega\right) = \sqrt{-i\omega}. \tag{11}$$

We now consider two substitutions and their product:

$$\begin{pmatrix}\alpha'' & \beta'' \\ \gamma'' & \delta''\end{pmatrix} = \begin{pmatrix}\alpha' & \beta' \\ \gamma' & \delta'\end{pmatrix}\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}.$$

If

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

then from the definition (7) one obtains

$$E\left(\begin{matrix}\alpha'' & \beta'' \\ \gamma'' & \delta''\end{matrix}; \omega\right) = E\left(\begin{matrix}\alpha' & \beta' \\ \gamma' & \delta'\end{matrix}; \omega'\right) E\left(\begin{matrix}\alpha & \beta \\ \gamma & \delta\end{matrix}; \omega\right). \tag{12}$$

Two special cases are obtained from this by putting

$$\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix} = \begin{pmatrix}1 & 0 \\ \pm 1 & 1\end{pmatrix}, \begin{pmatrix}0 & 1 \\ -1 & 0\end{pmatrix},$$

[5] Liouville's Journal, Ser. II, vol. III, 1858; *Oeuvres de Charles Hermite*, p. 487.

[6] Explanations to no. XXVIII of Riemann's works, second edition, and "Über die elliptischen Modulfunktionen", Crelle's Journal, vol. 83, p. 265. Compare also the author's paper "Zur Theorie der elliptischen Funktionen", Acta Mathematica, vol. 6, pp. 341 ff.

[7] Mertens, "Zur linearen Transformation der ϑ-Reihen", Transactions of the American Mathematical Society, July 1901. Scheibner, "Zur linearen Transformation der Theta-Funktionen und elliptischen Modulfunktionen", Berichte der Sächsischen Gesellschaft der Wissenschaften, October 1906.

and then writing again $\alpha, \beta, \gamma, \delta$ in place of $\alpha', \beta', \gamma', \delta'$. Thus

$$E\begin{pmatrix}\alpha \pm \beta & \beta \\ \gamma \pm \delta & \delta\end{pmatrix}; \omega\Big) = e^{\pm \pi i/12} E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega \pm 1\Big), \tag{13}$$

$$E\begin{pmatrix}-\beta & \alpha \\ -\delta & \gamma\end{pmatrix}; \omega\Big) = \sqrt{-i\omega}\, E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; -\frac{1}{\omega}\Big). \tag{14}$$

It was shown in § 30 that all linear transformations can be composed by repeated application of the two fundamental transformations

$$\begin{pmatrix}1 & 0 \\ 1 & 1\end{pmatrix}, \qquad \begin{pmatrix}0 & 1 \\ -1 & 0\end{pmatrix}$$

and of their inverse transformations. Therefore, by (12), the formulae (8) to (14) completely define the symbol E.

Thus, if we know an expression satisfying these conditions, it must agree with E. In order to set up such an expression, we distinguish two cases. Since α, β are relatively prime, one of them is certainly odd. We put:

1. α odd and positive:
$$E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega\Big) = \left(\frac{\beta}{\alpha}\right) i^{(\alpha-1)/2} e^{\frac{\pi i}{12}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]} \sqrt{\alpha + \beta\omega},$$

2. β odd and positive:
$$E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; \omega\Big) = \left(\frac{\alpha}{\beta}\right) i^{(1-\beta)/2} e^{\frac{\pi i}{12}[\beta(\alpha+\delta)-(\beta^2-1)\alpha\gamma]} \sqrt{-i(\alpha + \beta\omega)}. \tag{15}$$

Here the following is still to be noted. The roots $\sqrt{\alpha + \beta\omega}$ and $\sqrt{-i(\alpha + \beta\omega)}$ are to be taken with positive real part. That one of them could be purely imaginary is excluded by the assumption that ω has positive imaginary part and that α, respectively β, is positive; for then $\alpha + \beta\omega$, respectively $-i(\alpha + \beta\omega)$, cannot be real and negative. The signs $\left(\frac{\beta}{\alpha}\right)$ and $\left(\frac{\alpha}{\beta}\right)$ denote the Legendre-Jacobi symbol from the theory of quadratic residues, with the extension that $\left(\frac{\beta}{1}\right)$ and $\left(\frac{0}{1}\right)$ are to be equal to 1. If in the first case α, or in the second β, is negative, then on the right all signs of $\alpha, \beta, \gamma, \delta$ are to be changed. If both α and β are odd, then both (15), 1 and (15), 2 can be applied, and both give, as one easily verifies by means of the reciprocity law for quadratic residues, the same result.

It is namely true, when α and β are odd, α is assumed positive, and the upper or lower sign is to be taken according as β is positive or negative, that

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\pm\alpha}{\pm\beta}\right) = \pm e^{-\pi i(\alpha+1)(\beta-1)/4},$$

$$\sqrt{\pm i(\alpha + \beta\omega)} = e^{\pm \pi i/4}\sqrt{\alpha + \beta\omega},$$

and the identity of (15), 1 and 2 then follows from the congruences

$$-3\alpha\beta + \beta(\alpha + \delta) - (\beta^2 - 1)\alpha\gamma \equiv \alpha(\gamma - \beta) - (\alpha^2 - 1)\beta\delta \pmod{24},$$
$$\alpha\gamma + \alpha\beta \equiv \beta\delta \pmod{8},$$

of which the second, if α and β are both not divisible by 3, and hence $\alpha^2 \equiv \beta^2 \equiv 1 \pmod 3$, is also valid for the modulus 24.

That (15) satisfies the formulae (8) to (11) is seen immediately; it remains to show that (13) and (14) are fulfilled. We begin with (14), where it may be assumed that α is odd and positive. If in (14) one interchanges ω with $-1/\omega$, α and $-\beta$ are interchanged, and these cannot both be even. It is

$$\sqrt{-i\omega}\sqrt{\alpha - \frac{\beta}{\omega}} = \sqrt{-i(-\beta + \alpha\omega)};$$

indeed, putting temporarily

$$-i\omega = re^{i\varphi}, \qquad \alpha - \frac{\beta}{\omega} = \rho e^{i\psi}, \qquad -i(-\beta + \alpha\omega) = r\rho e^{i(\varphi+\psi)},$$

the real parts of $-i\omega$ and $-i(-\beta + \alpha\omega)$ being positive imply

$$-\frac{\pi}{2} < \varphi < \frac{\pi}{2}, \quad -\pi < \psi < \pi, \quad -\frac{\pi}{2} < \varphi + \psi < \frac{\pi}{2},$$

so that

$$\sqrt{-i\omega} = \sqrt{r}e^{i\varphi/2}, \qquad \sqrt{\alpha - \frac{\beta}{\omega}} = \sqrt{\rho}e^{i\psi/2},$$

and the real part of their product is positive. Consequently (15), 1 gives

$$\sqrt{-i\omega}E\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}; -\frac{1}{\omega}\Bigg) = \left(\frac{\beta}{\alpha}\right) i^{(\alpha-1)/2} e^{\frac{\pi i}{12}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]} \times \sqrt{-i(-\beta + \alpha\omega)}.$$

The same formula is obtained, since

$$\left(\frac{-\beta}{\alpha}\right) = i^{\alpha-1}\left(\frac{\beta}{\alpha}\right),$$

from (15), 2 for

$$E\left(\begin{matrix}-\beta & \alpha \\ -\delta & \gamma\end{matrix}; \omega\right),$$

and so (14) is proved.

There remains formula (13). It is enough to consider only the upper signs, that is, to prove

$$E\left(\begin{matrix}\alpha+\beta & \beta \\ \gamma+\delta & \delta\end{matrix}; \omega\right) = e^{\pi i/12}E\left(\begin{matrix}\alpha & \beta \\ \gamma & \delta\end{matrix}; \omega + 1\right),$$

for the other case is reduced to this one by replacing α, γ, ω with $\alpha + \beta, \gamma + \delta, \omega + 1$. If first β is odd and positive, then (13) follows from (15), 2 by the congruence

$$(\beta^2 - 1)(1 - \beta\gamma - \alpha\delta - \beta\delta) \equiv -\beta(\beta^2 - 1)(2\gamma + \delta) \equiv 0 \pmod{24}.$$

If β is even, then α is odd. Suppose α positive; then $\alpha + \beta$ may be positive or negative. If in the first case the upper signs, in the second case the lower signs are valid, (15), 1 gives

$$e^{\pi i/12}E\left(\begin{matrix}\alpha & \beta \\ \gamma & \delta\end{matrix}; \omega + 1\right) = \left(\frac{\beta}{\alpha}\right) i^{(1-\alpha)/2} e^{\frac{\pi i}{12}[1+\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}\sqrt{\alpha + \beta + \beta\omega}, \tag{16}$$

whereas

$$E\left(\begin{matrix}\alpha+\beta & \beta \\ \gamma+\delta & \delta\end{matrix}; \omega\right) = \left(\frac{\beta}{\pm(\alpha+\beta)}\right) i^{\{1\mp(\alpha+\beta)\}/2} e^{\frac{\pi i}{12}\{(\alpha+\beta)(\gamma+\delta-\beta)-[(\alpha+\beta)^2-1]\beta\delta\}} \times \sqrt{\pm(\alpha + \beta + \beta\omega)}. \tag{17}$$

If the lower signs hold, β is negative, and on putting

$$-(\alpha + \beta + \beta\omega) = re^{i\varphi}, \qquad i(\alpha + \beta + \beta\omega) = re^{i(\varphi-\pi/2)},$$

we have to set

$$\sqrt{-(\alpha + \beta + \beta\omega)} = i\sqrt{\alpha + \beta + \beta\omega}.$$

Furthermore, by the reciprocity law for quadratic residues,[8]

$$\left(\frac{\beta}{\alpha+\beta}\right) = (-1)^{\beta(\alpha+1)/4}\left(\frac{\beta}{\alpha}\right),$$

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = (-1)^{(\alpha-1)/2}(-1)^{\beta(\alpha-1)/4}\left(\frac{\beta}{\alpha}\right).$$

It follows that the two expressions (16), (17) agree, and consequently (13) is correct, by the congruence

$$\beta(3\alpha - 2\gamma + \beta - \delta + \beta^2\delta + 2\alpha\beta\delta) \equiv 0 \pmod{24},$$

which, since β is assumed even, follows from $\alpha\delta - \beta\gamma = 1$. Thus the formulae (15) are proved.

If we put

$$E\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}; \omega\Big) = \varepsilon\sqrt{\alpha+\beta\omega}, \tag{18}$$

then ε is a twenty-fourth root of unity, whose product with $\sqrt{\alpha+\beta\omega}$ is completely determined by (15), and the transformation of the η-function is

$$\eta\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varepsilon\sqrt{\alpha+\beta\omega}\,\eta(\omega). \tag{19}$$

For ε^{12} one finds

$$\varepsilon^{12} = (-1)^{\alpha\beta+\gamma\delta+\beta\gamma}. \tag{20}$$

§ 39. Linear transformation of the ϑ-functions.

The transformation formulae of the ϑ-functions are consequences of the fundamental properties of the σ-function: the latter remains unchanged under the substitution

$$(\omega_1', \omega_2') = \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}(\omega_1, \omega_2).$$

From § 37, (4), replacing $\omega_1, \omega_2, \eta_1$ by $\omega_1', \omega_2', \eta_1'$, one obtains

$$\frac{\omega_1 e^{\eta_1 u^2/\omega_1}\vartheta_{11}\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2}{\omega_1}\right)} = \frac{\omega_1' e^{\eta_1' u^2/\omega_1'}\vartheta_{11}\left(\frac{u}{\omega_1'}, \frac{\omega_2'}{\omega_1'}\right)}{\vartheta_{11}'\left(0, \frac{\omega_2'}{\omega_1'}\right)}.$$

Since, by § 35, (10), (13), (14),

$$\frac{\eta_1}{\omega_1} - \frac{\eta_1'}{\omega_1'} = \frac{\omega_1'\eta_1 - \eta_1'\omega_1}{\omega_1\omega_1'} = \beta\frac{\omega_2\eta_1 - \omega_1\eta_2}{\omega_1\omega_1'} = \frac{\pi i\beta}{\omega_1\omega_1'}, \tag{1}$$

if we put

$$v = \frac{u}{\omega_1}, \qquad \omega = \frac{\omega_2}{\omega_1}, \qquad v' = \frac{u}{\omega_1'} = \frac{v}{\alpha+\beta\omega}, \qquad \omega' = \frac{\omega_2'}{\omega_1'} = \frac{\gamma+\delta\omega}{\alpha+\beta\omega}, \tag{2}$$

[8] By the reciprocity law, if $\beta = \pm 2^\lambda\beta'$ is put and β' is assumed odd and positive, then, when $\alpha+\beta$ is positive,

$$\left(\frac{\beta}{\alpha+\beta}\right) = \left(\frac{\pm 2^\lambda}{\alpha+\beta}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}}, \qquad \left(\frac{\beta}{\alpha}\right) = \left(\frac{\pm 2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

If $\lambda \geq 2$, these two values are equal; if $\lambda = 1$, they differ by the factor $(-1)^{(\alpha+1)/2}$, in agreement with the first of the preceding formulae. The correctness of the second formula, when $\alpha+\beta$ is negative, follows from

$$\left(\frac{-\beta}{-(\alpha+\beta)}\right) = \left(\frac{-2^\lambda}{-(\alpha+\beta)}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{(\alpha+\beta-1)(\beta'-1)}{4}}, \qquad \left(\frac{\beta}{\alpha}\right) = \left(\frac{2^\lambda}{\alpha}\right)\left(\frac{\alpha}{\beta'}\right)(-1)^{\frac{\alpha-1}{2}}(-1)^{\frac{(\alpha-1)(\beta'-1)}{4}}.$$

then
$$\frac{\vartheta_{11}(v',\omega')}{\vartheta'_{11}(0,\omega')} = \frac{e^{\pi i\beta vv'}}{\alpha+\beta\omega}\frac{\vartheta_{11}(v,\omega)}{\vartheta'_{11}(0,\omega)}.$$
Moreover
$$\vartheta'_{11}(0,\omega) = 2\pi\eta(\omega)^3, \qquad \vartheta'_{11}(0,\omega') = 2\pi\eta(\omega')^3,$$
and hence by § 38, (19)
$$\vartheta'_{11}(0,\omega') = \varepsilon^3(\alpha+\beta\omega)^{3/2}\vartheta'_{11}(0,\omega).$$
Thus finally
$$e^{-\pi i\beta vv'}\vartheta_{11}(v',\omega') = \varepsilon^3(\alpha+\beta\omega)^{1/2}\vartheta_{11}(v,\omega). \tag{3}$$
The transformation formulae for the other three functions $\vartheta_{00}, \vartheta_{01}, \vartheta_{10}$ differ in the six classes of § 36 and can be derived from formulae (5), § 36 in the same way. But the six cases can also be assembled into a single formula-system by replacing v in (3) by
$$v+\frac{\alpha+\beta\omega}{2}, \qquad v+\frac{\gamma+\delta\omega}{2}, \qquad v+\frac{\alpha+\gamma+(\beta+\delta)\omega}{2},$$
and hence replacing v' respectively by
$$v'+\frac{1}{2}, \qquad v'+\frac{\omega'}{2}, \qquad v'+\frac{1+\omega'}{2},$$
and then applying the formulae (2), (3), (8) of § 21 on both the right and left sides of (3). This gives
$$e^{-\pi i\beta vv'}\vartheta_{10}(v',\omega') = i^{\delta}e^{-\pi i\alpha\beta/4}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\delta,1-\alpha}(v,\omega), \tag{4}$$
$$e^{-\pi i\beta vv'}\vartheta_{01}(v',\omega') = i^{\delta-1}e^{-\pi i\gamma\delta/4}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\delta,1-\gamma}(v,\omega), \tag{5}$$
$$e^{-\pi i\beta vv'}\vartheta_{00}(v',\omega') = i^{\delta+\beta-\alpha\delta}e^{-\pi i(\alpha\beta+\gamma\delta)/4}\varepsilon^3\sqrt{\alpha+\beta\omega}\,\vartheta_{1+\beta+\delta,1-\alpha-\gamma}(v,\omega). \tag{6}$$
The fourth powers of these functions are expressible more simply:
$$e^{-4\pi i\beta vv'}\vartheta^4_{g_1,g_2}(v',\omega') = (-1)^{g_2\alpha\beta+g_1\gamma\delta+\beta\gamma}(\alpha+\beta\omega)^2\vartheta^4_{g'_1,g'_2}(v,\omega), \tag{7}$$
where, in the six classes, the characteristics (g'_1, g'_2) belonging to $(g_1, g_2) = (00), (01), (10)$ are to be taken from the last column of the table in § 36.

A simpler transformation formula is obtained from the σ-functions for the product of the three ϑ-functions (4), (5), (6). In fact, forming the product of the three functions § 37, (6), one has
$$\sigma_{00}(u)\sigma_{01}(u)\sigma_{10}(u) = e^{3\eta_1 u^2/\omega_1}\frac{\vartheta_{00}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)\vartheta_{01}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)\vartheta_{10}\left(\frac{u}{\omega_1},\frac{\omega_2}{\omega_1}\right)}{\vartheta_{00}\vartheta_{01}\vartheta_{10}},$$
a function which, by § 36, remains completely unchanged under linear transformation. Using in addition the relation
$$\eta'_1\omega_1 - \eta_1\omega'_1 = -\pi i\beta,$$
one obtains
$$\frac{\vartheta_{00}(v,\omega)\vartheta_{01}(v,\omega)\vartheta_{10}(v,\omega)}{\vartheta_{00}\vartheta_{01}\vartheta_{10}} = e^{-3\pi i\beta vv'}\frac{\vartheta_{00}(v',\omega')\vartheta_{01}(v',\omega')\vartheta_{10}(v',\omega')}{\vartheta_{00}(0,\omega')\vartheta_{01}(0,\omega')\vartheta_{10}(0,\omega')}. \tag{8}$$
Here one may still put, by (19) of the preceding paragraph,
$$\vartheta_{00}(0,\omega')\vartheta_{01}(0,\omega')\vartheta_{10}(0,\omega') = \varepsilon^3(\alpha+\beta\omega)^{3/2}\vartheta_{00}\vartheta_{01}\vartheta_{10}.$$
But from (8) we draw another conclusion. If one puts
$$v' = \frac{h}{n}, \qquad v = \frac{h(\alpha+\beta\omega)}{n},$$

and takes the product for $h = 1, 2, \ldots, (n-1)/2$, then, using the known relation

$$\sum_{h=1}^{(n-1)/2} h^2 = \frac{n(n^2-1)}{24}$$

and applying on the right-hand side of (8) the formula (23), § 32, one obtains

$$2^{(n-1)/2} \prod_{h=1}^{(n-1)/2} \vartheta_{00}\left(\frac{h(\alpha+\beta\omega)}{n}\right)\vartheta_{01}\left(\frac{h(\alpha+\beta\omega)}{n}\right)\vartheta_{10}\left(\frac{h(\alpha+\beta\omega)}{n}\right) = e^{-\frac{\pi i}{8}\frac{n^2-1}{n}\beta(\alpha+\beta\omega)}\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2} \tag{9}$$

where now α, β may be any relatively prime pair.

§ 40. Linear transformation of the functions $f(\omega), f_1(\omega), f_2(\omega)$.

By § 34, (9), the formulae for the linear transformation of the f-functions are a simple consequence of the transformation of the η-function. Let us first suppose, in the linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

that β is even, and therefore α, δ are odd. Then

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \tag{1}$$

that is,

$$2\frac{\gamma+\delta\omega}{\alpha+\beta\omega} = \frac{2\gamma+\delta\cdot 2\omega}{\alpha+\frac{1}{2}\beta\cdot 2\omega}.$$

But by § 34, (9),

$$f_2(\omega) = \sqrt[6]{2}\frac{\eta(2\omega)}{\eta(\omega)}.$$

If therefore the substitution $\omega, (\gamma+\delta\omega)/(\alpha+\beta\omega)$ is made, then with the notation of § 38 one obtains

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \frac{E\begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix}; 2\omega}{E\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}; \omega} f_2(\omega). \tag{2}$$

Here the E-functions are to be determined by § 38, (15), 1, from which

$$\frac{E\begin{pmatrix} \alpha & \frac{1}{2}\beta \\ 2\gamma & \delta \end{pmatrix}; 2\omega}{E\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}; \omega} = \left(\frac{2}{\alpha}\right) e^{\frac{\pi i}{12}\left(\alpha\gamma+\frac{\alpha\beta}{2}+\frac{(\alpha^2-1)\beta\delta}{2}\right)}.$$

But $\frac{1}{4} = \frac{3}{8} - \frac{1}{3}$, and

$$\begin{aligned} \alpha\gamma + \frac{\alpha\beta}{2} + \frac{(\alpha^2-1)\beta\delta}{2} &\equiv \frac{\alpha(2\gamma+\beta)}{2} \pmod 8, \\ &\equiv \alpha(\gamma-\beta) - (\alpha^2-1)\beta\delta \pmod 3. \end{aligned}$$

Thus, putting for abbreviation

$$\rho = e^{\frac{2\pi i}{3}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}, \tag{3}$$

we get from (2)

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\alpha}\right)\rho e^{\frac{3\pi i}{8}\alpha(2\gamma+\beta)} f_2(\omega), \qquad \beta \equiv 0 \pmod 2. \tag{4}$$

In the same way all other formulae of this kind can be derived. They are obtained more simply from (4) itself with the aid of the fundamental transformations § 34, (13), (14), (15). Thus, if in (4) one replaces ω by $-1 : \omega$ and then interchanges $\alpha, \beta, \gamma, \delta$ with $\beta, -\alpha, \delta, -\gamma$ (whereby ρ remains unchanged), one obtains

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\beta}\right)\rho e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)} f_1(\omega), \qquad \alpha \equiv 0 \pmod 2, \tag{5}$$

and if in this ω is replaced by $\omega + 3$ and γ, α by $\gamma - 3\delta, \alpha - 3\beta$, then

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{\frac{3\pi i}{8}\beta(2\delta-\alpha)} f(\omega), \qquad \alpha - \beta \equiv 0 \pmod 2. \tag{6}$$

If in (4), (5), (6) one puts

$$f_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = f_1\left(-\frac{\alpha+\beta\omega}{\gamma+\delta\omega}\right),$$

and then interchanges $\alpha, \beta, \gamma, \delta$ with $-\gamma, -\delta, \alpha, \beta$, one obtains

$$f_1\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \left(\frac{2}{\gamma}\right)\rho e^{-\frac{3\pi i}{8}\gamma(2\alpha-\delta)} f_2(\omega), \qquad \delta \equiv 0 \pmod 2, \tag{7}$$

$$= \left(\frac{2}{\delta}\right)\rho e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)} f_1(\omega), \qquad \gamma \equiv 0 \pmod 2, \tag{8}$$

$$= -\rho e^{-\frac{3\pi i}{8}\delta(2\beta+\gamma)} f(\omega), \qquad \gamma - \delta \equiv 0 \pmod 2. \tag{9}$$

From (9) and (6), replacing ω by

$$\frac{-\gamma+\alpha\omega}{\delta-\beta\omega}$$

and then $\alpha, \beta, \gamma, \delta$ by $\delta, -\beta, -\gamma, \alpha$, one obtains

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{-\frac{3\pi i}{8}\alpha(2\beta+\gamma)} f_1(\omega), \qquad \alpha - \gamma \equiv 0 \pmod 2, \tag{10}$$

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = -\rho e^{\frac{3\pi i}{8}\beta(2\alpha-\delta)} f_2(\omega), \qquad \beta - \delta \equiv 0 \pmod 2. \tag{11}$$

Finally, if in (10) ω is replaced by $\omega + 9$ and γ, α by $\gamma - 9\delta, \alpha - 9\beta$, then

$$f\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \rho\left(\frac{2}{\alpha-\beta}\right) e^{-\frac{3\pi i}{8}(\alpha-\beta)(\alpha+\beta+\gamma-\delta)} f(\omega), \qquad \alpha+\beta+\gamma-\delta \equiv 0 \pmod 2. \tag{12}$$

The functions introduced by Hermite, $\varphi(\omega), \psi(\omega), \chi(\omega)$, are connected with $f(\omega), f_1(\omega), f_2(\omega)$ by the equations

$$f(\omega) = \frac{\sqrt[6]{2}}{\chi(\omega)}, \qquad f_1(\omega) = \sqrt[6]{2}\frac{\psi(\omega)}{\chi(\omega)}, \qquad f_2(\omega) = \sqrt[6]{2}\frac{\varphi(\omega)}{\chi(\omega)}.$$

The transformation formulae of the f-functions can be reshaped in many ways with use of the relation $\alpha\delta - \beta\gamma = 1$. Thus the formulae given by Hermite are not immediately recognizable as identical with ours; a simple calculation, however, shows their agreement. The formulae given above have the advantage that, without losing simplicity, they combine two of the six transformation classes at a time in one expression.

Fourth Section.

The Elliptic Functions.

§ 41. Connection of the ϑ-functions with the elliptic integrals.

By § 21 there exist, between the squares of the four ϑ-functions, two mutually independent linear equations, and one can therefore express two of these squares by means of the two others, or also express all four by two independent variables ξ, η. Doing the latter, let $\xi_1, \eta_1; \xi_2, \eta_2; \xi_3, \eta_3; \xi_4, \eta_4$ denote constants, and put, in connection with the notation of Vol. I, § 67,

$$\begin{aligned}\vartheta_{01}^2(u) &= \xi\eta_1 - \eta\xi_1 = (\xi\eta_1),\\ \vartheta_{11}^2(u) &= \xi\eta_2 - \eta\xi_2 = (\xi\eta_2),\\ \vartheta_{10}^2(u) &= \xi\eta_3 - \eta\xi_3 = (\xi\eta_3),\\ \vartheta_{00}^2(u) &= \xi\eta_4 - \eta\xi_4 = (\xi\eta_4).\end{aligned} \tag{1}$$

Between the constants ξ_i, η_i and the values $\vartheta_{01}, \vartheta_{10}, \vartheta_{00}$ there are four relations, which may be derived from the equations (13) of § 21. With the notation (1) these equations may be written in the form

$$\begin{aligned}(\xi\eta_3)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{10}^2 - (\xi\eta_2)\vartheta_{00}^2,\\ (\xi\eta_4)\vartheta_{01}^2 &= (\xi\eta_1)\vartheta_{00}^2 - (\xi\eta_2)\vartheta_{10}^2.\end{aligned} \tag{2}$$

If in (2) one puts $\xi, \eta = \xi_1, \eta_1$ and $= \xi_2, \eta_2$, these relations can be given the form

$$\begin{aligned}(\xi_1\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2,\\ (\xi_2\eta_3)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2,\\ (\xi_1\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{10}^2,\\ (\xi_2\eta_4)\vartheta_{01}^2 &= (\xi_2\eta_1)\vartheta_{00}^2,\end{aligned} \tag{3}$$

to which one can still add, by putting in (2) $\xi, \eta = \xi_4, \eta_4$ and using (3) and § 21, (14),

$$(\xi_4\eta_3) = (\xi_2\eta_1). \tag{4}$$

From (3) and (4) it follows for the double ratios that

$$\frac{(\xi_2\eta_3)(\xi_1\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{5}$$

$$\frac{(\xi_1\eta_2)(\xi_3\eta_4)}{(\xi_1\eta_3)(\xi_2\eta_4)} = \frac{\vartheta_{01}^4}{\vartheta_{00}^4}. \tag{6}$$

If we now differentiate two of the equations (1), say the first two, we obtain

$$2\vartheta_{01}(u)\vartheta_{01}'(u)\,du = \eta_1\,d\xi - \xi_1\,d\eta = (\eta_1\,d\xi),$$

$$2\vartheta_{11}(u)\vartheta_{11}'(u)\,du = \eta_2\,d\xi - \xi_2\,d\eta = (\eta_2\,d\xi),$$

and from these, using (1),

$$\begin{aligned}2\vartheta_{01}(u)\vartheta_{11}(u)&\left[\vartheta_{01}'(u)\vartheta_{11}(u) - \vartheta_{11}'(u)\vartheta_{01}(u)\right]\,du\\ &= (\eta_1\,d\xi)\vartheta_{11}^2(u) - (\eta_2\,d\xi)\vartheta_{01}^2(u)\\ &= (\eta_1\,d\xi)(\xi\eta_2) - (\eta_2\,d\xi)(\xi\eta_1) = (\xi_2\eta_1)(\xi\,d\eta).\end{aligned}$$

One can obtain the last expression without calculation by regarding the preceding expression as a linear function of $d\xi$, $d\eta$, which vanishes for $d\xi : d\eta = \xi : \eta$ and is therefore divisible by $(\xi\,d\eta)$. The quotient $(\xi_2\eta_1)$ is obtained by putting $d\xi : d\eta = \xi_2 : \eta_2$. With the use of § 23, (6) one thus obtains

$$2\pi\vartheta_{01}^2\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{10}(u)\vartheta_{01}(u)\,du = (\xi_1\eta_2)(\xi\,d\eta). \tag{7}$$

If, according to (3), the first and last formula are used to introduce

$$(\xi_1\eta_2)\vartheta_{00}^2 = \sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\,\vartheta_{01}^2, \tag{8}$$

and if one substitutes for the ϑ-functions the expressions (1), then finally

$$2\pi\vartheta_{00}^2\,du = \frac{\sqrt{(\xi_1\eta_3)(\xi_2\eta_4)}\,(\xi\,d\eta)}{\sqrt{(\xi\eta_1)(\xi\eta_2)(\xi\eta_3)(\xi\eta_4)}}, \tag{9}$$

whereby du is represented as an elliptic differential of the first kind in homogeneous variables [§ 1, (5)].

By (1) the ratio $\xi : \eta$ is determined as a doubly periodic function of u. Likewise the ratios of the square roots

$$\sqrt{(\xi\eta_1)}, \qquad \sqrt{(\xi\eta_2)}, \qquad \sqrt{(\xi\eta_3)}, \qquad \sqrt{(\xi\eta_4)} \tag{10}$$

are defined as single-valued doubly periodic functions, and the sign of the square roots in (9) is thereby, and by (8), also determined uniquely.

If ω is the modulus of the ϑ-functions, then, as follows from (1) by making the four ϑ-functions vanish, the following values belong together:

$$\begin{array}{r|l} u = -\frac{1}{2}\omega & \xi : \eta = \xi_1 : \eta_1, \\ u = 0 & \xi : \eta = \xi_2 : \eta_2, \\ u = \frac{1}{2} & \xi : \eta = \xi_3 : \eta_3, \\ u = \frac{1+\omega}{2} & \xi : \eta = \xi_4 : \eta_4. \end{array} \tag{11}$$

§ 42. Jacobi's elliptic functions.

Since one can replace the variables ξ, η by two new variables by means of a linear substitution in which four coefficients are disposable, one can assign arbitrary values to four of the quantities ξ_i, η_i, or to three of their ratios, without impairing the generality.

We shall put

$$\xi_1 = 0, \qquad \eta_2 = 0, \qquad \xi_3 = \eta_3, \tag{1}$$

and further introduce $\varkappa^2, \varkappa'^2, \zeta$ by the equations

$$\xi_4 = \varkappa^2\eta_4, \qquad \varkappa'^2 = 1 - \varkappa^2, \qquad \frac{\eta}{\xi} = \zeta. \tag{2}$$

From § 41, (3), (5), (6), there results

$$\begin{gathered} \frac{\xi_2}{\eta_1} = -\frac{\vartheta_{10}^2}{\vartheta_{00}^2}, \qquad \frac{\eta_3}{\eta_1} = \frac{\vartheta_{10}^2}{\vartheta_{01}^2}, \qquad \frac{\eta_4}{\eta_1} = \frac{\vartheta_{00}^2}{\vartheta_{01}^2}, \\ \sqrt{\varkappa} = \frac{\vartheta_{10}}{\vartheta_{00}}, \qquad \sqrt{\varkappa'} = \frac{\vartheta_{01}}{\vartheta_{00}}. \end{gathered} \tag{3}$$

From (1) and (9) of § 41 one finds:

$$\begin{aligned} \frac{\vartheta_{00}\vartheta_{11}(u)}{\vartheta_{10}\vartheta_{01}(u)} &= \sqrt{\zeta}, \\ \frac{\vartheta_{01}\vartheta_{10}(u)}{\vartheta_{10}\vartheta_{01}(u)} &= \sqrt{1-\zeta}, \\ \frac{\vartheta_{01}\vartheta_{00}(u)}{\vartheta_{00}\vartheta_{01}(u)} &= \sqrt{1-\varkappa^2\zeta}, \end{aligned} \tag{4}$$

and

$$2\pi\vartheta_{00}^2 u = \int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}. \tag{5}$$

In this formula the path of integration and the meaning of the root signs are determined by the formulae (4) themselves when the transition from 0 to u in the u-plane is given. The path of integration in (5), however, may then also be changed, provided only that none of the singular points $\infty, 1, 1/\varkappa^2$ is crossed.

The equation (5) can be written in the following three forms:

$$\begin{aligned} d\sqrt{\zeta} &= \pi\vartheta_{00}^2\sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\ du,\\ d\sqrt{1-\zeta} &= -\pi\vartheta_{00}^2\sqrt{\zeta(1-\varkappa^2\zeta)}\ du,\\ d\sqrt{1-\varkappa^2\zeta} &= -\pi\vartheta_{00}^2\varkappa^2\sqrt{\zeta(1-\zeta)}\ du. \end{aligned} \tag{6}$$

We introduce a new variable v by the equation

$$\pi\vartheta_{00}^2 u = v. \tag{7}$$

Then the equations (5), (6) become

$$v = \frac{1}{2}\int_0^\zeta \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{8}$$

$$\begin{aligned} d\sqrt{\zeta} &= \sqrt{(1-\zeta)(1-\varkappa^2\zeta)}\ dv,\\ d\sqrt{1-\zeta} &= -\sqrt{\zeta(1-\varkappa^2\zeta)}\ dv,\\ d\sqrt{1-\varkappa^2\zeta} &= -\varkappa^2\sqrt{\zeta(1-\zeta)}\ dv. \end{aligned} \tag{9}$$

We now regard the three quantities $\sqrt{\zeta} = x$, $\sqrt{1-\zeta} = y$, $\sqrt{1-\varkappa^2\zeta} = z$ as functions of the variable v, defined by (4) and (7). By (4) they are single-valued, doubly periodic functions of v, continuous apart from isolated points at which they become infinite. Following Jacobi, they are called the sinus amplitudinis, cosinus amplitudinis, and Δ amplitudinis of v, and are denoted by $\sin\operatorname{am} v, \cos\operatorname{am} v, \Delta\operatorname{am} v$. We shall use here the shorter notation of Gudermann, already mentioned in § 14, namely $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$, whence

$$\begin{aligned} \frac{\vartheta_{11}(u)}{\vartheta_{01}(u)} &= \sqrt{\varkappa}\ \operatorname{sn} v,\\ \frac{\vartheta_{10}(u)}{\vartheta_{01}(u)} &= \sqrt{\frac{\varkappa}{\varkappa'}}\ \operatorname{cn} v,\\ \frac{\vartheta_{00}(u)}{\vartheta_{01}(u)} &= \frac{1}{\sqrt{\varkappa'}}\ \operatorname{dn} v. \end{aligned} \tag{10}$$

These functions satisfy, by (9), the differential equations

$$\frac{dx}{dv} = yz, \qquad \frac{dy}{dv} = -zx, \qquad \frac{dz}{dv} = -\varkappa^2 xy, \tag{11}$$

and the auxiliary conditions

$$\operatorname{sn} 0 = 0, \qquad \operatorname{cn} 0 = 1, \qquad \operatorname{dn} 0 = 1. \tag{12}$$

The relations between them are

$$y^2 = 1 - x^2, \qquad z^2 = 1 - \varkappa^2 x^2. \tag{13}$$

From the fundamental properties of the ϑ-functions the first properties of the elliptic functions follow:

$$\operatorname{sn} v = -\operatorname{sn}(-v), \qquad \operatorname{cn} v = \operatorname{cn}(-v), \qquad \operatorname{dn} v = \operatorname{dn}(-v),$$

that is, $\operatorname{sn} v$ is an odd function, while $\operatorname{cn} v$ and $\operatorname{dn} v$ are even functions.

Put further

$$\pi\vartheta_{00}^2 = 2K, \qquad \pi\vartheta_{00}^2\omega = 2iK', \tag{14}$$

and consequently

$$\pi\vartheta_{01}^2 = 2\varkappa' K, \qquad \pi\vartheta_{10}^2 = 2\varkappa K, \qquad \omega = \frac{iK'}{K}. \tag{15}$$

Then v assumes the values $K, iK', K+iK'$ when $u = \frac{1}{2}, \frac{\omega}{2}, \frac{1+\omega}{2}$; and from (4), with regard to (3) and § 21, (10), one obtains

$$\begin{aligned}
\operatorname{sn} K &= 1, & \operatorname{sn}(K+iK') &= \frac{1}{\varkappa},\\
\operatorname{cn} K &= 0, & \operatorname{cn}(K+iK') &= -\frac{i\varkappa'}{\varkappa},\\
\operatorname{dn} K &= \varkappa', & \operatorname{dn}(K+iK') &= 0,\\
\operatorname{sn} iK', \quad \operatorname{cn} iK', && \operatorname{dn} iK' &= \infty.
\end{aligned} \tag{16}$$

Now K, K' may be expressed by means of (8) by definite integrals. If v is displaced from a vertex of the parallelogram $0, K, K+iK', iK'$ to the following one along the periphery, and hence u along

$$0, \frac{1}{2}, \frac{1+\omega}{2}, \frac{\omega}{2},$$

one obtains

$$K = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{17}$$

$$iK' = \frac{1}{2}\int_1^{1/\varkappa^2} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{18}$$

$$-K = \frac{1}{2}\int_{1/\varkappa^2}^{\infty} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}, \tag{19}$$

$$-iK' = \frac{1}{2}\int_{\infty}^{0} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}}. \tag{20}$$

The values of $\sqrt{\zeta}, \sqrt{1-\zeta}, \sqrt{1-\varkappa^2\zeta}$ in these integrations are determined by the formulae (4). If, however, ω is purely imaginary, and consequently $\varkappa^2$ is a positive proper fraction and K, K' are real and positive, then the paths for v are parallel to the real and imaginary axes, and, with regard to the remarks at the end of § 25, the formulae (4) show the following:

in (17)	$\sqrt{\zeta},\ \sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (18)	$\sqrt{\zeta},\ i\sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (19)	$\sqrt{\zeta},\ i\sqrt{1-\zeta},\ i\sqrt{1-\varkappa^2\zeta}$	are real and positive,
in (20)	$-i\sqrt{\zeta},\ \sqrt{1-\zeta},\ \sqrt{1-\varkappa^2\zeta}$	are real and positive.

Moreover $\sqrt{\zeta}$ has no maximum or minimum on any of the paths of integration. Hence K, K' can be defined by the following four real integrals with positive square root:

$$K = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2\zeta)}} = \frac{1}{2}\int_{1/\varkappa^2}^{\infty} \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\varkappa^2\zeta-1)}}, \tag{21}$$

$$K' = \frac{1}{2}\int_{-\infty}^{0} \frac{d\zeta}{\sqrt{-\zeta(1-\zeta)(1-\varkappa^2\zeta)}} = \frac{1}{2}\int_1^{1/\varkappa^2} \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(1-\varkappa^2\zeta)}}. \tag{22}$$

§ 43. The Jacobi functions $\Theta(v), H(v)$.

If the ϑ-functions of the variable u are regarded as functions of v, one obtains the Jacobi Θ-functions. We put

$$\vartheta_{01}\left(\frac{v}{2K},\omega\right)=\Theta(v),\qquad \vartheta_{11}\left(\frac{v}{2K},\omega\right)=H(v). \tag{1}$$

Then $\Theta(v), H(v)$ are to be denoted as functions $t(v, 2K, 2K')$, and by § 25 one has

$$\begin{aligned}\Theta(v)&=1-2q\cos\frac{\pi v}{K}+2q^4\cos\frac{2\pi v}{K}-2q^9\cos\frac{3\pi v}{K}+\cdots,\\ H(v)&=2q^{1/4}\sin\frac{\pi v}{2K}-2q^{9/4}\sin\frac{3\pi v}{2K}+2q^{25/4}\sin\frac{5\pi v}{2K}-\cdots,\\ q&=e^{-\pi K'/K}.\end{aligned} \tag{2}$$

By § 21, (8),

$$\begin{aligned}H(v)&=-ie^{-\frac{\pi K'}{4K}+\frac{\pi i v}{2K}}\Theta(v+iK'),\\ \Theta(v)&=-ie^{-\frac{\pi K'}{4K}+\frac{\pi i v}{2K}}H(v+iK').\end{aligned} \tag{3}$$

Furthermore,

$$\vartheta_{00}\left(\frac{v}{2K},\omega\right)=\Theta(v+K),\qquad \vartheta_{10}\left(\frac{v}{2K},\omega\right)=H(v+K), \tag{4}$$

whence, by means of the formulae (3), (10), (15) of the preceding paragraph and (5), § 23,

$$\begin{gathered}\sqrt{\varkappa}=\frac{H(K)}{\Theta(K)},\qquad \sqrt{\varkappa'}=\frac{\Theta(0)}{\Theta(K)},\\ \Theta(K)=\sqrt{\frac{2K}{\pi}},\qquad \Theta(0)=\sqrt{\frac{2\varkappa' K}{\pi}},\qquad H(K)=\sqrt{\frac{2\varkappa K}{\pi}},\\ H'(0)=\frac{\Theta(0)H(K)}{\Theta(K)}=\sqrt{\frac{2\varkappa\varkappa' K}{\pi}}.\end{gathered} \tag{5}$$

Finally the representation of the elliptic functions is

$$\frac{H(v)}{\Theta(v)}=\sqrt{\varkappa}\,\operatorname{sn} v,\qquad \frac{H(v+K)}{\Theta(v)}=\sqrt{\frac{\varkappa}{\varkappa'}}\operatorname{cn} v,\qquad \frac{\Theta(v+K)}{\Theta(v)}=\frac{1}{\sqrt{\varkappa'}}\operatorname{dn} v. \tag{6}$$

The functions Θ, H have, as follows easily from (3), the periodicity expressed by the following equations:

$$\Theta(v+2nK)=\Theta(v),\qquad \Theta(v+2niK')=(-1)^n e^{\frac{n^2\pi K'}{K}-\frac{\pi i n v}{K}}\Theta(v), \tag{7}$$

$$H(v+2nK)=(-1)^n H(v),\qquad H(v+2niK')=(-1)^n e^{\frac{n^2\pi K'}{K}-\frac{\pi i n v}{K}}H(v). \tag{8}$$

It is sometimes useful to combine the equations (3), (7), (8) as follows:

$$\Phi(v+niK')=i^n e^{\frac{n^2\pi i K'}{4K}-\frac{\pi i n v}{2K}}\Psi(v), \tag{9}$$

where, if n is even, Φ and Ψ are both equal to Θ or both equal to H, while if n is odd, $\Phi=\Theta$, $\Psi=H$, or $\Phi=H$, $\Psi=\Theta$, is to be put.

From § 22, (2), (4), the addition formulae result:

$$\begin{aligned}\Theta^2(0)\Theta(u+v)\Theta(u-v)&=\Theta^2(u)\Theta^2(v)-H^2(u)H^2(v),\\ \Theta^2(0)H(u+v)H(u-v)&=H^2(u)\Theta^2(v)-\Theta^2(u)H^2(v),\end{aligned} \tag{10}$$

for which one may also write

$$\begin{aligned}\Theta^2(0)\Theta(u+v)\Theta(u-v)&=\Theta^2(u)\Theta^2(v)(1-\varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v),\\ \Theta^2(0)H(u+v)H(u-v)&=\Theta^2(u)\Theta^2(v)\varkappa(\operatorname{sn}^2 u-\operatorname{sn}^2 v).\end{aligned} \tag{11}$$

We now pass to transferring the theorems on the ϑ-functions to the elliptic functions, and begin with the addition theorem.

§ 44. Addition theorem of the elliptic functions.

From the four ϑ-functions one can form altogether twelve quotients, which by § 42 can be expressed by elliptic functions. If these quotients are formed for the argument $u+v$, then by § 22 one can express them by ϑ-functions of u and of v separately, and hence also by the corresponding elliptic functions, always in such a way that the denominator contains only squares of ϑ and is therefore expressed rationally by $\operatorname{sn}^2 u, \operatorname{sn}^2 v$. Let us, in order to carry out an example, take formula (5), § 22, and divide it first by (1) and then by (4), changing the first time in (5) v into $-v$; then

$$\frac{\vartheta_{01}}{\vartheta_{00}^2}\frac{\vartheta_{10}\vartheta_{11}(u+v)}{\vartheta_{00}(u+v)} = \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)+\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{00}^2(u)\vartheta_{00}^2(v)+\vartheta_{11}^2(u)\vartheta_{11}^2(v)},$$

$$\frac{\vartheta_{10}}{\vartheta_{01}}\frac{\vartheta_{00}(u+v)}{\vartheta_{11}(u+v)} = \frac{\vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v)-\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v)}{\vartheta_{11}^2(u)\vartheta_{01}^2(v)-\vartheta_{01}^2(u)\vartheta_{11}^2(v)}.$$

If u, v are replaced by $u : 2K, v : 2K$, everything can be expressed by elliptic functions according to § 42, (10). One obtains

$$\frac{\operatorname{sn}(u+v)}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{sn} u \operatorname{cn} v + \operatorname{dn} v \operatorname{sn} v \operatorname{cn} u}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{1}$$

$$\frac{\operatorname{dn}(u+v)}{\operatorname{sn}(u+v)} = \frac{\operatorname{dn} u \operatorname{sn} u \operatorname{cn} v - \operatorname{dn} v \operatorname{sn} v \operatorname{cn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}. \tag{2}$$

In the same way, from the formulae (6), (3), (4); (7), (2), (4); (8), (2), (3); (9), (1), (2); (10), (1), (3), of § 22, one obtains

$$\frac{\operatorname{sn}(u+v)}{\operatorname{cn}(u+v)} = \frac{\operatorname{cn} u \operatorname{sn} u \operatorname{dn} v + \operatorname{cn} v \operatorname{sn} v \operatorname{dn} u}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{3}$$

$$\frac{\operatorname{cn}(u+v)}{\operatorname{sn}(u+v)} = \frac{\operatorname{cn} u \operatorname{sn} u \operatorname{dn} v - \operatorname{cn} v \operatorname{sn} v \operatorname{dn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \tag{4}$$

$$\operatorname{sn}(u+v) = \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v + \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{5}$$

$$\frac{1}{\operatorname{sn}(u+v)} = \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v - \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \tag{6}$$

$$\operatorname{cn}(u+v) = \frac{\operatorname{cn} u \operatorname{cn} v - \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{7}$$

$$\frac{1}{\operatorname{cn}(u+v)} = \frac{\operatorname{cn} u \operatorname{cn} v + \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{8}$$

$$\operatorname{dn}(u+v) = \frac{\operatorname{dn} u \operatorname{dn} v - \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{9}$$

$$\frac{1}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v + \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{10}$$

$$\frac{\operatorname{dn}(u+v)}{\operatorname{cn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v \operatorname{cn} u \operatorname{cn} v + \varkappa'^2 \operatorname{sn} u \operatorname{sn} v}{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{11}$$

$$\frac{\operatorname{cn}(u+v)}{\operatorname{dn}(u+v)} = \frac{\operatorname{dn} u \operatorname{dn} v \operatorname{cn} u \operatorname{cn} v - \varkappa'^2 \operatorname{sn} u \operatorname{sn} v}{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \tag{12}$$

One can still derive many other formulae in the same way. Among them we record the following three, which follow by division of § 22, (4), (3), (1), by (2):

$$\operatorname{sn}(u+v)\operatorname{sn}(u-v) = \frac{\operatorname{sn}^2 u - \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{13}$$

$$\operatorname{cn}(u+v)\operatorname{cn}(u-v) = \frac{\operatorname{cn}^2 u \operatorname{cn}^2 v - \varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{14}$$

$$\operatorname{dn}(u+v)\operatorname{dn}(u-v) = \frac{\operatorname{dn}^2 u \operatorname{dn}^2 v + \varkappa^2\varkappa'^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \tag{15}$$

The most important among these formulae are (5), (7), (9); from them, although by rather lengthy calculations, the others can all be derived. For clarity we put them once more in order:

$$\begin{aligned}
\operatorname{sn}(u+v) &= \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v + \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{1-\varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v},\\
\operatorname{cn}(u+v) &= \frac{\operatorname{cn} u \operatorname{cn} v - \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{1-\varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v},\\
\operatorname{dn}(u+v) &= \frac{\operatorname{dn} u \operatorname{dn} v - \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{1-\varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}.
\end{aligned} \tag{16}$$

By combining two of these equations at a time, one can give them, among others, the forms

$$\begin{aligned}
\operatorname{cn}(u+v)\operatorname{dn} u \operatorname{dn} v - \operatorname{dn}(u+v)\operatorname{cn} u \operatorname{cn} v &= -\varkappa'^2 \operatorname{sn} u \operatorname{sn} v,\\
\operatorname{dn}(u+v)\operatorname{sn} u \operatorname{cn} v - \operatorname{sn}(u+v)\operatorname{dn} u &= -\operatorname{cn} u \operatorname{sn} v,\\
\operatorname{sn}(u+v)\operatorname{cn} u - \operatorname{cn}(u+v)\operatorname{sn} u \operatorname{dn} v &= \operatorname{dn} u \operatorname{sn} v,\\
\operatorname{cn}(u+v)\operatorname{cn} u + \operatorname{sn}(u+v)\operatorname{sn} u \operatorname{dn} v &= \operatorname{cn} v.
\end{aligned} \tag{17}$$

We first apply the formulae (16) to establishing the periodicity of the elliptic functions, which of course can also be derived from § 21. If in (16) one puts $v = \pm K$, $v = K + iK'$, then by § 42, (16),

$$\operatorname{sn}(u \pm K) = \pm\frac{\operatorname{cn} u}{\operatorname{dn} u}, \qquad \operatorname{cn}(u \pm K) = \mp\frac{\varkappa' \operatorname{sn} u}{\operatorname{dn} u}, \qquad \operatorname{dn}(u \pm K) = \frac{\varkappa'}{\operatorname{dn} u}, \tag{18}$$

$$\operatorname{sn}(u+K+iK') = \frac{1}{\varkappa}\frac{\operatorname{dn} u}{\operatorname{cn} u}, \qquad \operatorname{cn}(u+K+iK') = \frac{i\varkappa'}{\varkappa}\frac{1}{\operatorname{cn} u}, \qquad \operatorname{dn}(u+K+iK') = \frac{i\varkappa' \operatorname{sn} u}{\operatorname{cn} u}. \tag{19}$$

If in (19) u is changed into $u - K$, then

$$\operatorname{sn}(u+iK') = \frac{1}{\varkappa}\frac{1}{\operatorname{sn} u}, \qquad \operatorname{cn}(u+iK') = -\frac{i}{\varkappa}\frac{\operatorname{dn} u}{\operatorname{sn} u}, \qquad \operatorname{dn}(u+iK') = -i\frac{\operatorname{cn} u}{\operatorname{sn} u}. \tag{20}$$

The formulae (20) show that the three functions $\operatorname{sn} u, \operatorname{cn} u, \operatorname{dn} u$ become infinite for $u = iK'$. Forming the quotients of two at a time gives

$$\lim \frac{\operatorname{sn} u}{\operatorname{cn} u} = i, \qquad \lim \frac{\operatorname{sn} u}{\operatorname{dn} u} = i \qquad \text{for } u = iK'. \tag{21}$$

With the help of the relations (18), (19), (20), one can derive from each addition formula three others, by replacing u by $u+K$, $u+K+iK'$, $u+iK'$, and hence making the substitutions collected in the following table:

$\operatorname{sn} u,$	$\operatorname{cn} u,$	$\operatorname{dn} u,$
$\frac{\operatorname{cn} u}{\operatorname{dn} u},$	$-\frac{\varkappa' \operatorname{sn} u}{\operatorname{dn} u},$	$\frac{\varkappa'}{\operatorname{dn} u},$
$\frac{1}{\varkappa}\frac{\operatorname{dn} u}{\operatorname{cn} u},$	$\frac{i\varkappa'}{\varkappa \operatorname{cn} u},$	$\frac{i\varkappa' \operatorname{sn} u}{\operatorname{cn} u},$
$\frac{1}{\varkappa \operatorname{sn} u},$	$-\frac{i \operatorname{dn} u}{\varkappa \operatorname{sn} u},$	$-\frac{i \operatorname{cn} u}{\operatorname{sn} u}.$

The same substitutions are to be applied also to $\operatorname{sn}(u \pm v)$, $\operatorname{cn}(u \pm v)$, $\operatorname{dn}(u \pm v)$. Thus, for example, the twelve formulae (1) to (12) arise from the three formulae (16).

If these substitutions are applied to the relations (18), (19), (20) themselves, one obtains

$$\begin{aligned}
\operatorname{sn}(u+2K) &= -\operatorname{sn} u, &\qquad \operatorname{sn}(u+2iK') &= \operatorname{sn} u,\\
\operatorname{cn}(u+2K) &= -\operatorname{cn} u, &\qquad \operatorname{cn}(u+2iK') &= -\operatorname{cn} u,\\
\operatorname{dn}(u+2K) &= \operatorname{dn} u, &\qquad \operatorname{dn}(u+2iK') &= -\operatorname{dn} u.
\end{aligned} \tag{22}$$

The common periods of these three functions are therefore $4K, 4iK'$; but in addition $\operatorname{sn} u$ has the period $2iK'$, $\operatorname{dn} u$ the period $2K$, and $\operatorname{cn} u$ the period $2K + 2iK'$.

We now give the theorem on Θ-functions (§ 21) only another expression, by stating the following theorem: every doubly periodic function of v with the periods $2K, 2iK'$ which everywhere has the character of a rational function is, if it is an even function, a rational function of $\operatorname{sn}^2 v$, and if it is an odd function, the product of $\operatorname{sn} v \operatorname{cn} v \operatorname{dn} v$ with a rational function of $\operatorname{sn}^2 v$.

Thus every theorem which relates to homogeneous functions of degree zero of ϑ-functions can be transformed into a theorem on elliptic functions; and from this one again obtains corresponding theorems on elliptic integrals. One sees at once that the addition formulae (16) are no others than the formulae (17) of § 13 in the first section. In the same way the transformation theory of the ϑ-functions gives us a solution of Jacobi's transformation problem for the elliptic integrals, as we presented it in § 8.

We shall prove this for the two principal transformations of the second order, which in § 32 as well as in § 9 we called the Gaussian and Landen transformations. By § 32, for the Gaussian transformation,

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right) \vartheta_{11}\left(u, \frac{\omega}{2}\right)}{\vartheta_{00}\left(0, \frac{\omega}{2}\right) \vartheta_{01}\left(u, \frac{\omega}{2}\right)} = \frac{2\vartheta_{01}(u)\vartheta_{11}(u)}{\vartheta_{01}^2(u) + \vartheta_{11}^2(u)}, \tag{23}$$

$$\frac{\vartheta_{10}\left(0, \frac{\omega}{2}\right)^2}{\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2} = \frac{2\vartheta_{10}\vartheta_{00}}{\vartheta_{00}^2 + \vartheta_{10}^2} = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \tag{24}$$

$$\vartheta_{00}\left(0, \frac{\omega}{2}\right)^2 = \vartheta_{00}^2 + \vartheta_{10}^2 = (1+\varkappa)\vartheta_{00}^2. \tag{25}$$

According to § 42 we therefore obtain a relation between elliptic functions with two different moduli. If λ, L, L' arise from $\varkappa, K, K'$ by the interchange $(\omega, \omega/2)$, then by (23), (24), (25)

$$\lambda = \frac{2\sqrt{\varkappa}}{1+\varkappa}, \qquad L = (1+\varkappa)K, \qquad L' = (1+\varkappa)K', \tag{26}$$

$$\operatorname{sn}[(1+\varkappa)v, \lambda] = \frac{(1+\varkappa)\operatorname{sn} v}{1+\varkappa \operatorname{sn}^2 v}, \tag{27}$$

where, for clarity, the modulus is included under the function sign sn as a second argument.

For the Landen transformation,

$$\frac{\vartheta_{11}(2u, 2\omega)}{\vartheta_{01}(2u, 2\omega)} = \frac{\vartheta_{10}(u)\vartheta_{11}(u)}{\vartheta_{00}(u)\vartheta_{01}(u)},$$

$$\frac{\vartheta_{01}(0, 2\omega)^2}{\vartheta_{00}(0, 2\omega)^2} = \frac{2\vartheta_{00}\vartheta_{01}}{\vartheta_{00}^2 + \vartheta_{01}^2} = \frac{2\sqrt{\varkappa'}}{1+\varkappa'},$$

$$2\vartheta_{00}^2(0, 2\omega) = (1+\varkappa')\vartheta_{00}^2.$$

It follows, when λ, λ', L, L' arise from $\varkappa, \varkappa', K, K'$ by the interchange $(\omega, 2\omega)$, that

$$\lambda' = \frac{2\sqrt{\varkappa'}}{1+\varkappa'}, \qquad \sqrt{\lambda} = \frac{\varkappa}{1+\varkappa'}, \qquad \lambda = \frac{1-\varkappa'}{1+\varkappa'}, \tag{28}$$

$$2L = (1+\varkappa')K, \qquad L' = (1+\varkappa')K',$$

$$\operatorname{sn}[(1+\varkappa')v, \lambda] = \frac{(1+\varkappa')\operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \tag{29}$$

and in (26) to (29), according to § 42, one recognizes the formulae (2) and (4) of § 9.

§ 45. Linear transformation of the elliptic functions.

The action of linear transformation on the elliptic functions is best surveyed from their representation by the σ-functions. From (4) or (10) of § 42 and the formulae (4), (6) of § 37 one obtains

$$\frac{\omega_1}{2K}\operatorname{sn}\frac{2Ku}{\omega_1} = \frac{\sigma(u)}{\sigma_{01}(u)}, \qquad \operatorname{cn}\frac{2Ku}{\omega_1} = \frac{\sigma_{10}(u)}{\sigma_{01}(u)}, \qquad \operatorname{dn}\frac{2Ku}{\omega_1} = \frac{\sigma_{00}(u)}{\sigma_{01}(u)}, \tag{1}$$

or also, using the homogeneity of the σ-function [§ 37, (8)],

$$\operatorname{sn} v = \frac{\sigma(v, 2K, 2iK')}{\sigma_{01}(v, 2K, 2iK')}, \qquad \operatorname{cn} v = \frac{\sigma_{10}(v, 2K, 2iK')}{\sigma_{01}(v, 2K, 2iK')}, \qquad \operatorname{dn} v = \frac{\sigma_{00}(v, 2K, 2iK')}{\sigma_{01}(v, 2K, 2iK')}. \tag{2}$$

The σ-functions occurring on the right side of (2) depend only on the two variables v, ω, and it is first a matter of determining the change of these functions under application of the fundamental transformations

$$(\omega, \omega + 1), \qquad \left(\omega, -\frac{1}{\omega}\right).$$

Because

$$\pi\vartheta_{00}^2 = 2K, \qquad 2iK' = 2\omega K,$$

one obtains from § 31, (6), (11), the corresponding changes:

$$\begin{array}{ccccc} \omega, & 2K, & 2iK', & \varkappa, & \varkappa', \\ \omega+1, & 2\varkappa' K, & 2\varkappa'(K+iK'), & \frac{i\varkappa}{\varkappa'}, & \frac{1}{\varkappa'}, \\ -1/\omega, & 2K', & 2iK, & \varkappa', & \varkappa. \end{array}$$

Put temporarily

$$f(v, \omega) = \sigma(v, 2K, 2iK').$$

Then

$$\begin{aligned} f(v, \omega+1) &= \sigma[v, 2\varkappa' K, 2\varkappa'(K+iK')] \\ &= \sigma(v, 2\varkappa' K, 2\varkappa' iK') \qquad \text{[by § 35, (7)]} \\ &= \varkappa'\sigma\left(\frac{v}{\varkappa'}, 2K, 2iK'\right) \qquad \text{[by § 37, (8)]}, \\ f\left(v, -\frac{1}{\omega}\right) &= \sigma(v, 2K', 2iK) \\ &= \sigma(v, -2iK, 2K') \qquad \text{[by § 35, (7)]} \\ &= -i\sigma(iv, 2K, 2iK') \qquad \text{[by § 37, (8)]}. \end{aligned}$$

Doing the corresponding calculation for the other three σ-functions and taking § 36, (7), into account, one obtains the following corresponding substitutions:

$$\begin{array}{ccccc} \omega, & \sigma(v), & \sigma_{00}(v), & \sigma_{01}(v), & \sigma_{10}(v), \\ \omega+1, & \varkappa'\sigma\left(\frac{v}{\varkappa'}\right), & \sigma_{01}\left(\frac{v}{\varkappa'}\right), & \sigma_{00}\left(\frac{v}{\varkappa'}\right), & \sigma_{10}\left(\frac{v}{\varkappa'}\right), \\ -1/\omega, & -i\sigma(iv), & \sigma_{00}(iv), & \sigma_{10}(iv), & \sigma_{01}(iv), \end{array} \tag{3}$$

where the periods are $2K, 2iK'$. From this, by (2), follow the first two linear transformations of the elliptic functions:

$$\begin{aligned} \operatorname{sn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \varkappa'\frac{\operatorname{sn} v}{\operatorname{dn} v}, \\ \operatorname{cn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \frac{\operatorname{cn} v}{\operatorname{dn} v}, \\ \operatorname{dn}\left(\varkappa' v, \frac{i\varkappa}{\varkappa'}\right) &= \frac{1}{\operatorname{dn} v}, \end{aligned} \tag{4}$$

$$\begin{aligned}\operatorname{sn}(iv,\varkappa') &= i\frac{\operatorname{sn} v}{\operatorname{cn} v},\\ \operatorname{cn}(iv,\varkappa') &= \frac{1}{\operatorname{cn} v},\\ \operatorname{dn}(iv,\varkappa') &= \frac{\operatorname{dn} v}{\operatorname{cn} v}.\end{aligned} \tag{5}$$

The first formula (5) shows by § 44, (21), that $\operatorname{sn}(v,\varkappa') = 1$ when $v = K'$; hence by § 42, (17), the second frequently used representation for K' is obtained:

$$K' = \frac{1}{2}\int_0^1 \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa'^2\zeta)}}. \tag{6}$$

From (4), (5) one obtains the remaining cases of linear transformation by repeated application. Put in (4) $iv, \varkappa'$ in place of $v, \varkappa$ and apply (5); then

$$\begin{aligned}\operatorname{sn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= i\varkappa\frac{\operatorname{sn} v}{\operatorname{dn} v},\\ \operatorname{cn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{1}{\operatorname{dn} v},\\ \operatorname{dn}\left(i\varkappa v, \frac{i\varkappa'}{\varkappa}\right) &= \frac{\operatorname{cn} v}{\operatorname{dn} v}.\end{aligned} \tag{7}$$

Conversely, replace in (5) $v, \varkappa, \varkappa'$ by $\varkappa' v, \frac{i\varkappa}{\varkappa'}, \frac{1}{\varkappa'}$, and apply (4). Then

$$\begin{aligned}\operatorname{sn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= i\varkappa'\frac{\operatorname{sn} v}{\operatorname{cn} v},\\ \operatorname{cn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{\operatorname{dn} v}{\operatorname{cn} v},\\ \operatorname{dn}\left(i\varkappa' v, \frac{1}{\varkappa'}\right) &= \frac{1}{\operatorname{cn} v}.\end{aligned} \tag{8}$$

If in this one again replaces v by iv and $\varkappa, \varkappa'$ by $\varkappa', \varkappa$, and applies (5), one finds

$$\begin{aligned}\operatorname{sn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \varkappa \operatorname{sn} v,\\ \operatorname{cn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{dn} v,\\ \operatorname{dn}\left(\varkappa v, \frac{1}{\varkappa}\right) &= \operatorname{cn} v,\end{aligned} \tag{9}$$

with which the six classes of linear transformations are exhausted, since more frequent repetition gives occasion for no new formulae.

§ 46. The Weierstrass $\wp$-function.

The linear transformations of the elliptic functions suggest that one should seek an elliptic function which, like the σ-function itself, is unchanged with respect to the linear transformations. In order to form such a function, we first put the equations (1) of the preceding paragraph into the following form:

$$\begin{aligned} \frac{\sigma_{10}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{\operatorname{cn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{00}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{\operatorname{dn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{01}(u)}{\sigma(u)} &= \frac{2K}{\omega_1}\frac{1}{\operatorname{sn} v}, \end{aligned} \tag{1}$$

where, for abbreviation,

$$v = \frac{2Ku}{\omega_1} \tag{2}$$

is put. From this, with the aid of the relations

$$\operatorname{cn}^2 v = 1 - \operatorname{sn}^2 v, \qquad \operatorname{dn}^2 v = 1 - \varkappa^2 \operatorname{sn}^2 v,$$

$$\frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{10}^2(u)}{\sigma^2(u)} = \frac{4K^2}{\omega_1^2}, \qquad \frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{00}^2(u)}{\sigma^2(u)} = \frac{4K^2}{\omega_1^2}\varkappa^2,$$

one concludes that three quantities e_1, e_2, e_3, independent of u and therefore depending only on ω_1, ω_2, can be so chosen that

$$\frac{\sigma_{10}^2(u)}{\sigma^2(u)} + e_1 = \frac{\sigma_{00}^2(u)}{\sigma^2(u)} + e_2 = \frac{\sigma_{01}^2(u)}{\sigma^2(u)} + e_3 = \wp(u), \tag{3}$$

where $\wp(u)$ is a doubly periodic function newly defined by (3). The quantities e_1, e_2, e_3 may be arbitrary, provided only that they satisfy the conditions

$$e_1 - e_3 = \frac{4K^2}{\omega_1^2}, \qquad e_2 - e_3 = \frac{4K^2}{\omega_1^2}\varkappa^2. \tag{4}$$

It is therefore still open to us, for their complete determination, to add the condition

$$e_1 + e_2 + e_3 = 0. \tag{5}$$

Then the following expressions for e_1, e_2, e_3 result:

$$\begin{aligned} e_1 &= \frac{4K^2}{\omega_1^2}\frac{1+\varkappa'^2}{3}, \\ e_2 &= -\frac{4K^2}{\omega_1^2}\frac{\varkappa'^2-\varkappa^2}{3}, \\ e_3 &= -\frac{4K^2}{\omega_1^2}\frac{1+\varkappa^2}{3}. \end{aligned} \tag{6}$$

Consequently the function $\wp(u)$, which has the periods ω_1, ω_2, is expressed by (1) as

$$\wp(u) = \frac{4K^2}{\omega_1^2}\left(\frac{1}{\operatorname{sn}^2 v} - \frac{1+\varkappa^2}{3}\right). \tag{7}$$

The function $\wp(u)$ has the desired property of invariance under linear transformations, as is seen from the expression

$$\wp(u) = \frac{1}{3}\frac{\sigma_{00}^2(u) + \sigma_{01}^2(u) + \sigma_{10}^2(u)}{\sigma^2(u)}. \tag{8}$$

In consequence of (3), the e_1, e_2, e_3 are interchanged under a linear transformation in the same way as the $\sigma_{10}, \sigma_{00}, \sigma_{01}$; a symmetric function of them is therefore unchanged under a linear transformation and is called an invariant. There are two fundamental invariants; we denote them by g_2, g_3:

$$g_2 = -4(e_2e_3 + e_3e_1 + e_1e_2) = 2(e_1^2 + e_2^2 + e_3^2) = \frac{64}{3}\frac{K^4}{\omega_1^4}(1 - \varkappa^2\varkappa'^2), \tag{9}$$

$$g_3 = 4e_1e_2e_3 = \frac{2^8}{27}\frac{K^6}{\omega_1^6}(2 + \varkappa^2\varkappa'^2)(\varkappa'^2 - \varkappa^2). \tag{10}$$

We also adjoin the function known by the name of the discriminant:

$$G = (e_2 - e_3)^2(e_3 - e_1)^2(e_1 - e_2)^2 = \frac{1}{16}(g_2^3 - 27g_3^2) = \frac{(2K)^{12}}{\omega_1^{12}}\varkappa^4\varkappa'^4, \tag{11}$$

for which, by § 43, (3), (15) and § 34, (2), one can also put

$$G = \frac{2^8\pi^{12}\eta(\omega)^{24}}{\omega_1^{12}}. \tag{12}$$

$\wp(u), g_2, g_3, G$ are homogeneous functions, as the following relations show, where λ is an arbitrary factor:

$$\begin{aligned} \wp(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \lambda^{-2}\wp(u, \omega_1, \omega_2), \\ \wp'(\lambda u, \lambda\omega_1, \lambda\omega_2) &= \lambda^{-3}\wp'(u, \omega_1, \omega_2), \\ g_2(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-4}g_2(\omega_1, \omega_2), \\ g_3(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-6}g_3(\omega_1, \omega_2), \\ G(\lambda\omega_1, \lambda\omega_2) &= \lambda^{-12}G(\omega_1, \omega_2). \end{aligned} \tag{13}$$

If, as in § 45, (2), we put

$$\omega_1 = 2K, \qquad \omega_2 = 2\mathrm{i}K',$$

then

$$\begin{aligned} &e_1 = \frac{1 + \varkappa'^2}{3}, \qquad e_2 = -\frac{\varkappa'^2 - \varkappa^2}{3}, \qquad e_3 = -\frac{1 + \varkappa^2}{3}, \\ &g_2 = \frac{4}{3}(1 - \varkappa^2\varkappa'^2), \quad g_3 = \frac{4}{27}(\varkappa'^2 - \varkappa^2)(2 + \varkappa^2\varkappa'^2), \\ &G = \varkappa^4\varkappa'^4. \end{aligned} \tag{14}$$

If we eliminate ω_1 from the general expressions (9), (10), (11) for g_2, g_3, G, we obtain homogeneous functions of order zero, hence functions of ω alone, which remain unchanged under linear transformations. Such functions are $g_2^3 : G$, $g_3^2 : G$. Among these functions, all of which can be reduced to one another, we single out one, which we denote by $j(\omega)$ and simply call the invariant, and define it thus:

$$j(\omega) = 2^8\frac{(1 - \varkappa^2\varkappa'^2)^3}{\varkappa^4\varkappa'^4}, \tag{15}$$

from which we get

$$\frac{4 \cdot 27\, g_2^3}{G} = j(\omega), \qquad \frac{4 \cdot 27 \cdot 27\, g_3^2}{G} = j(\omega) - 27 \cdot 64 = \frac{64(2 + \varkappa^2\varkappa'^2)^2(\varkappa'^2 - \varkappa^2)^2}{\varkappa^4\varkappa'^4}. \tag{16}$$

Considered as a function of $\varkappa^2$, the function $j(\omega)$ has the property of remaining unchanged when $\varkappa^2$ is replaced by one of the six values

$$\varkappa^2, \qquad \varkappa'^2, \qquad \frac{1}{\varkappa^2}, \qquad \frac{1}{\varkappa'^2}, \qquad -\frac{\varkappa^2}{\varkappa'^2}, \qquad -\frac{\varkappa'^2}{\varkappa^2}.$$

If, according to (7), we form the differential quotient of the function $\wp(u)$, we obtain

$$\wp'(u) = -\frac{16K^3}{\omega_1^3}\frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn}^3 v} = -2\frac{\sigma_{00}(u)\sigma_{10}(u)\sigma_{01}(u)}{\sigma(u)^3}, \tag{17}$$

and from this, by (3),

$$\begin{aligned}\wp'(u)^2 &= 4[\wp(u) - e_1][\wp(u) - e_2][\wp(u) - e_3] \\ &= 4\wp(u)^3 - g_2\wp(u) - g_3,\end{aligned} \tag{18}$$

or finally

$$du = \frac{d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}}, \tag{19}$$

from which one sees that the function $\wp(u)$ stands in the same relation to the Weierstrass normal form of the elliptic differential as the function $\operatorname{sn} v$ stands to the Legendre form.

§ 47. The elliptic transcendents of the second kind.

Jacobi introduced as transcendent of the second kind the function

$$Z(v) = \frac{\Theta'(v)}{\Theta(v)} = \frac{d \log \Theta(v)}{dv}. \tag{1}$$

The relation of this function to the elliptic integrals of the second kind results from formula (16) of § 23; it should immediately be observed that entirely similar considerations could be attached to the formulae (15), (17), (18) there, but that these would not lead to essentially new results.

If there one puts $v : 2K$ in place of v, it follows from § 42 and § 43 that

$$\begin{aligned}\frac{d^2 \log \Theta(v)}{dv^2} &= \frac{1}{4K^2}\frac{\vartheta_{01}''}{\vartheta_{01}} - \varkappa^2 \operatorname{sn}^2 v \\ &= \frac{1}{4K^2}\frac{\vartheta_{01}''}{\vartheta_{01}} - 1 + \operatorname{dn}^2 v.\end{aligned} \tag{2}$$

We put

$$1 - \frac{\vartheta_{01}''}{4K^2\vartheta_{01}} = \frac{E}{K}, \tag{3}$$

and obtain by integration of (2)

$$\frac{d \log \Theta(v)}{dv} = Z(v) = -\frac{E}{K}v + \int_0^v \operatorname{dn}^2 v \; dv,$$

or, putting

$$E(v) = \int_0^v \operatorname{dn}^2 v \; dv, \tag{4}$$

$$Z(v) = E(v) - \frac{E}{K}v, \qquad \frac{dZ(v)}{dv} = \operatorname{dn}^2 v - \frac{E}{K}. \tag{5}$$

The functions $\Theta(v)$ and $\Theta(v + K)$ are even functions, and therefore $\Theta'(0)$ and $\Theta'(K)$ are equal to zero. Hence, if in (5) we put $v = K$, it follows that

$$E = \int_0^K \operatorname{dn}^2 v \; dv. \tag{6}$$

If one also introduces for dv the differential

$$\frac{1}{2}\frac{d\xi}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}$$

[§ 42, (8)], then one obtains for $E(v)$ an elliptic integral of the second kind (§ 11):

$$E(v) = \frac{1}{2}\int_0^\xi \frac{d\xi(1-\varkappa^2\xi)}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}, \tag{7}$$

$$E = \frac{1}{2}\int_0^1 \frac{d\xi(1-\varkappa^2\xi)}{\sqrt{\xi(1-\xi)(1-\varkappa^2\xi)}}, \tag{8}$$

where the latter integral is to be taken in the same sense as in § 42, (17).

$E(v)$ and $Z(v)$ are odd functions of the argument. From the addition theorem of the Θ-function [§ 43, (11)] one obtains, by logarithmic differentiation,

$$Z(u+v) + Z(u-v) = 2Z(u) - \frac{2\varkappa^2 \operatorname{sn}^2 v \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u}{1-\varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \tag{9}$$

$$Z(u+v) - Z(u-v) = 2Z(v) - \frac{2\varkappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1-\varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \tag{10}$$

Here Z can also be replaced by E, and by addition one obtains [§ 44, (16)]

$$E(u) + E(v) - E(u+v) = \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u+v). \tag{11}$$

From this one obtains the periodicity properties of the function $E(u)$ if one puts $v = \pm K$, $K + \mathrm{i}K'$. But one also arrives at them in the following way.

From the first equation § 43, (3), it follows that

$$\frac{d \log H(v)}{dv} = Z(v + \mathrm{i}K') + \frac{\mathrm{i}\pi}{2K},$$

and hence from § 43, (6) and § 42, (11),

$$\begin{aligned} Z(v+\mathrm{i}K') &= Z(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} - \frac{\mathrm{i}\pi}{2K}, \\ Z(v+K) &= Z(v) - \frac{\varkappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \end{aligned} \tag{12}$$

and, if in the first of these formulae one puts $v = K$,

$$Z(K + \mathrm{i}K') = -\frac{\mathrm{i}\pi}{2K}. \tag{13}$$

By applying the formulae (12) twice [taking account of § 44, (18), (20)],

$$Z(v + 2\mathrm{i}K') = Z(v) - \frac{\mathrm{i}\pi}{K}, \qquad Z(v+2K) = Z(v). \tag{14}$$

Transferred to the function $E(v)$, these equations give

$$\begin{aligned} E(v+\mathrm{i}K') &= E(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} + \frac{\mathrm{i}EK'}{K} - \frac{\mathrm{i}\pi}{2K}, \\ E(v+K) &= E(v) + E - \frac{\varkappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \end{aligned} \tag{15}$$

$$E(v+2\mathrm{i}K') = E(v) + \frac{2\mathrm{i}EK'}{K} - \frac{\mathrm{i}\pi}{K}, \qquad E(v+2K) = E(v) + 2E. \tag{16}$$

If in (15) we put $v = K$ and define a quantity E' by the equation

$$E(K + \mathrm{i}K') = E' + \mathrm{i}(K' - E'), \tag{17}$$

then one obtains the formula known under the name of Legendre's relation,

$$EK' + KE' - KK' = \frac{\pi}{2}. \tag{18}$$

The meaning of the quantity E' introduced here is recognized if one derives Legendre's relation by a second path, using one of the linear transformations. It results, if in § 31, (11), one puts

$$u = \frac{v}{2K}, \qquad \omega = \frac{\mathrm{i}K'}{K}, \qquad \frac{u}{\omega} = \frac{v}{2\mathrm{i}K'},$$

that, by § 43, (1), (4),

$$\sqrt{K}\, e^{-\frac{\pi v^2}{4KK'}}\,\Theta(\mathrm{i}v, \varkappa') = \sqrt{K'}\, H(v + K), \tag{19}$$

and from this, by logarithmic differentiation,

$$\mathrm{i}Z(\mathrm{i}v, \varkappa') = \frac{\pi v}{2KK'} + \frac{H'(v + K)}{H(v + K)}, \tag{20}$$

or, by § 43, (6),

$$\mathrm{i}Z(\mathrm{i}v, \varkappa') = \frac{\pi v}{2KK'} + Z(v) + \frac{d \log \operatorname{cn} v}{dv}. \tag{21}$$

But by (5)

$$\mathrm{i}\frac{dZ(\mathrm{i}v, \varkappa')}{dv} = -\operatorname{dn}^2(\mathrm{i}v, \varkappa') + \frac{E'}{K'},$$

if E' now has the meaning

$$E' = \int_0^{K'} \operatorname{dn}^2(v, \varkappa')\; dv, \tag{22}$$

that is, arises from E by interchanging $\varkappa$ with $\varkappa'$. If, however, in (21) one puts $v = 0$ after having first differentiated, then the relation (18) again results; from this it follows that E' is in both cases the same quantity.

§ 48. The elliptic transcendents of the third kind.

If we integrate the formula (10) of the preceding paragraph,

$$\frac{\Theta'(u + v)}{\Theta(u + v)} - \frac{\Theta'(u - v)}{\Theta(u - v)} = 2Z(v) - \frac{2\varkappa^2 \operatorname{sn}^2 u \,\operatorname{sn} v \,\operatorname{cn} v \,\operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v},$$

with respect to u, we get

$$\frac{1}{2} \log \frac{\Theta(u - v)}{\Theta(u + v)} + uZ(v) = \int_0^u \frac{\varkappa^2 \operatorname{sn}^2 u \,\operatorname{sn} v \,\operatorname{cn} v \,\operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}\; du, \tag{1}$$

and we can also add the formula, to be derived in the same way [from § 43, (11)],

$$\frac{1}{2} \log \frac{H(v - u)}{H(v + u)} + uZ(v) = \int_0^u \frac{\operatorname{sn} v \,\operatorname{cn} v \,\operatorname{dn} v \; du}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \tag{2}$$

which, moreover, can also be derived from (1). We now put, with Jacobi,

$$\Pi(u, v) = \int_0^u \frac{\varkappa^2 \operatorname{sn}^2 u \,\operatorname{sn} v \,\operatorname{cn} v \,\operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}\; du, \tag{3}$$

and call $\Pi(u, v)$ the transcendent of the third kind with argument u and parameter v. If one replaces du by the expression

$$\frac{d\xi}{2\sqrt{\xi(1 - \xi)(1 - \varkappa^2 \xi)}}$$

and $\operatorname{sn}^2 u$ by ξ, then both (1) and (2) give an elliptic integral of the third kind, as we have learned it in § 11.

From (1) follows first

$$\Pi(u,v) - uZ(v) = \Pi(v,u) - vZ(u), \tag{4}$$

or Jacobi's theorem on the interchange of argument and parameter. The function $\Pi(u,v)$ vanishes when $v = K$ or $v = K + \mathrm{i}K'$, because for the former value $\operatorname{cn} v$, and for the latter $\operatorname{dn} v$, is equal to zero. If therefore we insert these values for u in (4), it follows that

$$\begin{aligned} \Pi(K,v) &= KZ(v), \\ \Pi(K+\mathrm{i}K',v) &= (K+\mathrm{i}K')Z(v) + \frac{\mathrm{i}\pi v}{2K} \quad [\S\,47, (13)], \end{aligned} \tag{5}$$

whereby the complete integrals of the third kind are led back to the second kind.

We still wish to derive the addition theorem for the Π-function, which Jacobi gives in the *Fundamenta nova*, art. 53–55, in various forms that can be reduced to one another only with difficulty. First, from the definition (1), if we now denote the parameter by a, we obtain

$$\begin{aligned} &\Pi(u+v,a) - \Pi(u,a) - \Pi(v,a) \\ &\qquad = \frac{1}{2}\log\frac{\Theta(u+v-a)\Theta(u+a)\Theta(v+a)}{\Theta(u+v+a)\Theta(u-a)\Theta(v-a)}, \end{aligned} \tag{6}$$

and it remains only to represent the expression standing under the logarithm by elliptic functions.

This is first done easily, as at the cited place of the *Fundamenta*, by means of the formula [§ 43, (11)]

$$\Theta^2(0)\Theta(u+v)\Theta(u-v) = \Theta^2(u)\Theta^2(v)[1 - \varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v],$$

if therein one puts, for u, v, first $u \pm a, v \pm a$ and then $\pm a, u+v \pm a$. One obtains thus

$$\Theta^2(0)\Theta(u+v\pm 2a)\Theta(u-v) = \Theta^2(u\pm a)\Theta^2(v\pm a)[1 - \varkappa^2\operatorname{sn}^2(u\pm a)\operatorname{sn}^2(v\pm a)],$$

$$\Theta^2(0)\Theta(u+v\pm 2a)\Theta(u+v) = \Theta^2(a)\Theta^2(u+v\pm a)[1 - \varkappa^2\operatorname{sn}^2 a\operatorname{sn}^2(u+v\pm a)],$$

from which follows

$$\frac{\Theta(u+v-a)\Theta(u+a)\Theta(v+a)}{\Theta(u+v+a)\Theta(u-a)\Theta(v-a)} = \sqrt{\frac{[1-\varkappa^2\operatorname{sn}^2(u-a)\operatorname{sn}^2(v-a)][1-\varkappa^2\operatorname{sn}^2 a\operatorname{sn}^2(u+v+a)]}{[1-\varkappa^2\operatorname{sn}^2(u+a)\operatorname{sn}^2(v+a)][1-\varkappa^2\operatorname{sn}^2 a\operatorname{sn}^2(u+v-a)]}}. \tag{7}$$

A second expression is obtained from the addition formula (13), § 22, which can be represented in the form

$$\Theta(0)\Theta(u\pm a)\Theta(v\pm a)\Theta(u+v) = \Theta(u)\Theta(v)\Theta(a)\Theta(u+v\pm a)[1 \pm \varkappa^2\operatorname{sn} a\operatorname{sn} u\operatorname{sn} v\operatorname{sn}(u+v\pm a)],$$

and hence, by division,

$$\frac{\Theta(u+v-a)\Theta(u+a)\Theta(v+a)}{\Theta(u+v+a)\Theta(u-a)\Theta(v-a)} = \frac{1+\varkappa^2\operatorname{sn} a\operatorname{sn} u\operatorname{sn} v\operatorname{sn}(u+v+a)}{1-\varkappa^2\operatorname{sn} a\operatorname{sn} u\operatorname{sn} v\operatorname{sn}(u+v-a)}. \tag{8}$$

Jacobi gives still a third expression for the Θ-quotients in (6):

$$\frac{\left\{1-\varkappa^2\operatorname{sn}^2\left(\frac{u-v}{2}\right)\operatorname{sn}^2\left(\frac{u+v}{2}+a\right)\right\}\left\{1-\varkappa^2\operatorname{sn}^2\left(\frac{u+v}{2}\right)\operatorname{sn}^2\left(\frac{u+v}{2}-a\right)\right\}}{\left\{1-\varkappa^2\operatorname{sn}^2\left(\frac{u-v}{2}\right)\operatorname{sn}^2\left(\frac{u+v}{2}-a\right)\right\}\left\{1-\varkappa^2\operatorname{sn}^2\left(\frac{u+v}{2}\right)\operatorname{sn}^2\left(\frac{u+v}{2}+a\right)\right\}}. \tag{9}$$

The direct conversion of the three expressions (7), (8), (9) into one another succeeds most simply if one starts from the formula last given by Jacobi,

$$1 - \varkappa^2\operatorname{sn}(a+u)\operatorname{sn}(a-u)\operatorname{sn}(a+v)\operatorname{sn}(a-v) = \frac{(1-\varkappa^2\operatorname{sn}^4 a)(1-\varkappa^2\operatorname{sn}^2 u\operatorname{sn}^2 v)}{(1-\varkappa^2\operatorname{sn}^2 a\operatorname{sn}^2 u)(1-\varkappa^2\operatorname{sn}^2 a\operatorname{sn}^2 v)}. \tag{10}$$

The verification of this formula is easy because, with the aid of § 44, (13), the right and left sides become rational functions of $\operatorname{sn}^2 u, \operatorname{sn}^2 v$, the identity of which is immediately evident.

If in this formula one replaces

$$u, \qquad v, \qquad a$$

by

$$\frac{u-v}{2}, \qquad \frac{u+v}{2} \pm a, \qquad \frac{u+v}{2},$$

one obtains

$$\begin{aligned} & 1 \pm \varkappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{sn} a \operatorname{sn}(u+v \pm a) \\ &= \frac{\left[1-\varkappa^2 \operatorname{sn}^4\left(\frac{u+v}{2}\right)\right]\left[1-\varkappa^2 \operatorname{sn}^2\left(\frac{u-v}{2}\right) \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right)\right]}{\left[1-\varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2}\right) \operatorname{sn}^2\left(\frac{u-v}{2}\right)\right]\left[1-\varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2}\right) \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right)\right]}, \end{aligned}$$

and, if the two formulae contained herein are divided by one another, the agreement of (8) and (9) results.

Putting $v=u$ in (10) gives

$$1-\varkappa^2 \operatorname{sn}^2(a+u) \operatorname{sn}^2(a-u)=\frac{(1-\varkappa^2 \operatorname{sn}^4 a)(1-\varkappa^2 \operatorname{sn}^4 u)}{(1-\varkappa^2 \operatorname{sn}^2 a \operatorname{sn}^2 u)^2}. \tag{11}$$

If herein one replaces first

$$a, \qquad u$$

by

$$\frac{u+v}{2} \pm a, \qquad -\frac{u+v}{2},$$

and then by

$$\frac{u+v}{2} \pm a, \qquad \frac{u-v}{2},$$

four formulae result:

$$1-\varkappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u+v \pm a)=\frac{\left[1-\varkappa^2 \operatorname{sn}^4\left(\frac{u+v}{2} \pm a\right)\right]\left(1-\varkappa^2 \operatorname{sn}^4 \frac{u+v}{2}\right)}{\left[1-\varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right) \operatorname{sn}^2 \frac{u+v}{2}\right]^2},$$

$$1-\varkappa^2 \operatorname{sn}^2(u \pm a) \operatorname{sn}^2(v \pm a)=\frac{\left[1-\varkappa^2 \operatorname{sn}^4\left(\frac{u+v}{2} \pm a\right)\right]\left(1-\varkappa^2 \operatorname{sn}^4 \frac{u-v}{2}\right)}{\left[1-\varkappa^2 \operatorname{sn}^2\left(\frac{u+v}{2} \pm a\right) \operatorname{sn}^2 \frac{u-v}{2}\right]^2},$$

from which the agreement of (9) with (7) results.

§ 49. The transcendents of the second and third kind of Weierstrass.

It remains to set up the transcendents of the second and third kind in the Weierstrass form. We reach the former similarly as in § 47, by twice differentiating $\log \sigma(u)$. Thus, from

$$\sigma(u)=\omega_1 e^{\eta_1 u^2/\omega_1} \frac{\vartheta_{11}(u/\omega_1)}{\vartheta_{11}'} \qquad [\S\, 37, (4)], \tag{1}$$

one obtains

$$\begin{aligned} \frac{d^2 \log \sigma(u)}{du^2} &= \frac{2\eta_1}{\omega_1}+\frac{d^2 \log \vartheta_{11}(u/\omega_1)}{du^2}, \\ &= \frac{2\eta_1}{\omega_1}+\frac{1}{\omega_1^2} \frac{\vartheta_{00}''}{\vartheta_{00}}-\frac{1}{\omega_1^2} \frac{\vartheta_{11}'^2}{\vartheta_{00}^2} \frac{\vartheta_{00}^2(u/\omega_1)}{\vartheta_{11}^2(u/\omega_1)} \qquad [\S\, 23, (18)], \end{aligned}$$

and therefore, if the ϑ-quotient is expressed by § 42 through elliptic functions, and these by § 46 through the $\wp$-function,

$$\frac{d^2 \log \sigma(u)}{du^2}=A-\wp(u),$$

where A is a constant. This, however, is found if the equation is written according to § 46, (8), in the form

$$\sigma''(u)\sigma(u) - \sigma'(u)\sigma'(u) = A\sigma^2(u) - \frac{1}{3}\left[\sigma_{00}^2(u) + \sigma_{01}^2(u) + \sigma_{10}^2(u)\right].$$

If one differentiates this twice and then puts $u = 0$, there results $A = 0$ [§ 35, (12), § 37, (5), (7)], and we get

$$\frac{d^2 \log \sigma(u)}{du^2} = -\wp(u), \tag{2}$$

a formula which, for Weierstrass, serves for the definition of the $\wp$-function.

If therefore

$$\frac{\sigma'(u)}{\sigma(u)} = \zeta(u) \tag{3}$$

is put, then $\zeta(u)$ is a single-valued function of u, and indeed an elliptic integral of the second kind:

$$\zeta(u) = -\int \wp(u)\ du = -\int \frac{\wp\ d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}} \qquad [§ 46, (19)], \tag{4}$$

where the additive constant is determined by the requirement that $\zeta(u)$ be an odd function. The periodicity of the ζ-function results from the period equations of the σ-function [§ 35, (9)]:

$$\zeta(u + \omega_1) - \zeta(u) = 2\eta_1, \qquad \zeta(u + \omega_2) - \zeta(u) = 2\eta_2. \tag{5}$$

Thus $2\eta_1$ and $2\eta_2$ are represented as complete integrals of the second kind, analogous to E, E':

$$2\eta_1 = -\int_u^{u+\omega_1} \wp(u)\ du, \qquad 2\eta_2 = -\int_u^{u+\omega_2} \wp(u)\ du, \tag{6}$$

with the equation corresponding to Legendre's relation

$$\eta_1\omega_2 - \eta_2\omega_1 = \pi\mathrm{i} \qquad [§ 35, (10)]. \tag{7}$$

To form the addition theorems for the Weierstrass transcendents, we start from the σ-function. For it, from (1), with the use of the addition formula for ϑ_{11}, § 22, (4), we obtain

$$\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)\sigma^2(v)} = \wp(v) - \wp(u), \tag{8}$$

and hence, by logarithmic differentiation,

$$\begin{aligned} \zeta(u+v) + \zeta(u-v) - 2\zeta(u) &= \frac{\wp'(u)}{\wp(u) - \wp(v)}, \\ \zeta(u+v) - \zeta(u-v) - 2\zeta(v) &= -\frac{\wp'(v)}{\wp(u) - \wp(v)}, \\ \zeta(u+v) &= \zeta(u) + \zeta(v) + \frac{1}{2}\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}. \end{aligned} \tag{9–10}$$

From this, by differentiating once more, one derives the addition theorem for the $\wp$-function. Differentiating (9) with respect to u and v gives

$$\wp(u+v) + \wp(u-v) - 2\wp(u) = -\frac{\wp''(u)}{\wp(u) - \wp(v)} + \frac{\wp'(u)^2}{[\wp(u) - \wp(v)]^2},$$

$$2\wp(u+v) - 2\wp(u-v) = -\frac{2\wp'(u)\wp'(v)}{[\wp(u) - \wp(v)]^2},$$

$$\wp(u+v) + \wp(u-v) - 2\wp(v) = \frac{\wp''(v)}{\wp(u) - \wp(v)} + \frac{\wp'(v)^2}{[\wp(u) - \wp(v)]^2},$$

and, adding,

$$4\wp(u+v)-2\wp(u)-2\wp(v)=\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2-\frac{\wp''(u)-\wp''(v)}{\wp(u)-\wp(v)}.$$

But

$$\wp'(u)^2=4\wp(u)^3-g_2\wp(u)-g_3,\qquad \wp''(u)=6\wp(u)^2-\frac{1}{2}g_2,$$

$$\wp''(u)-\wp''(v)=6[\wp^2(u)-\wp^2(v)],$$

from which one obtains

$$\wp(u+v)+\wp(u)+\wp(v)=\frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2, \tag{11}$$

in agreement with formula (21), § 13.

Finally one obtains, by integration of the second equation (9) with respect to u,

$$\frac{1}{2}\log\frac{\sigma(v-u)}{\sigma(v+u)}+u\zeta(v)=\frac{1}{2}\int_0^u\frac{\wp'(v)\,du}{\wp(u)-\wp(v)}, \tag{12}$$

whereby an integral of the third kind with argument u and parameter v is expressed by the σ-function. Another form of the integral of the third kind is obtained by integration of (10):

$$\log\frac{\sigma(u+v)}{\sigma(u)\sigma(v)}-u\zeta(v)=\frac{1}{2}\int\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\,du. \tag{13}$$

§ 50. Expansions of the elliptic functions.

If in the expansions of the theta-quotients derived in § 26 and collected in Table II one puts $a=0$, then, if one first omits the formulae which contain the factor $\vartheta_{11}(a)$ in the denominator, one obtains the partial-fraction expansions of twelve elliptic functions:

$$\begin{array}{lll} \operatorname{sn}2Ku, & \operatorname{cn}2Ku, & \operatorname{dn}2Ku,\\ \frac{1}{\operatorname{sn}2Ku}, & \frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku}, & \frac{\operatorname{dn}2Ku}{\operatorname{sn}2Ku},\\ \frac{1}{\operatorname{cn}2Ku}, & \frac{\operatorname{sn}2Ku}{\operatorname{cn}2Ku}, & \frac{\operatorname{dn}2Ku}{\operatorname{cn}2Ku},\\ \frac{1}{\operatorname{dn}2Ku}, & \frac{\operatorname{sn}2Ku}{\operatorname{dn}2Ku}, & \frac{\operatorname{cn}2Ku}{\operatorname{dn}2Ku}. \end{array}$$

Thus, from formula (2), Table II, one obtains

$$\frac{\vartheta_{11}\vartheta_{10}(u)}{\pi\vartheta_{11}(u)\vartheta_{10}}=\operatorname{cotg}\pi u-2\mathrm{i}\sum(-1)^{m/2}q^m\left(\frac{1}{e^{-2\pi\mathrm{i}u}-q^m}-\frac{1}{e^{2\pi\mathrm{i}u}-q^m}\right)$$

$$=\operatorname{cotg}\pi u+4\sin2\pi u\sum\frac{(-1)^{m/2}q^m}{1-2q^m\cos2\pi u+q^{2m}},$$

where m runs through the sequence of positive even numbers $2,4,6,8,\ldots$. But by § 42, (4),

$$\frac{\vartheta_{10}(u)}{\vartheta_{11}(u)}=\frac{\vartheta_{00}}{\vartheta_{01}}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku},$$

and from this it follows that

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku}=\operatorname{cotg}\pi u+4\sin2\pi u\sum(-1)^{m/2}q^m over 1-2q^m\cos2\pi u+q^{2m}. \tag{1}$$

If the same procedure is applied in formula (2) of Table III, one obtains

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku} = \operatorname{cotg}\pi u + 4\sum(-1)^{m'/2}q^{mm'/2}\sin\pi m u.$$

Here m and m' run independently through the sequences of positive even numbers, and the summation with respect to m' can still be carried out:

$$\sum(-1)^{m'/2}q^{mm'/2} = -\frac{q^m}{1+q^m},$$

therefore

$$\frac{2K}{\pi}\frac{\operatorname{cn}2Ku}{\operatorname{sn}2Ku} = \operatorname{cotg}\pi u - 4\sum\frac{q^m}{1+q^m}\sin\pi m u. \tag{2}$$

While the first two formulae converge for all values of u, the convergence of the third is restricted to the region in which the imaginary part of u is, in absolute value, smaller than the imaginary part of ω. The formula is therefore valid for real u in any case.

In Table IV the expansions for the twelve functions are collected.

Among the sixteen formulae of Tables II and III there are four which contain the factor $\vartheta_{11}(a)$ in the denominator, so that one cannot at once put $a=0$ in them. These are the formulae (1), (5), (9), (13). If one expands in these formulae according to ascending powers of a, the expansions on the right and on the left begin with a^{-1}; equating on both sides the terms independent of a gives the required expansions.

One can proceed, for example, by changing a into $-a$ in formula (1) and taking the arithmetic mean of the two formulae. Then on the right the infinite term $1:\operatorname{cotg}\pi a$ cancels out and on the left one obtains

$$\frac{\vartheta_{11}'[\vartheta_{11}(v+a)-\vartheta_{11}(v-a)]}{2\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \frac{\vartheta_{11}''(v)}{\pi\vartheta_{11}(v)} \qquad (a=0).$$

On the right side of (1) in Table II one obtains

$$\operatorname{cotg}\pi v - 4\mathrm{i}\sum\left(\frac{q^m}{e^{-2\pi\mathrm{i}v}-q^m} - \frac{q^m}{e^{2\pi\mathrm{i}v}-q^m}\right) = \operatorname{cotg}\pi v + 4\sum\frac{q^m\sin 2\pi v}{1-2q^m\cos 2\pi v+q^{2m}},$$

and on the right side of formula (1) of Table III one obtains

$$\operatorname{cotg}\pi v + 4\sum\frac{q^{m/2}}{1-q^m}\sin m\pi v.$$

If one also introduces Jacobi's H-function and writes u for v, one obtains

$$\frac{2K}{\pi}\frac{d\log H(2Ku)}{d(2Ku)} = \operatorname{cotg}\pi u + 4\sin 2\pi u\sum\frac{q^m}{1-2q^m\cos 2\pi u+q^{2m}} = \operatorname{cotg}\pi u + 4\sum\frac{q^m}{1-q^m}\sin m\pi u. \tag{3}$$

On the left stands a transcendent of the second kind, which by § 47, (12), can be expressed through the Z-function:

$$\frac{2K}{\pi}Z(2Ku) + \frac{2K}{\pi}\frac{\operatorname{cn}2Ku\ \operatorname{dn}2Ku}{\sin 2Ku},$$

for which one can also put

$$\frac{2K}{\pi}Z(2Ku) + \frac{d\log\sin 2Ku}{\pi\, du}.$$

In Table V the four formulae which result in this way are collected. If one subtracts formula (3) of this table from the three remaining ones, expansions result for the logarithmic derivatives of the three elliptic fundamental functions; of these we shall cite one:

$$\frac{d\log\operatorname{sn}2Ku}{\pi\, du} = \frac{d\log\sin\pi u}{\pi\, du} + 4\sin 2\pi u\sum_{h=1}^{\infty}\frac{(-1)^h q^h}{1-2q^h\cos 2\pi u+q^{2h}} = \frac{d\log\sin\pi u}{\pi\, du} - 4\sum_{h=1}^{\infty}\frac{q^h\sin 2h\pi u}{1+q^h}. \tag{4}$$

Expansions for the transcendents of the third kind result when one integrates the formulae of the fifth table between the limits $u+v$, $u-v$. Thus, for example, from formula (3) of this table one obtains

$$\frac{1}{2}\log\frac{\vartheta[2K(u-v)]}{\vartheta[2K(u+v)]} = -4\sum\frac{q^{m/2}}{m(1-q^m)}\sin m\pi u\sin m\pi v. \tag{5}$$

Putting $u=0$ in the formulae of Tables IV and V gives the following expansions:

$$\frac{2K}{\pi} = 1+4\sum\frac{q^{m/2}}{1+q^m} = 1+4\sum\frac{(-1)^{(n-1)/2}q^n}{1-q^n},$$

$$\frac{2\varkappa K}{\pi} = 4\sum\frac{q^{n/2}}{1+q^n} = 4\sum\frac{(-1)^{(n-1)/2}q^{n/2}}{1-q^n},$$

$$\frac{2\varkappa' K}{\pi} = 1+4\sum\frac{(-1)^{m/2}q^{m/2}}{1+q^m} = 1-4\sum\frac{(-1)^{(n-1)/2}q^n}{1+q^n}.$$

Finally one obtains an expansion for the complete integral of the second kind E from the formula

$$Z(v) = E(v) - \frac{E}{K}v.$$

If herein, after differentiation, one puts $v=0$, it follows that

$$Z'(0) = \frac{K-E}{K},$$

and hence, by Table V, formula (3),

$$K-E = \frac{2\pi^2}{K}\sum\frac{q^n}{(1-q^n)^2} = \frac{\pi^2}{K}\sum\frac{mq^{m/2}}{1-q^m}.$$

Fifth Section.

The Modular Functions.

§ 51. The elliptic differential equations.

The elliptic functions defined up to this point are functions of the two variables v,ω, and the moduli $\varkappa,\varkappa'$, furthermore $j(\omega), K, K'$, depend on the period-ratio ω, which must have a positive imaginary part. Likewise the function $\wp(u)$ is a function of the three variables u,ω_1,ω_2, and the invariants g_2,g_3 depend on ω_1,ω_2. But the problem of the inversion of the elliptic integrals, as we formulated it in § 14, presupposes that in the elliptic functions $\varkappa^2$ (or, in the Weierstrass normal form, g_2,g_3) have arbitrarily given values; the complete solution of this problem therefore requires that not ω, but $\varkappa^2$ (respectively g_2,g_3), appear as the second independent variable, and that ω be expressed by means of these. The next considerations will be devoted to this problem.

We start from the problem of integrating a system of differential equations, which we call the elliptic differential equations:

$$\frac{dx}{dv} = yz, \qquad \frac{dy}{dv} = -zx, \qquad \frac{dz}{dv} = -\varkappa^2 xy, \tag{1}$$

under the subsidiary conditions:

$$\text{for } v=0 \text{ one shall have } x=0, \quad y=1, \quad z=1. \tag{2}$$

We have seen in the preceding section that, when

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{3}$$

this problem is solved by the elliptic functions, and indeed in such a way that x, y, z become single-valued and, except where they are infinite, continuous functions of v in the whole v-plane. But if $\varkappa^2$ is now given, the question arises whether for every value of $\varkappa^2$ there is a value of ω satisfying the condition (3), and whether there are several such values.

We first prove that the functions x, y, z are uniquely determined by the differential equations (1) together with the subsidiary conditions.

It follows first from (1) that $x^2 + y^2$ and $\varkappa^2 x^2 + z^2$ are constant, and from the subsidiary conditions one obtains

$$y^2 = 1 - x^2, \qquad z^2 = 1 - \varkappa^2 x^2. \tag{4}$$

Suppose there exists a second system of functions x_1, y_1, z_1 satisfying the same conditions. Then first

$$y_1^2 = 1 - x_1^2, \qquad z_1^2 = 1 - \varkappa^2 x_1^2,$$

and a simple differentiation gives that

$$\frac{x y_1 z_1 - x_1 y z}{1 - \varkappa^2 x^2 x_1^2}$$

is constant. (The addition theorem of the elliptic integrals, from which one obtains the form of this expression, is confirmed immediately by differentiation.) From the subsidiary conditions, however, it follows that this constant vanishes, hence

$$x y_1 z_1 = x_1 y z,$$

from which one concludes easily, by squaring both sides,

$$x = x_1, \qquad y = y_1, \qquad z = z_1.$$

§ 52. The independent variable $\varkappa^2$. Linear differential equation for K.

We now first show that, and how, for an arbitrarily given value of $\varkappa^2$, at least one value of ω satisfying the condition

$$\varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4} \tag{1}$$

can be found. This question is answered most completely by reduction to a linear differential equation of the second order.

Such a differential equation is already contained in the formulae of § 23. It follows first from § 23, (14):

$$d \log \varkappa^2 = \pi \mathrm{i}\, \vartheta_{01}^4 \; d\omega,$$

or, with regard to

$$\pi\vartheta_{00}^2 = 2K, \qquad \pi\vartheta_{10}^2 = 2\varkappa K, \qquad \pi\vartheta_{01}^2 = 2\varkappa' K \quad [\S\, 42, (3), (15)], \tag{2}$$

$$\pi \; d(\varkappa^2) = 4\mathrm{i}\varkappa^2 \varkappa'^2 K^2 \; d\omega, \tag{3}$$

and from § 23, (19):

$$\frac{d}{d\omega} \frac{1}{K^2} \frac{dK}{d\omega} = -\frac{4}{\pi^2} \varkappa^2 \varkappa'^2 K^3.$$

If, by means of (3), one introduces the variable $\varkappa^2$ in place of ω, there follows

$$\frac{d}{d(\varkappa^2)} \left(\varkappa^2 \varkappa'^2 \frac{dK}{d(\varkappa^2)} \right) = \frac{1}{4} K, \tag{4}$$

and this is the desired differential equation.

Put further

$$-\mathrm{i}\omega = \frac{K'}{K}, \tag{5}$$

$$-\mathrm{i}K^2\, d\omega = K\, dK' - K'\, dK. \tag{6}$$

Then by (3) one obtains

$$\varkappa^2\varkappa'^2\left(K\frac{dK'}{d(\varkappa^2)} - K'\frac{dK}{d(\varkappa^2)}\right) = -\frac{\pi}{4}, \tag{7}$$

and therefore, by differentiating once more,

$$\frac{1}{K}\frac{d}{d(\varkappa^2)}\left(\varkappa^2\varkappa'^2\frac{dK}{d(\varkappa^2)}\right) = \frac{1}{K'}\frac{d}{d(\varkappa^2)}\left(\varkappa^2\varkappa'^2\frac{dK'}{d(\varkappa^2)}\right),$$

from which one sees that K' is the second particular integral of the differential equation (4).

This differential equation can be integrated by hypergeometric series (Gauss, *Disq. circa seriem infinitum*, Werke, Vol. III, p. 125).

If for the moment we put $\varkappa^2 = x$, $K = y$, then the differential equation (4) reads

$$x(1-x)\frac{d^2y}{dx^2} + (1-2x)\frac{dy}{dx} - \frac{1}{4}y = 0, \tag{8}$$

and it arises from the general Gaussian differential equation

$$(x - x^2)\frac{d^2y}{dx^2} + [\gamma - (\alpha+\beta+1)x]\frac{dy}{dx} - \alpha\beta y = 0 \tag{9}$$

when one sets $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$. Hence one of its particular integrals is

$$\begin{aligned} F\left(\frac{1}{2},\frac{1}{2},1,x\right) &= 1 + \sum_{\nu=1}^{\infty}\left(\frac{1\cdot 3\cdot 5\cdots(2\nu-1)}{2\cdot 4\cdots 2\nu}\right)^2 x^\nu \\ &= \sum_{\nu=0}^{\infty}\frac{\Pi(2\nu)^2}{\Pi(\nu)^4}\left(\frac{x}{16}\right)^\nu, \end{aligned} \tag{10}$$

and this series is convergent so long as the absolute value of x is smaller than 1. As the second particular integral one may choose

$$F\left(\frac{1}{2},\frac{1}{2},1,1-x\right), \tag{11}$$

but it has another convergence domain. If x is represented as a complex variable in a plane, then (10) converges in a circle described around the origin with radius 1, and (11) in an equal circle described around the point 1, so that the common convergence domain consists of a lune containing the part of the real axis lying between 0 and 1. But one can also, which is important for us, set up the second particular integral in a form which converges in the same circle as the series (10). In doing this, one should observe that the second particular integral becomes infinite for $x = 0$.

This second particular integral is obtained by the following passage to the limit. If we denote the hypergeometric series according to Gauss by

$$\begin{aligned} F(\alpha,\beta,\gamma,x) &= 1 + \frac{\alpha\beta}{1\cdot\gamma}x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1\cdot 2\cdot\gamma(\gamma+1)}x^2 + \cdots \\ &= \frac{\Pi(\gamma-1)}{\Pi(\alpha-1)\Pi(\beta-1)}\sum_{\nu=0}^{\infty}\frac{\Pi(\alpha+\nu-1)\Pi(\beta+\nu-1)}{\Pi(\gamma+\nu-1)\Pi(\nu)}x^\nu, \end{aligned}$$

then in general one obtains the two particular integrals of (9):

$$y_1 = F_1(x) = F(\alpha,\beta,\gamma,x),$$

$$y_2 = F_2'(x) = x^{1-\gamma}(1-x)^{\gamma-\alpha-\beta}F(1-\alpha, 1-\beta, 2-\gamma, x).$$

Since these, however, become identical for $\alpha = \beta = \frac{1}{2}$, $\gamma = 1$, one first puts $\alpha = \beta = \frac{1}{2}$, $\gamma = 1 + \varepsilon$, hence

$$y_1 = F\left(\frac{1}{2}, \frac{1}{2}, 1+\varepsilon, x\right), \qquad y_2 = \left(\frac{1-x}{x}\right)^{\varepsilon} F\left(\frac{1}{2}, \frac{1}{2}, 1-\varepsilon, x\right).$$

But

$$F\left(\frac{1}{2}, \frac{1}{2}, 1\pm\varepsilon, x\right) = \frac{\Pi(\pm\varepsilon)}{\Pi\left(-\frac{1}{2}\right)^2} \sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu - \frac{1}{2}\right)^2}{\Pi(\nu\pm\varepsilon)\Pi(\nu)} x^{\nu},$$

and, adding a constant factor, one may take the following also as the second particular integral:

$$\frac{1}{2\varepsilon} \sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu-\frac{1}{2}\right)^2}{\Pi(\nu)} \left[-\frac{1}{\Pi(\nu+\varepsilon)} + \left(\frac{x}{1-x}\right)^{-\varepsilon} \frac{1}{\Pi(\nu-\varepsilon)}\right] x^{\nu}.$$

If this is developed according to powers of ε and then $\varepsilon = 0$ is put, it follows that

$$\sum_{\nu=0}^{\infty} \frac{\Pi\left(\nu-\frac{1}{2}\right)^2}{\Pi(\nu)^2} x^{\nu} \left[\frac{\Pi'(\nu)}{\Pi(\nu)} - \frac{1}{2}\log\frac{x}{1-x}\right],$$

for which, by use of Gauss's formula

$$\frac{\Pi\left(\nu-\frac{1}{2}\right)\Pi(\nu)}{\Pi(2\nu)} = \sqrt{\pi}\, 2^{-2\nu},$$

one may, omitting a constant factor, also put

$$\sum_{\nu=0}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(\frac{x}{16}\right)^{\nu} \left[\frac{\Pi'(\nu)}{\Pi(\nu)} - \frac{1}{2}\log\frac{x}{1-x}\right]. \tag{12}$$

But for an integral positive ν one has

$$\frac{\Pi'(\nu)}{\Pi(\nu)} = \Pi'(0) + 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{\nu},$$

and if, therefore, for abbreviation we put

$$\begin{aligned} F(x) &= \sum_{\nu=0}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(\frac{x}{16}\right)^{\nu}, \\ G(x) &= 2\sum_{\nu=1}^{\infty} \frac{\Pi(2\nu)^2}{\Pi(\nu)^4} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{\nu}\right) \left(\frac{x}{16}\right)^{\nu}, \end{aligned} \tag{13}$$

then the two particular solutions of the differential equation (8) are obtained if for the second one takes a linear combination of (10) and (12):

$$y_1 = F(x), \qquad y_2 = \frac{1}{2}G(x) - \frac{1}{2}\log\frac{x}{16(1-x)}F(x). \tag{14}$$

But in order to represent the particular integrals K and K' by these functions, we must examine the behavior of K, K' for $q = 0$, for which at the same time $\varkappa^2 = 0$.

For $q = 0$, however, by § 25 and § 42 one has

$$q^{-1}\varkappa^2 = 16, \qquad 2K = \pi,$$

hence

$$\lim\left(\log\frac{\varkappa^2}{16}-\mathrm{i}\pi\omega\right)=0;$$

on the other hand, $2\mathrm{i}K'=\pi\vartheta_{00}^2\omega$ (§ 42), or

$$2K'+\log\frac{\varkappa^2}{16}=\left(\log\frac{\varkappa^2}{16}-\mathrm{i}\pi\omega\right)+\mathrm{i}\pi\omega(1-\vartheta_{00}^2),$$

from which

$$\lim_{\varkappa^2=0}\left(K'+\frac{1}{2}\log\frac{\varkappa^2}{16}\right)=0. \tag{15}$$

Since now K and K' are particular integrals of (8) (for $x=\varkappa^2$), they both have the form

$$K=a_1y_1+a_2y_2, \qquad K'=a_1'y_1+a_2'y_2,$$

where a_1, a_2, a_1', a_2' are constant. Since K remains finite for $x=0$ and assumes the value $\frac{1}{2}\pi$, one has $a_2=0$, $a_1=\frac{1}{2}\pi$, and in order that $K'+\frac{1}{2}\log x$ remain finite and, according to (15), assume the value $\log 4$, one must have $a_2'=1$ and $a_1'=0$. Thus one obtains, if $x=\varkappa^2$, $1-x=\varkappa'^2$ is put:

$$K=\frac{\pi}{2}F(\varkappa^2), \qquad K'=\frac{1}{2}G(\varkappa^2)-\frac{1}{2}\log\frac{\varkappa^2}{16\varkappa'^2}F(\varkappa^2), \tag{16}$$

and

$$q=e^{-\pi K'/K}=\frac{\varkappa^2}{16\varkappa'^2}e^{-\frac{G(\varkappa^2)}{F(\varkappa^2)}}. \tag{17}$$

Here, as already remarked, the series $G(\varkappa^2)$, $F(\varkappa^2)$ are convergent so long as the absolute value of $\varkappa^2$ is a proper fraction. In the differential equation (4) there are, to be sure, the means of continuing the expressions found for K, K', q beyond this convergence domain. But one arrives more simply by applying Landen's transformation.

We bound the plane of the complex values $\varkappa^2$ by making two cuts along the real axis, from 0 to $-\infty$ and from 1 to ∞. In the plane thus bounded all roots out of $\varkappa^2, \varkappa'^2$ are then uniquely determined by the condition that for positive proper fractional $\varkappa^2$ they shall be real and positive. Now, by the formulae of Landen's transformation [§ 44,(28)], if with $\varkappa_1, \varkappa_1', K_1, K_1'$ we denote the functions of ω which arise from $\varkappa, \varkappa', K, K'$ when ω is replaced by 2ω, then

$$\varkappa_1=\frac{1-\varkappa'}{1+\varkappa'}, \qquad \varkappa_1'=\frac{2\sqrt{\varkappa'}}{1+\varkappa'}, \qquad 2K_1=(1+\varkappa')K, \qquad K_1'=(1+\varkappa')K'. \tag{18}$$

Therefore, if in (17) one replaces ω by 2ω and takes the square root, there follows

$$q=\frac{1}{4}\frac{\varkappa_1}{\varkappa_1'}e^{-\frac{1}{2}\frac{G(\varkappa_1^2)}{F(\varkappa_1^2)}}. \tag{19}$$

Here the convergence domain now extends over the unit circle situated in the plane $\varkappa_1^2$, and this corresponds to the whole plane $\varkappa^2$. For the real part of $\varkappa'$ is positive in the whole plane of $\varkappa^2$, except on the two banks of the cut running from $+1$ to $+\infty$, on which $\varkappa'$ is purely imaginary. From this it follows that the absolute value of $\varkappa_1$ is smaller than 1, and that the circumference of the unit circle in the plane $\varkappa_1$ corresponds to the two banks of this cut.

But one can still improve the convergence of these expressions by applying Landen's transformation once more. If we denote the result of a second doubling of ω by the index 2, there follows

$$\sqrt{\varkappa_2}=\frac{1-\sqrt{\varkappa'}}{1+\sqrt{\varkappa'}}, \qquad 4K_2=(1+\sqrt{\varkappa'})^2K, \qquad K_2'=(1+\sqrt{\varkappa'})^2K', \qquad q=\sqrt{\frac{\varkappa_2}{4\varkappa_2'}e^{-\frac{1}{4}\frac{G(\varkappa_2^2)}{F(\varkappa_2^2)}}}, \tag{20}$$

where now the convergence domain covers the plane of $\varkappa^2$ twice (with a ramification at the point $\varkappa^2 = 1$).

This operation can be continued, by deriving from $\varkappa_2$, by doubling ω, a new quantity $\varkappa_3$, from this likewise $\varkappa_4$, and so forth, according to the formula (17). One then obtains a series of expressions for q, whose convergence becomes better and better. For an arbitrary ν one has

$$q = \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu} e^{-\frac{1}{2^\nu}\frac{G(\varkappa_\nu^2)}{F(\varkappa_\nu^2)}}}. \tag{21}$$

The first equation (18),

$$\varkappa_1 = \frac{\varkappa^2}{(1+\varkappa')^2},$$

shows that so long as the absolute value of $\varkappa^2$ is smaller than 1, and consequently the absolute value of $1+\varkappa'$ is greater than 1, the absolute value of $\varkappa_1$ is smaller than that of $\varkappa^2$, and consequently $\varkappa_\nu$ approaches the limit zero as ν increases without bound. From this one obtains for q the representation

$$q = \lim_{\nu=\infty} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4\varkappa'_\nu}} = \lim_{\nu=\infty} \sqrt[2^{\nu-1}]{\frac{\varkappa_\nu}{4}}, \tag{22}$$

which is suitable for calculation at real values of $\varkappa$.

From each of the formulae (21) one can derive an expansion of q according to the ascending powers of $\varkappa_\nu$, whose convergence increases with increasing ν. If in particular one applies formula (20), one obtains an expansion communicated by Weierstrass in the monthly reports of the Berlin Academy for the year 1883, whose convergence domain fills the plane $\varkappa^2$ twice.

The coefficients of this expansion are most simply computed from

$$\sqrt{\varkappa_2} = \frac{\vartheta_{10}(0,4\omega)}{\vartheta_{00}(0,4\omega)} = \frac{2q + 2q^9 + 2q^{25} + \cdots}{1 + 2q^4 + 2q^{16} + \cdots}, \tag{23}$$

and in this way one obtains, by the method of undetermined coefficients,

$$q = \frac{\sqrt{\varkappa_2}}{2} + 2\left(\frac{\sqrt{\varkappa_2}}{2}\right)^5 + 15\left(\frac{\sqrt{\varkappa_2}}{2}\right)^9 + 150\left(\frac{\sqrt{\varkappa_2}}{2}\right)^{13} + 1707\left(\frac{\sqrt{\varkappa_2}}{2}\right)^{17} + \cdots. \tag{24}$$

Likewise, from (23), conversely, $\sqrt{\varkappa_2}$ can be developed in a series proceeding according to powers of q:

$$\frac{1}{2}\sqrt{\varkappa_2} = q - 2q^5 + 5q^9 - 10q^{13} + 18q^{17} \cdots, \tag{25}$$

which one also obtains by inversion of the series (24).

With this the question posed at the beginning of this paragraph is completely answered.

With regard to the application of the formulae developed here to the numerical calculation of q, it should also be noted that, when $\varkappa^2$ lies nearer to 1, one obtains a better-converging procedure by interchanging $\varkappa^2$ with $\varkappa'^2$. The calculation then first gives not q, but

$$q' = e^{-\pi i/\omega},$$

from which, however, one obtains q from the formula

$$\log q \log q' = \pi^2.$$

§ 53. The solutions of the equation $j(\omega) = j(\omega')$.

The results of § 51 are sufficient first to determine the relations of the different values of ω which lead to the same value of $\varkappa^2$. Thus let

$$\frac{\vartheta_{10}^4(0,\omega')}{\vartheta_{00}^4(0,\omega')} = \frac{\vartheta_{10}^4(0,\omega)}{\vartheta_{00}^4(0,\omega)} = \varkappa^2. \tag{1}$$

Put

$$v = \pi\vartheta_{00}^2(0,\omega)u = \pi\vartheta_{00}^2(0,\omega')u'. \tag{2}$$

Then (by § 42) both the functions

$$\frac{\vartheta_{00}(0,\omega)\vartheta_{11}(u,\omega)}{\vartheta_{10}(0,\omega)\vartheta_{01}(u,\omega)}, \quad \frac{\vartheta_{01}(0,\omega)\vartheta_{10}(u,\omega)}{\vartheta_{10}(0,\omega)\vartheta_{01}(u,\omega)}, \quad \frac{\vartheta_{01}(0,\omega)\vartheta_{00}(u,\omega)}{\vartheta_{00}(0,\omega)\vartheta_{01}(u,\omega)}, \tag{3}$$

and also

$$\frac{\vartheta_{00}(0,\omega')\vartheta_{11}(u',\omega')}{\vartheta_{10}(0,\omega')\vartheta_{01}(u',\omega')}, \quad \frac{\vartheta_{01}(0,\omega')\vartheta_{10}(u',\omega')}{\vartheta_{10}(0,\omega')\vartheta_{01}(u',\omega')}, \quad \frac{\vartheta_{01}(0,\omega')\vartheta_{00}(u',\omega')}{\vartheta_{00}(0,\omega')\vartheta_{01}(u',\omega')}, \tag{4}$$

when put for x, y, z, satisfy the differential equations (1) of the preceding paragraph, and the corresponding functions among these are therefore identical. If now $u' = \frac{1}{2}$, then $\vartheta_{10}(u',\omega')$ vanishes, and therefore $\vartheta_{10}(u,\omega)$ must also vanish. Hence by (2), with regard to § 21,

$$\vartheta_{00}^2(0,\omega') = \vartheta_{00}^2(0,\omega)(\alpha + \beta\omega), \tag{5}$$

where α, β are whole numbers which satisfy the condition

$$\alpha \equiv 1, \qquad \beta \equiv 0 \pmod 2. \tag{6}$$

If $u' = \omega'/2$, then $\vartheta_{01}(u',\omega')$ and $\vartheta_{01}(u,\omega)$ vanish, and therefore, as above,

$$\vartheta_{00}^2(0,\omega')\omega' = \vartheta_{00}^2(0,\omega)(\gamma + \delta\omega), \tag{7}$$

where

$$\gamma \equiv 0, \qquad \delta \equiv 1 \pmod 2. \tag{8}$$

From this follows

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{9}$$

But since in the same way one can conclude

$$\vartheta_{00}^2(0,\omega) = \vartheta_{00}^2(0,\omega')(\alpha' + \beta'\omega'),$$

$$\vartheta_{00}^2(0,\omega)\omega = \vartheta_{00}^2(0,\omega')(\gamma' + \delta'\omega'),$$

where $\alpha', \beta', \gamma', \delta'$ are likewise whole numbers, substitution of (5) in these equations gives

$$1 = (\alpha + \beta\omega)(\alpha' + \beta'\omega'), \qquad \omega = (\alpha + \beta\omega)(\gamma' + \delta'\omega').$$

If in these equations ω' is expressed by ω according to (9), one obtains

$$1 = \alpha'(\alpha + \beta\omega) + \beta'(\gamma + \delta\omega),$$

$$\omega = \gamma'(\alpha + \beta\omega) + \delta'(\gamma + \delta\omega),$$

and since ω is not real, each of these equations splits into two others:

$$\alpha\alpha' + \gamma\beta' = 1, \qquad \beta\alpha' + \delta\beta' = 0,$$

$$\alpha\gamma' + \gamma\delta' = 0, \qquad \beta\gamma' + \delta\delta' = 1.$$

Hence

$$(\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma') = 1,$$

and therefore, since ω, ω' must both have a positive imaginary part, and hence by (9) the determinant $\alpha\delta - \beta\gamma$ must be positive,

$$\alpha\delta - \beta\gamma = 1, \qquad \alpha = \delta', \quad \delta = \alpha', \quad \gamma = -\gamma', \quad \beta = -\beta'.$$

1. Thus ω, ω' are connected with one another by a linear transformation, and indeed, by (6), (8), by one of the first class (§ 36). Conversely, it has already been proved above that two such values ω, ω' give the same value $\varkappa^2$.

A similar theorem results as an immediate consequence of this for the invariant $j(\omega)$ (§ 46).

2. The equation

$$j(\omega') = j(\omega) \tag{10}$$

is satisfied if and only if ω' is connected with ω by some linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

that is, if

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}, \qquad \alpha\delta - \beta\gamma = 1. \tag{11}$$

For if we denote by λ^2, λ'^2 the values of $\varkappa^2, \varkappa'^2$ belonging to ω', then equation (10) can be written

$$\frac{(1 - \lambda^2\lambda'^2)^3}{\lambda^4\lambda'^4} = \frac{(1 - \varkappa^2\varkappa'^2)^3}{\varkappa^4\varkappa'^4},$$

and with respect to λ^2 it is an equation of the sixth degree, whose roots are

$$\varkappa^2, \quad \varkappa'^2, \quad \frac{1}{\varkappa^2}, \quad \frac{1}{\varkappa'^2}, \quad -\frac{\varkappa'^2}{\varkappa^2}, \quad -\frac{\varkappa^2}{\varkappa'^2}.$$

With regard to § 45, therefore, one of the following relations holds:

$$\lambda^2 = \varkappa^2(\omega), \quad \varkappa^2\left(-\frac{1}{\omega}\right), \quad \varkappa^2(\omega + 1), \quad \varkappa^2\left(\frac{\omega - 1}{\omega}\right), \quad \varkappa^2\left(-\frac{1}{\omega + 1}\right), \quad \varkappa^2\left(\frac{\omega}{1 - \omega}\right),$$

from which the correctness of the second theorem follows from the first.

The variable ω is here always restricted to a positive imaginary part, and if we call two numbers ω, ω' connected by an equation (11) equivalent, then all numbers equivalent to ω have a positive imaginary part. We can also express our theorem 2 as follows:

3. The function $j(\omega)$ has one and the same value for all equivalent values ω, and only for these.[9]

To every value of ω belongs a definite value of $j(\omega)$, and to every value of $j(\omega)$ a value of ω and the totality of equivalent values ω'. Therefore if a single-valued function $\Phi(\omega)$ of ω satisfies the condition

$$\Phi\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = \Phi(\omega),$$

then $\Phi(\omega)$ is a single-valued function of $j(\omega)$; and if we therefore assume that $\Phi(\omega)$ becomes infinite for only a finite number of values $j(\omega)$, and becomes infinite only to finite order, it follows that $\Phi(\omega)$ is a rational function of $j(\omega)$.

The supposition is fulfilled, for example, whenever $\Phi(\omega)$ stands in an algebraic connection with $j(\omega)$.

[9]This theorem was first proved by Dedekind (Crelle, Vol. 83). Accordingly Dedekind calls the function $\mathrm{val}(\omega) = 3^{-3}2^{-6}j(\omega)$ the valence of ω.

§ 54. The modular functions.

Let $\psi(\omega)$ be a single-valued function of ω. If one applies to ω a linear transformation

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

then $\psi(\omega)$ goes over into another function

$$\psi\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right), \tag{1}$$

which we shall denote by $\psi \mid S$.

The function $\psi(\omega)$ may possibly remain unchanged under certain transformations S (for instance, always under the identical transformation). All transformations which leave a function $\psi(\omega)$ unchanged form a group $\mathfrak{G}$. For if

$$\psi \mid S = \psi, \qquad \psi \mid S_1 = \psi,$$

then also

$$\psi \mid SS_1 = \psi \mid S_1 = \psi.$$

The group $\mathfrak{G}$ is a divisor of the group $\mathfrak{L}$ of all linear transformations.

If $\mathfrak{G}$ is a divisor of $\mathfrak{L}$ of finite index $(\mathfrak{L}, \mathfrak{G})$, and $\psi(\omega)$ is a function of ω with the property that conversely

$$\psi(\omega') = \psi(\omega)$$

implies a linear connection

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

in which

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

is a transformation belonging to $\mathfrak{G}$, then $\psi(\omega)$ is called a modular function belonging to the group $\mathfrak{G}$.

We restrict ourselves to the consideration of such modular functions as stand in an algebraic connection with the modulus $\varkappa^2$, and hence also with $j(\omega)$.

Under this supposition the following general theorem can be stated:

If $\psi(\omega)$ belongs to the group $\mathfrak{G}$, and $\chi(\omega)$ remains unchanged under the transformations of $\mathfrak{G}$, then $\chi(\omega)$ can be expressed rationally through $\psi(\omega)$.

For first $\chi(\omega)$ is an algebraic function of $\psi(\omega)$, and $\psi(\omega)$, as an algebraic function of $j(\omega)$, can assume every value. But by supposition only one value of $\chi(\omega)$ belongs to every value of $\psi(\omega)$, and therefore $\chi(\omega)$, as a single-valued algebraic function of $\psi(\omega)$, is rational.

The study of the groups contained in $\mathfrak{L}$ and of the modular functions belonging to them forms the main subject of the great work of Klein and Fricke, *Vorlesungen über die Theorie der elliptischen Modulfunktionen* (2 vols., Leipzig, Teubner, 1890, 1892). Here we consider only some quite special cases of these groups and functions which we have encountered in the course of our investigations.

A large class of divisors of the group $\mathfrak{L}$ with finite index is formed by the so-called congruence groups, which, if m is any whole number, are characterized by the four congruences

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{m}.$$

The modular functions belonging to these groups are called congruence moduli of the m-th stage. For $m = 2$ this is the group $\mathfrak{A}$ (§ 36). We shall make use of the following theorems:

1. The modulus $\varkappa^2$ is, by § 52, a modular function belonging to this group; since, by § 36, all these transformations can be composed from the two

$$\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}, \qquad \begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix},$$

we can state the theorem:

Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega+2), \qquad \left(\omega, \frac{\omega}{1+2\omega}\right),$$

is a rational function of $\varkappa^2$.

2. The function $\varkappa^2\varkappa'^2$ belongs to the group $\mathfrak{A}'$ formed in common from the first and second classes of § 36. This group can be derived by repetition of the two transformations

$$\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}, \qquad \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix},$$

and hence we can state the theorem:

Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega+2), \qquad \left(\omega, -\frac{1}{\omega}\right),$$

is a rational function of $\varkappa^2\varkappa'^2$. From this there further follows:

3. Every modular function which remains unchanged under $(\omega, \omega+2)$ and changes its sign under $\left(\omega, -\frac{1}{\omega}\right)$ is the product of $(\varkappa'^2 - \varkappa^2)$ with a rational function of $\varkappa^2\varkappa'^2$.

4. The invariant $j(\omega)$ belongs to the group $\mathfrak{L}$ (§ 53), and hence the theorem:

Every modular function which remains unchanged under the two interchanges

$$(\omega, \omega+1), \qquad \left(\omega, -\frac{1}{\omega}\right)$$

is a rational function of $j(\omega)$.

Besides these we introduce two other modular functions. From the definition of the functions $f(\omega), f_1(\omega), f_2(\omega)$ [§ 34,(1),(10)] it follows, if the two equations § 42,(3) are multiplied by one another, that

$$f(\omega) = \frac{\sqrt[16]{2}}{\sqrt[12]{\varkappa\varkappa'}}, \qquad f_1(\omega) = \sqrt[16]{2}\,\sqrt[12]{\frac{\varkappa'^2}{\varkappa}}, \qquad f_2(\omega) = \sqrt[16]{2}\,\sqrt[12]{\frac{\varkappa^2}{\varkappa'}}, \tag{2}$$

$$\frac{f_1(\omega)}{f(\omega)} = \sqrt[4]{\varkappa'}, \qquad \frac{f_2(\omega)}{f(\omega)} = \sqrt[4]{\varkappa}. \tag{3}$$

On the basis of § 46,(15),(16) we define two functions $\gamma_2(\omega), \gamma_3(\omega)$:

$$\begin{aligned}\gamma_2(\omega) &= \sqrt[3]{j(\omega)} = \sqrt[3]{2^8}\,\frac{1-\varkappa^2\varkappa'^2}{\sqrt[3]{\varkappa^4\varkappa'^4}},\\ \gamma_3(\omega) &= \sqrt{j(\omega) - 27\cdot 64} = \frac{8(2+\varkappa^2\varkappa'^2)(\varkappa'^2-\varkappa^2)}{\varkappa^2\varkappa'^2},\end{aligned} \tag{4}$$

which, by (2) and (3), can be represented as single-valued functions of ω as follows:

$$\gamma_2(\omega) = \frac{f(\omega)^{24} - 16}{f(\omega)^8}, \qquad \gamma_3(\omega) = \frac{\left[f(\omega)^{24}+8\right]\left[f_1(\omega)^8 - f_2(\omega)^8\right]}{f(\omega)^8}, \tag{5}$$

for which, according to the fundamental formulae for the f-functions [§ 34,(11),(12)], one can also put

$$\gamma_2(\omega) = f(\omega)^8 f_1(\omega)^8 + f(\omega)^8 f_2(\omega)^8 - f_1(\omega)^8 f_2(\omega)^8, \tag{6}$$

$$\gamma_3(\omega) = \frac{1}{2}\left[f(\omega)^8 - f_1(\omega)^8\right]\left[f(\omega)^8 + f_2(\omega)^8\right]\left[f_1(\omega)^8 - f_2(\omega)^8\right]. \tag{7}$$

Thus $x = f^8, -f_1^8, -f_2^8$ are the roots of the cubic equation

$$x^3 - \gamma_2 x - 16 = 0, \tag{8}$$

and $4\gamma_3^2$ is the discriminant of this equation.

Also noteworthy are the differential equations:

$$d\varkappa^2 = -\,d\varkappa'^2 = \pi\mathrm{i}\vartheta_{00}^4 \varkappa^2\varkappa'^2\ d\omega \quad [\S\,52,(3)], \tag{9}$$

$$d\varkappa^2\varkappa'^2 = -\pi\mathrm{i}\vartheta_{00}^4(\varkappa^2 - \varkappa'^2)\varkappa^2\varkappa'^2\ d\omega, \tag{10}$$

$$df(\omega)^8 = \frac{\pi\mathrm{i}}{3}\vartheta_{00}^4\left[f_2(\omega)^8 - f_1(\omega)^8\right] d\omega, \tag{11}$$

$$d\gamma_2(\omega) = -\frac{2\pi\mathrm{i}}{3}\gamma_3(\omega)\eta(\omega)^4\ d\omega \quad [\text{by (5) and } \S\,34,(10)], \tag{12}$$

$$dj(\omega) = -2\pi\mathrm{i}\,\gamma_2(\omega)^2\gamma_3(\omega)\eta(\omega)^4\ d\omega. \tag{13}$$

Next one obtains for the linear fundamental transformations, from § 34,(13),(14),(15),

$$\gamma_2(\omega+1) = e^{-\frac{2\pi\mathrm{i}}{3}}\gamma_2(\omega), \qquad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega), \tag{14a}$$

$$\gamma_3(\omega+1) = -\gamma_3(\omega), \qquad \gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega), \tag{14b}$$

and from § 40 one finds in general

$$\gamma_2\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = \varrho\,\gamma_2(\omega), \qquad \gamma_3\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = (-1)^{\alpha\gamma+\beta\gamma+\beta\delta}\gamma_3(\omega), \tag{15}$$

where ϱ has the meaning given in § 40,(3):

$$\varrho = e^{-\frac{2\pi\mathrm{i}}{3}(\alpha\gamma+\beta\delta-\alpha\beta-\alpha^2\beta\delta)}.$$

These are therefore modular functions, and the groups to which they belong are characterized by the two congruences

$$-\alpha\beta + \alpha\gamma + \beta\delta - \alpha^2\beta\gamma \equiv 0 \pmod 3,$$
$$\alpha\gamma + \beta\gamma + \beta\delta \equiv 0 \pmod 2.$$

Accordingly we give our theorem 4 the following supplements, which follow from formulae (14).

5. A modular function which changes its sign under the two substitutions

$$(\omega, \omega+1), \qquad \left(\omega, -\frac{1}{\omega}\right),$$

is the product of $\gamma_3(\omega)$ with a rational function of $j(\omega)$.

6. A modular function which remains unchanged under the substitution $\left(\omega, -\frac{1}{\omega}\right)$ and under $(\omega, \omega+1)$ assumes the factor $e^{\pm 2\pi\mathrm{i}/3}$ is the product of $\gamma_2(\omega)^{\pm 1}$ with a rational function of $j(\omega)$.

7. A modular function which changes its sign under the substitutions $\left(\omega, -\frac{1}{\omega}\right)$ and under $(\omega, \omega+1)$ assumes the factor $-e^{\pm 2\pi\mathrm{i}/3}$ is the product of $\gamma_2(\omega)\gamma_3(\omega)^{\pm 1}$ with a rational function of $j(\omega)$.

We shall occupy ourselves more thoroughly with the modular functions in the second part.

§ 55. Representation of the elliptic functions by v and $\varkappa^2$.

If we conceive the inversion problem of the elliptic integrals, as in § 51, as the problem of integrating the elliptic differential equations, then it requires the representation of the three functions $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$ by the two independent variables $v, \varkappa^2$. This problem is solved by § 42, but only implicitly with respect to the variable $\varkappa^2$. One can also find representations in which $\varkappa^2$ occurs explicitly, namely by series which proceed according to powers of v, whose coefficients are rational functions of $\varkappa^2$. The coefficients of these series, to be sure, cannot be represented by a general law, but can only be computed successively. These representations are due principally to Weierstrass.[10]

The σ-functions, since they are everywhere finite and continuous functions of u, can be developed in unconditionally convergent power series according to u; and if we therefore develop the σ-functions occurring in § 45,(2) in this way, we obtain a solution of our problem. Accordingly we put

$$\begin{aligned}
\sigma(v, 2K, 2\mathrm{i}K') &= \sum_{\nu=0}^{\infty} A_\nu \frac{v^{2\nu+1}}{(2\nu+1)!}, \\
\sigma_{00}(v, 2K, 2\mathrm{i}K') &= \sum_{\nu=0}^{\infty} B_\nu \frac{v^{2\nu}}{(2\nu)!}, \\
\sigma_{01}(v, 2K, 2\mathrm{i}K') &= \sum_{\nu=0}^{\infty} C_\nu \frac{v^{2\nu}}{(2\nu)!}, \\
\sigma_{10}(v, 2K, 2\mathrm{i}K') &= \sum_{\nu=0}^{\infty} D_\nu \frac{v^{2\nu}}{(2\nu)!},
\end{aligned} \tag{1}$$

and the $A_\nu, B_\nu, C_\nu, D_\nu$ are the functions to be determined of $\varkappa^2$ or of ω. If we first regard them as functions of ω, then the linear transformations [§ 45,(3)] give

$$\begin{aligned}
A_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu A_\nu(\omega), \\
B_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu B_\nu(\omega), \\
C_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu D_\nu(\omega), \\
D_\nu\left(-\frac{1}{\omega}\right) &= (-1)^\nu C_\nu(\omega),
\end{aligned} \tag{2}$$

$$\begin{aligned}
\varkappa'^{2\nu} A_\nu(\omega+1) &= A_\nu(\omega), \\
\varkappa'^{2\nu} B_\nu(\omega+1) &= C_\nu(\omega), \\
\varkappa'^{2\nu} C_\nu(\omega+1) &= B_\nu(\omega), \\
\varkappa'^{2\nu} D_\nu(\omega+1) &= D_\nu(\omega),
\end{aligned} \tag{3}$$

$$A_\nu(\omega+2) = A_\nu(\omega), \quad B_\nu(\omega+2) = B_\nu(\omega), \quad C_\nu(\omega+2) = C_\nu(\omega), \quad D_\nu(\omega+2) = D_\nu(\omega). \tag{4}$$

From this we conclude the characteristic properties of these coefficients as functions of $\varkappa^2$. First it is clear from the meaning of the coefficients that for every value of $\varkappa^2$, with the possible exception of $\varkappa^2 = 0, 1, \infty$, the coefficients $A_\nu(\varkappa^2), B_\nu(\varkappa^2), C_\nu(\varkappa^2), D_\nu(\varkappa^2)$ are finite and continuous functions of $\varkappa^2$.

But even for $\varkappa^2 = 0$ these functions remain finite, and one can find their values easily by letting q, and hence $\varkappa^2$, pass into zero in the representations of § 37. One then obtains from the developments in § 25, with regard to the fact that by § 34 and § 42 for a vanishing q

$$\eta(\omega) = q^{1/12}, \qquad 2K = \pi,$$

[10] On the Weierstrass theory of elliptic functions one should compare especially the *Formeln und Lehrsätze zum Gebrauch der elliptischen Funktionen*, edited by H. A. Schwarz. On the series expansions one should also compare Königsberger, *Vorlesungen über die Theorie der elliptischen Funktionen*, Lecture 25.

that

$$
\begin{aligned}
e^{v^2/6}\sin v &= \sum_{\nu=0}^{\infty} A_\nu(0)\frac{v^{2\nu+1}}{(2\nu+1)!},\\
e^{v^2/6} &= \sum_{\nu=0}^{\infty} B_\nu(0)\frac{v^{2\nu}}{(2\nu)!},\\
e^{v^2/6} &= \sum_{\nu=0}^{\infty} C_\nu(0)\frac{v^{2\nu}}{(2\nu)!},\\
e^{v^2/6}\cos v &= \sum_{\nu=0}^{\infty} D_\nu(0)\frac{v^{2\nu}}{(2\nu)!}.
\end{aligned}
\tag{5}
$$

The formulae (2) now give

$$
\begin{aligned}
A_\nu(\varkappa'^2) &= (-1)^\nu A_\nu(\varkappa^2),\\
B_\nu(\varkappa'^2) &= (-1)^\nu B_\nu(\varkappa^2),\\
C_\nu(\varkappa'^2) &= (-1)^\nu D_\nu(\varkappa^2),\\
D_\nu(\varkappa'^2) &= (-1)^\nu C_\nu(\varkappa^2),
\end{aligned}
\tag{6}
$$

from which it follows that these functions also remain finite for $\varkappa^2 = 1$, namely

$$
\begin{aligned}
A_\nu(1) &= (-1)^\nu A_\nu(0),\\
B_\nu(1) &= (-1)^\nu B_\nu(0),\\
C_\nu(1) &= (-1)^\nu D_\nu(0),\\
D_\nu(1) &= (-1)^\nu C_\nu(0).
\end{aligned}
\tag{7}
$$

The system (3) may further be written in the form

$$
\begin{aligned}
\varkappa'^{2\nu} A_\nu\left(-\frac{\varkappa^2}{\varkappa'^2}\right) &= A_\nu(\varkappa^2),\\
\varkappa'^{2\nu} B_\nu\left(-\frac{\varkappa^2}{\varkappa'^2}\right) &= C_\nu(\varkappa^2),\\
\varkappa'^{2\nu} C_\nu\left(-\frac{\varkappa^2}{\varkappa'^2}\right) &= B_\nu(\varkappa^2),\\
\varkappa'^{2\nu} D_\nu\left(-\frac{\varkappa^2}{\varkappa'^2}\right) &= D_\nu(\varkappa^2),
\end{aligned}
\tag{8}
$$

whence for an infinite $\varkappa^2$ one obtains

$$
\begin{aligned}
A_\nu(\varkappa^2) &= \varkappa^{2\nu} A_\nu(0),\\
B_\nu(\varkappa^2) &= \varkappa^{2\nu} D_\nu(0),\\
C_\nu(\varkappa^2) &= \varkappa^{2\nu} B_\nu(0),\\
D_\nu(\varkappa^2) &= \varkappa^{2\nu} C_\nu(0),
\end{aligned}
\tag{9}
$$

and from this it is concluded that the coefficients $A_\nu, B_\nu, C_\nu, D_\nu$ are integral rational functions of $\varkappa^2$ of degree ν; from § 54, 2, 3, it further follows that A_ν, B_ν for even ν are expressed rationally through $\varkappa^2\varkappa'^2$, while for odd ν they are equal to the product of $(\varkappa'^2 - \varkappa^2)$ with a rational function of $\varkappa^2\varkappa'^2$.

From B_ν one finds C_ν by means of the formula

$$
C_\nu(\varkappa^2) = \varkappa'^{2\nu} B_\nu\left(-\frac{\varkappa^2}{\varkappa'^2}\right), \tag{10}
$$

and D_ν by application of (6) and (8):

$$D_\nu(\varkappa^2) = (-1)^\nu C_\nu(\varkappa'^2) = (-1)^\nu \varkappa^{2\nu} B_\nu\left(-\frac{\varkappa'^2}{\varkappa^2}\right). \tag{11}$$

Since from (5) one can easily determine the coefficients $A_\nu(0), B_\nu(0), C_\nu(0), D_\nu(0)$, the coefficients $A_\nu, B_\nu, C_\nu, D_\nu$ result from the system of formulae developed here, without further calculation, up to and including $\nu = 3$.

One finds:

$$A_0 = 1, \quad A_1 = 0, \quad A_2 = -\frac{2}{3}(1 - \varkappa^2\varkappa'^2),$$

$$B_0 = 1, \quad B_1 = \frac{1}{3}(\varkappa'^2 - \varkappa^2), \quad B_2 = \frac{1}{3}(1 + 2\varkappa^2\varkappa'^2),$$

$$C_0 = 1, \quad C_1 = \frac{1}{3}(1 + \varkappa^2), \quad C_2 = \frac{1}{3}(\varkappa'^4 - 2\varkappa^2),$$

$$D_0 = 1, \quad D_1 = -\frac{1}{3}(1 + \varkappa'^2), \quad D_2 = \frac{1}{3}(\varkappa^4 - 2\varkappa'^2),$$

$$A_3 = -\frac{8}{9}(\varkappa'^2 - \varkappa^2)(2 + \varkappa^2\varkappa'^2),$$

$$B_3 = \frac{1}{9}(\varkappa'^2 - \varkappa^2)(5 - 2\varkappa^2\varkappa'^2),$$

$$C_3 = \frac{1}{9}(1 + \varkappa^2)(5\varkappa'^4 + 2\varkappa^2),$$

$$D_3 = -\frac{1}{9}(1 + \varkappa'^2)(5\varkappa^4 + 2\varkappa'^2).$$

Further coefficients can be computed on the same way, although at greater length, by developing the expressions for the σ-functions in § 32 according to powers of q and, as here the lowest power of q was used, using the higher powers. Weierstrass makes use, for the recurrent computation of the coefficients, of certain partial differential equations, similar to that which we shall derive in the next paragraph for the function σ.

§ 56. Power series for the Weierstrass functions $\wp(u), \sigma(u)$.

The function $\sigma(u)$, considered as a function of u, can, as already remarked, be developed into a series according to powers of u convergent in the whole u-plane; and by § 35,(8),(12) this development has the following form:

$$\sigma(u) = u + \alpha u^3 + \alpha_1 u^5 + \alpha_2 u^7 + \alpha_3 u^9 + \cdots. \tag{1}$$

The function

$$\wp(u) = -\frac{d^2 \log \sigma(u)}{du^2} = \frac{\sigma'(u)^2 - \sigma(u)\sigma''(u)}{\sigma(u)^2} \tag{2}$$

can likewise be developed according to ascending powers of u, but no longer in a series convergent everywhere; rather this series has a convergence circle. It has the form

$$\wp(u) = \frac{1}{u^2} + a + a_1 u^2 + a_2 u^4 + a_3 u^6 + \cdots, \tag{3}$$

and one can determine the coefficients $a, a_1, a_2, a_3, \ldots$ successively, as functions of g_2, g_3, from the differential equation [§ 46,(18)]

$$\wp'(u)^2 = 4\wp(u)^3 - g_2\wp(u) - g_3. \tag{4}$$

First one has

$$\wp'(u) = -\frac{2}{u^3} + 2a_1 u + 4a_2 u^3 + 6a_3 u^5 + \cdots,$$

$$\wp'(u)^2 = \frac{4}{u^6} - \frac{8a_1}{u^2} - 16a_2 + (4a_1^2 - 24a_3)u^2 + \cdots,$$

and since $\wp'(u)^2$ contains no term with u^{-4}, while in $\wp(u)^3$ the term $3au^{-4}$ occurs, a must be zero. After this there results

$$\wp(u)^3 = \frac{1}{u^6} + \frac{3a_1}{u^2} + 3a_2 + 3(a_3 + a_1^2)u^2 + \cdots,$$

$$\wp(u) = \frac{1}{u^2} + a_1 u^2 + \cdots.$$

And if one inserts this in equation (4), there follows

$$a_1 = \frac{g_2}{20}, \qquad a_2 = \frac{g_3}{28}, \qquad a_3 = \frac{g_2^2}{1200}. \tag{5}$$

By integration of (2) one finds

$$\sigma'(u) = \left(\frac{1}{u} - \frac{a_1 u^3}{3} - \frac{a_2 u^5}{5} - \frac{a_3 u^7}{7} \cdots\right)\sigma(u),$$

and from this equation follows

$$\alpha = 0, \qquad \alpha_1 = -\frac{g_2}{240}, \qquad \alpha_2 = -\frac{g_3}{840}, \qquad \alpha_3 = -\frac{g_2^2}{161280}. \tag{6}$$

That $\alpha = 0$ follows also from the $\sigma'''(u) = 0$ proved in § 35; there, however, this was proved by a rather circuitous path. Thus we have the following beginning of the development of $\sigma(u)$, in which, for the sake of clarity, the numerical denominators are decomposed into their prime factors:

$$\sigma(u) = u - \frac{g_2}{2^4 \cdot 3 \cdot 5}u^5 - \frac{g_3}{2^3 \cdot 3 \cdot 5 \cdot 7}u^7 - \frac{g_2^2}{2^9 \cdot 3^2 \cdot 5 \cdot 7}u^9 - \cdots. \tag{7}$$

For the recurrent calculation of the coefficients in the power series for $\sigma(u)$ there serves a partial differential equation which may be regarded as a transformation of the partial differential equation for the ϑ-functions. We derive it in the following way.

The function $\sigma(u)$ was determined up to a factor independent of u by the two period equations § 35,(9)

$$\sigma(u + \omega_1) = -e^{\eta_1(2u+\omega_1)}\sigma(u), \qquad \sigma(u + \omega_2) = -e^{\eta_2(2u+\omega_2)}\sigma(u), \qquad \eta_1\omega_2 - \eta_2\omega_1 = \pi \mathrm{i}. \tag{8}$$

By differentiating the last equation (8), however, one obtains

$$\frac{\partial \eta_1}{\partial \omega_1}\omega_2 - \frac{\partial \eta_2}{\partial \omega_1}\omega_1 - \eta_2 = 0,$$

$$\frac{\partial \eta_1}{\partial \omega_2}\omega_2 - \frac{\partial \eta_2}{\partial \omega_2}\omega_1 + \eta_1 = 0,$$

hence

$$\omega_2\left(\eta_1\frac{\partial \eta_1}{\partial \omega_1} + \eta_2\frac{\partial \eta_1}{\partial \omega_2}\right) = \omega_1\left(\eta_1\frac{\partial \eta_2}{\partial \omega_1} + \eta_2\frac{\partial \eta_2}{\partial \omega_2}\right),$$

and we therefore introduce a quantity r, which we define thus:

$$\eta_1\frac{\partial \eta_1}{\partial \omega_1} + \eta_2\frac{\partial \eta_1}{\partial \omega_2} = -r\omega_1, \qquad \eta_1\frac{\partial \eta_2}{\partial \omega_1} + \eta_2\frac{\partial \eta_2}{\partial \omega_2} = -r\omega_2. \tag{9}$$

Now by differentiating the period equations (8) one proves that the function

$$S(u) = \frac{\partial^2\sigma}{\partial u^2} + 4\eta_1\frac{\partial\sigma}{\partial\omega_1} + 4\eta_2\frac{\partial\sigma}{\partial\omega_2} + 4ru^2\sigma$$

satisfies the period equations (8), and hence is equal to $C\sigma(u)$, where C is independent of u. But the development of $S(u)$ according to powers of u begins with u^3, whereas $\sigma(u)$ begins with u; consequently $C = 0$.

By (7) the developments of $\partial\sigma/\partial\omega_1$ and $\partial\sigma/\partial\omega_2$ begin only with u^5, and the first terms of $\partial^2\sigma/\partial u^2$ and $4ru^2\sigma$ are $-g_2u^3/12$ and $4ru^3$, whence $4r = g_2/12$ follows, and accordingly the differential equation for σ assumes the form

$$\frac{\partial^2\sigma}{\partial u^2} + 4\eta_1\frac{\partial\sigma}{\partial\omega_1} + 4\eta_2\frac{\partial\sigma}{\partial\omega_2} + \frac{g_2}{12}u^2\sigma = 0, \tag{10}$$

and at the same time from (9) there result the relations

$$\eta_1\frac{\partial\eta_1}{\partial\omega_1} + \eta_2\frac{\partial\eta_1}{\partial\omega_2} = -\frac{g_2}{48}\omega_1, \qquad \eta_1\frac{\partial\eta_2}{\partial\omega_1} + \eta_2\frac{\partial\eta_2}{\partial\omega_2} = -\frac{g_2}{48}\omega_2. \tag{11}$$

Now in the differential equation (10) the variables g_2, g_3 are to be introduced in place of the variables ω_1, ω_2; if therefore we put

$$\frac{\partial\sigma}{\partial\omega_1} = \frac{\partial\sigma}{\partial g_2}\frac{\partial g_2}{\partial\omega_1} + \frac{\partial\sigma}{\partial g_3}\frac{\partial g_3}{\partial\omega_1}, \qquad \frac{\partial\sigma}{\partial\omega_2} = \frac{\partial\sigma}{\partial g_2}\frac{\partial g_2}{\partial\omega_2} + \frac{\partial\sigma}{\partial g_3}\frac{\partial g_3}{\partial\omega_2},$$

then (10) assumes the form

$$\frac{\partial^2\sigma}{\partial u^2} + h_2\frac{\partial\sigma}{\partial g_2} + h_3\frac{\partial\sigma}{\partial g_3} + \frac{g_2}{12}u^2\sigma = 0, \tag{12}$$

where

$$h_2 = 4\left(\eta_1\frac{\partial g_2}{\partial\omega_1} + \eta_2\frac{\partial g_2}{\partial\omega_2}\right), \qquad h_3 = 4\left(\eta_1\frac{\partial g_3}{\partial\omega_1} + \eta_2\frac{\partial g_3}{\partial\omega_2}\right).$$

The quantities h_2, h_3, expressed by g_2, g_3, now result from the differential equation itself if one inserts for σ the development (7). According to this, up to the seventh power of u,

$$\frac{\partial^2\sigma}{\partial u^2} = -\frac{g_2}{12}u^3 - 12g_3\frac{u^5}{2^4\cdot 3\cdot 5} - \frac{3}{8}g_2^2\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},$$

$$\frac{\partial\sigma}{\partial g_2} = -\frac{u^5}{2^4\cdot 3\cdot 5}, \qquad \frac{\partial\sigma}{\partial g_3} = -\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},$$

$$\frac{g_2}{12}u^2\sigma = \frac{g_2}{12}u^3 - \frac{7}{24}g_2^2\frac{u^7}{2^3\cdot 3\cdot 5\cdot 7},$$

and if this is inserted in (12), and the coefficients of u^5 and u^7 are set equal to zero, there results

$$h_2 = -12g_3, \qquad h_3 = -\frac{2}{3}g_2^2,$$

and we obtain the differential equation

$$\frac{\partial^2\sigma}{\partial u^2} - 12g_3\frac{\partial\sigma}{\partial g_2} - \frac{2}{3}g_2^2\frac{\partial\sigma}{\partial g_3} + \frac{g_2}{12}u^2\sigma = 0. \tag{13}$$

If one writes the series development for σ with undetermined coefficients as

$$\sigma = \sum A_\nu\frac{u^{2\nu+1}}{(2\nu+1)!}, \tag{14}$$

then from (13) one obtains the recursion formula

$$A_\nu = 12g_3\frac{\partial A_{\nu-1}}{\partial g_2} + \frac{2}{3}g_2^2\frac{\partial A_{\nu-1}}{\partial g_3} - \frac{(2\nu-1)(2\nu-2)}{12}g_2A_{\nu-2}, \tag{15}$$

from which it is to be concluded that the A_ν are integral rational functions of g_2 and g_3, whose coefficients are rational numbers that can have only powers of 2 and of 3 in the denominator. The series (14) therefore consists of terms of the form

$$a_{m,n}\left(\frac{1}{2}g_2\right)^m(2g_3)^n\frac{u^{2\nu+1}}{(2\nu+1)!},$$

and from the homogeneity of the function σ [§ 37,(8), § 46,(13)] it follows that

$$2m+3n=\nu. \tag{16}$$

For the coefficients $a_{m,n}$ one then obtains from (15):

$$\begin{aligned} a_{m,n} = 3(m+1)a_{m+1,n-1} &+ \frac{16}{3}(n+1)a_{m-2,n+1} \\ &- \frac{1}{3}(2m+3n-1)(4m+6n-1)a_{m-1,n}, \end{aligned} \tag{17}$$

where $a_{0,0}=1$, and those a for which one of the two indices is negative are to be put equal to zero.

From this one can find $a_{m,n}$ if the earlier $a_{m,n}$, that is, those for which $2m+3n$ is smaller than ν, have already been calculated. It is seen from this that the $a_{m,n}$ contain only powers of 3 in the denominator. In the *Formeln und Lehrsätzen zum Gebrauch der elliptischen Funktionen*, edited by H. A. Schwarz, after lectures by Weierstrass (2nd edition, 1893), the $a_{m,n}$ have been calculated up to $\nu=17$. It appears there that these coefficients are not only whole numbers, but also contain, as ν increases, increasing powers of 3 as factors. I am not able to give a proof of this.

Sixth Section.

Multiplication and Division of the Elliptic Functions.

§ 57. Multiplication of the elliptic functions.

By the multiplication of the elliptic functions one understands the representation of the functions $\operatorname{sn} nv$, $\operatorname{cn} nv$, $\operatorname{dn} nv$, for an integral n, as rational functions of $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$; this is a problem which, as is clear from the addition theorem, can always be solved.

The form of the solution follows easily from the consideration of the ϑ-functions.

As is seen immediately from § 21, the functions $\vartheta_{g_1,g_2}(nu)$ are ϑ-functions of order n^2, whose characteristic is equal to $(0,0)$ when n is even, and equal to (g_1,g_2) when n is odd. Moreover $\vartheta_{11}(nu)$ is an odd function of u, whereas $\vartheta_{00}(nu)$, $\vartheta_{10}(nu)$, $\vartheta_{01}(nu)$ are even functions of u. These functions can therefore be represented rationally by the ϑ-functions, and the theorems of § 21 give the form of these expressions. If we again denote by $F^{(\nu)}(x,y)$ an integral rational homogeneous function of order ν, we obtain:

for even n:

$$\begin{aligned} \vartheta_{11}(nu) &= \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{11}(u)F^{\frac{n^2-4}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right], \\ \vartheta_{g_1,g_2}(nu) &= F^{\frac{n^2}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right], \qquad (g_1,g_2)=(10),(01),(00), \end{aligned} \tag{1}$$

for odd n:

$$\vartheta_{g_1,g_2}(nu) = \vartheta_{g_1,g_2}(u)F^{\frac{n^2-1}{2}}\left[\vartheta_{11}^2(u),\vartheta_{01}^2(u)\right]. \tag{2}$$

If we divide these formulae by $\vartheta_{01}(u)^{n^2}$, then the right hand sides can be represented as integral rational functions of the elliptic functions $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$ (§ 42). Thus, putting

$$v = \pi\vartheta_{00}^2 u, \qquad \operatorname{sn} v = x, \qquad \operatorname{cn} v = y, \qquad \operatorname{dn} v = z, \tag{3}$$

we can write (1), (2), after suitable determination of a constant factor, as follows:

I. for even n:

$$\begin{aligned}
\frac{\vartheta_{01}^{n^2}\vartheta_{00}}{\vartheta_{10}}\frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} &= xyz\,A(x^2), \quad & A(0) &= n,\\
\frac{\vartheta_{01}^{n^2}}{\vartheta_{10}}\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} &= B(x^2), & B(0) &= 1,\\
\frac{\vartheta_{01}^{n^2}}{\vartheta_{00}}\frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} &= C(x^2), & C(0) &= 1,\\
\vartheta_{01}^{n^2-1}\frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} &= D(x^2), & D(0) &= 1.
\end{aligned}$$

II. for odd n:

$$\begin{aligned}
\frac{\vartheta_{00}\vartheta_{01}^{n^2-1}}{\vartheta_{10}}\frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} &= x\,A(x^2), \quad & A(0) &= n,\\
\frac{\vartheta_{01}^{n^2}}{\vartheta_{10}}\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} &= y\,B(x^2), & B(0) &= 1,\\
\frac{\vartheta_{01}^{n^2}}{\vartheta_{00}}\frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} &= z\,C(x^2), & C(0) &= 1,\\
\vartheta_{01}^{n^2-1}\frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} &= D(x^2), & D(0) &= 1,
\end{aligned}$$

where A, B, C, D are integral rational functions of x^2, whose degrees are obtained from (1), (2) as follows:

	$A(x^2)$	$B(x^2)$	$C(x^2)$	$D(x^2)$	
degree:	$\frac{n^2}{2}-2$	$\frac{n^2}{2}$	$\frac{n^2}{2}$	$\frac{n^2}{2}$	n even,
	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	n odd.

From the definitions I, II one immediately obtains the values of $A(0), B(0), C(0), D(0)$, that is, the terms independent of n, as they have been added above.

Numerous relations between these functions are still obtained in the following way. To carry out one example here somewhat more fully, replace v by $v+K$, and therefore u by $u+\frac{1}{2}$. Then x, y, z go, by § 44, (18), into $y/z, -\varkappa' x/z, \varkappa'/z$, and by § 21, (8), from the first formula II for an odd n one obtains

$$\frac{\vartheta_{00}\vartheta_{01}^{n^2-1}}{\vartheta_{10}}\frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}}\left(\frac{\vartheta_{01}(u)}{\vartheta_{00}(u)}\right)^{n^2} = (-1)^{\frac{n-1}{2}}\frac{y}{z}A\left(\frac{y^2}{z^2}\right).$$

On the other hand, by the second equation II and by § 42, the left hand side is equal to

$$\frac{y}{z}\left(\frac{\sqrt{\varkappa'}}{z}\right)^{n^2-1} B(x^2),$$

and consequently

$$B(x^2) = (-1)^{\frac{n-1}{2}}\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} A\left(\frac{y^2}{z^2}\right).$$

Putting in this $x=1$, $y=0$, $z=\varkappa'$, or $x=0$, $y=1$, $z=1$, gives

$$B(1) = (-1)^{\frac{n-1}{2}} n\sqrt{\varkappa'}^{\,n^2-1}, \qquad A(1) = (-1)^{\frac{n-1}{2}}\sqrt{\varkappa'}^{\,n^2-1}.$$

One can derive a whole system of such formulae by making in I and II the following substitutions:

$$\begin{array}{c|ccc} u & x & y & z \\ \hline u+\frac{1}{2} & \frac{y}{z} & -\frac{\varkappa' x}{z} & \frac{\varkappa'}{z} \\ u+\frac{\omega}{2} & \frac{1}{\varkappa x} & -\frac{iz}{\varkappa x} & -\frac{iy}{x} \\ u+\frac{1+\omega}{2} & \frac{z}{\varkappa y} & \frac{i\varkappa'}{\varkappa y} & \frac{i\varkappa' x}{y}. \end{array}$$

Thus one obtains:

$$\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-4} A\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}-1} A(x^2), \quad A(1) = (-1)^{\frac{n}{2}-1} n\sqrt{\varkappa'}^{\,n^2-4}, \tag{4a}$$

$$\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} B\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}} B(x^2), \quad B(1) = (-1)^{\frac{n}{2}} \sqrt{\varkappa'}^{\,n^2},$$

$$\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} C\left(\frac{y^2}{z^2}\right) = C(x^2), \quad C(1) = \sqrt{\varkappa'}^{\,n^2},$$

$$\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2} D\left(\frac{y^2}{z^2}\right) = D(x^2), \quad D(1) = \sqrt{\varkappa'}^{\,n^2}, \qquad n \text{ even.}$$

Likewise,

$$\begin{aligned} \left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} A\left(\frac{y^2}{z^2}\right) &= (-1)^{\frac{n-1}{2}} B(x^2), & A(1) &= (-1)^{\frac{n-1}{2}} \sqrt{\varkappa'}^{\,n^2-1}, \\ \left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} B\left(\frac{y^2}{z^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & B(1) &= (-1)^{\frac{n-1}{2}} n\sqrt{\varkappa'}^{\,n^2-1}, \\ \left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} C\left(\frac{y^2}{z^2}\right) &= D(x^2), & C(1) &= \sqrt{\varkappa'}^{\,n^2-1}, \\ \left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} D\left(\frac{y^2}{z^2}\right) &= C(x^2), & D(1) &= \sqrt{\varkappa'}^{\,n^2-1}, \end{aligned} \tag{5}$$

for n odd.

Further,

$$\begin{aligned} (\sqrt{\varkappa}\,x)^{n^2-4} A\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n}{2}-1} A(x^2), & A(\infty) &= (-1)^{\frac{n}{2}-1} n(\sqrt{\varkappa})^{\,n^2-4}, \\ (\sqrt{\varkappa}\,x)^{n^2} B\left(\frac{1}{\varkappa^2 x^2}\right) &= B(x^2), & B(\infty) &= \sqrt{\varkappa}^{\,n^2}, \\ (\sqrt{\varkappa}\,x)^{n^2} C\left(\frac{1}{\varkappa^2 x^2}\right) &= C(x^2), & C(\infty) &= \sqrt{\varkappa}^{\,n^2}, \\ (\sqrt{\varkappa}\,x)^{n^2} D\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n}{2}} D(x^2), & D(\infty) &= (-1)^{\frac{n}{2}} \sqrt{\varkappa}^{\,n^2}, \end{aligned} \tag{6}$$

for n even, and

$$\begin{aligned} (\sqrt{\varkappa}\,x)^{n^2-1} A\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n-1}{2}} D(x^2), & A(\infty) &= (-1)^{\frac{n-1}{2}} \sqrt{\varkappa}^{\,n^2-1}, \\ (\sqrt{\varkappa}\,x)^{n^2-1} B\left(\frac{1}{\varkappa^2 x^2}\right) &= C(x^2), & B(\infty) &= \sqrt{\varkappa}^{\,n^2-1}, \\ (\sqrt{\varkappa}\,x)^{n^2-1} C\left(\frac{1}{\varkappa^2 x^2}\right) &= B(x^2), & C(\infty) &= \sqrt{\varkappa}^{\,n^2-1}, \\ (\sqrt{\varkappa}\,x)^{n^2-1} D\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & D(\infty) &= (-1)^{\frac{n-1}{2}} n\sqrt{\varkappa}^{\,n^2-1}, \end{aligned} \tag{7}$$

for n odd.

Here $A(\infty), B(\infty), C(\infty), D(\infty)$ are in each case to mean the coefficients of the highest power of x in these functions. In (7) the two last formulae can be derived from the first by interchanging x with $1/\varkappa x$.

The third system of formulae can be obtained either in the same way, or also by replacing x by $1/\varkappa x$ in (4), (5) and applying (6), (7):

$$\begin{aligned}
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-4} A\left(\frac{z^2}{\varkappa^2 y^2}\right) &= A(x^2), & A\left(\frac{1}{\varkappa^2}\right) &= n\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2} B\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n}{2}} B(x^2), & B\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2} C\left(\frac{z^2}{\varkappa^2 y^2}\right) &= C(x^2), & C\left(\frac{1}{\varkappa^2}\right) &= \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{2} D\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n}{2}} D(x^2), & D\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-4},
\end{aligned} \tag{8}$$

for n even, and

$$\begin{aligned}
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1} A\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n-1}{2}} C(x^2), & A\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n-1}{2}}\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1} B\left(\frac{z^2}{\varkappa^2 y^2}\right) &= D(x^2), & B\left(\frac{1}{\varkappa^2}\right) &= \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1} C\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & C\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n-1}{2}} n\sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-1} D\left(\frac{z^2}{\varkappa^2 y^2}\right) &= B(x^2), & D\left(\frac{1}{\varkappa^2}\right) &= \sqrt{\frac{\varkappa'}{\varkappa}}^{\,n^2-1},
\end{aligned} \tag{9}$$

for n odd.

According to these formulae, in the case of odd n one can reduce the four polynomials $A(x^2), B(x^2), C(x^2), D(x^2)$ to one of them, for example to $A(x^2)$.

If in the formulae I, II one divides the first three formulae by the last, then one obtains the multiplication formulae for the elliptic functions (§ 42):

$$\text{III. For even } n: \quad \begin{aligned} \operatorname{sn} nv &= \frac{xyz\,A(x^2)}{D(x^2)}, \\ \operatorname{cn} nv &= \frac{B(x^2)}{D(x^2)}, \\ \operatorname{dn} nv &= \frac{C(x^2)}{D(x^2)}, \end{aligned}$$

$$\text{IV. For odd } n: \quad \begin{aligned} \operatorname{sn} nv &= \frac{x\,A(x^2)}{D(x^2)}, \\ \operatorname{cn} nv &= \frac{y\,B(x^2)}{D(x^2)}, \\ \operatorname{dn} nv &= \frac{z\,C(x^2)}{D(x^2)}. \end{aligned}$$

The coefficients of A, B, C, D still depend on ω or on $\varkappa^2$. The manner of this dependence is revealed by the recursion formulae which follow from the addition theorem for the successive calculation of

A, B, C, D. If the value of the multiplier n to which these functions belong is indicated by an index, then first of all we have

$$A_1 = 1, \qquad B_1 = 1, \qquad C_1 = 1, \qquad D_1 = 1; \tag{10}$$

further, from the addition formulae [§ 44, (16), for $u = v$],

$$\begin{aligned}
\operatorname{sn} 2v &= \frac{2 \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \varkappa^2 \operatorname{sn}^4 v},\\
\operatorname{cn} 2v &= \frac{\operatorname{cn}^2 v - \operatorname{sn}^2 v \operatorname{dn}^2 v}{1 - \varkappa^2 \operatorname{sn}^4 v},\\
\operatorname{dn} 2v &= \frac{\operatorname{dn}^2 v - \varkappa^2 \operatorname{sn}^2 v \operatorname{cn}^2 v}{1 - \varkappa^2 \operatorname{sn}^4 v},
\end{aligned} \tag{11}$$

and hence

$$A_2 = 2, \qquad B_2 = 1 - 2x^2 + \varkappa^2 x^4, \qquad C_2 = 1 - 2\varkappa^2 x^2 + \varkappa^2 x^4, \qquad D_2 = 1 - \varkappa^2 x^4. \tag{12}$$

If in (11) we replace v by nv, then

$$\begin{aligned}
A_{2n} &= 2 A_n B_n C_n D_n,\\
B_{2n} &= B_n^2 D_n^2 - x^2 y^2 z^2 A_n^2 C_n^2, \quad n \text{ even},\\
&= y^2 B_n^2 D_n^2 - x^2 z^2 A_n^2 C_n^2, \quad n \text{ odd},\\
C_{2n} &= C_n^2 D_n^2 - \varkappa^2 x^2 y^2 z^2 A_n^2 B_n^2, \quad n \text{ even},\\
&= z^2 C_n^2 D_n^2 - \varkappa^2 x^2 y^2 A_n^2 B_n^2, \quad n \text{ odd},\\
D_{2n} &= D_n^4 - \varkappa^2 x^4 y^4 z^4 A_n^4, \quad n \text{ even},\\
&= D_n^4 - \varkappa^2 x^4 A_n^4, \quad n \text{ odd}.
\end{aligned} \tag{13}$$

If in the addition formulae of § 44 one puts $u = nv$, $v = (n+1)v$, then

$$\begin{aligned}
A_{2n+1} &= y^2 z^2 A_n D_n B_{n+1} C_{n+1} + A_{n+1} D_{n+1} C_n B_n, \quad n \text{ even},\\
&= A_n D_n B_{n+1} C_{n+1} + y^2 z^2 A_{n+1} D_{n+1} C_n B_n, \quad n \text{ odd},\\
B_{2n+1} &= B_n B_{n+1} D_n D_{n+1} - x^2 z^2 A_n A_{n+1} C_n C_{n+1},\\
C_{2n+1} &= C_n C_{n+1} D_n D_{n+1} - \varkappa^2 x^2 y^2 A_n A_{n+1} B_n B_{n+1},\\
D_{2n+1} &= D_n^2 D_{n+1}^2 - \varkappa^2 x^4 y^2 z^2 A_n^2 A_{n+1}^2.
\end{aligned} \tag{14}$$

From these formulae one concludes that the A, B, C, D are integral rational functions of x^2, and that the numerical coefficients are integral rational numbers. For this property holds by (10), (12) for $n = 1, 2$, and hence in general by (13), (14).

Concerning the degree with respect to x^2 one may also conclude that the coefficients of $x^{2\nu}$ do not exceed degree ν with respect to $\varkappa^2$. For if this rule is correct for n and $n+1$, then its correctness follows for $2n$ and $2n+1$; and it is true for $n = 1, n = 2$.

From the fundamental equations between the three elliptic functions,

$$1 = \operatorname{cn}^2 v + \operatorname{sn}^2 v = \operatorname{dn}^2 v + \varkappa^2 \operatorname{sn}^2 v,$$

there also result the relations

$$\begin{aligned}
D^2 &= B^2 + x^2 y^2 z^2 A^2 = C^2 + \varkappa^2 x^2 y^2 z^2 A^2, \quad n \text{ even},\\
D^2 &= y^2 B^2 + x^2 A^2 = z^2 C^2 + \varkappa^2 x^2 A^2, \quad n \text{ odd},
\end{aligned} \tag{15}$$

with whose help, from (14), one can express B_{2n} and C_{2n} in terms of A_n and D_n alone:

$$\left.\begin{aligned}
B_{2n} &= D_n^4 - 2x^2 y^2 z^2 A_n^2 D_n^2 + \varkappa^2 x^4 y^4 z^4 A_n^4,\\
C_{2n} &= D_n^4 - 2\varkappa^2 x^2 y^2 z^2 A_n^2 D_n^2 + \varkappa^2 x^4 y^4 z^4 A_n^4,
\end{aligned}\right\} \quad n \text{ even}, \tag{16a}$$

$$\left.\begin{aligned}
B_{2n} &= D_n^4 - 2x^2 A_n^2 D_n^2 + \varkappa^2 x^4 A_n^4,\\
C_{2n} &= D_n^4 - 2\varkappa^2 x^2 A_n^2 D_n^2 + \varkappa^2 x^4 A_n^4,
\end{aligned}\right\} \quad n \text{ odd}. \tag{16b}$$

§ 58. Multiplication of the function $\wp(u)$.

A very elegant form is taken by the multiplication theory for the Weierstrass function $\wp(u)$ with the two periods ω_1, ω_2.

Consider the doubly periodic function

$$\wp(nu) - \wp(u), \tag{1}$$

where n may be any positive integer. This function becomes infinite for $u = 0$, and in such a way that

$$\wp(nu) - \wp(u) + \frac{n^2-1}{n^2}\frac{1}{u^2} \tag{2}$$

remains finite for $u = 0$. But it also becomes infinite for all values u^* of u satisfying

$$nu^* \equiv 0 \pmod{\omega_1, \omega_2},$$

that is, values of the form

$$u^* = \frac{\nu_1\omega_1 + \nu_2\omega_2}{n}, \tag{3}$$

where ν_1, ν_2 denote arbitrary integers. We obtain all incongruent values contained in (3) by letting ν_1 and ν_2 each run through a complete system of residues modulo n.

The function (1) vanishes for all non-zero values u^0 of u which satisfy

$$\pm nu^0 \equiv u^0 \pmod{\omega_1, \omega_2},$$

that is, for

$$u^0 = \frac{\nu_1\omega_1 + \nu_2\omega_2}{n \pm 1}, \tag{4}$$

again with arbitrary integers ν_1, ν_2. The zeros are of first order and the poles of second order.

We therefore introduce a function $\psi_n(u)$, defined as follows:

$$\psi_n(u)^2 = n^2 \prod^{\nu_1,\nu_2} \left[\wp(u) - \wp\left(\frac{\nu_1\omega_1 + \nu_2\omega_2}{n}\right)\right], \tag{5}$$

where ν_1, ν_2 each pass through a complete residue system modulo n, with only the pair $0, 0$ excluded, and we add the further determination that $\nu_1 = 1$ is to hold.

When n is odd, each linear factor occurs twice in the product (5), since the two incongruent values

$$u = \pm\frac{\nu_1\omega_1 + \nu_2\omega_2}{n}$$

give the same value of $\wp(u)$. But when n is even, all linear factors again occur twice in (5), with the exception of the three

$$\wp(u) - \wp\left(\frac{\omega_1}{2}\right), \qquad \wp(u) - \wp\left(\frac{\omega_2}{2}\right), \qquad \wp(u) - \wp\left(\frac{\omega_1+\omega_2}{2}\right),$$

which occur only simply. If one observes now that, by § 41,

$$\wp'(u)^2 = 4\left[\wp(u) - \wp\left(\frac{\omega_1}{2}\right)\right]\left[\wp(u) - \wp\left(\frac{\omega_2}{2}\right)\right]\left[\wp(u) - \wp\left(\frac{\omega_1+\omega_2}{2}\right)\right],$$

then the result is

$$n \text{ odd}: \ \psi_n = P_n, \qquad n \text{ even}: \ \psi_n = \wp'(u)P_n, \tag{6}$$

where P_n denotes an integral rational function of $\wp(u)$, of degree $\frac{1}{2}(n^2-1)$ for odd n and of degree $\frac{1}{2}(n^2-4)$ for even n, and whose highest term is respectively

$$n\wp(u)^{\frac{n^2-1}{2}}, \qquad -\frac{n}{2}\wp(u)^{\frac{n^2-4}{2}}.$$

This also determines the sign which was still undetermined in (5). With this determination of the signs the initial term in the expansion of $\psi_n(u)$ in ascending powers of u is in both cases (§ 56)

$$\frac{n}{u^{n^2-1}}. \tag{7}$$

Now observe that the poles of the function (1) coincide with the zeros of $\psi_n(u)$, and the zeros of (1) with the zeros of $\psi_{n+1}(u)$; consequently

$$\frac{\psi_n(u)^2[\wp(nu)-\wp(u)]}{\psi_{n+1}(u)\psi_{n-1}(u)} \tag{8}$$

is an everywhere finite doubly periodic function, and hence a constant, whose value at $u=0$ is found to be -1. Thus

$$\wp(nu)=\wp(u)-\frac{\psi_{n+1}(u)\psi_{n-1}(u)}{\psi_n(u)^2}, \tag{9}$$

from which the two following formulae follow:

$$\begin{aligned}\wp(nu)&=\wp(u)-\frac{P_{n+1}P_{n-1}}{\wp'(u)^2P_n^2}, \quad n \text{ even},\\ &=\wp(u)-\frac{\wp'(u)^2P_{n+1}P_{n-1}}{P_n^2}, \quad n \text{ odd}.\end{aligned} \tag{10}$$

We first determine the function P_n in the first cases $n=1,2,3,4$. For this purpose, by the addition theorem of the $\wp$-function [§ 49, (11)], we form

$$4[\wp(u)+\wp(v)+\wp(u+v)]=\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2,$$

and put $u=v$, determining the limit on the right by differentiation:

$$\wp(2u)+2\wp(u)=\frac{1}{4}\left(\frac{\wp''(u)}{\wp'(u)}\right)^2=\frac{\frac{1}{4}(6\wp(u)^2-\frac{1}{2}g_2)^2}{4\wp(u)^3-g_2\wp(u)-g_3},$$

or

$$\wp(2u)=\wp(u)-\frac{3\wp(u)^4-\frac{3}{2}g_2\wp(u)^2-3g_3\wp(u)-\frac{g_2^2}{16}}{\wp'(u)^2}. \tag{11}$$

Therefore

$$\psi_1=1, \qquad \psi_2=-\wp'(u), \qquad \psi_3=P_3,$$

and

$$P_1=1, \qquad P_2=-1, \qquad P_3=3\wp(u)^4-\frac{3}{2}g_2\wp(u)^2-3g_3\wp(u)-\frac{g_2^2}{16}. \tag{12}$$

Furthermore, from the addition formula (§ 49),

$$\wp(v+u)-\wp(v-u)=-\frac{\wp'(u)\wp'(v)}{[\wp(v)-\wp(u)]^2},$$

by putting $v=2u$ we obtain, using (9),

$$\wp(3u)-\wp(u)=-\frac{\wp'(u)\wp'(2u)}{[\wp(2u)-\wp(u)]^2}=-\frac{\wp'(u)^5\wp'(2u)}{\psi_3(u)^2}=-\frac{\psi_2(u)\psi_4(u)}{\psi_3(u)^2},$$

hence

$$\psi_4(u)=\wp'(u)P_4=-\wp'(u)^4\wp'(2u).$$

If now the value $\wp'(2u)$ is formed from (11), one obtains

$$\begin{aligned} P_4 = & -2\wp(u)^6 + \frac{5}{2}g_2\wp(u)^4 + 10g_3\wp(u)^3 + \frac{5}{8}g_2^2\wp(u)^2 \\ & + \frac{1}{2}g_2g_3\wp(u) + g_3^2 - \frac{g_2^3}{32}. \end{aligned} \tag{13}$$

For larger values of n we derive recursion formulae. To this end we apply (9) to two different numbers m, n and obtain

$$\wp(u) - \wp(mu) = \frac{\psi_{m+1}(u)\psi_{m-1}(u)}{\psi_m(u)^2}, \qquad \wp(u) - \wp(nu) = \frac{\psi_{n+1}(u)\psi_{n-1}(u)}{\psi_n(u)^2}.$$

From this one sees that the right hand sides become equal for those values u^0 of u which are different from zero and satisfy

$$mu^0 \equiv \pm nu^0 \pmod{\omega_1, \omega_2},$$

or

$$(m \pm n)u^0 \equiv 0 \pmod{\omega_1, \omega_2}.$$

For the same values u^0 the functions

$$\psi_{m\pm n}(u)$$

also vanish, so that the two functions

$$\psi_{m+1}(u)\psi_{m-1}(u)\psi_n(u)^2 - \psi_{n+1}(u)\psi_{n-1}(u)\psi_m(u)^2, \qquad \psi_{m+n}(u)\psi_{m-n}(u),$$

which by (6) are integral rational functions of $\wp(u)$, vanish for the same finite values of $\wp(u)$. They therefore differ only by a constant factor, which is obtained from (7) as equal to 1. We have, therefore,

$$\psi_{m+n}(u)\psi_{m-n}(u) = \psi_{m+1}(u)\psi_{m-1}(u)\psi_n(u)^2 - \psi_{n+1}(u)\psi_{n-1}(u)\psi_m(u)^2. \tag{14}$$

We apply this formula to two special cases by putting $n+1, n$ or $n+1, n-1$ in place of m, n, and obtain

$$\begin{aligned} \psi_{2n+1}(u) &= \psi_{n+2}(u)\psi_n(u)^3 - \psi_{n+1}(u)^3\psi_{n-1}(u), \\ \wp'(u)\psi_{2n}(u) &= -\psi_n(u)\big[\psi_{n+2}(u)\psi_{n-1}(u)^2 - \psi_{n+1}(u)^2\psi_{n-2}(u)\big]. \end{aligned}$$

Hence, by (6), the recursion formulae for P_n are

$$\begin{aligned} P_{2n+1} &= \wp'(u)^4 P_{n+2}P_n^3 - P_{n+1}^3 P_{n-1}, \quad n \text{ even}, \\ &= P_{n+2}P_n^3 - \wp'(u)^4 P_{n+1}^3 P_{n-1}, \quad n \text{ odd}, \end{aligned} \tag{15}$$

$$P_{2n} = -P_n\big(P_{n+2}P_{n-1}^2 - P_{n+1}^2 P_{n-2}\big). \tag{16}$$

Since all the functions P_n can thus be calculated from P_1, P_2, P_3, P_4, it follows that the coefficients of P_n are composed rationally from g_2, g_3 and rational numbers.

Our further considerations on division, however, will be attached to the multiplication formulae of the functions $\operatorname{sn} u, \operatorname{cn} u, \operatorname{dn} u$, whose advantages will come into full force only in the fourth part.

§ 59. Division by 2.

Turning first to the simplest case, we put v in place of $2v$ in the equations (5) of § 57. If

$$x = \operatorname{sn}\frac{v}{2}, \qquad y = \operatorname{cn}\frac{v}{2}, \qquad z = \operatorname{dn}\frac{v}{2}, \tag{1}$$

then

$$\begin{aligned} \operatorname{sn} v &= \frac{2xyz}{1-\varkappa^2 x^4}, \\ \operatorname{cn} v &= \frac{y^2 - x^2 z^2}{1-\varkappa^2 x^4}, \\ \operatorname{dn} v &= \frac{z^2 - \varkappa^2 x^2 y^2}{1-\varkappa^2 x^4}. \end{aligned} \tag{2}$$

The last two of these equations are quadratic with respect to x^2, but they have only one common root; for the roots of the second equation (2) are

$$\operatorname{sn}^2\frac{v}{2}, \qquad \operatorname{sn}^2\left(\frac{v}{2} + K + iK'\right),$$

and those of the third are

$$\operatorname{sn}^2\frac{v}{2}, \qquad \operatorname{sn}^2\left(\frac{v}{2} + K\right).$$

From (2) one now easily finds

$$1 + \operatorname{cn} v = \frac{2y^2}{1-\varkappa^2 x^4}, \qquad 1 + \operatorname{dn} v = \frac{2z^2}{1-\varkappa^2 x^4},$$

$$1 - \operatorname{cn} v = \frac{2x^2 z^2}{1-\varkappa^2 x^4}, \qquad 1 - \operatorname{dn} v = \frac{2\varkappa^2 x^2 y^2}{1-\varkappa^2 x^4},$$

$$\operatorname{dn} v + \operatorname{cn} v = \frac{2y^2 z^2}{1-\varkappa^2 x^4}.$$

Thus

$$\begin{aligned} x &= \sqrt{\frac{1-\operatorname{cn} v}{1+\operatorname{dn} v}} = \frac{1}{\varkappa}\sqrt{\frac{1-\operatorname{dn} v}{1+\operatorname{cn} v}}, \\ y &= \sqrt{\frac{\operatorname{dn} v + \operatorname{cn} v}{1+\operatorname{dn} v}}, \qquad z = \sqrt{\frac{\operatorname{dn} v + \operatorname{cn} v}{1+\operatorname{cn} v}}. \end{aligned} \tag{3}$$

Between the signs of these three quantities there is still, by the first equation (2), a relation, so that one obtains only four different systems of values, having the following meaning:

$$\begin{array}{ccc} \operatorname{sn}\left(\frac{v}{2}\right), & \operatorname{cn}\left(\frac{v}{2}\right), & \operatorname{dn}\left(\frac{v}{2}\right), \\ \operatorname{sn}\left(\frac{v}{2} + 2K\right), & \operatorname{cn}\left(\frac{v}{2} + 2K\right), & \operatorname{dn}\left(\frac{v}{2} + 2K\right), \\ \operatorname{sn}\left(\frac{v}{2} + 2iK'\right), & \operatorname{cn}\left(\frac{v}{2} + 2iK'\right), & \operatorname{dn}\left(\frac{v}{2} + 2iK'\right), \\ \operatorname{sn}\left(\frac{v}{2} + 2K + 2iK'\right), & \operatorname{cn}\left(\frac{v}{2} + 2K + 2iK'\right), & \operatorname{dn}\left(\frac{v}{2} + 2K + 2iK'\right). \end{array}$$

We also give the special cases following from (3):

$$\operatorname{sn}\frac{K}{2} = \frac{1}{\sqrt{1+\varkappa'}}, \qquad \operatorname{sn}\frac{iK'}{2} = \frac{i}{\sqrt{\varkappa}}, \qquad \operatorname{sn}\frac{K+iK'}{2} = \sqrt{\frac{\varkappa + i\varkappa'}{\varkappa}},$$

$$\operatorname{cn}\frac{K}{2} = \sqrt{\frac{\varkappa'}{1+\varkappa'}}, \qquad \operatorname{cn}\frac{iK'}{2} = \sqrt{\frac{1+\varkappa}{\varkappa}}, \qquad \operatorname{cn}\frac{K+iK'}{2} = \sqrt{\frac{-i\varkappa'}{\varkappa}},$$

$$\operatorname{dn}\frac{K}{2} = \sqrt{\varkappa'}, \qquad \operatorname{dn}\frac{iK'}{2} = \sqrt{1+\varkappa}, \qquad \operatorname{dn}\frac{K+iK'}{2} = \sqrt{\frac{\varkappa}{\varkappa + i\varkappa'}}.$$

It follows now that division by 2, and consequently also by every power of 2, can be carried out through a chain of square roots. If therefore the problem of division by an odd number is assumed solved, then division by an even number has been reduced to square roots. In what follows we shall concern ourselves exclusively with division by odd numbers.

§ 60. Division by an odd number.

If n is an odd number and

$$x = \operatorname{sn}\frac{v}{n}, \qquad y = \operatorname{cn}\frac{v}{n}, \qquad z = \operatorname{dn}\frac{v}{n}, \tag{1}$$

then from equations IV, § 57, we get

$$D(x^2)\operatorname{sn} v - xA(x^2) = 0, \qquad D(x^2)\operatorname{cn} v - yB(x^2) = 0, \qquad D(x^2)\operatorname{dn} v - zC(x^2) = 0. \tag{2}$$

The first of these equations is of degree n^2 with respect to the unknown x; by the second and third, y and z are expressed rationally in x (and in $\operatorname{cn} v, \operatorname{dn} v$). Thus there arise n^2 systems of values for the three unknowns x, y, z, with the meaning

$$\begin{aligned} x_{\mu,\mu'} &= \operatorname{sn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right), \\ y_{\mu,\mu'} &= \operatorname{cn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right), \\ z_{\mu,\mu'} &= \operatorname{dn}\left(\frac{v}{n} + \frac{4\mu K + 4\mu' iK'}{n}\right), \end{aligned} \tag{3}$$

where μ, μ' each run through a complete system of residues modulo n.

The first equation (2),

$$D(x^2)\operatorname{sn} v - xA(x^2) = 0, \tag{4}$$

of degree n^2, is called the general division equation. It is irreducible in the sense that it cannot be decomposed into factors which are rational with respect to $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$ and have arbitrary coefficients independent of v.

For, first, the n^2 values $x_{\mu,\mu'}$ are all different from one another; and if any rational equation

$$F\left(\operatorname{sn}\frac{v}{n}, \operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v\right) = 0$$

holds, then the variable v in it can be replaced by $v + 4\mu K + 4\mu' iK'$. Because of the periodicity of $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$, it follows that

$$F(x, \operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v) = 0$$

is satisfied for all roots of (4). To determine its Galois group, we first fix the field of rationality. It is to comprise the following quantities:

1. rational numbers,
2. rational functions of $\varkappa^2$,
3. the three functions $\operatorname{sn} v, \operatorname{cn} v, \operatorname{dn} v$,
4. the quantities $\operatorname{sn}\left(\frac{4\mu K'}{n} + \frac{4\mu' iK'}{n}\right)$.

Then from (2) it follows that

$$\operatorname{cn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right), \qquad \operatorname{dn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right)$$

also belong to the field of rationality.

We note now that, by the addition theorem, each of the quantities $x_{\mu,\mu'}$ can be expressed rationally by any other among them. If in particular

$$x_{\mu,\mu'} = F_{\mu,\mu'}(x_{0,0}), \tag{5}$$

then

$$x_{\mu+\nu,\mu'+\nu'} = F_{\mu,\mu'}(x_{\nu,\nu'}) = F_{\mu+\nu,\mu'+\nu'}(x_{0,0}). \tag{6}$$

It follows that the Galois group of our equation consists of the n^2 substitutions $S_{\nu,\nu'}$ obtained by increasing the indices μ, μ' in the roots $x_{\mu,\mu'}$ all by the same pair of numbers ν, ν' (where each index is to be taken modulo n).

Indeed, by (5) every rational function of the roots $x_{\mu,\mu'}$ can be expressed rationally by $x_{0,0}$ in the form

$$\Phi(x_{0,0})$$

and hence, by (6), goes under the substitution $S_{\nu,\nu'}$ into $\Phi(x_{\nu,\nu'})$. If it remains unaltered by this substitution, it is therefore rational. Conversely, it follows from irreducibility that every rational equation between the roots, since it can be put into the form

$$\psi(x_{0,0}) = 0,$$

and consequently implies

$$\psi(x_{\nu,\nu'}) = 0,$$

admits the substitution $S_{\nu,\nu'}$. But the group of the $S_{\nu,\nu'}$ is, because of

$$S_{\nu,\nu'}S_{\mu,\mu'} = S_{\nu+\mu,\nu'+\mu'},$$

an Abelian group, which, according to the formula

$$S_{\nu,\nu'} = S_{1,0}^{\nu} S_{0,1}^{\nu'},$$

is represented by the basis $S_{1,0}, S_{0,1}$; hence the general division equation is an Abelian equation and therefore algebraically solvable (Vol. I, § 169 f.).

§ 61. Division of the periods.

The problem still to be solved consists now in determining algebraically the quantities

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right). \tag{1}$$

One obtains all values of this quantity by letting μ, μ', independently of one another, each run through a complete system of residues modulo n. These values, however, are also all different from one another; for $\operatorname{sn} v$ can be equal to $\operatorname{sn} v'$ only when v' is congruent to v, or congruent to $2K - v$, modulo $4K, 2iK'$, and both cannot occur for two different arguments of (1).

The n^2 quantities (1) are the roots of the equation

$$xA(x^2) = 0, \tag{2}$$

and from the formulae

$$y = \frac{D(x^2)}{B(x^2)}, \qquad z = \frac{D(x^2)}{C(x^2)} \tag{3}$$

the quantities

$$y_{\mu,\mu'} = \operatorname{cn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right), \qquad z_{\mu,\mu'} = \operatorname{dn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right)$$

can also be rationally expressed by $x_{\mu,\mu'}$, if the field of rationality consists of rational numbers and rational functions of $\varkappa^2$.

From the roots x^2 of the equation $A = 0$, the roots of $B = 0$, $C = 0$, $D = 0$ can be rationally derived according to § 60, (2), and § 44, (18), (19), (20). Namely, if x^2 is a root of $A = 0$, then

$$\frac{1-x^2}{1-\varkappa^2 x^2}, \qquad \frac{1-\varkappa^2 x^2}{\varkappa^2(1-x^2)}, \qquad \frac{1}{\varkappa^2 x^2}$$

are the roots of $B = 0$, $C = 0$, $D = 0$. The meaning of these latter roots is

$$\operatorname{sn}\left(\frac{(4\mu+1)K + 4\mu' iK'}{n}\right)^2,$$

$$\operatorname{sn}\left(\frac{(4\mu+1)K + (4\mu'+1) iK'}{n}\right)^2,$$

$$\operatorname{sn}\left(\frac{4\mu K + (4\mu'+1) iK'}{n}\right)^2.$$

Thus, by solving the equation $A = 0$, all quantities of the form

$$\operatorname{sn}\left(\frac{\mu K + \mu' iK'}{n}\right)^2,$$

where μ, μ' are arbitrary integers, are rationally determined.

§ 62. Abelian relations.

For a closer investigation of the algebraic nature of the period-division equation, a system of relations between its roots is of great importance; we now turn to its derivation.[11]

We consider the sum

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(u + \frac{2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(u + \frac{2\nu}{n}\right)}, \tag{1}$$

where ν is taken over a complete system of residues for the (odd) modulus n. Here ν' is an arbitrary integer and g_1, g_2 is one of the three even characteristics $(0,0), (0,1), (1,0)$. This sum is independent of the particular residue system chosen for ν.

The common denominator of the fractions occurring in (1),

$$\prod^{\nu} \vartheta_{g_1,g_2}\left(u + \frac{2\nu}{n}\right) = \pm \prod^{\nu} \vartheta_{g_1,g_2}\left(u + \frac{\nu}{n}\right),$$

is, apart from a constant factor, equal by the last formulae of § 33 to

$$\vartheta_{g_1,g_2}(nu, n\omega).$$

Thus, if we put the sum (1) equal to

$$\frac{\Phi(u)}{\vartheta_{g_1,g_2}(nu, n\omega)}, \tag{2}$$

[11] These relations originate with Abel (Oeuvres complètes, vol. I, p. 523; vol. II, p. 251 of the new edition). The proof of these relations given above follows Hermite (Crelle's Journal, vol. 32, p. 283). Also to be mentioned are Sylow, Christiania Videnskabsselskabs Forhandlinger 1864 and 1871; Kronecker, Reports of the Berlin Academy, 19 July 1875; Engel, Reports of the Saxon Society of Sciences, 31 July 1884.

then $\Phi(u)$ is an integral transcendental function of u. Now

$$\frac{\vartheta_{11}\left(u+\frac{2\nu+1}{n}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu+1}{n}\right)} = (-1)^{g_1+1}\frac{\vartheta_{11}\left(u+\frac{2\nu+n+1}{n}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu+n+1}{n}\right)}$$

and, simultaneously with ν, the integer $\nu+\frac{1}{2}(n+1)$ runs through a complete system of residues modulo n. Hence from (1) and (2) there follows

$$\Phi\left(u+\frac{1}{n}\right) = -e^{-\frac{4\nu'\pi i}{n}}\Phi(u), \tag{3}$$

and similarly

$$\Phi(u+\omega) = -e^{-n\pi i(2u+\omega)}\Phi(u). \tag{4}$$

It follows that $\Phi(u)$ is a t-function with periods $1/n, \omega$, determined up to a factor independent of u by the conditions (3), (4), and expressible by a ϑ-function (§ 19). It is easy to see that the conditions (3), (4) are satisfied by the function

$$e^{-4\pi i\nu' u}\vartheta_{11}(nu-4\nu'\omega, n\omega),$$

and hence

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(u+\frac{2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(u+\frac{2\nu}{n}\right)} = Ce^{-4\pi i\nu' u}\frac{\vartheta_{11}(nu-2\nu'\omega, n\omega)}{\vartheta_{g_1,g_2}(nu, n\omega)}. \tag{5}$$

The constant C can be determined easily if one multiplies on the right and on the left by $\vartheta_{g_1,g_2}(u)$ and then puts

$$u = \frac{(1-g_2)+(1-g_1)\omega}{2}.$$

Formula (2) of § 21 then gives

$$C = (-1)^{(g_1+1)(g_2+1)\frac{n-1}{2}}\, n\, \frac{\vartheta_{g_1,g_2}(0,\omega)}{\vartheta_{g_1,g_2}(2\nu'\omega, n\omega)}\frac{\vartheta'_{11}(0,n\omega)}{\vartheta'_{11}(0,\omega)}. \tag{6}$$

The Abelian relations are obtained simply by putting $nu = 2\nu'\omega$ in (5), whereby the right hand side vanishes:

$$\sum^{\nu} e^{\frac{2\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2\nu'\omega+2\nu}{n}\right)}{\vartheta_{g_1,g_2}\left(\frac{2\nu'\omega+2\nu}{n}\right)} = 0. \tag{7}$$

If a linear transformation

$$\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}, \qquad \alpha\delta-\beta\gamma = 1$$

is applied to the ϑ-functions occurring here, then by § 39 one obtains the general formula

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2(\nu\alpha+\nu'\gamma)+2(\nu\beta+\nu'\delta)\omega}{n}\right)}{\vartheta_{g_1,g_2}\left(\frac{2(\nu\alpha+\nu'\gamma)+2(\nu\beta+\nu'\delta)\omega}{n}\right)} = 0. \tag{8}$$

If now $g_1, g_2 = 0, 1$ is put and the notation of § 42 is introduced, these equations pass immediately into relations between the roots $x_{\nu,\nu'}$ (§ 61):

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu\alpha+\nu'\gamma,\nu\beta+\nu'\delta} = 0, \tag{9}$$

and, corresponding to the two special cases

$$\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \qquad \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix},$$

one obtains

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu,\nu'} = 0, \tag{10}$$

$$\sum^{\nu'} e^{-\frac{8\nu'\nu\pi i}{n}} x_{\nu,\nu'} = 0. \tag{11}$$

From (5), (6) one can also determine the value of these sums when the nth root of unity

$$e^{\frac{8\pi i}{n}}$$

is replaced by another; this is done by replacing ν' in (5) by another symbol, m, and then putting $nu = 2\nu'\omega$. We restrict ourselves to the case $(g_1, g_2) = (0, 1)$ and multiply (5) also by $\vartheta_{00} : \vartheta_{10}$; then

$$\sum^{\nu} e^{\frac{8m\nu\pi i}{n}} x_{\nu,\nu'} = (-1)^{\frac{n-1}{2}} n e^{-\frac{4\pi i m\nu'\omega}{n}} \frac{\vartheta_{00}\vartheta_{01}\vartheta_{11}'(0, n\omega)\vartheta_{11}(2(\nu'-m)\omega, n\omega)}{\vartheta_{10}\vartheta_{11}'\vartheta_{01}(2m\omega, n\omega)\vartheta_{01}(2\nu'\omega, n\omega)}, \tag{12}$$

where the left hand side vanishes then, but also only then, when $m \equiv \nu' \pmod{n}$.

§ 63. The Galois group of the division equation.

In the thirteenth section of the first volume it was shown how the algebraic nature of an equation finds its simplest expression in the Galois group of the equation.

Let us briefly recall here the definition of the Galois group.

After the domain of rationality Ω has been fixed, the Galois group of an equation $F(x) = 0$, of which it is only assumed that it has no equal roots, is defined as the group $\mathfrak{G}$ of the permutations among the roots of this equation which has the following double property:

a) Every rational equation in Ω which holds between the roots of $F(x)$ remains true when the roots are subjected to any permutation of this group.

b) Every rational function in Ω of the roots of $F(x)$ which admits all permutations of the group is contained in Ω.

Since we now seek to determine the group $\mathfrak{G}$ of the division equation, we first assume as domain of rationality the totality of all rational numbers and rational functions of $\varkappa^2$.

By the addition theorem and the multiplication theorem (compare § 57, 61), if f and φ_m denote rational functions which are independent of μ, μ', ν, ν', one has

$$x_{\mu+\nu,\mu'+\nu'} = f(x_{\mu,\mu'}, x_{\nu,\nu'}), \tag{1}$$

$$x_{m\mu,m\mu'} = \varphi_m(x_{\mu,\mu'}). \tag{2}$$

Let now S be any substitution of the group $\mathfrak{G}$, by which, if a, b, c, ∂ are integers taken modulo n,

$$x_{1,0},\ x_{0,1} \quad \text{go into} \quad x_{\partial,-c},\ x_{-b,a}.$$

Then we may apply S to each of the formulae (1), (2). But by (2)

$$x_{\mu,0} = \varphi_\mu(x_{1,0}),$$

and from this it follows that under the substitution S

$$x_{\mu,0} \quad \text{goes into} \quad \varphi_\mu(x_{\partial,-c}) = x_{\partial\mu,-c\mu}.$$

Similarly one finds that under S

$$x_{0,\mu'} \quad \text{goes into} \quad \varphi_{\mu'}(x_{-b,a}) = x_{-b\mu',a\mu'}.$$

Applying this to the equation following from (1),

$$x_{\mu,\mu'} = f(x_{\mu,0}, x_{0,\mu'}),$$

one obtains that under S

$$x_{\mu,\mu'} \quad \text{goes into} \quad x_{\partial\mu-b\mu',-c\mu+a\mu'}. \tag{3}$$

From this we first conclude that the numbers a, b, c, ∂ must be so constituted that their determinant

$$m = a\partial - bc \tag{4}$$

has no divisor in common with n. For if such a divisor were present, then μ, μ' could be so chosen, without both being divisible by n, that

$$\partial\mu - b\mu' \equiv 0, \qquad -c\mu + a\mu' \equiv 0 \pmod{n},$$

and several distinct roots $x_{\mu,\mu'}$ would pass under S into one and the same root, which is impossible. For first, neither a and b, nor c and ∂, can have a divisor in common with n, since otherwise, by taking $a\mu', b\mu'$, or $c\mu, \partial\mu$, divisible by n, all $x_{\mu',0}$ or all $x_{0,\mu}$ would pass into $x_{0,0}$. If one then puts

$$m\mu = n(ha + h'b), \qquad m\mu' = n(hc + h'\partial),$$

and determines h and h' so that $ha + h'b$ has no divisor in common with n, then, when n and m have a common divisor, μ, μ' need not both be divisible by n, and not only $x_{0,0}$ but also $x_{\mu,\mu'}$ passes, by (3), into $x_{0,0}$.

If we denote the interchange (3), in abbreviated form, by

$$(x_{\mu,\mu'}, x_{\nu,\nu'}),$$

then

$$\nu \equiv \partial\mu - b\mu', \qquad \nu' \equiv -c\mu + a\mu' \pmod{n},$$

and from this, by solving, using the notation of § 28, (4),

$$m(\mu, \mu') \equiv \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} (\nu, \nu'). \tag{5}$$

It is therefore

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} \tag{6}$$

a convenient sign for the interchange S, and from (5) one sees that two such interchanges

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}, \qquad S' = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix}$$

are composed exactly according to the rule given in § 28:

$$SS' = S'' = \begin{pmatrix} aa' + bc' & ab' + b\partial' \\ ca' + \partial c' & cb' + \partial\partial' \end{pmatrix}, \tag{7}$$

so that the interchange S'' of the roots arises when first the interchange S, and then the interchange S', is performed among the roots of the division equation. It must always be observed, however, that here only the residues modulo n of the numbers a, b, c, ∂ come into account, so that the number of the interchanges S is always finite.[12]

[12] If, as at first sight might seem more natural, one chose the notation $S = \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix}$, the composition would have to be denoted by the formula $S'S = S''$, hence conversely to the usual composition of permutations.

1. The totality of all substitutions $\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$ forms a group, which we denote by $\mathfrak{A}$, and in it, as we have proved, the group $\mathfrak{G}$ of the division equation is contained.

In $\mathfrak{A}$ there is contained, as a divisor, a group $\mathfrak{B}$, consisting of all substitutions

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{8}$$

which satisfy the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n}. \tag{9}$$

Every substitution S can be derived by composition of

$$\begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix}$$

with a substitution T; for, in order that

$$S = \begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix} T,$$

$\alpha, \beta, \gamma, \delta$ are simply to be determined from the congruences

$$a \equiv \alpha m, quad b \equiv \beta m, \qquad c \equiv \gamma, \qquad \partial \equiv \delta \pmod{n}.$$

We now consider

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' \mathrm{i} K'}{n}\right) = \frac{\vartheta_{00}\,\vartheta_{11}\left(\frac{2\mu+2\mu'\omega}{n}\right)}{\vartheta_{10}\,\vartheta_{01}\left(\frac{2\mu+2\mu'\omega}{n}\right)}, \qquad \varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{10}$$

as functions of ω, and apply to them a linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{8}$$

by replacing ω by

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{11}$$

Then from the formulae § 39, (3), (5), and (6), it follows that

$$x_{\mu,\mu'} \quad \text{goes into} \quad x_{\alpha\mu+\gamma\mu',\beta\mu+\delta\mu'},$$

that is, the $x_{\mu,\mu'}$ undergo a substitution which, by (3), (6), is to be denoted by

$$T = \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix},$$

while $\varkappa^2$ remains unchanged.

In this way, conversely, every one of the substitutions T can arise; for if any substitution T is given, then by adding suitable multiples of the odd number n to $\alpha, \beta, \gamma, \delta$ one can always satisfy the conditions

$$\alpha \equiv 1, \qquad \delta \equiv 1, \qquad \beta \equiv 0, \qquad \gamma \equiv 0 \pmod{8}.$$

Any rational equation between the roots $x_{\mu,\mu'}$ and $\varkappa^2$, even if arbitrary constant, that is, ω-independent, numerical coefficients occur in it, passes, by (10), into an identity; and one may therefore substitute for ω the expression

$$\frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

that is, one may apply every substitution T to the rational equation between the $x_{\mu,\mu'}$. Hence it follows that the whole group $\mathfrak{B}$ is contained in $\mathfrak{G}$ (Vol. I, § 156), and even in the group $\mathfrak{G}'$ which arises from the group $\mathfrak{G}$ of the division equation by adjunction of arbitrary numbers.

If we enlarge the domain of rationality of the rational numbers by adjoining the nth roots of unity, then the Abelian relations of the preceding paragraph also belong to the rational equations. But to these relations, for instance to

$$\sum^{\nu} e^{\frac{8\nu'\nu\pi i}{n}} x_{\nu,\nu'} = 0,$$

none of the substitutions

$$M = \begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix}$$

in which m differs from 1 is applicable. For by this substitution, if $mm' \equiv 1 \pmod{n}$, there passes

$$\sum^{\nu} e^{\frac{8\nu'\nu\pi i}{n}} x_{\nu,\nu'} \quad \text{into} \quad \sum^{\nu} e^{\frac{8m'\nu'\nu\pi i}{n}} x_{\nu,\nu'},$$

which by § 62, (12), is different from zero if m' and therefore also m is not congruent to 1 modulo n. Thus the following theorem results:

2. After adjunction of the nth roots of unity (and of arbitrary other constants), $\mathfrak{B}$ is the Galois group of the division equation. It will also be called the monodromy group of the division equation.

3. We shall now still prove that, in the original domain of rationality, hence without adjunction of the nth roots of unity or of other irrational numbers, the group of the division equation is identical with the group $\mathfrak{A}$.

It has been shown in Vol. I, § 174, that the primitive nth roots of unity ρ are roots of an irreducible equation

$$\Phi(\rho) = 0, \tag{12}$$

and that this equation also cannot decompose when the independent variable $\varkappa^2$ is adjoined to the domain of rationality of the rational numbers.

The degree of this equation is $\varphi(n)$, that is, equal to the number of residues modulo n which are relatively prime to n.

Let now

$$F(r) = 0 \tag{13}$$

be a Galois resolvent of degree μ of the division equation in the original domain of rationality (so that all roots $x_{\nu,\nu'}$ are rationally representable by r, Vol. I, § 152). This equation must decompose after adjunction of a root of (12); for if this were not so, every rational relation

$$\psi(r, \rho) = 0 \tag{14}$$

would be satisfied for all roots of (13), and $\psi(t, \rho)$ would be divisible by $F(t)$, for a variable t. The same would still hold if ρ were replaced by any other root ρ^m of (12). This, however, is not the case by § 62, (12), if in place of (14) one puts one of the Abelian relations.

Let now, after adjunction of ρ,

$$F(r, \rho) = 0 \tag{15}$$

be the Galois resolvent of degree ν of the division equation, so that ν is, by 2., equal to the degree of the group $\mathfrak{B}$. Then $F(t)$ is algebraically divisible by $F(t, \rho)$, and therefore, by the irreducibility of (12), also by each of the functions $F(t, \rho^m)$. Since the functions $F(t, \rho^m)$ are irreducible, two of them can have a common divisor only when they are entirely identical. But if

$$F(t, \rho) = F(t, \rho^m)$$

were true, then it would follow from every equation of the form (14) that $\psi(t, \rho)$ is divisible by $F(t, \rho) = F(t, \rho^m)$, and it would follow that in (14) the interchange (ρ, ρ^m) is admissible, which, again by the Abelian relations, is not the case. Consequently the $\varphi(n)$ functions $F(t, \rho^m)$ are all different from one another, and $F(t)$ is divisible by their product. Thus the degree μ of $F(t)$ is at least equal to $\nu\varphi(n)$. But it also cannot be larger than $\nu\varphi(n)$, since $\nu\varphi(n)$ is the degree of the group $\mathfrak{A}$, and since the group $\mathfrak{G}$, of degree μ, is surely contained in $\mathfrak{A}$. It follows from this that

$$\mu = \nu\varphi(n), \tag{16}$$

and at the same time that $\mathfrak{G}$ is identical with $\mathfrak{A}$.

But from this it also follows that $F(t)$ is equal to the product of all factors $F(t, \rho^m)$, hence

$$F(t) = \prod^{m} F(t, \rho^m), \tag{17}$$

and, since $F(t) = 0$ has no equal roots, that $F(r, \rho)$ indeed vanishes, but none of the other factors $F(r, \rho^m)$. The equations

$$F(r, t) = 0, \qquad \Phi(t) = 0 \tag{18}$$

therefore have only the one root $t = \rho$ in common, and by seeking their greatest common divisor one finds ρ rationally expressed by r, that is, by the roots of the division equation.

These expressions for ρ do not change their value when a substitution of the group $\mathfrak{B}$ is applied to r, whereas, by (12), ρ passes by a substitution of $\mathfrak{A}$ into a power of ρ.

The Abelian relations show that under the substitution contained in $\mathfrak{A}$

$$\begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix} \tag{19}$$

ρ goes into ρ^m. For if we put

$$e^{\frac{8\pi i}{n}} = \rho^\lambda,$$

one of the Abelian relations is

$$\sum^{\nu} \rho^{\lambda\nu\nu'} x_{\nu,\nu'} = 0, \tag{20}$$

and, when the expression through r is put for ρ, all substitutions of $\mathfrak{A}$, hence also (19), are applicable. By § 62, (12), however, (20) remains correct under this substitution only when ρ passes into ρ^m.

Since ρ is not changed by the substitutions in $\mathfrak{B}$, and since the group $\mathfrak{A}$ arises from $\mathfrak{B}$ by composition with (19), it follows that under any substitution in $\mathfrak{A}$

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$$

ρ passes into ρ^m, where m is congruent to the determinant $(a\partial - bc)$ modulo n.

4. We finally determine still the number ν, that is, the degree of the group $\mathfrak{B}$.

As is known, $\varphi(n)$ has the expression (Vol. I, § 140)

$$\varphi(n) = n \prod \left(1 - \frac{1}{p}\right), \tag{21}$$

where the product sign extends over all distinct prime numbers p contained in n.

The number ν is equal to the number of systems of numbers $\alpha, \beta, \gamma, \delta$, not congruent modulo n, which satisfy the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n}. \tag{22}$$

We ask first for the number of pairs α, β which have no common divisor with n, and denote it by $\chi(n)$.

If $n = n'n''$ and n' is relatively prime to n'', then from every combination of a number-pair α', β' belonging to n' with a number-pair α'', β'' belonging to n'' one can derive a pair belonging to n by

$$\alpha = n''\alpha' + n'\alpha'', \qquad \beta = n''\beta' + n'\beta'',$$

and in this way one obtains all number-pairs α, β, and each only once. Hence

$$\chi(n) = \chi(n')\chi(n''). \tag{23}$$

It remains to determine $\chi(p^\pi)$, that is, $\chi(n)$ for a prime-power $n = p^\pi$.

If one first sets for α, β all numbers distinct modulo p^π, then their number is $p^{2\pi}$. But one must subtract all those pairs in which both α and β are divisible by p; their number is $p^{2\pi-2}$. Thus

$$\chi(p^\pi) = p^{2\pi} - p^{2\pi-2},$$

and consequently, by (23), in general

$$\chi(n) = n^2 \prod \left(1 - \frac{1}{p^2}\right), \tag{24}$$

where p runs through the prime numbers contained in n.

Each number-pair α, β can be transformed, by increasing by multiples of n, into one whose numbers are relatively prime among themselves, and then γ, δ can be determined so that

$$\alpha\delta - \beta\gamma = 1.$$

Herein γ, δ can be replaced by $\gamma + h\alpha, \delta + h\beta$, and letting h run through a complete residue system modulo n shows that to each number-pair α, β there belong n number-pairs γ, δ satisfying (22). Consequently the order of the group $\mathfrak{B}$ is

$$\nu = n^3 \prod \left(1 - \frac{1}{p^2}\right),$$

or, if we introduce also the numerical function

$$\psi(n) = n \prod \left(1 + \frac{1}{p}\right), \tag{25}$$

then

$$\nu = n\varphi(n)\psi(n), \tag{26}$$

and the order of the group $\mathfrak{A}$ is

$$\mu = n\varphi(n)^2\psi(n). \tag{27}$$

§ 64. The irreducible factors of the division equation.

If

$$\nu = \partial\mu - b\mu', \qquad \nu' = -c\mu + a\mu',$$

and $a\partial - bc$ is relatively prime to n, then the greatest common divisor of μ, μ', n is at the same time the greatest common divisor of ν, ν', n. Thus the roots $x_{\mu,\mu'}$ of the division equation decompose, according to the greatest common divisor of μ, μ', n, into systems which can only ever pass into one another by the substitutions of the group $\mathfrak{A}$; that is, the group $\mathfrak{G}$ is intransitive, and the systems of intransitivity are the systems $x_{\mu,\mu'}$ in which μ, μ', n have one and the same greatest common divisor. The division equation is therefore reducible (except when n is a prime number), and the roots of one of these systems of imprimitivity satisfy a rational equation (Vol. I, § 157).

By the preceding paragraph there are $\varphi(n)\psi(n)$ number-pairs μ, μ' whose greatest common divisor with n is equal to 1, and the roots $x_{\mu,\mu'}$ corresponding to these number-pairs therefore satisfy a rational equation of degree $\varphi(n)\psi(n)$. We call this equation the proper division equation for the divisor n, because only when μ, μ', n have no common divisor is $x_{\mu,\mu'}$ not at the same time a root of a division equation for a smaller divisor. In contrast to this, we call the equation whose roots are all of the $x_{\mu,\mu'}$ the general division equation for the divisor n.

Under any substitution of the group $\mathfrak{A}$

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix},$$

$(1,0), (0,1)$ pass into $(\partial, -c), (-b, a)$; and if S is not the identical substitution, then certainly at least one of the two roots $x_{1,0}, x_{0,1}$ is changed by S. It follows from this that the Galois group of the proper division equation is exactly the same as that of the general one, only applied to the case in which μ, μ', n have no common divisor, namely, according as the nth roots of unity are adjoined or not, $\mathfrak{B}$ or $\mathfrak{A}$.

From this it follows further that the proper division equation is irreducible, even after adjunction of arbitrary constants.

For if γ, δ are any two numbers having no common divisor with n, then one can determine α, β according to the congruence

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n},$$

and apply the substitution contained in $\mathfrak{B}$

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

to every rational equation between the roots $x_{\mu,\mu'}$. Thus if $x_{1,0}$ satisfies a rational equation, with arbitrary constant coefficients,

$$\Phi(x_{1,0}) = 0,$$

then every other root $x_{\delta,-\gamma}$ of the proper division equation satisfies the same equation, and from this the irreducibility of the latter follows.

§ 65. Reduction of the division equation to transformation equations.

The roots of the proper division equation can be arranged in rows in the following manner. Choose arbitrarily one of the roots

$$x_{\mu_1,\mu_1'} = \operatorname{sn}\left(\frac{4\mu_1 K + 4\mu_1' \mathrm{i} K'}{n}\right) = \operatorname{sn}\Omega_1.$$

Among the roots of the division equation there also occur the $\varphi(n)$ quantities

$$(R_1) \qquad \operatorname{sn} h\Omega_1,$$

which, when h runs through a complete system of incongruent numbers prime to n, are all different from one another. We shall call the system (R_1) the first row of the roots.

If now $\operatorname{sn}\Omega_2$ is a root not contained in (R_1), then the $\varphi(n)$ quantities

$$(R_2) \qquad \operatorname{sn} h\Omega_2,$$

which are different both among themselves and from the roots (R_1), form a second row; and in this way one can continue until all $\varphi(n)\psi(n)$ roots of the proper division equation have been distributed into $\psi(n)$ rows of $\varphi(n)$ terms each.

By the multiplication theorem every other root of the same row can be rationally expressed through one root in the form

$$\operatorname{sn} h\Omega = f_h(\operatorname{sn}\Omega), \tag{1}$$

where f_h is a rational function depending only on the multiplier h, and not on the choice of Ω.

If the two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ belong to the same row, then h must be choosable so that

$$h\mu \equiv \nu, \qquad h\mu' \equiv \nu' \pmod{n}, \tag{2}$$

and conversely, if this is the case, the two roots belong to the same row. But from (2) the congruence

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{n} \tag{3}$$

follows, and conversely the congruences (2) can again be derived from this.

For since μ, μ', n are relatively prime, one can determine α, β so that

$$\alpha\mu - \beta\mu' \equiv 1 \pmod{n},$$

and from this, with the help of (3), it follows that

$$\nu \equiv (\alpha\nu - \beta\nu')\mu, \qquad \nu' \equiv (\alpha\nu - \beta\nu')\mu' \pmod{n}.$$

Thus, if $\alpha\nu - \beta\nu' = h$ is put, the congruences (2) result.

The congruence (3) is therefore the necessary and sufficient condition that the two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ of the proper division equation belong to the same row.

It follows from this that the division into rows is entirely independent of the arbitrariness in the assumption about the roots $\operatorname{sn}\Omega_1, \operatorname{sn}\Omega_2, 7\ldots$.

But still more follows from this, namely that the substitutions of the group $\mathfrak{A}$ do not tear the rows apart, but only interchange them among one another. The group of the division equation is therefore imprimitive, and the individual rows are the systems of imprimitivity (Vol. I, § 158).

For if one applies simultaneously to (μ, μ') and (ν, ν') the substitution

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix},$$

then the congruence (3) remains preserved.

If one selects, among the substitutions of the group $\mathfrak{A}$, those substitutions which only interchange among themselves the roots of a row, one obtains a group, and indeed a divisor of $\mathfrak{A}$. Thus to every row there belongs such a divisor of $\mathfrak{A}$. Let $R_{\mu,\mu'}$ denote the row in which the root $x_{\mu,\mu'}$ occurs, and let $\mathfrak{A}_{\mu,\mu'}$ denote the divisor of $\mathfrak{A}$ belonging to this row. Then, by (3) [compare § 63, (3)], the substitutions

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$$

of $\mathfrak{A}_{\mu,\mu'}$ are characterized by the congruence

$$(\partial\mu - b\mu')\mu' + (c\mu - a\mu')\mu \equiv 0 \pmod{n}. \tag{4}$$

Similarly one obtains a group $\mathfrak{B}_{\mu,\mu'}$ if one seeks the substitutions in $\mathfrak{B}$ which only interchange among themselves the roots of the row $R_{\mu,\mu'}$; their substitutions

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

are characterized by the two congruences

$$\begin{matrix} (\delta\mu - \beta\mu')\mu' + (\gamma\mu - \alpha\mu')\mu \equiv 0, \\ \alpha\delta - \beta\gamma \equiv 1 \end{matrix} \quad \pmod{n}. \tag{5}$$

For example, the groups $\mathfrak{A}_{1,0}$, $\mathfrak{B}_{1,0}$ consist of the substitutions

$$\begin{pmatrix} a & b \\ 0 & \partial \end{pmatrix}, \qquad \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \qquad \alpha\delta \equiv 1 \pmod{n}.$$

The groups $\mathfrak{A}_{\mu,\mu'}$ are conjugate divisors of $\mathfrak{A}$ (Vol. II, § 3), for if $x_{1,0}$ passes under S into $x_{\mu,\mu'}$, and hence $R_{1,0}$ into $R_{\mu,\mu'}$, then under the group

$$S^{-1}\mathfrak{A}_{1,0}S$$

the roots of $R_{\mu,\mu'}$ are interchanged only among themselves, and hence

$$S^{-1}\mathfrak{A}_{1,0}S = \mathfrak{A}_{\mu,\mu'}. \tag{6}$$

Likewise, if T is a substitution in $\mathfrak{B}$ under which $x_{1,0}$ passes into $x_{\mu,\mu'}$, then

$$T^{-1}\mathfrak{B}_{1,0}T = \mathfrak{B}_{\mu,\mu'}. \tag{7}$$

The greatest common divisor $\mathfrak{A}_0$ of all the groups $\mathfrak{A}_{\mu,\mu'}$ is determined by (4) as

$$b \equiv 0, \qquad c \equiv 0, \qquad a \equiv \partial \pmod{n},$$

and therefore consists of the substitutions

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix},$$

where a is any number prime to n. $\mathfrak{A}_0$ is identical with the greatest common divisor of three of the groups $\mathfrak{A}_{\mu,\mu'}$ which belong to different rows. For $\mathfrak{A}_0$ is the greatest common divisor of $\mathfrak{A}_{\mu,\mu'}, \mathfrak{A}_{\nu,\nu'}, \mathfrak{A}_{\rho,\rho'}$ when the three congruences

$$\begin{aligned} c\mu^2 - (\partial - a)\mu\mu' - b\mu'^2 &\equiv 0, \\ c\nu^2 - (\partial - a)\nu\nu' - b\nu'^2 &\equiv 0, \quad \pmod{n} \\ c\rho^2 - (\partial - a)\rho\rho' - b\rho'^2 &\equiv 0 \end{aligned}$$

are fulfilled only under the assumption

$$\partial \equiv a, \qquad b \equiv 0, \qquad c \equiv 0 \pmod{n}.$$

This occurs when the determinant

$$\begin{vmatrix} \mu^2 & \mu\mu' & \mu'^2 \\ \nu^2 & \nu\nu' & \nu'^2 \\ \rho^2 & \rho\rho' & \rho'^2 \end{vmatrix} = -(\nu\rho' - \rho\nu')(\rho\mu' - \mu\rho')(\mu\nu' - \nu\mu')$$

is relatively prime to n.

Similarly, by (5), the greatest common divisor $\mathfrak{B}_0$ of all $\mathfrak{B}_{\mu,\mu'}$ is the totality of the substitutions

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \tag{8}$$

with the condition

$$\alpha^2 \equiv 1 \pmod{n}. \tag{9}$$

The congruence (9) has, if k is the number of distinct prime numbers contained in n, 2^k incongruent solutions, and this is therefore the degree of the group $\mathfrak{B}_0$.

A rational function ξ of the roots of one row, say the row $R_{1,0}$, can by (1) be represented rationally by one of these roots. If this function has the property of remaining unchanged when this one root is replaced by another of the same row, for example if ξ is a symmetric function of the roots of one row, then ξ obtains, under application of the substitutions of $\mathfrak{A}$ (or also of $\mathfrak{B}$), only

$$\nu = \psi(n) \tag{10}$$

different values

$$\xi_1, \xi_2, \ldots, \xi_\nu; \tag{11}$$

and if these values are all different from one another, then ξ belongs to the group $\mathfrak{A}_{1,0}$, and all other symmetric functions of the roots of $R_{1,0}$ are rationally representable by ξ (Vol. I, § 162).

The ν quantities (11) are the roots of an irreducible rational equation of the νth degree, which we call a transformation equation.

If only rational numerical coefficients occur in ξ, the transformation equation itself has rational numerical coefficients. It remains irreducible, however, even when nth roots of unity, or indeed any constants whatever, are adjoined; this follows from the fact that by substitutions of the group $\mathfrak{B}$ every root of the division equation can be carried into every other, and hence every row into every other row.

By adjunction of one root ξ of the transformation equation, say the one belonging to the group $\mathfrak{A}_{1,0}$, the group of the division equation is reduced to $\mathfrak{A}_{1,0}$. This latter group, however, is no longer transitive, but interchanges among themselves the $\varphi(n)$ roots $x_{h,0}$, where h runs through a complete system of incongruent numbers prime to n. The division equation therefore becomes reducible and has a factor of degree $\varphi(n)$, whose roots are the $x_{h,0}$. The influence of a substitution of the group $\mathfrak{A}_{1,0}$,

$$\begin{pmatrix} a & b \\ 0 & \partial \end{pmatrix},$$

on $x_{h,0}$ consists in $x_{h,0}$ passing into $x_{ah,0}$; and the group of this equation of degree $\varphi(n)$ therefore consists of the interchanges

$$(x_{h,0}, x_{ah,0}),$$

where a may be any number prime to n. This group is Abelian, and therefore the roots $x_{h,0}$ are algebraically determinable by radicals after adjunction of ξ.

If, however, we adjoin not merely one, but all roots ξ of the transformation equation, or, what amounts to the same thing, the three belonging to $R_{1,0}, R_{0,1}, R_{1,1}$, then the group of the division equation is reduced to $\mathfrak{A}_0$; and if we still adjoin nth roots of unity, to $\mathfrak{B}_0$. The group $\mathfrak{B}_0$ is an Abelian group of degree 2^k, containing only elements whose degree is 2. Consequently the division equation is solvable by square roots.

In order to find the form of this solution, put

$$n = p^\pi p'^{\pi'} \cdots,$$

where $p, p', \ldots$ are distinct prime numbers and $\pi, \pi', \ldots$ positive exponents. Determine the numbers $c, c', \ldots$ from the congruences

$$\begin{array}{llll} c \equiv -1 & (\mathrm{mod}\ p^\pi), & c' \equiv -1 & (\mathrm{mod}\ p'^{\pi'}), \ldots, \\ c \equiv +1 & (\mathrm{mod}\ np^{-\pi}), & c' \equiv +1 & (\mathrm{mod}\ np'^{-\pi'}), \ldots. \end{array}$$

Then every solution α of the congruence

$$\alpha^2 \equiv 1 \pmod n$$

is obtained, and each only once, in the form

$$\alpha \equiv c^\varepsilon c'^{\varepsilon'} \cdots \pmod n, \tag{12}$$

where $\varepsilon, \varepsilon', \ldots$ take the values $0, 1$.

If now Ω is any one of the values

$$\frac{4\mu K + 4\mu' \mathrm{i} K'}{n},$$

so that $\operatorname{sn}\Omega$ is any root of the proper division equation, then the sum, extended over all α,

$$\sum (\pm 1)^{\varepsilon} (\pm 1)^{\varepsilon'} \cdots \operatorname{sn} \alpha\Omega = \psi(\Omega), \tag{13}$$

in which the k signs ± 1 can be chosen arbitrarily, has the property that for every α determined by (12),

$$\psi(\alpha\Omega) = (\pm 1)^{\varepsilon} (\pm 1)^{\varepsilon'} \cdots \psi(\Omega), \tag{14}$$

and the square of $\psi(\Omega)$ remains unchanged by the substitutions of the group $\mathfrak{B}_0$, hence is rationally expressible by nth roots of unity and the roots of a transformation equation. Thus $\psi(\Omega)$ is the square root $\sqrt{A}$ of such an expression A. But the number of the expressions (13) is 2^k, and by adding all the equations so formed there results

$$2^k \operatorname{sn}\Omega = \sum \sqrt{A}. \tag{15}$$

It remains to be observed that one half of the quantities ψ, and hence also of the A, vanish. For

$$-1 \equiv cc' \cdots \pmod{n},$$

and if one therefore puts $\alpha = -1$ in (14), it follows that

$$\psi(-\Omega) = (\pm 1)(\pm 1) \cdots \psi(\Omega),$$

whereas on the other hand

$$\psi(-\Omega) = -\psi(\Omega).$$

Consequently $\psi(\Omega)$ always vanishes when among the k quantities ± 1 occurring in (14) an even number of negative units is contained.

The roots $x_{\mu,\mu'}$ can therefore be expressed linearly by 2^{k-1} square roots.

If n is a prime number or a power of a prime number, then the squares of the roots of the division equation are rationally expressible by the roots of the transformation equation.[13]

If n is a composite number, then the division of the roots $x_{\mu,\mu'}$ into rows can be carried still further. If p is a prime number contained in n, then one takes two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ into the same row or into different rows according as

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{p}$$

or not. If, according to this, $x_{\nu,\nu'}$ and $x_{\nu_1,\nu_1'}$ belong to one row with $x_{\mu,\mu'}$, then they also belong among themselves to the same row. The number of rows so formed is, as is easily shown, $p+1$, and in this way one obtains, as the first resolvent of the division equation, a transformation equation belonging to the divisor p. We shall later show by another path how the transformation equations for composite divisors can be reduced to those for prime divisors.

Seventh Section.

Theory of the Transformation Equations.

[13] Compare, on this theorem, Kronecker, Monatsbericht der Berliner Akademie, 19 July 1875. Kronecker there points out an oversight which occurs in a paper of Jacobi concerning this subject (Crelle, vol. 47 and vol. 50).

§ 66. Formation of transformation equations.

After the division equations have now been reduced to transformation equations, we pass to a more exact study of these latter equations.

We assume n odd, understand by p the prime numbers which divide n, put

$$\nu = \psi(n) = n \prod \left(1 + \frac{1}{p}\right), \tag{1}$$

and denote the ν rows of roots of the division equation by $R_1, R_2, \ldots, R_\nu$.

From each of these rows, for

$$\Omega_{\mu,\mu'} = \frac{4\mu K + 4\mu' \mathrm{i} K'}{n}, \tag{2}$$

we take one representative $\Omega_1, \Omega_2, \ldots, \Omega_\nu$; and we obtain the $\varphi(n)$ roots of one row when in

$$\operatorname{sn} m\Omega$$

m runs through a complete system of mutually incongruent numbers relatively prime to n.

The simplest expressions which can be introduced as roots of transformation equations are the products

$$\prod_{h=1}^{n-1} \Phi(\operatorname{sn} h\Omega), \tag{3}$$

where Φ is an arbitrary rational function and h runs through the sequence of numbers $1, 2, \ldots, n-1$.

Every such function is rationally expressible through $\operatorname{sn}\Omega$ and plainly remains unchanged if Ω is replaced by any $m\Omega$; thus it remains only to arrange that the ν values of (3), corresponding to the ν rows, are distinct from one another in order to make (3) the root of an irreducible transformation equation.[14]

We assume that the function $\Phi(x)$ is either an even or an odd function of x, and accordingly make the following distinction.

If $\Phi(x)$ is an even function, then among the factors of the product (3) two at a time, namely

$$\Phi(\operatorname{sn} h\Omega), \qquad \Phi[\operatorname{sn}(n-h)\Omega], \tag{4}$$

are equal to one another. We therefore put

$$\Pi(\Omega) = \prod_{h=1}^{(n-1)/2} \Phi(\operatorname{sn} h\Omega). \tag{5}$$

Then, if m is any number relatively prime to n,

$$\Pi(m\Omega) = \Pi(\Omega) \tag{6}$$

for, among the $\frac{1}{2}(n-1)$ numbers hm, no two have a sum or difference divisible by n; hence, apart from sign, the $\operatorname{sn} h\Omega$ are the same as the $\operatorname{sn} hm\Omega$. Thus $\Pi(\Omega)$ is the root of a transformation equation. This class of transformation equations we call *modular equations*.

If $\Phi(x)$ is an odd function, then the two quantities (4) are opposite, and (6) is no longer generally correct; rather one has

$$\Pi(m\Omega) = \pm\Pi(\Omega). \tag{7}$$

Accordingly, at least in general, not $\Pi(\Omega)$ but only $\Pi(\Omega)^2$ is the root of a transformation equation. This kind of transformation equation we call a *multiplier equation*. In order to recognize the cases in which $\Pi(\Omega)$ itself is a root of a transformation equation, the sign in (7) must be determined.

[14] Expressions such as (3) are still roots of transformation equations when h takes only values relatively prime to n. Such transformation equations have hitherto been little used. We shall later become acquainted with an example.

This succeeds on the basis of a theorem of number theory derived in Vol. I, § 145.

The theorem is as follows.

If m, n are any two relatively prime numbers, the latter odd, and if μ denotes the number among

$$m,\ 2m,\ 3m, \dots, \frac{n-1}{2}m$$

whose absolutely least residues modulo n are negative, then

$$(-1)^{\mu} = \left(\frac{m}{n}\right), \tag{8}$$

where $\left(\frac{m}{n}\right)$ is the Legendre-Jacobi symbol from the theory of quadratic residues.[15]

This theorem now leads directly to the determination of the sign in (7). For if in (5) the function $\Phi(x)$ is odd, then in passing from Ω to $m\Omega$ exactly μ factors in (5) change sign, and we conclude

$$\Pi(m\Omega) = \left(\frac{m}{n}\right)\Pi(\Omega). \tag{9}$$

If n is not a square, m can always be chosen so that $\left(\frac{m}{n}\right) = -1$. For, if $n = p^{\lambda}n'$, λ odd, n' not divisible by p, and β a quadratic non-residue of p, it is enough to determine m from the congruences

$$m \equiv \beta \pmod{p^{\lambda}}, \qquad m \equiv 1 \pmod{n'}$$

to find such a number m. Then $\Pi(m\Omega) = -\Pi(\Omega)$, and not $\Pi(\Omega)$ but only $\Pi(\Omega)^2$ is a root of a transformation equation. If, however, n is a square, then $\Pi(m\Omega) = \Pi(\Omega)$ for every m, and therefore $\Pi(\Omega)$ itself is a root of a transformation equation. Thus the following theorem results:

For an odd function Φ, $\Pi(\Omega)^2$, and only when n is a square $\Pi(\Omega)$ itself, is the root of a transformation equation.[16]

§ 67. Special transformation equations.

The roots of the transformation equations shall now be represented by ϑ-functions. For this purpose we put

$$\Omega = \frac{4\mu K + 4\mu' \mathrm{i}K'}{n}, \qquad \mathrm{i}K' = K\omega, \qquad \varpi = \frac{\mu + \mu'\omega}{n}, \qquad \Omega = 4K\varpi, \tag{1}$$

and consider the products

$$\prod_{h=1}^{(n-1)/2} \vartheta_{00}(2h\varpi), \qquad \prod_{h=1}^{(n-1)/2} \vartheta_{01}(2h\varpi), \qquad \prod_{h=1}^{(n-1)/2} \vartheta_{10}(2h\varpi), \qquad \prod_{h=1}^{(n-1)/2} \vartheta_{11}(2h\varpi).$$

The following notation is expedient:

$$\begin{aligned}
P_{00}\vartheta_{00}^{(n-1)/2} &= e^{\frac{\pi \mathrm{i}}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{h=1}^{(n-1)/2} \vartheta_{00}(2h\varpi),\\
P_{10}\vartheta_{10}^{(n-1)/2} &= -e^{\frac{\pi \mathrm{i}}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{h=1}^{(n-1)/2} \vartheta_{10}(2h\varpi),\\
P_{01}\vartheta_{01}^{(n-1)/2} &= -e^{\frac{\pi \mathrm{i}}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}} \prod_{h=1}^{(n-1)/2} \vartheta_{01}(2h\varpi).
\end{aligned} \tag{2}$$

[15] Compare on this theorem Schering and Kronecker in the Monthly Report of the Berlin Academy of June 22, 1876; Schering, *Acta Mathematica* 1.

[16] This simplification of the multiplier equation in the case where n is a square was discovered by Joubert, but proved by him in a way entirely different from ours. *Sur les equations, qui se rencontrent dans la theorie de la transformation des fonctions elliptiques*, Paris, 1876.

Formula § 39, (9), is now to be applied, but in a form somewhat different for the present purpose.

If in § 39, (8), one puts $v' = 2h/n$, $v = 2h(\alpha+\beta\omega)/n$, and again takes the product over $h = 1, 2, \dots, (n-1)/2$, then, using § 32, (24), one obtains

$$2^{(n-1)/2}\prod_{h=1}^{(n-1)/2}\vartheta_{00}\left(\frac{2h(\alpha+\beta\omega)}{n}\right)\vartheta_{01}\left(\frac{2h(\alpha+\beta\omega)}{n}\right)\vartheta_{10}\left(\frac{2h(\alpha+\beta\omega)}{n}\right)$$
$$= (-1)^{(n^2-1)/8}e^{\frac{\pi i}{2}\frac{n^2-1}{n}\beta(\alpha+\beta\omega)}\vartheta_{00}^{(n-1)/2}\vartheta_{10}^{(n-1)/2}\vartheta_{01}^{(n-1)/2}.$$

Therefore, by multiplying the three equations (2), one obtains

$$(-1)^{(n^2-1)/8}2^{(n-1)/2}P_{00}P_{10}P_{01} = 1. \tag{3}$$

In addition we put

$$P_{11}\eta(\omega)^{(n-1)/2} = e^{\frac{\pi i}{6}\mu'(\mu+\mu'\omega)\frac{n^2-1}{n}}\prod_{h=1}^{(n-1)/2}\vartheta_{11}(2h\varpi). \tag{4}$$

These quantities P can be expressed through the roots of the division equation if, using (3), the quotients

$$\frac{P_{00}^2}{P_{01}P_{10}},\qquad \frac{P_{10}^2}{P_{01}P_{00}},\qquad \frac{P_{01}^2}{P_{10}P_{00}},\qquad \frac{P_{11}^3}{P_{00}P_{01}P_{10}}$$

are formed and elliptic functions are introduced by means of the formulae of § 42.

One obtains in this way

$$\begin{aligned}(-1)^{(n^2-1)/8}2^{(n-1)/2}P_{00}^3 &= \prod_{h=1}^{(n-1)/2}\frac{\mathrm{dn}^2\,h\Omega}{\mathrm{cn}\,h\Omega},\\ (-1)^{(n^2-1)/8}2^{(n-1)/2}P_{10}^3 &= \prod_{h=1}^{(n-1)/2}\frac{\mathrm{cn}^2\,h\Omega}{\mathrm{dn}\,h\Omega},\\ (-1)^{(n^2-1)/8}2^{(n-1)/2}P_{01}^3 &= \prod_{h=1}^{(n-1)/2}\frac{1}{\mathrm{cn}\,h\Omega\ \mathrm{dn}\,h\Omega},\end{aligned} \tag{5}$$

and

$$(-1)^{(n^2-1)/8}P_{11}^3 = (\varkappa\varkappa')^{(n-1)/2}\prod_{h=1}^{(n-1)/2}\frac{\mathrm{sn}^3\,h\Omega}{\mathrm{cn}\,h\Omega\ \mathrm{dn}\,h\Omega}. \tag{6}$$

These expressions now show that P_{00}^3, P_{10}^3, P_{01}^3 are roots of transformation equations, namely modular equations. P_{11}^6 is the root of a multiplier equation, and if n is a square, then P_{11}^3 also is the root of a multiplier equation.

The functions P may be combined with one another in manifold ways in order to obtain new transformation equations. We give the following:

$$\frac{P_{10}}{P_{00}} = \prod_{h=1}^{(n-1)/2}\frac{\mathrm{cn}\,h\Omega}{\mathrm{dn}\,h\Omega},\qquad \frac{P_{01}}{P_{00}} = \prod_{h=1}^{(n-1)/2}\frac{1}{\mathrm{dn}\,h\Omega}, \tag{7}$$

$$\begin{aligned}(-1)^{(n^2-1)/8}\frac{2^{(n-1)/3}P_{00}^2P_{11}}{(\sqrt{\varkappa\varkappa'})^{(n-1)/3}} &= \prod_{h=1}^{(n-1)/2}\frac{\mathrm{sn}\,h\Omega\ \mathrm{dn}\,h\Omega}{\mathrm{cn}\,h\Omega},\\ (-1)^{(n^2-1)/8}\frac{2^{(n-1)/3}P_{10}^2P_{11}}{(\sqrt{\varkappa\varkappa'})^{(n-1)/3}} &= \prod_{h=1}^{(n-1)/2}\frac{\mathrm{sn}\,h\Omega\ \mathrm{cn}\,h\Omega}{\mathrm{dn}\,h\Omega},\\ (-1)^{(n^2-1)/8}\frac{2^{(n-1)/3}P_{01}^2P_{11}}{(\sqrt{\varkappa\varkappa'})^{(n-1)/3}} &= \prod_{h=1}^{(n-1)/2}\frac{\mathrm{sn}\,h\Omega}{\mathrm{cn}\,h\Omega\ \mathrm{dn}\,h\Omega}.\end{aligned} \tag{8}$$

An important result is still obtained by applying the multiplication theorem. If in the last formula II of § 57 one puts $u = 2h\varpi$, one finds

$$e^{4\pi i\omega h^2\mu'^2}\vartheta_{01}(2h\varpi)^{n^2} = \frac{\vartheta_{01}^{n^2}}{D(\operatorname{sn}^2 h\Omega)}, \tag{9}$$

where, as we recall, D is an integral rational function whose coefficients are formed rationally from $\varkappa^2$ and integers.

If we take the product of these expressions for $h = 1, 2, \ldots, (n-1)/2$, then from the last equation (2) follows, because, as is known, $\sum h^2 = n(n^2-1)/24$,

$$P_{01}^{n^2} = \prod_{h=1}^{(n-1)/2} \frac{1}{D(\operatorname{sn}^2 h\Omega)}. \tag{10}$$

From this we conclude that $P_{01}^{n^2}$ also is the root of a transformation equation. This is nothing new when n is divisible by 3, but is new when n is not divisible by 3, so that n^2-1 is divisible by 3. In that case, namely, from (10), using (5), (7), (8), we obtain

$$\begin{aligned}
P_{01} &= 2^{(n-1)(n^2-1)/6}\prod_{h=1}^{(n-1)/2}\frac{(\operatorname{cn} h\Omega)^{(n^2-1)/3}(\operatorname{dn} h\Omega)^{(n^2-1)/3}}{D(\operatorname{sn}^2 h\Omega)},\\
P_{00} &= 2^{(n-1)(n^2-1)/6}\prod_{h=1}^{(n-1)/2}\frac{(\operatorname{cn} h\Omega)^{(n^2-1)/3}(\operatorname{dn} h\Omega)^{(n^2+2)/3}}{D(\operatorname{sn}^2 h\Omega)},\\
P_{10} &= 2^{(n-1)(n^2-1)/6}\prod_{h=1}^{(n-1)/2}\frac{(\operatorname{cn} h\Omega)^{(n^2+2)/3}(\operatorname{dn} h\Omega)^{(n^2-1)/3}}{D(\operatorname{sn}^2 h\Omega)}.
\end{aligned} \tag{11}$$

Furthermore,

$$(-1)^{(n^2-1)/8}2^{n^2(n-1)/3}P_{11} = (\sqrt{\varkappa\varkappa'})^{(n-1)/3}\prod_{h=1}^{(n-1)/2}\frac{\operatorname{sn} h\Omega\, D(\operatorname{sn}^2 h\Omega)^2}{(\operatorname{cn} h\Omega)^{(2n^2+1)/3}(\operatorname{dn} h\Omega)^{(2n^2+1)/3}}. \tag{12}$$

If the rationality domain is to remain that of rational numbers and rational functions of $\varkappa^2$, one writes the last equation better in the form, compare § 54, (4),

$$\begin{aligned}
P_{11}\gamma_2(\omega)^{(n-1)/2} &= (-1)^{(n^2-1)/8}2^{-(n^2-4)(n-1)/3}(\varkappa\varkappa')^{-(n-1)/2}(1-\varkappa^2\varkappa'^2)^{(n-1)/2}\\
&\quad\times\prod_{h=1}^{(n-1)/2}\frac{\operatorname{sn} h\Omega\, D(\operatorname{sn}^2 h\Omega)^2}{(\operatorname{cn} h\Omega)^{(2n^2+1)/3}(\operatorname{dn} h\Omega)^{(2n^2+1)/3}},
\end{aligned} \tag{13}$$

and attaches to it the following theorem:

If n is not divisible by 3, then P_{00}, P_{01}, P_{10}, $P_{11}\gamma_2(\omega)^{n-1}$, and, if n is a square, also $P_{11}\gamma_2(\omega)^{(n-1)/2}$ are roots of transformation equations.

§ 68. Second representation of the roots of the transformation equations.

If we first take into view one of the rows, namely that to which the root $x_{1,0}$ belongs, so that $\mu = 1$, $\mu' = 0$, then we can apply to (2), (3) of the preceding paragraph the formulae (20), (21), (10), § 34. If there, when v is odd, one puts $n - v$ in place of v and uses the formula

$$\vartheta_{g_1g_2}\left(\frac{n-v}{n}\right) = (-1)^{g_1(g_2+1)}\vartheta_{g_1g_2}\left(\frac{v}{n}\right),$$

then only the even v occur in these formulae, and if one puts $v = 2h$, the formulae (20), (21) occurring there can also be written

$$\sqrt{n}\,\eta(n\omega)\eta(\omega)^{(n-3)/2} = \prod_{h=1}^{(n-1)/2} \vartheta_{11}\left(\frac{2h}{n}\right),$$

$$f(n\omega)\eta(\omega)^{(n-1)/2} = f(\omega) \prod_{h=1}^{(n-1)/2} \vartheta_{00}\left(\frac{2h}{n}\right),$$

$$f_1(n\omega)\eta(\omega)^{(n-1)/2} = f_1(\omega) \prod_{h=1}^{(n-1)/2} \vartheta_{01}\left(\frac{2h}{n}\right),$$

$$f_2(n\omega)\eta(\omega)^{(n-1)/2} = \left(\frac{2}{n}\right) f_2(\omega) \prod_{h=1}^{(n-1)/2} \vartheta_{10}\left(\frac{2h}{n}\right),$$

where $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$. If now the functions P corresponding to the case $\mu = 1$, $\mu' = 0$ are denoted by P^0, it follows from § 67, (2), and (4), with regard to § 34, (10), that

$$\begin{aligned} P^0_{00} &= \frac{f(n\omega)}{f(\omega)^n}, \\ P^0_{01} &= \frac{f_1(n\omega)}{f_1(\omega)^n}, \\ P^0_{10} &= \left(\frac{2}{n}\right) \frac{f_2(n\omega)}{f_2(\omega)^n}, \end{aligned} \tag{1}$$

and

$$P^0_{11} = \sqrt{n}\,\frac{\eta(n\omega)}{\eta(\omega)}. \tag{2}$$

In order to represent by such formulae all roots of the transformation equations, we determine a linear transformation as follows:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{16}, \tag{3}$$

$$\alpha \equiv \mu, \qquad \beta \equiv \mu' \pmod{n}, \tag{4}$$

and in (1), (2) replace ω by

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{5}$$

Then the formulae (3) to (6) of § 39 and (19) of § 38 can be applied to the left hand side, and give

$$\begin{aligned} P_{00} &= \frac{f(n\omega')}{f(\omega')^n}, \\ P_{01} &= \frac{f_1(n\omega')}{f_1(\omega')^n}, \\ P_{10} &= \left(\frac{2}{n}\right) \frac{f_2(n\omega')}{f_2(\omega')^n}, \\ P_{11} &= \varepsilon^{1-n}\sqrt{n}\,\frac{\eta(n\omega')}{\eta(\omega')}, \end{aligned} \tag{6}$$

where ε is the twenty-fourth root of unity defined exactly in § 33, (15), (18).

Now two transformations

$$\begin{pmatrix}\alpha' & \beta' \\ \gamma' & \delta'\end{pmatrix}, \qquad \begin{pmatrix}a & 0 \\ c & \partial\end{pmatrix}, \qquad a\partial = n,$$

the first linear, the other of the nth order, may be so determined that

$$\begin{pmatrix}\alpha' & \beta' \\ \gamma' & \delta'\end{pmatrix}\begin{pmatrix}a & 0 \\ c & \partial\end{pmatrix} = \begin{pmatrix}1 & 0 \\ 0 & n\end{pmatrix}\begin{pmatrix}\alpha & \beta \\ \gamma & \delta\end{pmatrix}. \tag{7}$$

For if we write (7) in the form

$$\begin{pmatrix}\delta' & -\beta' \\ -\gamma' & \alpha'\end{pmatrix}\begin{pmatrix}1 & 0 \\ 0 & n\end{pmatrix} = \begin{pmatrix}a & 0 \\ c & \partial\end{pmatrix}\begin{pmatrix}\delta & -\beta \\ -\gamma & \alpha\end{pmatrix},$$

then one obtains from this simply

$$\delta' = a\delta, \qquad \gamma' = \partial\gamma - c\delta, \qquad \partial\beta' = \beta, \qquad n\alpha' = \partial\alpha - c\beta, \qquad \alpha\alpha' = \alpha - c\beta'. \tag{8}$$

Thus first ∂ is determined as the greatest common divisor of β and n, or, because of (4), of μ' and n. For if ∂ were not the greatest common divisor of $n = \partial a$ and $\beta = \partial\beta'$, then a and β' would have a common divisor, and hence so would β and $\alpha = \alpha\alpha' + c\beta$, which cannot be. Thereby at the same time $a = n/\partial$ is determined, and c must satisfy the congruence

$$\partial\alpha - c\beta \equiv 0 \pmod{n}, \tag{9}$$

by which, however, c is determined not absolutely, but only modulo a. One may therefore still attach to c, for example, the condition that it be divisible by some power of 2, or, if a is not divisible by 3, by some power of 3, which we shall sometimes do later.

The three numbers a, ∂, c can have no common divisor; for such a divisor, by (8), would also divide α and β, which is impossible. If, therefore, we denote the greatest common divisor of a and ∂ by e, then c must be relatively prime to e.

Because of (4) and (8), the numbers a, ∂, c, the latter modulo a, are completely determined by the two numbers μ, μ', and do not change when μ, μ' are multiplied by a common factor relatively prime to n, that is, replaced by another pair in the same row. If n is decomposed in any way into two factors a, ∂, then c can still assume

$$\frac{a}{e}\varphi(e)$$

different values modulo a relatively prime to e. Each of these assumptions leads to a pair of numbers μ, μ' by the congruences

$$\mu \equiv \alpha \equiv a\alpha' + c\beta', \qquad \mu' \equiv \beta \equiv \partial\beta' \pmod{n}, \tag{10}$$

where β' denotes an arbitrary number relatively prime to a, and α' is to be so determined that α, β become relatively prime. It is also easy to see, by § 65, (3), that so long as a, ∂, c remain the same, the pairs μ, μ' determined by (10) belong to the same row.

We call the numbers a, c, ∂ the transformation numbers, as in § 27, and n the transformation degree.

The system of numbers a, c, ∂ is therefore completely characteristic for one row, and the number of rows is equal to the number of these systems of numbers. Hence for the number $\psi(n)$ of § 63 follows

$$\psi(n) = \sum \frac{a}{e}\varphi(e), \tag{11}$$

where the sum extends over all divisors a of n.

It is of interest to prove this relation directly as well; in doing so the restriction to an odd n may be dropped. If we now regard $\psi(n)$ as a sign for the sum (11), then, first, when n' and n'' are relatively prime,

$$\psi(n')\psi(n'') = \psi(n'n''),$$

and it remains only to determine the sum $\psi(n)$ in the case in which $n = p^\pi$ is a power of a prime. In this case e is equal to the smaller of the two divisors a, ∂, and when $a = \partial$ it is $e = a$. If, therefore, we separate off the first and last terms of the sum $\psi(n)$, we obtain

$$\psi(p^\pi) = 1 + p^\pi + \sum_{1<a\leq\sqrt{n}} \varphi(a) + \sum_{\sqrt{n}<a<n} \frac{a}{\partial}\varphi(\partial).$$

But

$$\varphi(a) = a\frac{p-1}{p}, \qquad \varphi(\partial) = \partial\frac{p-1}{p},$$

so that

$$\psi(p^\pi) = 1 + p^\pi + \frac{p-1}{p}\sum_{1<a<n} a = 1 + p^\pi + (p-1)(1 + p + p^2 + \cdots + p^{\pi-2}) = p^\pi\left(1 + \frac{1}{p}\right),$$

which gives the expression for $\psi(n)$ as in § 63.[17]

According to (7), in the formulae (6), one must now put for $n\omega'$:

$$\frac{\gamma' + \delta'\frac{c+\partial\omega}{a}}{\alpha' + \beta'\frac{c+\partial\omega}{a}},$$

and the linear transformations of the f-functions, § 40, (4), (8), (11), can be applied. We assume here

$$c \equiv 0 \pmod{16}, \tag{12}$$

so that by (3) and (8)

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \equiv \begin{pmatrix} a^3 & 0 \\ 0 & a \end{pmatrix} \pmod{16}.$$

If we denote by ρ the third root of unity

$$\rho = e^{-\frac{2\pi i}{3}[\alpha'(\gamma'-\beta')-(\alpha'^2-1)\beta'\delta']}e^{\frac{2n\pi i}{3a}[\alpha(\gamma-\beta)-(\alpha^2-1)\beta\delta]}, \tag{13}$$

then we obtain

$$\begin{aligned} P_{00} &= \rho\,\frac{f\left(\frac{c+\partial\omega}{a}\right)}{f(\omega)^n}, \\ P_{01} &= \rho\left(\frac{2}{a}\right)\frac{f_1\left(\frac{c+\partial\omega}{a}\right)}{f_1(\omega)^n}, \\ P_{10} &= \rho\left(\frac{2}{\partial}\right)\frac{f_2\left(\frac{c+\partial\omega}{a}\right)}{f_2(\omega)^n}. \end{aligned} \tag{14}$$

The quotient

$$\frac{P_{10}}{P_{00}} = \left(\frac{2}{\partial}\right)\frac{f_2\left(\frac{c+\partial\omega}{a}\right)}{f\left(\frac{c+\partial\omega}{a}\right)}\left(\frac{f(\omega)}{f_2(\omega)}\right)^n \tag{15}$$

[17] Compare Dedekind, *Über die elliptischen Modulfunktionen*, Journal für die reine und angewandte Mathematik, vol. 83.

gives, by § 54, (3), if one puts

$$u(\omega) = \sqrt[3]{\varkappa},$$

the quantities

$$\left(\frac{2}{\partial}\right) \frac{u\left(\frac{c+\partial\omega}{a}\right)}{u(\omega)^n} \tag{16}$$

as roots of a modular equation; and this is the Jacobi modular equation.[18]

If n is not divisible by 3, we assume

$$c \equiv 0 \pmod 3, \tag{17}$$

whereby ρ obtains the value 1.

In the same way, one can apply the transformation of the η-function, § 38, (15), (18), (19), to the last of equations (6), and obtains

$$P_{11} = \left(\frac{\beta}{\alpha}\right)\left(\frac{\beta'}{\alpha'}\right) \rho\, \mathrm{i}^{(a-1)/2} \sqrt{\partial}\, \frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}. \tag{18}$$

Here it would already suffice if c were divisible by 8. If n is not divisible by 3 and c is divisible by 3, then here also $\rho = 1$ is to be put.

The determination of the sign in (18) is of interest for us only in the case where n is a square. But by (8)

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\beta'}{\alpha'}\right) = \left(\frac{\partial}{\alpha}\right)\left(\frac{\beta'}{\alpha\alpha'}\right) = \left(\frac{\partial}{\alpha}\right)\left(\frac{\beta'}{a}\right) = \left(\frac{\alpha}{\partial}\right)\left(\frac{\beta'}{a}\right),$$

the last equality by the reciprocity law of quadratic residues, since $\alpha \equiv 1 \pmod 4$. If now n is a square, then $a : e$ and $\partial : e$ are also squares, and there results

$$\left(\frac{\alpha}{\partial}\right)\left(\frac{\beta'}{a}\right) = \left(\frac{\alpha\beta'}{e}\right) = \left(\frac{c}{e}\right),$$

so that in this case

$$P_{11} = \rho\, \mathrm{i}^{(a-1)/2} \left(\frac{c}{e}\right) \sqrt{\partial}\, \frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}.$$

The formulae (10) set up for the characterization of a row are a special case of a more general system obtained by applying a linear transformation to α', β' in (10). One then obtains the following:

If a, b, c, ∂ are four integers without common divisor which satisfy

$$a\partial - bc = n,$$

then the pairs of numbers μ, μ' corresponding to one row are obtained by letting α', β' in

$$\mu \equiv a\alpha' + c\beta', \qquad \mu' \equiv b\alpha' + \partial\beta' \pmod n \tag{19}$$

run through all and only such values for which μ, μ' have no common divisor with n.

That two pairs of values μ, μ' corresponding to the congruences (19) really belong to the same row follows immediately from § 65, (3); and likewise it is self-evident that all pairs of numbers of one row are contained in this form, since α', β' can be replaced by $h\alpha', h\beta'$ if h is relatively prime to n. That one obtains all rows in this way is shown by the formulae (10).

[18] If M is the Jacobi multiplier, then

$$\frac{(\varkappa\varkappa')^{(n-1)/3}}{M} = 4^{-(n-1)/3} P_{11}^2 P_{00}^4.$$

§ 69. The invariant equation.

Among the transformation equations, one deserves special interest, namely that whose roots are the $\psi(n)$ quantities

$$j\left(\frac{c+\partial\omega}{a}\right), \tag{1}$$

or, according to determination (19), § 68, the identical quantities

$$j\left(\frac{c+\partial\omega}{a+b\omega}\right), \tag{2}$$

where $j(\omega)$ is the invariant defined in § 46.

This equation is called the invariant equation.

By § 54, the function $j(\omega)$ can be represented rationally through $f(\omega)^{24}$, namely

$$j(\omega) = \frac{[f(\omega)^{24} - 16]^3}{f(\omega)^{24}}, \tag{3}$$

so that, by the results of the preceding paragraph, the quantities (1) are the roots of an equation whose coefficients are rational functions of $\varkappa^2$.

But if in (3) one substitutes for $f(\omega)$ one of the earlier expansions, for example § 24, (11),

$$f(\omega) = q^{-1/24}\prod_{\nu=1}^{\infty}(1+q^{2\nu-1}),$$

then one recognizes that $j(\omega)$ can be developed in a series progressing in powers of q^2, which has the form

$$j(\omega) = q^{-2} + a_1 + a_2 q^2 + a_3 q^4 + \cdots, \tag{4}$$

where $a_1, a_2, a_3, \ldots$ are integral rational numbers which may be computed successively; for example $a_1 = 744$, $a_2 = 196884$. The developments of the quantities (1) therefore begin with

$$e^{-2\pi i c/a} e^{-2\pi i\partial\omega/a}, \tag{5}$$

and are consequently all different from one another. Thus the invariant equation is irreducible by § 65.

There is, however, a second way of arriving at this equation, and indeed at transformation equations generally, which we now wish first to learn in the case of the invariant equation. This derivation rests upon the theorems of § 54 on modular functions and is valid also for an even n. Here it is theorem 4 of § 54 that is applied:

I. Every modular function which remains unchanged under the two transformations

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \tag{6}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{7}$$

is a rational function of $j(\omega)$.

We first prove that by application of the substitutions (6), (7), the ν quantities (1) are permuted among one another. We have the composition

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ c_1 & \partial \end{pmatrix}, \tag{8}$$

if

$$c_1 = c + \partial - \lambda a \tag{9}$$

is put. Thus, by the transformation (6), since $j(\omega)$ remains unchanged under every linear transformation,

$$j\left(\frac{c+\partial\omega}{a}\right) \quad \text{passes into} \quad j\left(\frac{c_1+\partial\omega}{a}\right).$$

Next determine a_2, ∂_2, c_2 so that

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}\begin{pmatrix} a_2 & 0 \\ c_2 & \partial_2 \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1. \tag{10}$$

For this it is necessary that

$$\alpha a_2 + \beta c_2 = 0, \qquad \gamma a_2 + \delta c_2 = -\partial, \tag{11}$$

$$\beta\partial_2 = a, \qquad \delta\partial_2 = c. \tag{12}$$

Thus ∂_2 is determined as the greatest common divisor of a and c, and hence, because $a_2\partial_2 = n$, also a_2. From the two equations (12) one now knows β, δ as relatively prime numbers and can determine α, γ from the Diophantine equation

$$\alpha\delta - \beta\gamma = 1.$$

Once this is done, the two equations (11) give

$$a_2 = \partial\beta, \qquad c_2 = -\partial\alpha, \tag{13}$$

whereby c_2 is also determined, and at the same time it appears that ∂ is the greatest common divisor of a_2, c_2. If α, γ are replaced by another solution $\alpha + h\beta$, $\gamma + h\delta$, then only c_2 changes by a multiple of a_2.

Under the transformation (7), therefore,

$$j\left(\frac{c+\partial\omega}{a}\right) \quad \text{passes into} \quad j\left(\frac{c_2+\partial_2\omega}{a_2}\right).$$

If now we form a symmetric function of all quantities (1), for example, for an indeterminate x, the product

$$\prod \left[x - j\left(\frac{c+\partial\omega}{a}\right)\right],$$

then this function is unchanged by the transformations (6), (7), and is therefore, by the theorem mentioned above, a rational function of $j(\omega)$. Moreover it is an integral rational function of degree ν in x, with initial term x^ν, and we denote it by $F_n[x, j(\omega)]$. The equation

$$F_n[x, j(\omega)] = 0 \tag{14}$$

has the quantities $j\left(\frac{c+\partial\omega}{a}\right)$ as roots and is the invariant equation, whose most important properties we shall now derive.

1. The invariant equation is irreducible if the rationality domain is taken to be the totality of all rational functions of $j(\omega)$ with arbitrary numerical coefficients. For if any rational equation

$$\Phi[j(n\omega), j(\omega)] = 0 \tag{15}$$

exists, then any arbitrary linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

may be applied to it. By § 29, (5), if

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$$

is any transformation of order n, the linear transformations

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$$

can always be so determined that

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{16}$$

From this, however, it follows that equation (9) is satisfied by each of the quantities

$$j\left(\frac{c + \partial\omega}{a + b\omega}\right)$$

and hence also by each of the quantities (1); therefore, because these quantities (1) are distinct, the irreducibility of (14) follows.

2. The function

$$F_n[x, j(\omega)] = \prod \left[x - j\left(\frac{c + \partial\omega}{a}\right)\right] \tag{17}$$

has, for every finite value of ω with positive imaginary part, and hence also for every finite value of $j(\omega)$, a finite value, and is consequently an integral rational function of $j(\omega)$.

If, moreover, from (3) one seeks the leading term in the expansion of the function (17) in powers of q, one obtains

$$(-1)^\nu e^{-2\pi \mathrm{i} \sum c/a} e^{-2\pi \mathrm{i}\omega \sum \partial/a} = Cq^{-2\nu},$$

where C is a finite constant. Hence

$$j(\omega)^\nu F_n[x, j(\omega)]$$

is finite for an infinite $j(\omega)$; that is, the degree of $F_n[x, j(\omega)]$ with respect to $j(\omega)$ is likewise the νth.

3. It is

$$F_n[j(n\omega), j(\omega)] = 0,$$

and if we replace $n\omega$ by ω,

$$F_n\left[j(\omega), j\left(\frac{\omega}{n}\right)\right] = 0.$$

Since $j\left(\frac{\omega}{n}\right)$ is likewise a root of the invariant equation, it follows that the two equations of degree ν

$$F_n[x, j(\omega)] = 0, \qquad F_n[j(\omega), x] = 0$$

have one root in common, and consequently, because of the irreducibility of the first and equality of the degrees, all roots in common. It is therefore

$$F_n(x, y) = CF_n(y, x)$$

for indeterminate values of the variables x, y and a constant C. Hence also, by interchanging x and y,

$$F_n(y, x) = CF_n(x, y),$$

so that $C^2 = 1$, or

$$F_n(x, y) = \pm F_n(y, x).$$

If now we put $x = y$, it follows
$$F_n(y, y) = \pm F_n(y, y),$$
and, if the lower sign holds,
$$F_n(y, y) = 0.$$
But this is possible only when $n = 1$, where $F_1(x, y) = x - y$, since otherwise $F_n(x, y)$ would have to be divisible by $x - y$, contrary to irreducibility. Hence, whenever $n > 1$,
$$F_n(x, y) = F_n(y, x). \tag{18}$$

4. Let
$$n = n'n'', \qquad \nu' = \psi(n'), \qquad \nu'' = \psi(n''),$$
where n' and n'' have no common divisor, and let $x_1, x_2, \ldots, x_{\nu'}$ be the roots of the equation
$$F_{n'}[x, j(\omega)] = 0. \tag{19}$$
The product
$$F_{n''}(x, x_1)F_{n''}(x, x_2) \cdots F_{n''}(x, x_{\nu'}), \tag{20}$$
whose degree with respect to x is
$$\nu = \psi(n) = \psi(n')\psi(n''),$$
depends, as a symmetric function of the roots of (19), rationally upon $j(\omega)$ and vanishes for
$$x = j\left(\frac{\omega}{n'n''}\right),$$
that is, for a root of the equation $F_n[x, j(\omega)] = 0$. Because of the irreducibility of the latter equation, the equality of degrees, and the coefficient of x^ν, it is therefore
$$F_n[x, j(\omega)] = F_{n''}(x, x_1)F_{n''}(x, x_2) \cdots F_{n''}(x, x_{\nu'}). \tag{21}$$

5. If $n = p^\pi$ is a prime power, then the degree of the function $F_n[x, j(\omega)]$ is $p^{\pi-1}(p + 1)$, and the equation
$$F_{p^{\pi-1}}[x, j(\omega)] = 0 \tag{22}$$
is of degree
$$\nu' = p^{\pi-2}(p + 1).$$
Let its roots be denoted by $x_1, x_2, \ldots, x_{\nu'}$.

The product
$$P = F_p(x, x_1)F_p(x, x_2) \cdots F_p(x, x_{\nu'})$$
has with respect to x the degree $\nu'(p + 1) = p^{\pi-2}(p + 1)^2$; it depends symmetrically upon the roots of (22), hence rationally upon $j(\omega)$, and vanishes for
$$x = j\left(\frac{\omega}{p^\pi}\right).$$
Thus P is divisible by $F_n[x, j(\omega)]$.

But the degree of P is higher than that of F_n. If we take
$$x_1 = j\left(\frac{p^{\pi-2}c + \omega}{p^{\pi-1}}\right),$$
where c may have each of the values $0, 1, 2, \ldots, p - 1$, then $F_p(x, x_1)$ vanishes for
$$x = j\left(\frac{\omega}{p^{\pi-2}}\right);$$

that is, p of the factors of P have a definite factor in common with $F_{p^{\pi-2}}[x, j(\omega)]$. It follows that P is divisible by

$$\{F_{p^{\pi-2}}[x, j(\omega)]\}^p,$$

and consequently, as comparison of the degrees and highest terms teaches,

$$F_{p^\pi}[x, j(\omega)] = \frac{F_p(x, x_1)F_p(x, x_2)\cdots F_p(x, x_{\nu'})}{\{F_{p^{\pi-2}}[x, j(\omega)]\}^p}. \tag{23}$$

This formula is subject to an exception for $\pi = 2$, because in that case the degree of the function on the right hand side is still too high. In this case, however, each of the $p+1$ factors of the product P has the divisor $x - j(\omega)$, since precisely every x_i is a root of the equation $F_p[x, j(\omega)] = 0$, and in place of (23) the formula

$$F_{p^2}[x, j(\omega)] = \frac{F_p(x, x_1)F_p(x, x_2)\cdots F_p(x, x_{p+1})}{[x - j(\omega)]^{p+1}} \tag{24}$$

occurs.

By this the solution of the invariant equation $F_n = 0$ is reduced to the successive solution of such cases in which n is a prime number.

6. Whereas in the derivation of the transformation equations from the division equations it is fixed from the outset that the numerical coefficients in these equations are rational numbers, the second derivation at first teaches us nothing about the numerical coefficients. But we can prove afterwards that these coefficients are not only rational but also integral numbers, and at the same time we arrive at an important theorem concerning the divisibility of these coefficients.

If we can prove that, when p is a prime, the coefficients in $F_p(x, y)$ are integral numbers, then by 4. and 5. the same follows for every composite n.

By (4), $j(\omega)$ may be developed into a series of the form

$$j(\omega) = q^{-2}\sum_{h=0}^{\infty} a_h q^{2h}, \qquad a_0 = 1, \tag{25}$$

whose coefficients a_h are integral numbers. If from this, when p is a prime, we form the pth power and take account of Fermat's theorem valid for every integer a,

$$a^p \equiv a \pmod{p},$$

then

$$j(\omega)^p = q^{-2p}\sum_{h=0}^{\infty} a_h q^{2hp} + pq^{-2(p-1)}\sum_{h=0}^{\infty} b_h q^{2h}, \tag{26}$$

where the b_h likewise are integral numbers. On the other hand, if in (25) ω is replaced by $p\omega$,

$$j(p\omega) = q^{-2p}\sum_{h=0}^{\infty} a_h q^{2hp}. \tag{27}$$

Putting now

$$j(\omega) = u, \qquad j(p\omega) = v,$$

one obtains

$$u^p - v = pq^{-2(p-1)}\sum_{h=0}^{\infty} b_h q^{2h},$$

which may also be written

$$(u^p - v)(u - v^p) = pq^{-2(p^2+p-1)}\sum_{h=0}^{\infty} c_h q^{2h}, \tag{28}$$

where c_h is a third system of integers.

Now in $F_p(x, y)$ the highest terms with respect to x, y, namely x^{p+1} and y^{p+1}, occur, and we therefore put

$$F_p(x, y) = (x^p - y)(x - y^p) - \sum_{h,k=0}^{p} c_{h,k} x^h y^k, \tag{29}$$

where $c_{h,k}$ are the coefficients to be determined and, by 3., satisfy the condition

$$c_{h,k} = c_{k,h}.$$

To determine them, we put in (29)

$$x = u, \qquad y = v,$$

whereby F_p vanishes, and obtain

$$(u^p - v)(u - v^p) = \sum c_{h,k} u^h v^k. \tag{30}$$

From this it follows first that

$$c_{p,p} = 0$$

must be, since by (28) in the expansion of the left hand side in powers of q the power $q^{-2(p^2+p)}$ cannot occur. We can therefore now write (29) in the form

$$(u^p - v)(u - v^p) = \sum_{h=0}^{p} \sum_{k=0}^{h-1} c_{h,k}(u^h v^k + u^k v^h) + \sum_{h=0}^{p-1} c_{h,h} u^h v^h. \tag{31}$$

Here the expansions, arising from (25) and (27), of

$$u^h v^k + u^k v^h, \qquad u^h v^h$$

in powers of q are to be inserted; their leading terms are

$$q^{-2(hp+k)}, \qquad q^{-2h(p+1)},$$

with coefficient 1.

On the right hand side of (31) no two terms occur whose expansions begin with the same power of q; for from

$$hp + k = h'p + k'$$

it follows that $k \equiv k' \pmod{p}$, and hence, since k and k' are less than p, that $k = k'$ and $h = h'$.

If therefore the series which represent the two sides of (31) are ordered according to ascending powers of q, and coefficients of equal powers are then set equal to one another, one obtains a sequence of linear equations for the unknowns $c_{h,k}$, of which each following equation contains only one new unknown, with coefficient 1. The known terms of these equations arising from the left hand side are, by (28), integral numbers all divisible by p; consequently the $c_{h,k}$ also yield integral numerical values divisible by p. Accordingly we have

$$F_p(x, y) = (x^p - y)(x - y^p) - p \sum_{h,k=0}^{p} a_{h,k} x^h y^k, \tag{32}$$

where $a_{h,k}$ are integral numbers satisfying the conditions

$$a_{h,k} = a_{k,h}, \qquad a_{p,p} = 0.$$

Volume III. Seventh Section.

Theory of the Transformation Equations.

§ 70. Transformation equations of the first stage.

The invariant equation is of great theoretical importance, partly on account of its general validity (also for even n), and partly on account of the ease with which the linear transformations can be applied to $j(\omega)$. The actual calculation of this equation, however, proves to be so complicated, and the numerical coefficients are so large, that the calculation has so far been carried through only in the simplest case $p = 2$. By contrast, by using other modular functions one can obtain far simpler transformation equations.

On the principle to be applied here we make the following remarks.

Suppose there is given some system of ν functions of ω, corresponding to the ν systems of transformation numbers a, c, ∂, which we shall denote by

$$\Phi_{a,c,\partial} \tag{1}$$

and which are permuted among one another, isomorphically with the functions

$$j\left(\frac{c+\partial\omega}{a}\right), \tag{2}$$

by the substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \qquad (\omega, \omega+1). \tag{3}$$

Then (1) is rationally expressible through (2) and through $j(\omega)$; for the function

$$F_n[x, j(\omega)] \sum_{a,c,\partial} \frac{\Phi_{a,c,\partial}}{x - j\left(\frac{c+\partial\omega}{a}\right)} = \Psi[x, j(\omega)] \tag{4}$$

remains unchanged under the substitutions (3), and is therefore, for an indeterminate x, a rational function of $j(\omega)$, and moreover an integral rational function of x, at most of degree $\nu - 1$. Here $F_n[x, j(\omega)]$ has the same meaning as in (17) of the preceding paragraph, and $\nu = \psi(n)$ is the degree of this function with respect to x. From (4), however, by putting

$$x = j\left(\frac{c+\partial\omega}{a}\right),$$

one obtains

$$\Phi_{a,c,\partial} = \frac{\Psi\left[j\left(\frac{c+\partial\omega}{a}\right), j(\omega)\right]}{F_n'\left[j\left(\frac{c+\partial\omega}{a}\right)\right]}. \tag{5}$$

Thus $\Phi_{a,c,\partial}$ belongs to the algebraic field which consists of the rational functions of $j((c+\partial\omega)/a)$ and $j(\omega)$, and we can apply the theorems of Vol. I, § 151. From these it follows that the ν quantities $\Phi_{a,c,\partial}$ are the roots of an equation of degree ν whose coefficients depend rationally on $j(\omega)$. If the ν quantities $\Phi_{a,c,\partial}$ are mutually distinct, then this equation is irreducible. Such an equation we call a transformation equation belonging to the transformation degree n and of the first stage.[19] Every other quantity of the field can be expressed rationally through such a $\Phi_{a,c,\partial}$ and through $j(\omega)$.

If the functions $\Phi_{a,c,\partial}$ have the property of remaining finite for every finite value of ω with positive imaginary part, hence for every finite $j(\omega)$, then the function $\Psi[x, j(\omega)]$ in (4) is also an

[19] In the first edition I called these equations "invariant transformation equations." Klein objected to this expression (*Vorlesungen über ausgewählte Kapitel der Zahlentheorie*, autographed notebook, Göttingen 1897). I follow Klein's suggestion by now calling these equations "transformation equations of the first stage," without here going more closely into the reason for this expression.

integral function with respect to $j(\omega)$. The formula (5) could therefore fail only for those special values of ω for which two roots of the invariant equation become equal.

If the ν quantities $\Phi_{a,c,\partial}$ are not all mutually distinct, then they decompose into rows of equal numbers of quantities equal to one another, and the equation of degree ν derived from (5) is a power of an irreducible equation, which we shall occasionally also call a transformation equation (Vol. I, § 151, 2).

If the quantities $\Phi_{a,c,\partial}$ are chosen so that they do not become infinite for any finite ω with positive imaginary component, then they remain finite for every finite $j(\omega)$; from this it follows that the coefficients in the function of degree ν

$$\prod(x - \Phi_{a,c,\partial}) \tag{6}$$

are integral rational functions of $j(\omega)$, that is, the $\Phi_{a,c,\partial}$ are integral algebraic functions of $j(\omega)$.

If one arranges the functions $\Phi_{a,c,\partial}$, for instance by a suitable determination of constants still available in them, so that they also remain finite for an infinitely imaginary ω, that is, for $q = 0$, then these functions are also finite for an infinite $j(\omega)$, and the coefficients in (6) are constant. But this is possible only because the $\Phi_{a,c,\partial}$ are all equal to one and the same constant C. If $\Phi_{a,c,\partial}$ is known as a rational function of another quantity $\Psi_{a,c,\partial}$, then $\Phi_{a,c,\partial} = C$ gives either an identity or a transformation equation for $\Psi_{a,c,\partial}$.

§ 71. The transformation equations for γ_2 and γ_3.

Transformation equations of the first stage are obtained first from consideration of the functions § 54, (4), (5):

$$\gamma_2(\omega) = \sqrt[3]{j(\omega)}, \qquad \gamma_3(\omega) = \sqrt{j(\omega) - 1728}.$$

If n is not divisible by 3, then we can choose the numbers a, c, ∂ so that c is always divisible by 3.

Now, according to § 54, (15), a linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

exerts on the function $\gamma_2(\omega)$ the influence

$$\gamma_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = e^{-\frac{2\pi i}{3}(\alpha\gamma + \beta\delta - \alpha_3 - \alpha_2^3\delta)}\gamma_2(\omega).$$

In the compositions (8), (10) of § 69, one has

$$\lambda \equiv n, \qquad \delta \equiv 0 \pmod 3,$$

and then α can still be so determined that α too, and consequently c_2, become divisible by 3.

It follows that under the two substitutions

$$(\omega, \omega + 1), \qquad \left(\omega, -\frac{1}{\omega}\right)$$

the ν functions

$$\gamma_2\left(\frac{c + \partial\omega}{a}\right)\gamma_2(\omega)^{-n} \tag{1}$$

are only permuted among one another and hence are the roots of a transformation equation of the first stage.

Secondly, if n is an odd number, take c even.

For the function $\gamma_3(\omega)$, according to § 54, (15), one has

$$\gamma_3\left(\frac{\gamma+\delta\omega}{\alpha+\beta\omega}\right) = (-1)^{\alpha\gamma+\beta\delta+\gamma^3\delta}\gamma_3(\omega),$$

and in the compositions (8), (10), § 69, one has

$$\lambda \equiv 1, \qquad \alpha \equiv \delta \equiv 0 \pmod 2.$$

The ν quantities

$$\gamma_3\left(\frac{c+\partial\omega}{a}\right)\gamma_3(\omega) \tag{2}$$

therefore permute among one another and consequently likewise are the roots of a transformation equation of the first stage.

In order to form the first of these equations for the simplest case $n = 2$, one observes the relation § 54, (5):

$$\gamma_2(\omega) = \frac{f(\omega)^{24}-16}{f(\omega)^8} = \frac{f_1(\omega)^{24}+16}{f_1(\omega)^8} = \frac{f_2(\omega)^{24}+16}{f_2(\omega)^8}, \tag{3}$$

from which, because

$$f_1(2\omega)f_2(\omega) = \sqrt{2} \qquad [§\,34, (16)],$$

it follows that

$$\begin{aligned} \gamma_2(2\omega) &= \frac{2^8+f_2(\omega)^{24}}{f_2(\omega)^{16}}, \\ \gamma_2\left(\frac{\omega}{2}\right) &= \frac{2^8+f_1(\omega)^{24}}{f_1(\omega)^{16}}, \\ \gamma_2\left(\frac{\omega+3}{2}\right) &= \frac{2^8-f(\omega)^{24}}{f(\omega)^{16}}. \end{aligned} \tag{4}$$

Denoting these three quantities by x, x_0, x_1, one obtains from the relations (6), (8), § 54,

$$x + x_0 + x_1 = \gamma_2(\omega)^2,$$

$$\begin{aligned} xx_0 + xx_1 + x_0x_1 &= 495\,\gamma_2(\omega), \\ xx_0x_1 &= -j(\omega) + 2^4\cdot 3^3\cdot 5^3, \end{aligned}$$

so that x, x_0, x_1 are the roots of the equation

$$x^3 - \gamma_2(\omega)^2x^2 + 5\cdot 9\cdot 11\,\gamma_2(\omega)x + j(\omega) - 2^4\cdot 3^3\cdot 5^3 = 0. \tag{5}$$

The equation whose roots are the cubes of the roots of (5) is the invariant equation for $n = 2$, and can be calculated from it without difficulty.

§ 72. Multiplier equations of the first stage.

Among the multiplier equations we shall consider here those whose roots are the different powers of P_{11}, multiplied by powers of $\gamma_2(\omega)$, $\gamma_3(\omega)$, whose coefficients depend rationally on $j(\omega)$. These equations we call multiplier equations of the first stage.[20]

[20]These equations were studied especially thoroughly by Kiepert, first in several papers in Crelle's Journal, vols. 57, 88, 95, and most fully in papers in the *Mathematische Annalen*, vols. 26, 33. Compare also F. Klein, *Mathematische Annalen*, vols. 14, 15, and the above-mentioned autographed lectures.

We consider the quantities § 68, (16):

$$P_{c,\partial,a} = \mathrm{i}^{\frac{a-1}{2}} \left(\frac{c}{e}\right) \sqrt{n}\, \frac{\eta\left(\frac{c+\partial\omega}{a}\right)}{\eta(\omega)}, \tag{1}$$

and the influence exerted on them by the transformations

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This influence is determined by the formulae (8) to (13), § 69, according to which, since $c_1 \equiv c \pmod e$,

$$P_{c,\partial,a} \quad \text{under } (\omega, \omega+1) \quad \text{goes into} \quad e^{\frac{\pi i}{12}(\lambda-1)} P_{c_1,\partial,a}, \tag{2}$$

and under $(\omega, -1/\omega)$ into

$$\sqrt{\frac{\partial}{c_2}}\, \frac{\mathrm{i}^{(a-a_2)/2}}{\sqrt{-i\omega}} \left(\frac{c}{e}\right)\left(\frac{c_2}{e_2}\right) E\left(\alpha, \beta, c_2 + \hat{c}_2 \omega over a_2\right) P_{c_2,\partial_2,a_2},$$

where E has the meaning assigned in § 38, (15). But by § 69, (11), (12),

$$\sqrt{\partial}\sqrt{\alpha + \beta \frac{c_2 + \hat{c}_2\omega}{a_2}} = \sqrt{\partial_2 \omega},$$

and therefore, if we put

$$E\left(\alpha, \beta, \frac{c_2 + \hat{c}_2\omega}{a_2}\right) = r\sqrt{-i\omega}\sqrt{\frac{c_2}{\partial}},$$

then

$$r = \left(\frac{\alpha}{\beta}\right) \mathrm{i}^{\frac{1-\beta}{2}} e^{\frac{\pi i}{12}\left[\partial(\alpha+\partial) - (\partial^2-1)\alpha\gamma\right]}$$

is a twenty-fourth root of unity, and under the interchange

$$\left(\omega, -\frac{1}{\omega}\right)$$

$P_{c,\partial,a}$ goes into

$$r\, \mathrm{i}^{(a-a_2)/2} \left(\frac{c}{e}\right)\left(\frac{c_2}{e_2}\right) P_{c_2,\partial_2,a_2}.$$

From this follows, as above, the conclusion:

1. For every arbitrary n, the quantities

$$P_{c,\partial,a}^{24}$$

are the roots of a transformation equation.

If n is not divisible by 3, then c, c_2, c_1 can be assumed divisible by 3. Then

$$\lambda \equiv n, \quad \alpha \equiv \partial \equiv 0 \pmod 3.$$

Thus, as follows from § 38, (15), r is an eighth root of unity; if one further observes

$$\gamma_2(\omega+1) = e^{-2\pi i/3}\gamma_2(\omega), \qquad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega), \qquad [\S\, 54, (14)],$$

then we have the theorem:

2. If n is not divisible by 3 and c is divisible by 3, then the quantities

$$P_{c,\partial,a}^{8}\gamma_2(\omega)^{n-1}$$

are roots of a transformation equation.

To make the application to the simplest case $n = 2$, we put

$$x = 16\left(\frac{\eta(2\omega)}{\eta(\omega)}\right)^8, \qquad x_0 = \frac{\eta\left(\frac{\omega}{2}\right)^8}{\eta(\omega)^8}, \qquad x_1 = \frac{\eta\left(\frac{\omega+3}{2}\right)^8}{\eta(\omega)^8}, \tag{3}$$

and then x, x_0, x_1 are nothing other than our functions

$$f_2(\omega)^8, \quad f_1(\omega)^8, \quad -f(\omega)^8. \qquad [\S\,34, (9)]$$

These, as we have already seen earlier, § 54, (8), are the roots of the cubic equation

$$x^3 - \gamma_2(\omega)x + 16 = 0. \tag{4}$$

The circumstance that f_2^8, f_1^8, $-f^8$ themselves are roots of a transformation equation for transformation degree 2 explains the phenomenon that upon adjoining these quantities, hence also upon adjoining α_2, the transformation equations belonging to even n become reducible.

If n is an odd number, take c divisible by 4; then by § 69, (9), (11), (12),

$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \partial \equiv 0, \quad \beta \equiv a\partial_2, \quad \gamma \equiv -a_2\alpha \pmod 4,$$

and therefore

$$r^6 = (-1)^{(\beta-1)/2} = (-1)^{(n-1)/2}(-1)^{(a-a_2)/2}.$$

With regard to

$$\gamma_3(\omega+1) = -\gamma_3(\omega), \qquad \gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega), \qquad [\S\,54, (14)],$$

it follows that

3. If n is odd and c divisible by 4, then the quantities

$$P^6_{c,\partial,a}\gamma_3(\omega)^{(n-1)/2}$$

are roots of a multiplier equation. If $n \equiv 1 \pmod 4$, the same holds for $P^6_{c,\partial,a}$.

As example choose $n = 3$ and put

$$x = 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6, \quad x_0 = -\left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad x_1 = -\left(\frac{\eta\left(\frac{4+\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad x_2 = -\left(\frac{\eta\left(\frac{8+\omega}{3}\right)}{\eta(\omega)}\right)^6. \tag{5}$$

The coefficients of the equation which has these quantities as roots are rational functions of $\gamma_3(\omega)$; and since none of the quantities (5) becomes infinite for a finite value of γ_3, they are integral functions of γ_3. According to 3. we can therefore set this equation in the form

$$x^4 + A_1\gamma_3x^3 + A_2x^2 + A_3\gamma_3x + A_4 = 0,$$

where A_1, A_2, A_3, A_4 are integral rational functions of $j(\omega)$. First one obtains A_4 as the product $xx_0x_1x_2$, which cannot vanish for any value of $j(\omega)$ and therefore must be constant. From the initial terms of the expansion § 24,

$$x = 27q\cdots, \quad x_0 = -q^{-1/3}\cdots, \quad x_1 = -e^{2\pi i/3}q^{-1/3}\cdots, \quad x_2 = -e^{-2\pi i/3}q^{-1/3}\cdots, \quad \gamma_3 = -q^{-1}\cdots, \tag{6}$$

one therefore finds

$$A_4 = -27.$$

Furthermore,

$$-A_1\gamma_3 = x + x_0 + x_1 + x_2.$$

Since the right hand side is not even infinite of the first order for an infinite γ_3, that is, for $q = 0$, A_1 must vanish.

From

$$-A_3\gamma_3 = xx_0x_1x_2\left(\frac{1}{x}+\frac{1}{x_0}+\frac{1}{x_1}+\frac{1}{x_2}\right)$$

one obtains from (6), for A_3, the constant value 1. In order, however, to determine A_2, we must go one term further in the expansion. We put in

$$x_0^4 + \gamma_3 x_0 - 27 = -A_2 x_0^2$$

for x_0 the expansion

$$x_0 = -q^{-1/3} + 6q^{1/3} + \cdots,$$

and find $A_2 = 18$, so that the sought equation is

$$x^4 + 18x^2 + \gamma_3 x - 27 = 0. \tag{7}$$

If n is odd and not divisible by 3, take c divisible by 12. Then

$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \partial \equiv 0, \quad \beta \equiv a\partial_2, \quad \gamma \equiv -a_2\alpha \pmod{12},$$

and one obtains

$$r^2 = (-1)^{(\beta-1)/2} = (-1)^{(n-1)/2}(-1)^{(a-a_2)/2},$$

hence the theorem:

4. If n is relatively prime to 6 and c divisible by 12, then the quantities

$$P^2_{c,\partial,a}\gamma_2(\omega)^{n-1}\gamma_3(\omega)^{(n-1)/2}$$

are roots of a multiplier equation of the first stage.

Since the functions P^2 become neither infinite nor zero for any finite value of $j(\omega)$, we conclude that the coefficients of the equation whose roots are the P^2 are integral rational functions of γ_2, γ_3, and that the last coefficient is a constant.

If n is a prime number p, then the roots of this equation are

$$x = p\left(\frac{\eta(p\omega)}{\eta(\omega)}\right)^2, \qquad x_h = (-1)^{(p-1)/2}e^{2\pi i h/p}\left(\frac{\eta\left(\frac{12h+\omega}{p}\right)}{\eta(\omega)}\right)^2,$$

where $h = 0, 1, \ldots, p-1$, and the initial terms of the expansions are the following:

$$x = pq^{(p-1)/6}\cdots, \qquad x_h = (-1)^{(p-1)/2}e^{2\pi i h/p}q^{-(p-1)/6p}\cdots,$$

from which the value $(-1)^{(p-1)/2}p$ is obtained for the last coefficient. From the fact that the first term in the expansion of $j(\omega)$ by powers of q has coefficient 1, one readily concludes that the numerical coefficients in these equations are rational integers.

For $p = 5$ the equation in question has the form

$$x^6 + A_1\gamma_2^2x^5 + A_2\gamma_2x^4 + A_3x^3 + A_4\gamma_2^2x + 5 = 0,$$

where A_1, A_2, A_3, A_4 are integral rational functions of $j(\omega)$ and, as is easy to see, constants.

From the initial terms one obtains immediately

$$A_1 = 0, \qquad A_2 = 0, \qquad A_4 = -1.$$

But in order to determine A_3, one goes in the expansion of x_0 to the next following term:

$$x_0 = q^{-7/15}(1 - 2q^{2/5} + \cdots),$$

from which $A_3 = 10$ is obtained; thus, for $n = 5$, the multiplier equation is

$$x^6 + 10x^3 - \gamma_2 x + 5 = 0. \tag{8}$$

In the same way the equations for $p = 7$, $p = 11$ are calculated, by using the expansions of x_0:

$$p = 7: \quad x_0 = -q^{-1/7}(1 - 2q^{2/7} - q^{4/7} + 2q^{6/7} \cdots),$$

$$p = 11: \quad x_0 = -q^{-5/33}(1 - 2q^{4/11} - q^{6/11} + 2q^{8/11} + q^{10/11} + 2q^{12/11} + \cdots),$$

while from $\gamma_2(\omega), \gamma_3(\omega)$ only the first terms are ever needed. Thus one finds

$$p = 7: \quad x^8 + 7 \cdot 2x^6 + 7 \cdot 9x^4 + 7 \cdot 10x^2 + \gamma_3 x - 7 = 0, \tag{9}$$

$$p = 11: \quad x^{12} - 11 \cdot 90x^8 + 11 \cdot 40\gamma_2 x^6 + 11 \cdot 15\gamma_3 x^5 + 11 \cdot 2\gamma_2^2 x^2 + \gamma_2\gamma_3 x - 11 = 0. \tag{10}$$

If n is an odd square, take c divisible by 8. Then $a : e$ and $\partial : e$ are likewise squares. Further,

$$\alpha \equiv \partial \equiv 0 \pmod 8, \qquad \lambda \equiv 1 \pmod 8, \tag{11}$$

$$a = \beta\partial_2, \quad a_2 \equiv \beta\partial, \quad a \equiv \partial \equiv e \equiv 1, \qquad c = \partial\hat{\partial}_2, \quad c_2 = -\alpha\partial, \quad a_2 \equiv \partial_2 \equiv e_2 \equiv 1 \pmod 8. \tag{12}$$

The root of unity r in (2) receives the value

$$r = \left(\frac{\alpha}{\beta}\right) \mathrm{i}^{(1-\beta)/2} e^{\frac{2\pi i}{3}(\alpha+\partial) - (\partial-1)\alpha\gamma/8}. \tag{13}$$

But by (12) β is divisible both by e and by e_2, and $\beta e e_2$ is a square; hence

$$\left(\frac{\alpha}{\beta}\right) = \left(\frac{\alpha}{ee_2}\right) = \left(\frac{\alpha}{e}\right)\left(\frac{\alpha}{e_2}\right) = \left(\frac{\partial}{e}\right)\left(\frac{\alpha}{e_2}\right) \quad (\text{because } \alpha\partial \equiv 1 \pmod{e}).$$

Further by (12)

$$\left(\frac{\partial}{e}\right) = \left(\frac{c}{e}\right)\left(\frac{\partial_2}{e}\right), \qquad \left(\frac{\alpha}{e_2}\right) = \left(\frac{c_2}{e_2}\right)\left(\frac{-\partial}{e_2}\right),$$

and

$$\left(\frac{c_2}{e}\right) = \left(\frac{c_2}{e_2}\right), \qquad \left(\frac{-\partial}{e_2}\right) = \left(\frac{-e}{e_2}\right), \qquad \left(\frac{e_2}{e}\right)\left(\frac{-e}{e_2}\right) = (-1)^{(e+1)(e_2-1)/4};$$

therefore

$$\left(\frac{\alpha}{\beta}\right) = (-1)^{(e+1)(e_2-1)/4}\left(\frac{c}{e}\right)\left(\frac{c_2}{e_2}\right),$$

and

$$\mathrm{i}^{(1-\beta)/2} = \mathrm{i}^{(1-e)/2}, \qquad \mathrm{i}^{(a-a_2)/2} = \mathrm{i}^{(e-e_2)/2}.$$

By the interchanges (2), $P_{c,\partial,a}$ therefore goes into

$$P_{c,\partial,a} \quad \text{under } (\omega, \omega{+}1) \text{ into } e^{\frac{2\pi i}{8}\frac{\lambda-1}{2}} P_{c_1,\partial,a}, \quad \text{and under } \left(\omega, -\frac{1}{\omega}\right) \text{ into } e^{\frac{\frac{2\pi i}{8}\{\beta(\alpha+\partial)-(\partial^2-1)\alpha\gamma}{4\}}} P_{c_2,\partial_2,a_2}. \tag{14}$$

If n is still not divisible by 3, take c divisible by 3, whereby

$$\lambda \equiv 1, \qquad \alpha \equiv \partial \equiv 0 \pmod 3,$$

and the third roots of unity occurring in (14) receive the value 1.

From this the following theorems result:

5. If n is an odd square and c divisible by 8, then the quantities

$$P^3_{c,\partial,a}$$

and, if n is an odd square not divisible by 3 and c is divisible by 24, then the quantities

$$P_{c,\partial,a}$$

are roots of multiplier equations of the first stage.

The first examples are $n = 9$, $n = 25$.

For the formation of these equations one can proceed in different ways. We shall here, following Kiepert,[21] take the path of expressing, by § 69, the roots of a transformation equation belonging to degree p rationally through those for degree p^2, and then inserting this expression into the transformation equation belonging to p.

According to 3., formula (7), the equation

$$x^4 + 18x^2 + \gamma_3(\omega)x - 27 = 0 \tag{15}$$

is satisfied by the two functions

$$-\left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)}\right)^6, \qquad 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6. \tag{16}$$

If we denote the first of these values by x, then the second passes, under the substitution $(\omega, \omega/3)$, into $-27/x$, and therefore x also satisfies the equation

$$27^3 + 18 \cdot 27x^2 - \gamma_3\left(\frac{\omega}{3}\right)x^3 - x^4 = 0. \tag{17}$$

Now put

$$y = \left(\frac{\eta\left(\frac{\omega}{9}\right)}{\eta(\omega)}\right)^3. \tag{18}$$

Then x passes under the substitution $(\omega, \omega/3)$ into y^2/x, so that one also obtains the equation

$$y^8 + 18y^4x^2 + \gamma_3\left(\frac{\omega}{3}\right)y^2x^3 - 27x^4 = 0, \tag{19}$$

and by eliminating $\gamma_3(\omega/3)$ from (17) and (19), after removing the factor $y^2 - 27$, one obtains

$$x^4 - 18x^2y^2 - y^2(y^4 - 27y^2 + 27^2) = 0. \tag{20}$$

Solving this quadratic equation for x^2, one obtains

$$x^2 = y^3 + 9y^2 + 27y = (y+3)^3 - 27, \tag{21}$$

where the sign is decided by inserting the initial terms of the expansions, most simply since this equation is also satisfied, according to § 69, by

$$x = 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6, \qquad y = 27\left(\frac{\eta(9\omega)}{\eta(\omega)}\right)^3,$$

putting

$$x = 27q + \cdots, \qquad y = 27q^2 + \cdots.$$

If in (15) the first power of x is isolated and the equation is squared, then insertion of (21) gives the sought multiplier equation for the ninth transformation degree. It receives a simpler form if one puts

$$t = y + 9, \tag{22}$$

[21] *Zur Transformationstheorie der elliptischen Funktionen*, Crelle's Journal, vols. 87, 88, 95.

namely

$$[j(\omega) - 27 \cdot 64](t - 27) = (t^2 - 36t + 27 \cdot 8)^2, \tag{23}$$

or

$$j(\omega)(t - 27) = t(t - 24)^3. \tag{24}$$

Quite similarly one can proceed for the twenty-fifth transformation degree.

If we put

$$x = \left(\frac{\eta\left(\frac{\omega}{5}\right)}{\eta(\omega)} \right)^2, \qquad y = \left(\frac{\eta\left(\frac{\omega}{25}\right)}{\eta(\omega)} \right), \tag{25}$$

then first, by 4., (8),

$$x^6 + 10x^3 - \gamma_2(\omega)x + 5 = 0, \tag{26}$$

from which, as above, the two equations

$$5^5 + 10 \cdot 5^2 x^3 - \gamma_2\left(\frac{\omega}{5}\right) x^5 + x^6 = 0,$$

$$y^{11} + 10y^6 x^3 - \gamma_2\left(\frac{\omega}{5}\right) y^2 x^5 + 5x^6 = 0,$$

and by eliminating $\gamma_2(\omega/5)$,

$$x^6 - 10x^3 y^2(5 + y^2) - y^2(y^8 + 5y^6 + 5^2 y^4 + 5^3 y^2 + 5^4) = 0.$$

Solving this equation for x^3 gives

$$x^3 = y^5 + 5y^4 + 15y^3 + 25y^2 + 25, \tag{27}$$

and if, for abbreviation, we denote the right hand side of this equation by $\chi(y)$, then from (26)

$$\gamma_2(\omega)y^2 = \chi(y)^2 + 10\chi(y) + 5^2. \tag{28}$$

This is the sought equation of degree 30 for y.

§ 73. The Schlaefli modular equations.

One obtains simpler transformation equations when to the domain of rationality, which until now consisted of the rational functions of $j(\omega)$, one adjoins the quantity $f(\omega)^{24}$. To this domain of rationality belong the functions of ω which remain unchanged by the two substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \qquad (\omega, \omega+2). \tag{1}$$

If, therefore, a system of ν functions $\Phi_{a,c,\partial}$ is only permuted among itself by the substitutions (1), then these functions are the roots of a transformation equation whose coefficients depend rationally on $f(\omega)^{24}$. For an odd n, which we shall now always suppose, this includes, as we have already shown before by another method, certain powers of the quantities

$$P_{a,c,\partial},$$

where, once for all, it is to be noted that c is supposed divisible by 16, and, if n is not divisible by 3, by 48.

The somewhat extended principles of § 70 lead comparatively simply to the calculation of these equations.

An extension is necessary for the following reason. In the preceding considerations we could conclude that, if a rational function of $j(\omega)$ remains finite for every finite ω with positive imaginary part, then it is an integral function of $j(\omega)$, since to every finite $j(\omega)$ there also belongs a finite non-real ω (§ 52). But for rational functions of $f(\omega)$ we can conclude from finiteness for every finite imaginary ω only that they are integral functions of $f(\omega)$ and $1/f(\omega)$, since only to every finite $f(\omega)$ except $f(\omega) = 0$ does there belong a finite imaginary ω.

To the substitution

$$\omega \mapsto \frac{\omega}{\omega+1} \tag{2}$$

there corresponds, however, the interchange

$$f(\omega) \mapsto \frac{2}{f(\omega)} \qquad [\S\, 34, (13)].$$

If, therefore, the functions $\Phi_{a,c,\partial}$ are so chosen that they too are only permuted among themselves by the substitution (2), then the coefficients of the equation whose roots they are depend rationally on

$$f(\omega)^{24} + \frac{2^{24}}{f(\omega)^{24}}.$$

They are integral functions of this combination if the $\Phi_{a,c,\partial}$ remain finite for every finite non-real ω; and they are constant if all $\Phi_{a,c,\partial}$ remain finite also for $q = 0$, that is, for $f(\omega) = 0$ and $f(\omega) = \infty$. In this case all the $\Phi_{a,c,\partial}$ are equal to one and the same constant.

Thus the following theorem results. Form integral rational functions $\Phi_{a,c,\partial}$ from

$$f\left(\frac{c+\partial\omega}{a}\right), \quad \frac{1}{f\left(\frac{c+\partial\omega}{a}\right)}, \quad f(\omega), \quad \frac{1}{f(\omega)},$$

which have the properties:

1. to be only permuted among themselves by the substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \qquad (\omega, \omega+2), \qquad \left(\omega, \frac{\omega}{\omega+1}\right);$$

2. not to become infinite for $q = 0$.

Then

$$\Phi_{a,c,\partial} = \text{constant}$$

is a transformation equation.

In order to satisfy these conditions, one must first investigate the effect of the substitutions (1) on the functions f. This follows from the compositions § 69, (8) to (13) and from the transformation formulae for the f-functions, § 40.

For abbreviation we introduce the notation

$$f(\omega) = u, \quad f_1(\omega) = u_1, \quad f_2(\omega) = u_2, \qquad f\Big(\frac{c+\partial\omega}{a}\Big) = v, \quad f_1\Big(\frac{c+\partial\omega}{a}\Big) = v_1, \quad f_2\Big(\frac{c+\partial\omega}{a}\Big) = v_2. \tag{3}$$

Then the corresponding changes are obtained, if in the notation no account is taken of the transformation of the numbers a, c, ∂ into a, c_1, ∂ or a_2, c_2, ∂_2, as characterized by the cited formulae of § 69:

substitution	u	u_1	u_2
ω	u	u_1	u_2
$-1/\omega$	u	u_2	u_1
$\omega+2$	u	$e^{\pi i/3}u_1$	$e^{\pi i/3}u_2$
$(\omega-1)/(\omega+1)$	$\lambda_1 u$	$\lambda_2 u_2$	$\lambda_3 u_1$

with analogous formulae for the v's. The factors λ_i are third roots of unity, and, when n is not divisible by 3, they are equal to 1.

In order further to determine the effect of the substitution (2), which arises from the transformation of second degree

$$\frac{\omega}{\omega+1},$$

on the functions u, v, we must determine the transformations of first and nth order so that

$$\frac{c+\partial\omega}{a} = \frac{\beta+\delta\omega'}{\alpha+\gamma\omega'}, \qquad \frac{c'+\partial'\omega'}{a'} \quad \text{with} \quad \omega' = \frac{\omega}{\omega+1}, \tag{5}$$

which leads to the equations

$$c' = \alpha c - \beta a + \gamma\partial + \delta c, \qquad a' = \alpha a + \beta c, \qquad \partial' = \gamma a + \delta\partial. \tag{6}$$

From these equations ∂' is first determined as the greatest common divisor of a and $c+\partial$; from $n = a'\partial'$ one obtains a', and then c' can be chosen divisible by 16, or by 48 if necessary. After applying the transformation formulae of § 40 one obtains the corresponding interchanges

$$\Big(\omega, \frac{\omega}{\omega+1}\Big): \qquad u \leftrightarrow \frac{2}{u}, \qquad v \leftrightarrow \frac{2}{v}, \tag{9}$$

up to third roots of unity which again are 1 when n is not divisible by 3.

We apply the interchanges (4), (5), (9) to the following functions:

$$A = \Big(\frac{u}{v}\Big)^r + \Big(\frac{v}{u}\Big)^r, \qquad B = (uv)^s + \Big(\frac{2}{n}\Big)^{r+s}\frac{2^s}{(uv)^s}, \tag{10}$$

where r, s are two integers satisfying

$$(n-1)r \equiv 0, \qquad (n+1)s \equiv 0 \pmod{12}, \tag{11}$$

so that, if n is divisible by 3, both are divisible by 3, and

$$\Big(\frac{2}{n}\Big) = (-1)^{(n^2-1)/8}.$$

Then the following corresponding interchanges result:

substitution	A	B
$-1/\omega$	A	B
$\omega+2$	$(-1)^{(n-1)r/12}A$	$(-1)^{(n+1)s/12}B$
$(\omega-1)/(\omega+1)$	$\left(\frac{2}{n}\right)^r A$	$\left(\frac{2}{n}\right)^s B$

(12)

If now from A, B one forms an integral rational function with numerical coefficients

$$\Phi_{a,c,\partial} = \sum M_{h,k} A^h B^k, \tag{13}$$

where, if sign changes occur in (12), the exponents h, k are to be arranged so that in all terms of (13) the same sign changes occur, then the system of functions $\Phi_{a,c,\partial}$, or at least $\Phi^2_{a,c,\partial}$, satisfies condition 1 of the theorem just stated. To satisfy condition 2 as well, and thus to obtain a transformation equation, the coefficients $M_{h,k}$ must be determined so that all ν values of $\Phi_{a,c,\partial}$ remain finite for $q = 0$.

This task is essentially simplified when n has no quadratic divisor, and still more when n is a prime number. If n has no quadratic divisor, all values $\Phi_{a,c,\partial}$ can be derived from $\Phi_{a,0,\partial}$ by increasing ω by certain integers; hence if, as required, no negative powers occur in the development of $\Phi_{a,0,\partial}$ in ascending powers of q, the same is true of all $\Phi_{a,c,\partial}$.

If, however, n is a prime number, it is enough to prove that $\Phi_{1,0,n}$ contains no negative powers of q, since by interchanging ω with ω/n the same follows for $\Phi_{n,0,1}$.

The constant value which the function $\Phi_{a,c,\partial}$ obtains is determined from one term of the development. In carrying out these calculations one uses the developments § 24, (11),

$$\begin{aligned} A &= q^{-(n-1)r/24} \prod_{h=1}^{\infty} \left(\frac{1+q^{n(2h-1)}}{1+q^{2h-1}} \right)^r + q^{(n-1)r/24} \prod_{h=1}^{\infty} \left(\frac{1+q^{2h-1}}{1+q^{n(2h-1)}} \right)^r, \\ B &= q^{-(n+1)s/24} \prod_{h=1}^{\infty} (1+q^{2h-1})^s (1+q^{n(2h-1)})^s \\ &\quad + \left(\frac{2}{n}\right)^{r+s} q^{(n+1)s/24} 2^s \prod_{h=1}^{\infty} \frac{1}{(1+q^{2h-1})^s (1+q^{n(2h-1)})^s}. \end{aligned} \tag{14}$$

The formulae are not always simplest when r, s are taken as small as possible; rather it proves most expedient to add the condition

$$\frac{(n-1)r}{12} \equiv \frac{(n+1)s}{12} \pmod 2. \tag{15}$$

For the first seven odd primes one thus obtains the following determination of A and B:

n	A	B
3	$\left(\frac{u}{v}\right)^6 + \left(\frac{v}{u}\right)^6$	$(uv)^3 - \frac{8}{(uv)^3}$
5	$\left(\frac{u}{v}\right)^3 + \left(\frac{v}{u}\right)^3$	$(uv)^2 - \frac{4}{(uv)^2}$
7	$\left(\frac{u}{v}\right)^4 + \left(\frac{v}{u}\right)^4$	$(uv)^3 + \frac{8}{(uv)^3}$
11	$\left(\frac{u}{v}\right)^6 + \left(\frac{v}{u}\right)^6$	$uv - \frac{2}{uv}$
13	$\frac{u}{v} + \frac{v}{u}$	$(uv)^6 - \frac{64}{(uv)^6}$
17	$\left(\frac{u}{v}\right)^3 + \left(\frac{v}{u}\right)^3$	$(uv)^4 + \frac{16}{(uv)^4}$
19	$\left(\frac{u}{v}\right)^2 + \left(\frac{v}{u}\right)^2$	$(uv)^3 - \frac{8}{(uv)^3}$.

The developments in powers of q for A and B follow from (14), as far as they are needed for the calculation, in this way:

$$\begin{aligned}
n=3:&\quad A=q^{-1/2}(1-6q+\cdots),\quad B=q^{-1/2}(1+\cdots);\\
n=5:&\quad A=q^{-1/2}(1-3q+\cdots),\quad B=q^{-1/2}(1+\cdots);\\
n=7:&\quad A=q^{-1}(1-4q+\cdots),\quad B=q^{-1}(1+3q+\cdots);\\
n=11:&\quad A=q^{-5/2}(1-6q+\cdots),\quad B=q^{-1/2}(1+q+\cdots);\\
n=13:&\quad A=q^{-1/2}(1-q+\cdots),\quad B=q^{-7/2}(1-6q+\cdots);\\
n=17:&\quad A=q^{-2}(1-3q+6q^2-13q^3+25q^4-39q^5+76q^6+\cdots),\\
&\quad B=q^{-3}(1+4q+4q^2+8q^3+14q^4+28q^5+49q^6+\cdots);\\
n=19:&\quad A=q^{-3/2}(1-2q-3q^2+5q^3+11q^4-13q^5+24q^6-28q^7+\cdots),\\
&\quad B=q^{-5/2}(1+3q+3q^2+4q^3+9q^4+4q^5+3q^6-27q^7+\cdots).
\end{aligned}$$

Eliminating the negative powers gives the desired equations between A and B:

$$\begin{aligned}
n=3:&\quad A+B=0,\\
n=5:&\quad A-B+1=0,\\
n=7:&\quad A^2-B+7=0,\\
n=11:&\quad A-B^5-572B=0,\\
n=13:&\quad B^2+A^7-24A-B=0,\\
n=17:&\quad A^3-B^2+17AB-544A^2+53B+116A-440=0,\\
n=19:&\quad A^5-B^3+19AB^2-95A^2B-109A^3+128B-128A=0.
\end{aligned}$$

These equations were first set up by Schlaefli. [22]

From this system of equations a second and a third system are derived for the functions f_1, f_2, by replacing ω first by $\omega+1$ and then by $-1/\omega$. These two systems have the same form; one time one must put

$$u=f_1(\omega),\qquad v=f_1\left(\frac{c+\partial\omega}{a}\right),$$

and the other time

$$u=f_2(\omega),\qquad v=f_2\left(\frac{c+\partial\omega}{a}\right).$$

[22] Journal fuer Mathematik, vol. 72.

From the interchanges (4) one obtains, in this way:

$$\text{II.}\quad n=3,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^6-\left(\tfrac{v_1}{u_1}\right)^6,\quad B_1=(u_1v_1)^3+\tfrac{8}{(u_1v_1)^3},$$
$$A_1+B_1=0,$$

$$n=5,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^3-\left(\tfrac{v_1}{u_1}\right)^3,\quad B_1=(u_1v_1)^2+\tfrac{4}{(u_1v_1)^2},$$
$$A_1+B_1=0,$$

$$n=7,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^4+\left(\tfrac{v_1}{u_1}\right)^4,\quad B_1=(u_1v_1)^3+\tfrac{8}{(u_1v_1)^3},$$
$$A_1-B_1-7=0,$$

$$n=11,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^6-\left(\tfrac{v_1}{u_1}\right)^6,\quad B_1=u_1v_1+\tfrac{2}{u_1v_1},$$
$$A_1+B_1^5+B_1^3-2B_1=0,$$

$$n=13,\quad A_1=\tfrac{u_1}{v_1}-\tfrac{v_1}{u_1},\quad B_1=(u_1v_1)^6+\tfrac{64}{(u_1v_1)^6},$$
$$A_1^7-6A_1^5+A_1^3+20A_1+B_1=0,$$

$$n=17,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^3+\left(\tfrac{v_1}{u_1}\right)^3,\quad B_1=(u_1v_1)^4+\tfrac{16}{(u_1v_1)^4},$$
$$A_1^3-B_1^2-17A_1B_1-34A_1^2-34B_1+116A_1+440=0,$$

$$n=19,\quad A_1=\left(\tfrac{u_1}{v_1}\right)^2-\left(\tfrac{v_1}{u_1}\right)^2,\quad B_1=(u_1v_1)^3+\tfrac{8}{(u_1v_1)^3},$$
$$A_1^5+B_1^3-19A_1B_1^2+95A_1^2B_1-109A_1^3+128B_1-128A_1=0.$$

§ 74. The form of the Schlaefli modular equations for a prime degree.

The form which we found in the preceding paragraph for the relations existing between

$$u=f(\omega),\qquad v=f\left(\frac{c+\partial\omega}{a}\right)\tag{1}$$

can, at least when the degree of transformation is a prime number p, easily be brought under a general law. Since in what follows much depends on this form, we shall enter here into somewhat more detail about it.

The determination of the numbers r, s according to (11) and (15) of the preceding paragraph depends on the behavior of p with respect to the modulus 24; and since we may exclude the case $p=3$, we have the following cases:

$p \pmod{24}$	r	s
1	1	12
5	3	2
7	4	3
11	6	1
13	1	6
17	3	4
19	2	3
23	12	1,

(2)

so that $r+s$ is always odd, and $p+1$ is divisible by $2r$, $p-1$ by $2s$. Accordingly

$$A=\left(\frac{u}{v}\right)^r+\left(\frac{v}{u}\right)^r,\qquad B=(uv)^s+\left(\frac{2}{p}\right)\frac{2^s}{(uv)^s}.\tag{3}$$

Now we know that the $p-1$ quantities v are the roots of an irreducible transformation equation

$$\Phi_p(v,u) = v^{p+1} + U_1 v^p + \cdots + U_{p+1} = 0, \tag{4}$$

in which the coefficients $U_1, \ldots, U_{p+1}$ are rational functions of u.

We conclude at once that they are integral rational functions of u. For first, when $u = \infty$, all roots of (4) become infinite, as is seen by putting $\omega = i\infty$, hence letting $q = 0$. Secondly, according to (9) of the preceding paragraph, by the interchange

$$\left(u, \frac{\sqrt{2}}{u}\right)$$

the totality of the roots v goes over into

$$\left(\frac{2}{p}\right)\frac{\sqrt{2}}{v}.$$

From this it follows that for $u = 0$ all roots v go over into zero, so that none of them becomes infinite. Hence one concludes not only that $U_1, U_2,$
$ldots, U_{p+1}$ are integral functions of u, but also that each of them must contain the factor u.

We conclude next, exactly as for the invariant equation (§ 69, 2., 3. and 6.), that

$$\Phi_p(v,u) = \Phi_p(u,v), \tag{5}$$

and that

$$\Phi_p(v,u) = (v^p - u)(v - u^p) + p\sum_{1,p}^{h,k} c_{h,k} u^h v^k, \tag{6}$$

where $c_{h,k} = c_{k,h}$ are integers.

We shall represent this function in the form

$$\Phi_p(v,u) = v^{p+1} + u^{p+1} + \sum_{1,p}^{h,k} a_{h,k} u^h v^k, \tag{7}$$

where therefore $a_{h,k} = a_{k,h}$ are likewise integers, whose divisibility by p is now no longer relevant, and in particular $a_{p,1} = -1$. If in the equation

$$\Phi_p[f(p\omega), f(\omega)] = 0$$

one replaces ω by $\omega + 2$, then, because of irreducibility and on the basis of the relation

$$f(\omega+2) = e^{-\pi i/12} f(\omega),$$

it follows that in (7) not all terms occur, but only those for which the exponents h, k satisfy the congruence

$$hp + k \equiv p + 1 \pmod{24}. \tag{8}$$

Now we know one more property of the functions $\Phi_p(v,u)$, which results from the interchange (9) of the preceding paragraph already used, where $\rho = 1$ is to be put; if, for abbreviation, we put

$$\varepsilon = \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8},$$

this property can be written

$$\Phi_p(v,u) = \left(\frac{uv}{\sqrt{2}}\right)^{p+1} \Phi_p\left(\frac{\varepsilon\sqrt{2}}{v}, \frac{\sqrt{2}}{u}\right). \tag{9}$$

From this one obtains the relations

$$\varepsilon^h 2^{(h+k-p-1)/2} a_{p+1-h,p+1-k} = a_{h,k}, \qquad a_{h,k} = a_{k,h}. \tag{10}$$

If, therefore, we combine in (7) the terms with equal coefficients $a_{h,k}$, we may restrict ourselves to the assumption that $h \geq k$, $h+k \geq p+1$; and, apart from the two terms v^{p+1} and u^{p+1}, Φ_p appears as an aggregate of terms of the following two forms (taking into account also that, by (8), $\varepsilon^h = \varepsilon^k$):

$$\begin{aligned} &v^h u^k + v^k u^h - \varepsilon^h 2^{(h+k-p-1)/2}(u^{p+1-h}v^{p+1-k} + u^{p+1-k}v^{p+1-h}) \\ &\quad = (uv)^{(p+1)/2}\left[\left(\frac{v}{u}\right)^{(h-k)/2} + \left(\frac{u}{v}\right)^{(h-k)/2}\right]\left[(uv)^{(h+k-p-1)/2} + \frac{\varepsilon^h 2^{(h+k-p-1)/2}}{(uv)^{(h+k-p-1)/2}}\right], \end{aligned} \tag{11}$$

and

$$\begin{aligned} &v^h u^h + \varepsilon^h 2^{h-(p+1)/2}(uv)^{p+1-h} \\ &\quad = (uv)^{(p+1)/2}\left[(uv)^{h-(p+1)/2} + \frac{\varepsilon^h 2^{h-(p+1)/2}}{(uv)^{h-(p+1)/2}}\right]. \end{aligned} \tag{12}$$

The coefficients of these terms in Φ_p are integers.

Now (8) gives

$$h-k \equiv (h-1)(p+1), \qquad h+k-p-1 \equiv -h(p-1) \pmod{24},$$

and hence $h-k$ is divisible by $2r$, and $h+k-p-1$ (where h may also be equal to k) by $2s$ [§ 74, (2)].

If, therefore, we put

$$\frac{h-k}{2} = r\alpha, \qquad \frac{h+k-p-1}{2} = s\beta,$$

then α and β are positive integers which lie between the bounds

$$0 \leq \alpha < \frac{p+1}{2r}, \qquad 0 \leq \beta \leq \frac{p-1}{2s}, \tag{13}$$

and in addition the congruence following from (8), $-h(p-1) \equiv 2s\beta \pmod{24}$, shows that at least in the cases where $\varepsilon = -1$ one has $h \equiv \beta \pmod 2$ [§ 74, (2)], and hence always $\varepsilon^h = \varepsilon^\beta$.

Thus (11) becomes

$$(uv)^{(p+1)/2}\left[\left(\frac{v}{u}\right)^{r\alpha} + \left(\frac{u}{v}\right)^{r\alpha}\right]\left[(uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}}\right],$$

and (12) becomes

$$(uv)^{(p+1)/2}\left[(uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}}\right].$$

Now the well-known formulae hold, when x, y are arbitrary quantities and

$$x + \frac{\gamma}{x} = y$$

is put:

$$x^2 + \frac{\gamma^2}{x^2} = y^2 - 2\gamma, \qquad x^3 + \frac{\gamma^3}{x^3} = y^3 - 3\gamma y, \ldots,$$

$$\left(x^n + \frac{\gamma^n}{x^n}\right)\left(x + \frac{\gamma}{x}\right) = x^{n+1} + \frac{\gamma^{n+1}}{x^{n+1}} + \gamma\left(x^{n-1} + \frac{\gamma^{n-1}}{x^{n-1}}\right),$$

from which one recognizes, by induction from n to $n+1$, that

$$x^n + \frac{\gamma^n}{x^n}$$

can, for arbitrary n, be represented as an integral rational function of degree n in y, which, if γ is an integer, has integral coefficients and leading coefficient 1.

The two quantities

$$\left(\frac{v}{u}\right)^{r\alpha} + \left(\frac{u}{v}\right)^{r\alpha} \quad \text{and} \quad (uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}}$$

can therefore be represented in this way as integral rational functions of A and B of degrees α and β, and we find, if we also separate off the term corresponding to $\beta = (p-1)/(2s)$, which has coefficient -1, and denote integral coefficients by $c_{\alpha,\beta}$:

$$\frac{\Phi_p(u,v)}{(uv)^{(p+1)/2}} = A^{(p+1)/(2r)} - B^{(p-1)/(2s)} + \sum c_{\alpha,\beta} A^\alpha B^\beta, \tag{14}$$

where α and β are bound by the limits

$$0 \leq \alpha < \frac{p+1}{2r}, \qquad 0 \leq \beta < \frac{p-1}{2s}.$$

This is the form which in the preceding paragraph we gave to the modular equations up to $p = 19$.

It remains to mention that the consideration carried out in § 69, 4., 5. for the invariant equation in the case of composite transformation degree applies unchanged also to the Schlaefli modular equations. Hence we may conclude that all these equations have rational numerical coefficients.

§ 75. The irrational forms of the modular equations.

By applying the same procedure, much simpler forms can be given to the transformation equations. The equation forms with which we shall now be concerned contain the three functions f, f_1, f_2 simultaneously; but since, by § 34, (11), (12), two of these functions can be expressed by the third, two of them can be eliminated, and one can thus arrive at the equations of the preceding paragraph. If one substitutes for f_1, f_2 their expressions through f, or for the three functions f, f_1, f_2 their expressions through k^2, radical signs occur; from this the name of these equations is explained.

We put, as above,

$$\begin{aligned} u &= f(\omega), & v &= f\left(\frac{c+\partial\omega}{a}\right), \\ u_1 &= f_1(\omega), & v_1 &= \left(\frac{2}{a}\right) f_1\left(\frac{c+\partial\omega}{a}\right), \\ u_2 &= f_2(\omega), & v_2 &= \left(\frac{2}{\partial}\right) f_2\left(\frac{c+\partial\omega}{a}\right), \\ uu_1u_2 &= \sqrt{2}, & vv_1v_2 &= \left(\frac{2}{n}\right)\sqrt{2}, \end{aligned} \tag{1}$$

and obtain the following associated interchanges:

$$\begin{array}{c|ccc} \omega & uv & u_1v_1 & u_2v_2 \\ -1/\omega & \varrho uv & \varrho u_2v_2 & \varrho u_1v_1 \\ \omega+1 & e^{-(n+1)\pi i/24}\sigma u_1v_1 & e^{-(n+1)\pi i/24}\sigma uv & e^{(n+1)\pi i/12}\sigma u_2v_2, \end{array} \tag{2}$$

where ϱ, σ are the third roots of unity defined above in § 73, (4).

We distinguish three different cases according to the behavior of n modulo 8.

1. $n+1 \equiv 0 \pmod 8$.

We put

$$\begin{aligned} 2A &= uv + (-1)^{(n+1)/8}(u_1v_1 + u_2v_2), \\ B &= uvu_1v_1 + uvu_2v_2 + (-1)^{(n+1)/8}u_1v_1u_2v_2 \\ &= \frac{2}{u_1v_1} + \frac{2}{u_2v_2} + (-1)^{(n+1)/8}\frac{2}{uv}, \end{aligned} \tag{3}$$

so that the associated interchanges become

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho A & \varrho^2 B \\ \omega+1 & e^{\pi i(n+1)/12}\sigma A & e^{-\pi i(n+1)/12}\sigma^2 B. \end{array} \tag{4}$$

A product of the form A^hB^k therefore takes, under the two interchanges

$$\left(\omega, -\frac{1}{\omega}\right), \qquad (\omega, \omega+1),$$

the factors

$$\varrho^{h-k}, \qquad e^{(h-k)(n+1)\pi i/12}\sigma^{h-k},$$

which are equal to 1 when $n+1$ is divisible by 3. We therefore now form, with numerical coefficients $M_{h,k}$, functions of the form

$$\Phi_{a,c,\partial} = \sum M_{h,k}A^hB^k, \tag{5}$$

where, if $n \equiv 0$ or $1 \pmod 3$, h and k may assume only integral values whose differences $h-k$ leave the same residue on division by 3, while if $n \equiv -1 \pmod 3$ they are not subject to this restriction. Such a function itself, or at least its third power, therefore satisfies a transformation equation of the first stage. It also remains finite for all finite values of $j(\omega)$, and consequently is an integral algebraic function of $j(\omega)$. [The transformation of the second order $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ need not be brought in here, since the totality of the $\Phi_{a,c,\partial}$ remains unchanged already under the substitution $(\omega, \omega+1)$, not only under $(\omega, \omega+2)$. Compare the remark at the beginning of § 73.]

If, therefore, we determine the constants in $\Phi_{a,c,\partial}$ so that no negative powers of q occur in the expansions of these functions, then all the $\Phi_{a,c,\partial}$ must be equal to one and the same constant.

To reach this goal it is sufficient here also, if n has no square divisor, that $\Phi_{1,0,n}$, and if n is a prime number, that $\Phi_{1,0,n}$, remain finite for $q = 0$.

In these calculations we use the expansions

$$\begin{aligned} uv &= q^{-(n+1)/24} \prod (1 + q^{n(2h+1)})(1 + q^{2h+1}), \\ u_1 v_1 &= q^{-(n+1)/24} \prod (1 - q^{n(2h+1)})(1 - q^{2h+1}), \\ u_2 v_2 &= 2q^{(n+1)/12} \prod (1 + q^{2hn})(1 + q^{2h}), \end{aligned} \tag{6}$$

from which, by expansion in powers of q,

$$\begin{aligned} uv &= q^{-(n+1)/24}(1 + q + q^3 + q^4 + q^5 + q^6 - q^7 + 2q^8 \cdots), \\ u_1 v_1 &= q^{-(n+1)/24}(1 - q - q^3 + q^4 - q^5 + q^6 - q^7 + 2q^8 \cdots), \\ u_2 v_2 &= 2q^{(n+1)/12}(1 + q^2 + q^4 + 2q^6 + 2q^8 + \cdots). \end{aligned} \tag{7}$$

The latter formulae are correct for $n > 7$; for $n = 3, 5, 7$ the terms from q^3, q^5, q^7 on are to be modified. For $n = 7$, $n = 23$ these expressions show that A itself remains finite for $q = 0$, whence for these cases the equations follow:

$$n = 7, \quad A = 0, \qquad n = 23, \quad A = 1. \tag{8}$$

By a simple calculation one further finds:

$$\begin{aligned} n &= 31, \quad (A^2 - B)^2 - A = 0, \\ n &= 47, \quad A^2 - A - B = 2, \\ n &= 71, \quad A^3 - 4A^2 + 2A - B = 1. \end{aligned} \tag{9}$$

The calculation is not quite so simple for composite n. For example, for $n = 15$ one must take account of the condition of finiteness for $q = 0$ not only for $\Phi_{1,0,15}$, but also for $\Phi_{3,0,5}$. These two, however, are sufficient. One obtains:

$$n = 15, \qquad A^3 - AB + 1 = 0. \tag{10}$$

2. If $n \equiv 3 \pmod 8$, the same conclusions are to be drawn if we put

$$\begin{aligned} 4A &= u^2 v^2 - u_1^2 v_1^2 - u_2^2 v_2^2, \\ B &= u^2 v^2 u_1^2 v_1^2 + u^2 v^2 u_2^2 v_2^2 - u_1^2 v_1^2 u_2^2 v_2^2, \end{aligned} \tag{11}$$

for which one obtains from (2) the associated interchanges

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho^2 A & \varrho B \\ \omega + 1 & e^{(n+1)\pi i/6} \sigma^2 A & e^{-(n+1)\pi i/6} \sigma B, \end{array} \tag{12}$$

and from this, in the same way, one derives the equations

$$n = 3, \quad A = 0, \qquad n = 11, \quad A = 1, \qquad n = 19, \quad A^5 - 7A^2 - B = 0. \tag{13}$$

3. If $n \equiv 1 \pmod 4$, the same procedure can be followed with the functions

$$\begin{aligned} 8A &= u^4v^4 - u_1^4v_1^4 - u_2^4v_2^4, \\ B &= u^4v^4u_1^4v_1^4 + u^4v^4u_2^4v_2^4 - u_1^4v_1^4u_2^4v_2^4, \end{aligned} \tag{14}$$

for which one obtains the associated interchanges

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho A & \varrho^2 B \\ \omega+1 & e^{(n+1)\pi i/3}\sigma A & e^{-(n+1)\pi i/3}\sigma^2 B. \end{array} \tag{15}$$

Only the first case, $n = 5$, leads here to a simple result:

$$n = 5, \qquad A = 1. \tag{16}$$

For $n = 5$ an even simpler form of the transformation equation can be obtained. If we apply the substitutions $(\omega, -1/\omega)$ and $(\omega, \omega+1)$ to the three functions

$$w = \frac{u_1^2v_2^2 + u_2^2v_1^2}{uv}, \qquad w_1 = \frac{u_2^2v^2 - v_2^2u^2}{u_1v_1}, \qquad w_2 = \frac{u_1^2v^2 - v_1^2u^2}{u_2v_2}, \tag{17}$$

then, under the assumption $n = 5 \pmod 8$, the associated interchanges become

$$\begin{array}{c|ccc} \omega & w & w_1 & w_2 \\ -1/\omega & w & w_2 & w_1 \\ \omega+1 & e^{-\pi i(n-5)/24}w_1 & e^{-\pi i(n-5)/24}w & e^{\pi i(n-5)/12}w_2. \end{array}$$

Thus, if one sets

$$\begin{aligned} A &= w^2 + w_1^2 + w_2^2, \\ B &= w^2w_1^2 + w^2w_2^2 + w_1^2w_2^2, \\ C &= ww_1w_2, \end{aligned}$$

one obtains

$$\begin{array}{c|ccc} \omega & A & B & C \\ -1/\omega & A & B & C \\ \omega+1 & e^{-\pi i(n-5)/12}A & e^{-\pi i(n-5)/6}B & C, \end{array}$$

so that two of these functions can be used just as A and B above. But for $n = 5$ it appears that no negative powers occur in the expansions of w, w_1, w_2 in powers of q, and therefore that A, B, C, and consequently also w, w_1, w_2 themselves, are constant.

Thus the modular equation for $n = 5$ can be put in each of the three forms

$$\begin{aligned} u_1^2v_2^2 + u_2^2v_1^2 &= 2uv, \\ u_2^2v^2 - v_2^2u^2 &= 2u_1v_1, \\ u_1^2v^2 - v_1^2u^2 &= 2u_2v_2. \end{aligned} \tag{18}$$

These equations can also be derived from the equation given by Jacobi (Fund. art. 30, collected works, vol. 1, p. 123).

We close these considerations by deriving, in the simplest cases, the Jacobian modular equations from these irrational forms.

We put [cf. § 54, (3)]

$$\frac{u_2}{u} = x = \sqrt[4]{k'}, \qquad \frac{u_1}{u} = x' = \sqrt[4]{k''}, \qquad \frac{v_2}{v} = y = \sqrt[4]{\lambda'}, \qquad \frac{v_1}{v} = y' = \sqrt[4]{\lambda''}, \tag{19a}$$

and eliminate the quantities u and v by means of the relations

$$x^8 + x'^8 = 1, \qquad xx' = \frac{\sqrt{2}}{u^3}, \qquad y^8 + y'^8 = 1, \qquad yy' = \frac{\sqrt{2}}{v^3}. \tag{19}$$

For $n = 3$ one obtains from (13)

$$x^2y^2 + x'^2y'^2 = 1, \tag{20}$$

a form of the modular equation which is due to Legendre. From it, by inserting for x', y' the values from (19),

$$x^8 + y^8 = 4x^2y^2 - 6x^4y^4 + 4x^6y^6,$$

or

$$(x^4 - y^4)^2 = 4(xy - x^3y^3)^2,$$

and then extracting the root and determining the sign by $q = 0$, one obtains

$$x^4 - y^4 = 2xy - 2x^3y^3, \qquad (n = 3), \tag{21}$$

which, apart from the sign of $\sqrt[4]{\lambda}$ not explained more fully there, agrees with § 10, (12).

For $n = 5$ the second equation (18) gives

$$(x^2 - y^2)^3 = 4xyx'^4y'^4,$$

and hence, by squaring,

$$(x^2 - y^2)^6 = 16x^2y^2(1 - x^8 - y^8 + x^8y^8),$$

which is easily brought into the form

$$(x^2 - y^2)^2[(x^2 - y^2)^2 + 8x^2y^2]^2 = 16x^2y^2(1 - x^4y^4)^2.$$

Extracting the root gives, as above,

$$x^6 - y^6 - 4xy(1 - x^4y^4) + 5x^2y^2(x^2 - y^2) = 0 \qquad (n = 5). \tag{22}$$

For $n = 7$, without extracting roots, one obtains from (8):

$$xy + x'y' = 1, \tag{23}$$

and from this

$$\begin{aligned} x^8 + y^8 - 8xy(1 + x^6y^6) + 28x^2y^2(1 + x^4y^4) \\ - 56x^3y^3(1 + x^2y^2) + 70x^4y^4 = 0 \qquad (n = 7). \end{aligned} \tag{24}$$

§ 76. Composite transformation degrees.

If the transformation degree is a composite number, then one can set up still simpler transformation equations than those obtained by the path of the preceding paragraph.

We carry out these considerations here only in the simplest cases.

Let the odd transformation degree n be decomposed into two factors

$$n = n'n'', \tag{1}$$

which are relatively prime to one another.

To each transformation of degree n of the form

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \tag{2}$$

there can then be assigned one and only one transformation of the degrees n', n'':

$$\begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \qquad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}, \tag{3}$$

determined by the following conditions:

$$a = a'a'', \qquad \partial = \partial'\partial'', \qquad \partial''c' \equiv c \pmod{a'}, \qquad \partial'c'' \equiv c \pmod{a''}, \tag{4}$$

and conversely, from every pair of transformations of the form (3), according to (4), one and only one transformation (2) follows. For by (4) ∂', ∂'' are first determined as the greatest common divisors of ∂ with n' and n''; then c' is determined modulo a', and c'' modulo a'', from the two last congruences (4).

It is now chiefly important to show that the congruences (4) remain preserved when the transformations (2) and (3), according to § 69, (8) to (12), are transformed by the two linear transformations

$$(\omega, \omega + 1), \qquad \left(\omega, -\frac{1}{\omega}\right).$$

According to the formulae just mentioned we set:

$$\begin{aligned}
\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ c_1 & \partial \end{pmatrix}, \\
\begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda' & 1 \end{pmatrix}\begin{pmatrix} a' & 0 \\ c_1' & \partial' \end{pmatrix}, \\
\begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda'' & 1 \end{pmatrix}\begin{pmatrix} a'' & 0 \\ c_1'' & \partial'' \end{pmatrix},
\end{aligned} \tag{5}$$

$$\begin{aligned}
\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}\begin{pmatrix} a_2 & 0 \\ c_2 & \partial_2 \end{pmatrix}, \\
\begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}\begin{pmatrix} a_2' & 0 \\ c_2' & \partial_2' \end{pmatrix}, \\
\begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha'' & \beta'' \\ \gamma'' & \delta'' \end{pmatrix}\begin{pmatrix} a_2'' & 0 \\ c_2'' & \partial_2'' \end{pmatrix}.
\end{aligned} \tag{6}$$

First, by § 69, (9), one has

$$c_1 \equiv c + \partial \pmod{a}, \qquad c_1' \equiv c' + \partial' \pmod{a'}, \qquad c_1'' \equiv c'' + \partial'' \pmod{a''},$$

whence, by (4),

$$\begin{aligned}
\partial''c_1' &\equiv \partial''c' + \partial \equiv c_1 \pmod{a'}, \\
\partial'c_1'' &\equiv \partial'c'' + \partial \equiv c_1 \pmod{a''},
\end{aligned}$$

in agreement with the congruences (4).

For the composition (6), according to § 69, (12), $\partial_2, \partial_2', \partial_2''$ are the greatest common divisors of a, c; of a', c'; and of a'', c''; since however ∂'' is relatively prime to a' and $\partial''c' \equiv c \pmod{a'}$, ∂_2' is also the greatest common divisor of a' and c, and for the same reasons ∂_2'' is the greatest common divisor of a'' and c. It follows that $\partial_2', \partial_2''$ are relatively prime:

$$\partial_2 = \partial_2'\partial_2'', \qquad a_2 = a_2'a_2''.$$

Further, by § 69, (11), (12), (13),

$$\partial_2\alpha = -c\gamma - a, \qquad c_2 = -\partial\alpha,$$

$$\partial_2'\alpha' = \alpha' c' - \partial' a', \qquad c_2' = -\partial'\alpha',$$
$$\partial_2''\alpha'' = \alpha'' c'' - \partial'' a'', \qquad c_2'' = -\partial''\alpha'',$$

and hence, by (4),

$$\partial_2' c_2'' - c_2\partial_2'' \equiv \alpha\alpha'(\partial c' - c\partial') \equiv 0 \pmod{a'},$$

therefore also

$$\partial_2' c_2'' - c_2\partial_2'' \equiv 0 \pmod{a_2'},$$

in agreement with (4); and in the same way

$$\partial_2'' c_2' - c_2\partial_2' \equiv 0 \pmod{a_2''}.$$

Thus the proof is completed that the association of the transformations expressed by (4),

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}, \quad \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix},$$

is not disturbed by applying any linear transformation to ω. Since here we suppose n odd, we can always assume c, c', c'' divisible by 16; and if n, and consequently also n', n'', are not divisible by 3, then c, c', c'' can also be assumed divisible by 3. But if n is divisible by 3, then one of the two factors n', n'', say n'', will be divisible by 3, the other n' not. Then c may still be assumed divisible by 3, but not c' and c''. In this case the dependence of c'' on c is to be determined more precisely by the congruence

$$\partial' c'' \equiv c \pmod{3a''}. \tag{7}$$

A solution of this congruence can always be derived from a solution of the congruence (4), $\partial' c'' \equiv c \pmod{a''}$, by adding to c'' a multiple of a''.

The congruence (7) then has, by § 69, (9) to (13), the consequence

$$\lambda \equiv \lambda'\lambda'' \pmod 3, \tag{8}$$

$$\alpha \equiv \alpha''\partial_2', \quad \beta \equiv \beta'' a_2', \quad \gamma \equiv \gamma'' a_2', \quad \delta \equiv \delta''\partial_2' \pmod 3. \tag{9}$$

Let now $v', v_1', v_2'; v'', v_1'', v_2''$ have the same meaning for the numbers n', n'' that v, v_1, v_2 had in § 73, (3) for the number n, namely

$$\begin{aligned} v' &= f\left(\frac{c' + \partial'\omega}{a'}\right), & v'' &= f\left(\frac{c'' + \partial''\omega}{a''}\right), \\ v_1' &= \left(\frac{2}{a'}\right) f_1\left(\frac{c' + \partial'\omega}{a'}\right), & v_1'' &= \left(\frac{2}{a''}\right) f_1\left(\frac{c'' + \partial''\omega}{a''}\right), \\ v_2' &= \left(\frac{2}{\partial'}\right) f_2\left(\frac{c' + \partial'\omega}{a'}\right), & v_2'' &= \left(\frac{2}{\partial''}\right) f_2\left(\frac{c'' + \partial''\omega}{a''}\right). \end{aligned} \tag{10}$$

We apply the interchange table (4), § 73, to these functions. Since n' is in all circumstances not divisible by 3, the cubic roots of unity ϱ', σ' are to be set equal to 1, while by the congruences (8), (9)

$$\varrho'' = \varrho^{n'}, \qquad \sigma'' = \sigma^{n'} \tag{11}$$

results. Thus we obtain the following associated interchanges:

$$\begin{array}{c|ccc} \omega & v' & v_1' & v_2' \\ -1/\omega & v' & v_2' & v_1' \\ \omega+1 & e^{-n'\pi i/24} v_1' & e^{-n'\pi i/24} v' & e^{n'\pi i/12} v_2' \\ \hline \omega & v'' & v_1'' & v_2'' \\ -1/\omega & \varrho^{n'} v'' & \varrho^{n'} v_2'' & \varrho^{n'} v_1'' \\ \omega+1 & \sigma^{n'} e^{-n''\pi i/24} v_1'' & \sigma^{n'} e^{-n''\pi i/24} v'' & \sigma^{n'} e^{n''\pi i/12} v_2'', \end{array} \tag{12}$$

which hold together with the interchanges (4), § 73.

We now distinguish two cases.

1. If

$$(n'+1)(n''+1) = 8\mu \equiv 0 \pmod 8, \tag{13}$$

then we put

$$U = uv'v'', \qquad U_1 = u_1 v_1' v_1'', \qquad U_2 = u_2 v_2' v_2'', \tag{14}$$

$$\begin{aligned} 2A &= U + (-1)^\mu (U_1 + U_2), \\ B &= UU_1 + UU_2 + (-1)^\mu U_1 U_2 \\ &= \frac{4}{U_1} + \frac{4}{U_2} + (-1)^\mu \frac{4}{U}. \end{aligned} \tag{15}$$

From (12) and § 73, (4), we then obtain the following associated interchanges:

$$\begin{array}{r|cc} \omega & A & B \\ -1/\omega & \varrho^{n'+1} A & \varrho^{-(n'+1)} B \\ \omega+1 & \sigma^{n'+1} e^{2\mu\pi i/3} A & \sigma^{-(n'+1)} e^{-2\mu\pi i/3} B. \end{array} \tag{16}$$

We apply our principle for deriving modular equations to these functions, and add the following remarks.

The third roots of unity occurring in (16) are equal to 1 if either n is not divisible by 3 and μ is divisible by 3, or n'' is divisible by 3 and n' leaves the residue 2. In these cases every rational function of A and B is a root of a transformation equation. In the other cases this property belongs to the cube of such a rational function of A and B in which the differences of the exponents of all terms are congruent to one another modulo 3.

If these rational functions are whole functions, and if in addition all their values are finite for $q = 0$, then they must be equal to a constant, and thereby we obtain transformation equations.

If n', n'' are prime numbers, then it is sufficient here also, as in § 72, that the negative powers of q disappear in the expansion of such a function in increasing powers of q in the one principal case, namely the case in which

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}, \quad \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}$$

are equal to

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & n' \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & n'' \end{pmatrix},$$

that is,

$$U = f(\omega) f(n'\omega) f(n''\omega) f(n'n''\omega);$$

for from the expansion for this one case the expansions for the remaining cases can be derived by replacing ω by

$$\frac{\omega}{n}, \quad \frac{\omega}{n'}, \quad \frac{\omega}{n''}, \tag{17}$$

and then also increasing ω by whole numbers, whereby no negative powers of q can be newly introduced.

For carrying out the calculation one uses the expansions

$$\begin{aligned} U &= q^{-(n'+1)(n''+1)/24} \prod (1+q^{2h-1})(1+q^{(2h-1)n'})(1+q^{(2h-1)n''})(1+q^{(2h-1)n}), \\ U_1 &= q^{-(n'+1)(n''+1)/24} \prod (1-q^{2h-1})(1-q^{(2h-1)n'})(1-q^{(2h-1)n''})(1-q^{(2h-1)n}), \\ U_2 &= 4q^{(n'+1)(n''+1)/12} \prod (1+q^{2h})(1+q^{2hn'})(1+q^{2hn''})(1+q^{2hn}), \end{aligned} \tag{18}$$

which, in the individual cases, give the power expansions of A, B, from which the negative powers of q are to be eliminated. In this way one calculates very simply the following equations:

$$\begin{array}{ll} n = 15, & A = 1, \\ n = 21, & (A^2 - B)^2 - A = 0, \\ n = 33, & A^2 - B - A = 4, \\ n = 35, & A^2 - B - A = 2, \\ n = 55, & A^3 - B - 4A^2 - A + 4 = 0. \end{array} \tag{19}$$

2. If

$$(n' - 1)(n'' - 1) = 8\mu \equiv 0 \pmod 8,$$

then we put

$$\begin{aligned} A &= \frac{uv}{v'v''} + (-1)^\mu \left(\frac{u_1 v_1}{v_1' v_1''} + \frac{u_2 v_2}{v_2' v_2''} \right), \\ B &= \frac{v'v''}{uv} + (-1)^\mu \left(\frac{v_1' v_1''}{u_1 v_1} + \frac{v_2' v_2''}{u_2 v_2} \right), \end{aligned} \tag{20}$$

and obtain from (12) the associated interchanges

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho^{1-n'} A & \varrho^{n'-1} B \\ \omega + 1 & \sigma^{1-n'} e^{2\mu\pi i/3} A & \sigma^{n'-1} e^{-2\mu\pi i/3} B. \end{array} \tag{21}$$

Therefore we can apply the same procedure as above, if we add the restriction that only symmetric functions of A and B, that is, rational functions of AB and $A + B$, are used; for only under this assumption do all remaining values of such a function follow from one of its values by the interchanges (17). One easily calculates the following examples:

$$\begin{array}{ll} n = 15, & AB + 1 = 0, \\ n = 35, & 2(A + B) - AB = 5, \\ n = 39, & 2(A + B) - AB = 3. \end{array} \tag{22}$$

The calculation presents no insurmountable difficulties even in still more complicated cases. Thus, in the paper in *Acta mathematica*, vol. II, p. 359, for the case $n = 105$, I communicated the following formula:

$$A = \sum \frac{f(3\omega)f(5\omega)f(7\omega)f(105\omega)}{f(\omega)f(15\omega)f(21\omega)f(35\omega)},$$

$$B = \sum \frac{f(\omega)f(15\omega)f(21\omega)f(35\omega)}{f(3\omega)f(5\omega)f(7\omega)f(105\omega)},$$

$$\begin{aligned} A^2 + B^2 - 4(A + B)^3 &+ 10AB(A + B) \\ &+ 4(A + B)^2 + 10AB + 14(A + B) + 5 = 0, \end{aligned}$$

where the sums $\sum$ in A and B refer to the three functions f, f_1, f_2.

§ 77. Geometric interpretation of the irrational modular equations as modular correspondences.

We shall make the content of the irrational forms of the modular equations more vivid by a geometric interpretation. First consider the case $n \equiv -1 \pmod 8$ and put

$$x = f(\omega), \qquad y = f_1(\omega), \qquad z = f_2(\omega). \tag{1}$$

Then, if x, y, z are regarded as Cartesian coordinates of a point P, a space curve is represented, which we may also express by the two equations

$$x^8 - y^8 - z^8 = 0, \qquad xyz = \sqrt{2}, \tag{2}$$

and which we shall call the fundamental curve. This curve is therefore of order 24. It can be mapped onto a plane curve of order 3 if one puts $x^8 = X$, $y^8 = Y$:

$$XY(X - Y) = 16. \tag{3}$$

Putting [§ 75, (1)]

$$\xi = f\left(\frac{c + \delta\omega}{a}\right), \qquad \eta = \left(\frac{2}{\delta}\right) f_1\left(\frac{c + \delta\omega}{a}\right), \qquad \zeta = \left(\frac{2}{a}\right) f_2\left(\frac{c + \delta\omega}{a}\right), \tag{4}$$

the quantities ξ, η, ζ also satisfy the equations (2), since

$$\left(\frac{2}{a}\right)\left(\frac{2}{\delta}\right) = \left(\frac{2}{n}\right) = \pm 1,$$

and the point Π whose coordinates are ξ, η, ζ therefore also lies on the fundamental curve.

An equation

$$\Phi(x, y, z, \xi, \eta, \zeta) = 0$$

means, when the point P is held fixed, a surface on which Π is to lie, and the intersection of this surface with the fundamental curve gives a certain number of points Π corresponding to the point P. Likewise a certain number of points P correspond to a fixed point Π, and this assignment is called a correspondence on the fundamental curve. If the point P coincides with one of the points Π corresponding to it, we obtain a double point or coincidence of the correspondence.

If equation (5) is not changed when P and Π are interchanged, the correspondence is reciprocal. Such correspondences are given by the equations (8), (9), (10), § 75.

Then by § 75, (3), one has to put

$$\begin{aligned} 2A &= x\xi + (-1)^{(n+1)/8}(y\eta + z\zeta), \\ B &= x\xi y\eta + x\xi z\zeta + (-1)^{(n+1)/8} y\eta z\zeta. \end{aligned} \tag{5}$$

A is of the first degree, and B of the second degree, with respect to the coordinates of each of the two points P and Π.

We shall determine the degree of these correspondences, that is, the number of points Π corresponding to one point P. For this purpose we must multiply the degree of the equation $\Phi = 0$ with respect to ξ, η, ζ by the degree of the fundamental curve, namely by 24, and one obtains:

$$\begin{array}{r|l} \text{for } n = 7, & m = 24, \\ n = 23, & m = 24, \\ n = 31, & m = 96, \\ n = 47, & m = 48, \\ n = 71, & m = 72, \\ n = 15, & m = 72. \end{array}$$

In the cases $23, 47, 71$, therefore, the degree of the correspondence is equal to the degree of the corresponding modular equation, that is, equal to the number of the transformations in question; in the cases $n = 7, 31, 15$ the degree is three times the degree of the modular equations.

What is the significance of these excess points? They are explained by the fact that in these cases, in which $n \equiv 0, 1 \pmod 3$, in the function $\Phi_{a,c,\delta}$ [§ 75, (5)], which according to the above equation (5) is equal to a constant, the exponents $h - 2k$ in all terms have the same residue modulo 3, and hence the equation $\Phi = 0$, and likewise the equations (2), remain satisfied when ξ, η, ζ are multiplied by any third root of unity. Thus every three points of the correspondence give the same transformation.

We might also, when x, y, z are determined by (1), have put instead of (4)

$$\xi = f\left(\frac{c+\delta\omega}{a}\right), \qquad \eta = \left(\frac{2}{\delta}\right) f_2\left(\frac{c+\delta\omega}{a}\right), \qquad \zeta = \left(\frac{2}{a}\right) f_1\left(\frac{c+\delta\omega}{a}\right), \tag{6}$$

and consequently for A, B:

$$\begin{aligned} 2A &= x\xi + (-1)^{(n+1)/8}(y\zeta + z\eta), \\ B &= x\xi y\zeta + x\xi z\eta + (-1)^{(n+1)/8} y\eta z\zeta. \end{aligned} \tag{7}$$

The fundamental curve and the degree of the correspondence would then have remained the same, but the coincidences would have acquired another, and indeed a simpler, meaning. In both cases the coincidences belong to the singular values of the modular functions, in which ω is an imaginary quadratic irrationality, as we shall see more precisely later.

In the case $n \equiv 3 \pmod 8$ we put

$$x = f^2(\omega), \qquad y = f_1^2(\omega), \qquad z = f_2^2(\omega),$$

$$\xi = f^2\left(\frac{c+\delta\omega}{a}\right), \qquad \eta = f_1^2\left(\frac{c+\delta\omega}{a}\right), \qquad \rho = f_2^2\left(\frac{c+\delta\omega}{a}\right), \tag{8}$$

and obtain a fundamental curve

$$x^4 - y^4 - z^4 = 0, \qquad xyz = 2, \tag{9}$$

which is only of order 12. We then have, further, by § 75,

$$\begin{aligned} 4A &= x\xi - y\eta - z\rho, \\ B &= x\xi y\eta + x\xi z\rho - y\eta z\rho, \end{aligned} \tag{10}$$

and for the degree m of the correspondence one obtains, according to § 75 (13),

$$\begin{array}{r|l} n = 3, & m = 12, \\ n = 11, & m = 12, \\ n = 19, & m = 60. \end{array}$$

Thus again, as in the preceding case, when $n \equiv 0, 1 \pmod 3$, a number three times too large is obtained.

Finally, if $n \equiv 1 \pmod 4$, one puts

$$x = f^4(\omega), \qquad y = f_1^4(\omega), \qquad z = f_2^4(\omega)$$

and obtains a fundamental curve of the sixth degree.

For $n = 5$ the correct number $m = 6$ results.

Eighth Section. The group of the transformation equations and the equation of the fifth degree.

§ 78. The Galois group of the transformation equations for a prime degree.

A more penetrating algebraic study of the transformation equations requires knowledge of their Galois group. Since we have derived the transformation equations from the division equations, whose group is known to us (§ 63), we can likewise form the group of the transformation equations.

The transformation equations arose (§ 65) because the roots of the division equations could be divided into series which were not torn apart by the interchanges of the group of the division equation, but only interchanged among themselves.

To each of these series there corresponds a definite root of a transformation equation, and hence the group of the latter consists of the totality of the interchanges produced by the group of the division equation among the series.

If the transformation degree n is an odd prime number p, this group admits a very elegant representation, suitable for further investigations, and we shall now retain this assumption.

In § 65, (3), the necessary and sufficient condition was already found that two roots of the division equation

$$x_{\mu,\mu'}, \qquad x_{\nu,\nu'} \tag{1}$$

belong to the same series R, namely the congruence

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{p}. \tag{2}$$

If now, as is customary in number theory (Gauss, *Disquisitiones arithmeticae*, art. 31), by the symbol

$$\frac{a}{b} \pmod{p}$$

we understand an integer [or residue class (mod p)] which, when multiplied by b, leaves the residue a upon division by p, then, provided that $\mu'\nu'$ are not divisible by p, the congruence (2) defining a series can also be written

$$\frac{\mu}{\mu'} \equiv \frac{\nu}{\nu'} \pmod{p}, \tag{3}$$

and we are therefore naturally led to designate the series R by the value of the ratio

$$\frac{\mu}{\mu'} \equiv z \pmod{p}, \tag{4}$$

which may be congruent to any of the numbers $0, 1, \dots, p-1$, and which is invariant for an entire series. At first the one series remains undesignated in which μ', and therefore all ν', are divisible by p; but this series also fits very well into the general notation if, in case $\mu' \equiv 0 \pmod{p}$, we put

$$\frac{\mu}{\mu'} \equiv \infty \pmod{p}, \tag{5}$$

so that we still obtain a $(p+1)$st series R_∞. The totality of the series, whose number is $p+1$, is therefore to be denoted by

$$R_\infty, R_0, R_1, \dots, R_{p-1}. \tag{6}$$

Correspondingly, the associated roots of a transformation equation are to be denoted by

$$v_\infty, v_0, v_1, \dots, v_{p-1}. \tag{7}$$

If, for example, we take the invariant equation (§ 69) as basis, then, according to the determinations of § 68, (10), about the numbers a, c, δ, one has to put

$$v_\infty = j(p\omega), \qquad v_z = j\left(\frac{z+\omega}{p}\right), \tag{8}$$

and similarly if any other transformation equation is chosen.

With this notation we are able to derive at once the group of the transformation equation from that of the division equation.

We first take as rationality domain the totality of the rational functions of x^2 with rational numerical coefficients.

According to § 63, 3., in this rationality domain the group of the division equation consists of all substitutions by which μ, μ' are transformed into

$$\delta\mu - b\mu', \qquad -c\mu + a\mu',$$

where a, b, c, δ are arbitrary integers taken modulo p, whose determinant

$$\Delta = a\delta - bc \tag{9}$$

is not divisible by p, and the number of all these substitutions is

$$p(p-1)(p^2-1) \qquad [\S\ 63, (27)]. \tag{10}$$

From this, however, by the notation (4), (5), the group of the transformation equation is obtained as consisting of all interchanges expressed by the symbol

$$\left(z, \frac{\delta z - b}{-cz + a}\right) \pmod p. \tag{11}$$

We denote a substitution (z, z'), if

$$z \equiv \frac{c + \delta z'}{a + bz'} \pmod p, \tag{12}$$

similarly as before by

$$\begin{pmatrix} a & b \\ c & \delta \end{pmatrix}, \tag{13}$$

and obtain for the composition of two such substitutions, if

$$z' \equiv \frac{c' + \delta' z''}{a' + b' z''} \pmod p,$$

the rule

$$\begin{aligned} (z, z')(z', z'') &= \begin{pmatrix} a & b \\ c & \delta \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & \delta' \end{pmatrix} \\ &\equiv \begin{pmatrix} aa' + bc' & ab' + b\delta' \\ ca' + \delta c' & cb' + \delta\delta' \end{pmatrix} \pmod p, \end{aligned} \tag{14}$$

in agreement with the rule for the composition of two transformations in § 28. The substitution (11) is therefore to be denoted by

$$\begin{pmatrix} \delta & c \\ b & a \end{pmatrix}.$$

The group formed by all substitutions of this form according to (14) will be denoted by $\mathfrak{L}$ (the group of linear substitutions).

The functions (12) of z', and hence also the substitutions (13), remain unchanged if the four numbers a, b, c, δ are all multiplied by one and the same factor not divisible by p. Consequently, by (10), the number of these functions, or the degree of the group $\mathfrak{L}$, is

$$p(p^2 - 1), \tag{15}$$

which is also easily found by direct counting.

If in one of the substitutions (13) the four numbers a, b, c, δ are multiplied by a common factor not divisible by p, then the determinant Δ is multiplied by the square of this factor. Hence it is not the determinant Δ which is preserved by this multiplication, but rather its quadratic character, that is, the value of the symbol

$$\left(\frac{\Delta}{p}\right).$$

If two of the substitutions (13) are composed, their determinants multiply, and from this it follows that all substitutions of the group $\mathfrak{L}$ in which Δ is a quadratic residue of p form a group among themselves, which we shall denote by $\mathfrak{L}_0$.

The group $\mathfrak{L}_0$ is a proper divisor of the group $\mathfrak{L}$ of index 2.

If the substitutions of $\mathfrak{L}_0$ are composed with any substitution $(z, \beta z) = \begin{pmatrix} 1 & 0 \\ 0 & \beta^{-1} \end{pmatrix}$, where β and hence also β^{-1} is a quadratic non-residue of p, then the whole group $\mathfrak{L}$ is obtained.

If Δ is a quadratic residue of p, then a common factor to be adjoined to a, b, c, δ can be chosen so that $\Delta = 1 \pmod p$, and hence we can represent the group $\mathfrak{L}_0$ also by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma = 1 \pmod p. \tag{16}$$

In one of these substitutions the numbers $\alpha, \beta, \gamma, \delta$ are determined modulo p up to the common sign. The degree of the group $\mathfrak{L}_0$ is

$$\frac{1}{2}p(p^2 - 1). \tag{17}$$

But one comes directly to the form (16) if, as group of the division equation, one considers not the group $\mathfrak{U}$, but the group $\mathfrak{B}$ of § 63, that is, if the pth roots of unity are adjoined to the rationality domain, whence the theorem follows:

The group $\mathfrak{L}_0$ is the group of the transformation equation when the pth roots of unity are adjoined to the rationality domain.

For the reduction of the group $\mathfrak{L}$ to the group $\mathfrak{L}_0$, however, the adjunction of a two-valued function is already sufficient; thus the adjunction of the pth roots of unity does too much. The theorems of §§ 63, 65 serve to recognize which irrationality must be adjoined.

In § 63, 3., we saw that the pth root of unity ϱ is rationally representable by the roots of the division equation, and that by a substitution of the group $\mathfrak{U}$ whose determinant is congruent to m, ϱ passes into ϱ^m; further, in § 65 we proved that by adjoining all roots of a transformation equation the group of the division equation is reduced to the group $\mathfrak{U}_0$, which consists of all substitutions of the form

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \tag{18}$$

where a is an arbitrary number not divisible by p. The determinants of the substitutions of $\mathfrak{U}_0$ are therefore squares and hence are congruent modulo p to one of the $\frac{1}{2}(p-1)$ quadratic residues of p.

The sum

$$A = \sum^{a} \varrho^a, \tag{19}$$

where a is to run through all quadratic residues of p, therefore remains unchanged under the substitutions of $\mathfrak{U}_0$ and consequently can be expressed rationally by the roots of the transformation equation. The sum A remains unchanged under the substitutions of the group $\mathfrak{B}$ and hence also under $\mathfrak{L}_0$, while under the substitutions of $\mathfrak{L}$, and hence also of $\mathfrak{U}$, it receives two different values, namely, if b runs through the series of non-residues,

$$A = \sum^{a} \varrho^a, \qquad B = \sum^{b} \varrho^b.$$

Therefore A is a function belonging to the group $\mathfrak{L}_0$, and by its adjunction $\mathfrak{L}$ is reduced to $\mathfrak{L}_0$.

The values of the sums A, B are known, however (Vol. I, § 179):

$$A = \frac{-1 + \sqrt{(-1)^{(p-1)/2} p}}{2}, \qquad B = \frac{-1 - \sqrt{(-1)^{(p-1)/2} p}}{2}.$$

The sign of the root depends on the choice of the root ϱ and can be determined, but this is not relevant here. We therefore have the theorem:

The group of the transformation equation is $\mathfrak{L}_0$ when $\sqrt{(-1)^{(p-1)/2} p}$ is adjoined to the rationality domain.

The group of the invariant equation can also be derived without division of the elliptic functions in the following simple way, though in this way one obtains only the monodromy group, that is, the group in the field of rational functions of $j(\omega)$, without regard to the constants to be adjoined. We restrict ourselves to the case of a prime degree p of the transformation, and put

$$v_c = j\left(\frac{c+\omega}{p}\right), \qquad v_\infty = j(p\omega), \qquad u = j(\omega), \tag{20}$$

where c is to be taken modulo p.

If a linear substitution

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

is applied to ω, then $j(\omega)$ remains unchanged and v_c passes into $v_{c'}$. To find c', one has to seek a second linear substitution

$$S' = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$$

such that

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} p & 0 \\ c' & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ c & 1 \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

or

$$\begin{aligned} p\alpha' + c'\beta' &= p\alpha, & \beta' &= p\beta, \\ p\gamma' + c'\delta' &= c\alpha + \gamma, & \delta' &= c\beta + \delta. \end{aligned}$$

From this, by the third equation, $c'\delta' = c\alpha + \gamma$; and with the help of the fourth,

$$c' = \frac{c\alpha + \gamma}{c\beta + \delta} \pmod{p}.$$

If $c\beta + \delta = 0$, then $c' = \infty$; and if $c = \infty$, then $c' = \alpha/\beta$.

In applying this to the determination of c, c', one may replace $\alpha, \beta, \gamma, \delta$ by congruent numbers a, b, c, d whose determinant $ad - bc$ is a quadratic residue of p, and it can also be shown that in this way every substitution of the group $\mathfrak{L}_0$ can be obtained. Thus every function of v which remains unchanged under the substitutions of the group $\mathfrak{L}_0$ also remains unchanged when a linear substitution is applied to ω, and is consequently a rational function of $j(\omega)$. But as to the nature of the coefficients in this function, this consideration teaches us nothing, and hence $\mathfrak{L}_0$ is recognized only as the monodromy group of the invariant equation.

Volume III. Eighth Section. The group of the transformation equations and the equation of the fifth degree.

§ 79. Examination of the group $\mathfrak{L}_0$.

In the tenth section of Volume II we investigated the congruence group and its divisors quite generally. Here we carry out this investigation once more in a special form, insofar as it has reference to the transformation problem.

The substitutions contained in $\mathfrak{L}_0$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \left(z, \frac{-c + az}{d - bz} \right), \tag{1}$$

in which

$$\Delta = ad - bc \tag{2}$$

is a quadratic residue of p, can, as already remarked above, be brought to the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \left(z, \frac{-\gamma + \alpha z}{\delta - \beta z} \right), \tag{3}$$

where

$$\alpha\delta - \beta\gamma = 1 \pmod{p}. \tag{4}$$

For we need only, if by $\sqrt{\Delta} \pmod{p}$ we understand a whole number whose square is congruent to Δ modulo p, to put

$$a \equiv \alpha\sqrt{\Delta}, \quad b \equiv \beta\sqrt{\Delta}, \quad c \equiv \gamma\sqrt{\Delta}, \quad d \equiv \delta\sqrt{\Delta} \pmod{p}.$$

In the form (3) the signs of $\alpha, \beta, \gamma, \delta$ can still be changed simultaneously.

We ask for those elements z which remain unchanged by a substitution of the form (3). These are determined by the congruence

$$z \equiv \frac{-\gamma + \alpha z}{\delta - \beta z} \pmod{p},$$

or

$$\beta z^2 + (\alpha - \delta)z - \gamma \equiv 0 \pmod{p}, \tag{5}$$

a congruence which, unless it is identical, has at most two incongruent roots. To obtain these, under the supposition first that β is not divisible by p, we write (5) as

$$\left(\beta z + \frac{\alpha - \delta}{2}\right)^2 \equiv \left(\frac{\alpha - \delta}{2}\right)^2 + \beta\gamma \pmod{p},$$

or, with the help of (4),

$$\left(\beta z + \frac{\alpha - \delta}{2}\right)^2 \equiv \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \pmod{p}. \tag{6}$$

Accordingly three different cases are to be distinguished:

$$\text{I.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \equiv 0 \pmod{p};$$

then the congruence (5) has one root; there is one and only one element z which remains unchanged.

$$\text{II.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \quad \text{is a quadratic residue of } p;$$

the congruence (5) has two different roots; there are two elements z which remain unchanged.

$$\text{III.} \qquad \left(\frac{\alpha+\delta}{2}\right)^2 - 1 \quad \text{is a quadratic non-residue of } p;$$

the congruence (5) has no root at all, and all elements z are permuted.

If $\beta \equiv 0$, then (5) always has the one root $z = \infty$; if then $\delta = \alpha$ and $\gamma \not\equiv 0 \pmod{p}$, there is only this one. But in this case

$$\alpha\delta \equiv 1, \quad \alpha \equiv \delta, \quad \left(\frac{\alpha+\delta}{2}\right)^2 \equiv 1 \pmod{p},$$

and condition I is fulfilled. If however at the same time $\gamma \equiv 0 \pmod{p}$, the substitution is the identical one.

If $\beta \equiv 0$, but $\alpha - \delta \not\equiv 0 \pmod{p}$, then (5) has still a second root; since now $\alpha\delta \equiv 1 \pmod{p}$, we have

$$\left(\frac{\alpha+\delta}{2}\right)^2 - 1 \equiv \left(\frac{\alpha-\delta}{2}\right)^2 \pmod{p},$$

therefore a quadratic residue, and condition II is fulfilled. We summarize these propositions thus:

In case I one element, or all elements, remain unchanged; in case II two elements remain unchanged; and in case III all elements are changed.

If we repeat one and the same substitution

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{7}$$

several times, the substitutions

$$A, A^2, A^3, \ldots$$

arise, and in their series the identical substitution $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ must occur once. If n is the least positive number for which

$$A^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

then n is called the degree of A (Vol. II, § 2).

If for an arbitrary m we put

$$A^m = \begin{pmatrix} \alpha_m & \beta_m \\ \gamma_m & \delta_m \end{pmatrix},$$

then for the computation of the numbers $\alpha_m, \beta_m, \gamma_m, \delta_m$ we obtain the following system of recurrent formulae:

$$\begin{aligned} \alpha_{m+1} &= \alpha\alpha_m + \beta\gamma_m, \quad & \beta_{m+1} &= \alpha\beta_m + \beta\delta_m, \\ \gamma_{m+1} &= \gamma\alpha_m + \delta\gamma_m, \quad & \delta_{m+1} &= \gamma\beta_m + \delta\delta_m. \end{aligned} \tag{8}$$

The numbers $\alpha_m, \beta_m, \gamma_m, \delta_m$ are computed especially simply from these formulae in case I.

In this case, since the signs of $\alpha, \beta, \gamma, \delta$ may all be simultaneously reversed, we can assume

$$\alpha + \delta = 2, \qquad \alpha\delta - \beta\gamma = 1 \pmod{p},$$

and from (8) obtain

$$\left.\begin{aligned} \alpha_2 &\equiv -1 + 2\alpha, & \beta_2 &\equiv 2\beta, \\ \delta_2 &\equiv -1 + 2\delta, & \gamma_2 &\equiv 2\gamma, \\ \alpha_3 &\equiv -2 + 3\alpha, & \beta_3 &\equiv 3\beta, \\ \delta_3 &\equiv -2 + 3\delta, & \gamma_3 &\equiv 3\gamma, \end{aligned}\right. \quad \pmod{p},$$

from which, by the induction from m to $m+1$, it follows that

$$\alpha_m \equiv -(m-1) + m\alpha, \quad \delta_m \equiv -(m-1) + m\delta, \quad \beta_m \equiv m\beta, \quad \gamma_m \equiv m\gamma \pmod{p}. \tag{9}$$

From this one sees that all the A^m again belong to case I, and that p is the degree of A.

The analogous consideration of the two other cases has been carried out by Serret, but appears unnecessary for our purpose.

On the other hand, we shall add here the theorem that the whole group $\mathfrak{L}_0$ can be composed from the following two special substitutions

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{10}$$

The proof follows from § 50, if one observes that to each number system $\alpha, \beta, \gamma, \delta$ satisfying the condition $\alpha\delta - \beta\gamma = 1 \pmod{p}$, the number system $\alpha', \beta', \gamma', \delta'$ can be chosen so that

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \equiv \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \pmod{p}, \qquad \alpha'\delta' - \beta'\gamma' = 1.$$

Thus all substitutions of $\mathfrak{L}_0$ are contained among those considered in § 30. But we can also here easily display the composition of an arbitrary substitution in $\mathfrak{L}_0$ actually from A and B, and so give the proof independently of § 30.

For if first c is an arbitrary number not divisible by p, then actual calculation gives easily

$$A^c = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} c & 0 \\ 0 & c^{-1} \end{pmatrix} = A^{-c^{-1}} B A^{-c} B A^{-c^{-1}} B, \tag{11}$$

and then, if $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ is any substitution in $\mathfrak{L}_0$ and β is different from 0,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ \delta\beta^{-1} - 1 & 1 \end{pmatrix} \begin{pmatrix} \beta & 0 \\ 0 & \beta^{-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha\beta^{-1} & 1 \end{pmatrix}, \tag{12}$$

and if $\beta = 0$,

$$\begin{pmatrix} \alpha & 0 \\ \gamma & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \gamma\alpha & 1 \end{pmatrix}. \tag{13}$$

It is to be concluded from this theorem that a group contained in $\mathfrak{L}_0$ which contains the two substitutions A, B must necessarily be identical with $\mathfrak{L}_0$.

§ 80. Normal divisors of the group $\mathfrak{L}_0$.

In order to learn the possible reductions of the transformation equation, it is above all necessary to investigate the divisors of the group $\mathfrak{L}_0$. We ask first about the existence of a normal divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ (Vol. II, § 3).[23]

Let

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{1}$$

be any substitution in $\mathfrak{K}$ different from the identical substitution. Then, by the nature of a normal divisor, every substitution

$$U = TST^{-1} \tag{2}$$

[23]According to Galois: "proper divisors".

is also contained in $\mathfrak{K}$, if

$$T = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \tag{3}$$

is an arbitrary substitution in $\mathfrak{L}_0$.

We set ourselves the problem of determining T and ξ so that

$$U = \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \tag{4}$$

shall hold. This condition can also be written

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix},$$

and leads to the congruences

$$\left.\begin{array}{ll} 1. & \alpha'\alpha + \beta'\gamma \equiv \gamma', \\ 2. & \alpha'\beta + \beta'\delta \equiv \delta', \\ 3. & \gamma'\alpha + \delta'\gamma \equiv -\alpha' + \gamma'\xi, \\ 4. & \gamma'\beta + \delta'\delta \equiv -\beta' + \delta'\xi, \end{array}\right\} \pmod{p}, \tag{5}$$

and from 1 and 2 there also follows

$$\alpha' \equiv \gamma'\delta - \delta'\gamma, \qquad \beta' \equiv -\gamma'\beta + \delta'\alpha \pmod{p}. \tag{6}$$

If these values are inserted in the third and fourth congruences of (5), there follows

$$\gamma'(\alpha + \delta - \xi) \equiv 0, \qquad \delta'(\alpha + \delta - \xi) \equiv 0 \pmod{p},$$

whence, since γ', δ' cannot both be divisible by p, it follows that

$$\xi \equiv \alpha + \delta \pmod{p}. \tag{7}$$

If ξ is so determined, then in (5) the congruences 3 and 4 follow from 1 and 2. If one now inserts α', β' from 1 and 2 in

$$\alpha'\delta' - \beta'\gamma' \equiv 1 \pmod{p}, \tag{8}$$

one obtains

$$\alpha'^2\beta + \alpha'\beta'(\delta - \alpha) - \beta'^2\gamma \equiv 1 \pmod{p}. \tag{9}$$

If α', β' are determined from this, all demands stated in (5), and hence also in (2), (4), are satisfied.

It remains therefore only to prove that the congruence (9) is always soluble.

This possibility is evident if β or $-\gamma$ is a quadratic residue of p; for then

$$\alpha' \equiv \sqrt{\beta^{-1}}, \quad \beta' = 0,$$

or

$$\alpha' = 0, \quad \beta' \equiv \sqrt{-\gamma^{-1}} \pmod{p}$$

satisfies the demand.

It is further to be distinguished whether the substitution S belongs to class I, II, or III of the preceding paragraph.

First, if

$$\text{I.} \qquad \left(\frac{\alpha + \delta}{2}\right)^2 \equiv 1 \pmod{p},$$

then, if S^m is put in place of S, by § 79, (9), β passes into $m\beta$ and γ into $m\gamma$; and m can always be so determined that $m\beta$ or $-m\gamma$ becomes a quadratic residue of p. This also includes the case in which $\beta = 0$ or $\gamma = 0$ and $\alpha \equiv \delta \pmod p$.

If, on the other hand, $\beta = 0$ and $\alpha - \delta$ is not divisible by p, then one can choose β' arbitrarily in (9) and then determine α' from a congruence of the first degree.

If β is not divisible by p, put (9) in the form

$$\left[\alpha'\beta + \beta'\frac{\delta-\alpha}{2}\right]^2 \equiv \beta + \beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \pmod p. \tag{10}$$

If now

$$\text{II.} \qquad \left(\frac{\alpha+\delta}{2}\right)^2 - 1 \quad \text{is a quadratic residue of } p,$$

then β' is determined from the congruence

$$\beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \equiv \left(\frac{\beta-1}{2}\right)^2,$$

whereby (10) passes into

$$\alpha'\beta + \beta'\frac{\delta-\alpha}{2} \equiv \pm\frac{\beta+1}{2} \pmod p,$$

and from this α' can be determined, whichever sign is chosen.

If

$$\text{III.} \qquad \left(\frac{\alpha+\delta}{2}\right)^2 - 1 \quad \text{is a quadratic non-residue of } p,$$

and at the same time β is a quadratic non-residue, then a non-residue ν can always be so determined that $\beta + \nu$ becomes a quadratic residue; for if ν in $\beta + \nu$ is allowed to run through the series of non-residues, not all non-residues can arise, since among them β too would have to occur.

Then β' can be so determined that

$$\beta'^2\left[\left(\frac{\alpha+\delta}{2}\right)^2 - 1\right] \equiv \nu \pmod p,$$

and then α' from

$$\alpha'\beta + \beta'\frac{\delta-\alpha}{2} \equiv \pm\sqrt{\beta+\nu} \pmod p.$$

From this follows:

In a normal divisor $\mathfrak{K}$ of the group $\mathfrak{L}_0$ there must certainly occur a substitution U of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix}. \tag{11}$$

Let first ξ be different from zero modulo p.

By the concept of the normal divisor, $\mathfrak{K}$ also contains the substitution

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \xi & 1 \\ -1 & 0 \end{pmatrix}, \tag{12}$$

and consequently also

$$\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix}\begin{pmatrix} \xi & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ \xi^2-1 & -1 \end{pmatrix} = A^{\xi^2-1} \quad [\text{§ } 79, (11)], \tag{13}$$

hence also A and all its powers. Therefore $\mathfrak{K}$ also contains

$$\begin{pmatrix} 1 & 0 \\ \eta & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & \xi+\eta \end{pmatrix} \tag{14}$$

for arbitrary η, that is, the substitution (11) for every arbitrary ξ, and hence also

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Accordingly, by the theorem of the preceding paragraph, the group $\mathfrak{K}$ is identical with $\mathfrak{L}_0$.

There remains still the possibility to discuss in which, in (11), $\xi = 0$.

In this case the group $\mathfrak{K}$ contains the substitution

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and consequently, for an arbitrary α not divisible by p, also

$$V = \begin{pmatrix} 0 & \alpha \\ -\alpha^{-1} & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -\alpha \\ \alpha^{-1} & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha^2 & 0 \\ 0 & \alpha^{-2} \end{pmatrix}.$$

From this one derives further, as contained in $\mathfrak{K}$,

$$W = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \alpha^{-2} & 0 \\ 0 & \alpha^2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \alpha^2 & 0 \\ 0 & \alpha^{-2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \alpha^4-1 & 1 \end{pmatrix}.$$

If now p is greater than 5, then α can be so taken that $\alpha^4 - 1$ is not divisible by p; then $\mathfrak{K}$ also contains the substitution A and its powers, and hence $\mathfrak{K}$ is identical with $\mathfrak{L}_0$.

If $p = 5$, then $\alpha^4 - 1$ is always divisible by p, so this conclusion is not applicable. But in this case it follows from the concept of the normal divisor that the following is contained in $\mathfrak{K}$:

$$\begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix},$$

$$\begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix},$$

that is, A^2, and hence all powers of A. From this one concludes as above that $\mathfrak{K}$ is identical with $\mathfrak{L}_0$. We have therefore the theorem:

If $p > 3$, the group $\mathfrak{L}_0$ has no normal divisor.

In the case $p = 3$, $\mathfrak{K}$ likewise contains

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and therefore also

$$\begin{pmatrix} \pm 1 & 0 \\ \pm 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \mp 1 & 1 \end{pmatrix} = \begin{pmatrix} \mp 1 & 1 \\ 1 & \pm 1 \end{pmatrix},$$

and in fact the four substitutions

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{15}$$

form a normal divisor of $\mathfrak{L}_0$ of index 3.

In the case $p = 3$, the group $\mathfrak{L}_0$ is isomorphic with the alternating group of interchanges of four elements. This is the group of an arbitrary equation of the fourth degree when the square root of the discriminant is adjoined to the domain of rationality. The divisor (15) of this group of index 3 then supplies the cubic resolvent of the biquadratic equation known in algebra.

A function belonging to the group (15) of the roots v_∞, v_0, v_1, v_2 of the transformation equation for the third transformation degree is, for example,

$$v_\infty v_0 + v_1 v_2$$

or

$$(v_\infty - v_0)(v_1 - v_2).$$

The difference between these two functions is that the first remains unchanged by the substitution of determinant -1

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

whereas the second changes its sign under it. The first will therefore lead to an equation of the third degree with rational coefficients, while in the cubic equation for the second $\sqrt{-3}$ still occurs (§ 78).

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§ 81. Non-normal divisors of $\mathfrak{L}_0$.

Leaving aside the case $p = 3$, we now ask for the non-normal divisors of the group $\mathfrak{L}_0$ whose index is smaller than $p + 1$; on the existence of such divisors depends the possibility of forming resolvents of the transformation equation whose degree is lower than $p + 1$.

First it can be shown that the index of a divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ can never be smaller than p.

Let

$$\mathfrak{K} = S_0, S_1, \ldots, S_{\nu-1} \tag{1}$$

be any divisor of $\mathfrak{L}_0$ of degree ν, and, if it is possible,

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{2}$$

a substitution of degree p which occurs in $\mathfrak{L}_0$ but not in $\mathfrak{K}$. Then, since p is prime, no lower power of T than the pth can occur in $\mathfrak{K}$. The cosets

$$\mathfrak{K}, \quad T\mathfrak{K}, \quad T^2\mathfrak{K}, \ldots, \quad T^{p-1}\mathfrak{K} \tag{3}$$

contain altogether mutually different elements $T^i S_k$, and hence νp is at most equal to the degree of the group $\mathfrak{L}_0$, that is:

$$\nu p \leq p\,\frac{p^2 - 1}{2}.$$

Thus the index of $\mathfrak{K}$, i.e. the quotient

$$p\,\frac{p^2 - 1}{2} : \nu,$$

is equal to p or greater than p.

A divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ whose index is smaller than p must therefore contain all substitutions of degree p, hence also (§ 79, I) all substitutions T in which

$$\alpha + \delta \equiv 2 \pmod p \tag{4}$$

holds. Consequently such a group $\mathfrak{K}$ contains first the substitution

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \tag{5}$$

and its powers; likewise the substitutions

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence also

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}. \tag{6}$$

But if the substitutions A, B are contained in $\mathfrak{K}$, then by § 79 $\mathfrak{K}$ must be identical with $\mathfrak{L}_0$. We therefore have the theorem:

The index of a divisor of $\mathfrak{L}_0$ cannot be smaller than p.

Our problem is therefore restricted to the question of the divisors of $\mathfrak{L}_0$ of index p, or of degree

$$\frac{p^2 - 1}{2}.$$

Let $\mathfrak{K}$ be such a divisor and let $S_0, S_1, \dots, S_{\nu-1}$ be its elements, so that the number ν has the value

$$\nu = \frac{p^2 - 1}{2}. \tag{7}$$

Since the degree of every element of a group is always a divisor of the degree of the group, no element of degree p can occur in $\mathfrak{K}$, since ν is not divisible by p. If therefore for T we take the substitution of degree p

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \tag{8}$$

then the group $\mathfrak{L}_0$ can be decomposed into the p cosets:

$$\mathfrak{L}_0 = \mathfrak{K}, \quad A\mathfrak{K}, \quad A^2\mathfrak{K}, \dots, \quad A^{p-1}\mathfrak{K}. \tag{9}$$

Together with $\mathfrak{K}$, all its conjugate divisors

$$T\mathfrak{K}T^{-1}$$

have the same degree and hence also the same index. Let now g be a primitive root of the prime number p, and

$$C = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \tag{10}$$

a substitution which plainly has degree $(p-1)/2$. Like every substitution of $\mathfrak{L}_0$, it must occur in one of the rows (9); that is, there is a substitution S in $\mathfrak{K}$ and an exponent λ for which

$$C = A^\lambda S, \qquad S = A^{-\lambda} C. \tag{11}$$

But γ can further be determined so that

$$A^\gamma C A^{-\gamma} = C; \tag{12}$$

for this one obtains the condition

$$\gamma(g - g^{-1}) \equiv \lambda g \pmod{p},$$

which, since $p > 3$, can always be satisfied; but then, by (11),

$$C = A^\gamma S A^{-\gamma}.$$

Thus C occurs in the divisor $A^\gamma \mathfrak{K} A^{-\gamma}$ conjugate to $\mathfrak{K}$, and therefore, by putting this group in place of $\mathfrak{K}$, we may assume without loss of generality that $\mathfrak{K}$ itself contains the substitution C.

Now in $\mathfrak{K}$ the substitution

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

may either occur or not occur. In the first case $\mathfrak{K}$ also contains all substitutions composed from B and C, which are all of one of the two forms

$$\begin{pmatrix} g^r & 0 \\ 0 & g^{-r} \end{pmatrix}, \tag{13}$$

$$\begin{pmatrix} 0 & g^r \\ -g^{-r} & 0 \end{pmatrix}, \qquad r = 0, 1, \dots, \frac{p-3}{2}, \tag{14}$$

and their number is $p-1$. Thus in the first case $\mathfrak{K}$ contains all substitutions of the forms (13), (14), and in the second all substitutions of the form (13) and none of the form (14).

Let now

$$V = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{15}$$

be a substitution contained in $\mathfrak{K}$ which is contained neither in the form (13) nor in the form (14), so that neither α and δ, nor β and γ, are simultaneously congruent to 0. Then all substitutions of the form

$$C^r V C^s = \begin{pmatrix} \alpha g^{r+s} & \beta g^{r-s} \\ \gamma g^{-r+s} & \delta g^{-r-s} \end{pmatrix}, \tag{16}$$

when r, s both run through the row of numbers

$$0, 1, 2, \ldots, \frac{p-3}{2},$$

are contained in $\mathfrak{K}$ and are all distinct from one another. For if two of the substitutions (16) were equal, then there would have to be one among them which, without r, s vanishing, is identical with V. This requires

$$\alpha = \pm\alpha g^{r+s}, \quad \beta = \pm\beta g^{r-s}, \quad \gamma = \pm\gamma g^{-r+s}, \quad \delta = \pm\delta g^{-r-s} \pmod p,$$

where in all four formulae either the upper or the lower signs hold; hence, since neither α and δ, nor β and γ, vanish simultaneously,

$$r + s \equiv 0, \qquad r - s \equiv 0 \pmod{p-1},$$

or

$$r + s \equiv \frac{p-1}{2}, \qquad r - s \equiv \frac{p-1}{2} \pmod{p-1},$$

and therefore in both cases

$$r = 0, \quad s = 0 \pmod{\frac{p-1}{2}}, \qquad \text{as was required to be proved.}$$

In the form (16) there are therefore $\left(\frac{p-1}{2}\right)^2$ different substitutions. If the group $\mathfrak{K}$ is not yet exhausted by these, one chooses a substitution V' not contained in (13), (14), (16), and forms in the same way the row

$$C^r V' C^s, \tag{17}$$

whose substitutions are different both among themselves and from those contained in (13), (14), (16), and one continues in this way until the whole group $\mathfrak{K}$ is exhausted.

If the number of rows so formed, (16), (17), ..., is denoted by q, then the degree of the group $\mathfrak{K}$ is obtained as follows:

1. if B does not occur in $\mathfrak{K}$:

$$= \frac{p-1}{2} + q\left(\frac{p-1}{2}\right)^2;$$

2. if B occurs in $\mathfrak{K}$:

$$= p - 1 + q\left(\frac{p-1}{2}\right)^2.$$

This number, according to the requirement of our problem, must be

$$= \frac{(p-1)(p+1)}{2}.$$

But from this it follows that case 1 is impossible; for in that case one would have to have

$$q\left(\frac{p-1}{2}\right) = p,$$

hence

$$q = p, \qquad p = 3,$$

which we have excluded.

In case 2, however, it follows for every arbitrary p that

$$q = 2.$$

Hence in any case the substitution B must occur in $\mathfrak{K}$, and the group sought is exhausted by (13), (14), (16), (17).

We first show that in the two substitutions V, V' of the group $\mathfrak{K}$ none of the numbers $\alpha, \beta, \gamma, \delta, \alpha', \beta', \gamma', \delta'$ can be congruent to 0. Namely, as we have already seen, there does not occur in $\mathfrak{K}$

$$\begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}$$

if γ is different from 0, and therefore also not

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha\gamma & 1 \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ \gamma & \alpha^{-1} \end{pmatrix},$$

for arbitrary α, and consequently also not

$$\begin{pmatrix} \alpha & 0 \\ \gamma & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \alpha \\ -\alpha^{-1} & \gamma \end{pmatrix},$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \alpha & 0 \\ \gamma & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \gamma & \alpha^{-1} \\ -\alpha & 0 \end{pmatrix},$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \alpha \\ -\alpha^{-1} & -\gamma \end{pmatrix} = \begin{pmatrix} \alpha^{-1} & \gamma \\ 0 & \alpha \end{pmatrix};$$

among these are contained all substitutions

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

in which one of the four numbers $\alpha, \beta, \gamma, \delta$ is congruent to 0.

We now denote the substitutions of the system (16) by

$$W = \begin{pmatrix} \alpha g^{r+s} & \beta g^{r-s} \\ \gamma g^{-r+s} & \delta g^{-r-s} \end{pmatrix},$$

and put

$$\xi \equiv \alpha g^{r+s}, \qquad \varrho \equiv \alpha^{-1}\beta g^{-2s} \pmod{p}, \tag{18}$$

$$\alpha\delta \equiv m, \qquad \beta\gamma \equiv m - 1. \tag{19}$$

The number m does not change when V is replaced by an arbitrary substitution of the system (16). The number m' is to have the corresponding meaning for the system (17); in either of the two systems ξ can assume each of the values

$$1, 2, \ldots, p - 1 \tag{20}$$

and ϱ runs through, according as $\alpha^{-1}\beta$ is a quadratic residue or non-residue, in one system the row of residues or that of non-residues. Changing the sign of ξ gives no new substitution W. Accordingly we can represent W in this way:

$$W = \begin{pmatrix} \xi & \xi\varrho \\ \xi^{-1}\varrho^{-1}(m-1) & \xi^{-1}m \end{pmatrix}. \tag{21}$$

Since now the square of W,

$$W^2 = \begin{pmatrix} \xi^2 + m - 1 & (\xi^2 + m)\varrho \\ \xi^{-2}\varrho^{-1}(m-1)(\xi^2+m) & [(m-1)\xi^2 + m^2]\xi^{-2} \end{pmatrix},$$

must belong to $\mathfrak{K}$, it is contained under one of the systems (13), (14), (16), (17), from which the following four possibilities result:

$$\begin{aligned} &1. \quad \xi^2 + m \equiv 0, \\ &2. \quad \xi^2 + m - 1 \equiv 0, \quad (m-1)\xi^2 + m^2 \equiv 0, \\ &3. \quad (\xi^2 + m - 1)[(m-1)\xi^2 + m^2] \equiv m'\xi^2, \\ &4. \quad (\xi^2 + m - 1)[(m-1)\xi^2 + m^2] \equiv m\xi^2, \end{aligned} \tag{22}$$

and each of the $p-1$ values (20), inserted for ξ, must satisfy one of these four cases. If one of the congruences (22), 2, is fulfilled, then the other must follow from it, which is possible only when

$$2m \equiv 1, \qquad \xi^2 \equiv m \pmod{p}. \tag{23}$$

Now a congruence with respect to a prime modulus has at most as many incongruent roots as the degree of the congruence; 1 and 2 can therefore be satisfied for at most two values each, and 3 and 4 for at most four values each, of ξ. Thus altogether there are at most twelve values of ξ satisfying one of the four congruences (22).

It follows that

$$p - 1 \leq 12, \qquad p \leq 13.$$

But if $p = 13$, then each of the congruences (22) must have the maximum number of roots; hence 2 must also be satisfied for two values of ξ; then by (23) it would have to be that $m \equiv 7$, $\xi^2 \equiv 7 \pmod{13}$, which is impossible, since 7 is a quadratic non-residue of 13.

A divisor of $\mathfrak{L}_0$ of index p therefore does not exist when $p > 11$.

§ 82. Divisors of $\mathfrak{L}_0$ of index p for $p = 5, 7, 11$.

There remains the possibility that for $p = 5, 7, 11$ there exist divisors of $\mathfrak{L}_0$ of index p.

1. $p = 5$. We first investigate the case $p = 5$. Since m and $m - 1$ cannot be congruent to 0, the assumptions for m that remain are

$$m \equiv 2, 3, 4 \pmod{p}.$$

But if we replace the substitution W of § 81, (21) by

$$WB = \begin{pmatrix} -\xi\varrho & \xi \\ -\xi^{-1}m & \xi^{-1}\varrho^{-1}(m-1) \end{pmatrix}, \tag{1}$$

whereby m passes into $1 - m$, then the case $m \equiv 4$ is reduced to the case $m \equiv 2$.

For $m \equiv 2$, however, (23) of § 81 is not fulfilled, and none of the congruences (22) is possible, since ± 2 are non-residues of 5 and the left-hand sides of (22), 3 and 4, reduce for $m = 2$ to $\xi^4 - 1$, which is divisible by 5 for every non-zero ξ. Thus only this remains:

$$m \equiv 3, \qquad m' \equiv 3 \pmod{5}, \tag{2}$$

and the two systems W, W' [§ 81, (16), (17)] can differ from one another only in that $\alpha^{-1}\beta$, and hence also ϱ, is in one a quadratic residue and in the other a quadratic non-residue of 5. The group sought therefore consists, if it exists, of the 12 substitutions contained in the three forms

$$\begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}, \quad \begin{pmatrix} 0 & \xi \\ -\xi^{-1} & 0 \end{pmatrix}, \quad \begin{pmatrix} \xi & \eta \\ 2\eta^{-1} & 3\xi^{-1} \end{pmatrix}, \tag{3}$$

where one is to put
$$\xi \equiv 1, 2, \qquad \eta \equiv \pm 1, \pm 2 \pmod 5.$$

2. $p = 7, 11$.

Since -1 for $p = 7, 11$ is to be found among the quadratic non-residues, the composed substitution WB, (1), belongs to the W' [because the quadratic character of the quantity denoted in (18), § 81 by ϱ is opposite in W and WB]. Accordingly, by (1) and § 81, (19), (21),
$$\beta'\gamma' \equiv -m \equiv m' - 1,$$
hence
$$m + m' \equiv 1 \pmod p. \tag{4}$$

For $p = 7$, since interchanging m and m' gives nothing new, the three following possibilities remain:
$$1.\ m \equiv 2,\ m' \equiv 6; \qquad 2.\ m \equiv 3,\ m' \equiv 5; \qquad 3.\ m \equiv 4,\ m' \equiv 4.$$
In case 1, of the congruences § 81, (22), 1 and 2 are impossible; hence for every ξ one of the two congruences 3, 4 would have to be fulfilled, which here read
$$(\xi^2 + 1)(\xi^2 + 4) \equiv 2\xi^2,\ 6\xi^2 \pmod 7;$$
neither of these is fulfilled for $\xi \equiv 1$.

In case 3, (22), 1 is not satisfiable and (22), 2 is fulfilled for $\xi^2 \equiv 4$; hence for $\xi = 1, 2$ the congruence (22), 3 or 4,
$$(\xi^2 + 3)(3\xi^2 + 2) \equiv 4\xi^2 \pmod 7$$
would have to be fulfilled, which again is not the case. Thus for $p = 7$ there remains only
$$m \equiv 3, \qquad m' \equiv 5 \pmod 7. \tag{5}$$

For $p = 11$, the possibility is excluded from the outset that the congruences (22), 2, be fulfilled; for in that case, in consequence of (4) and § 81, (23),
$$m \equiv m'$$
would have to hold; then (22), 3 and 4 would not be different, and the congruences (22) together could have at most eight roots and not ten, as they would have to. Thus the congruences (22), 1, 3, 4 must each have the maximum number of roots, and, in order that 1 be fulfilled, m, and for the same reasons also m', must be a quadratic non-residue of 11. But only the two numbers
$$m, m' \equiv 2, -1 \pmod{11} \tag{6}$$
satisfy this condition and the condition (4) simultaneously.

The groups sought therefore consist in these two cases, if they exist, of the following substitutions:
$$p = 7: \quad \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}, \begin{pmatrix} 0 & \xi \\ \xi^{-1} & 0 \end{pmatrix}, \begin{pmatrix} \xi & \xi\varrho \\ 2\xi^{-1}\varrho^{-1} & 3\xi^{-1} \end{pmatrix}, \begin{pmatrix} -\xi\varrho & \xi \\ -3\xi^{-1} & 2\xi^{-1}\varrho^{-1} \end{pmatrix}, \quad \xi \equiv 1, 2, 3 \pmod 7, \tag{7}$$

$$p = 11: \quad \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}, \begin{pmatrix} 0 & \xi \\ -\xi^{-1} & 0 \end{pmatrix}, \begin{pmatrix} \xi & \xi\varrho \\ \xi^{-1}\varrho^{-1} & 2\xi^{-1} \end{pmatrix}, \begin{pmatrix} -\xi\varrho & \xi \\ -2\xi^{-1} & \xi^{-1}\varrho^{-1} \end{pmatrix}, \quad \xi \equiv 1, 2, 3, 4, 5 \pmod{11}; \tag{8}$$
in (7) and (8) ϱ runs either through the row of quadratic residues or through that of non-residues, so that for $p = 7$ or 11 one obtains two groups of index p in each case.

That the substitution-systems assembled in (3), (7), (8) really constitute groups can be shown by direct composition in several ways. More simply, however, one reaches this proof by forming functions of the v_z, § 78, (7), which remain unchanged by the substitutions of these systems and which, under the substitutions of $\mathfrak{L}_0$, acquire altogether only p different values.

As is clear from their manner of origin, the systems (3), (7), (8) can be composed by repeated application of B, C, U, U', when U, U' are any two special substitutions W, W'. For $p = 5$, U^2 belongs to the W', and for $p = 7, 11$, UB belongs to the W', so that U' may still be omitted. If we choose U arbitrarily and take for the primitive root g in C, for $p = 5, 7, 11$ respectively, $2, -2, 2$, we can compose the group (3) from

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \quad (p = 5),$$

the group (7) from

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & \pm 1 \\ \pm 2 & 3 \end{pmatrix} \quad (p = 7),$$

and the group (8) from

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 2 \end{pmatrix} \quad (p = 11),$$

where in the last two cases the double signs give rise to the two different groups mentioned above.

We now assemble the interchanges of the indices z corresponding to these substitutions [§ 79, (3)].

For $p = 5$:

$$\begin{array}{rrrrrrr} z = & \infty, & 0, & 1, & 2, & 3, & 4, \\ B: & 0, & \infty, & 4, & 2, & 3, & 1, \\ C: & \infty, & 0, & 4, & 3, & 2, & 1, \\ U: & 4, & 1, & 2, & 0, & \infty, & 3. \end{array}$$

Thus by B, C, U the index-pairs

$$(\infty, 0), \quad (1, 4), \quad (2, 3)$$

are not torn apart, but only interchanged among one another. Moreover, in each case the elements in two of these pairs are interchanged. If therefore we form a function such as

$$V = (v_\infty - v_0)(v_1 - v_4)(v_2 - v_3), \tag{9}$$

then it remains unchanged by B, C, U and hence by the whole system (3), while by repeated application of the cyclic interchange $(0, 1, 2, 3, 4)$, that is, of the substitution

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

it assumes five different values. But from A and B [§ 79, (10)] the whole group $\mathfrak{L}_0$ can be composed, and therefore V obtains, under application of $\mathfrak{L}_0$, no more than five values. The five values of V are the roots of an equation of the fifth degree.

For $p = 7$ one has

$$B = \left(z, -\frac{1}{z}\right), \qquad C = (z, 4z), \qquad U = \left(z, \frac{\mp 2 + z}{3 \mp z}\right).$$

For better overview we further display B, C, U by the cyclic interchanges that they produce in the eight values of z:

$$\begin{aligned} B &= (0, \infty)(1, -1)(2, 3)(-2, -3), \\ C &= (1, -3, 2)(-1, 3, -2), \\ U &= (\infty, \mp 1, \pm 1, \pm 3)(0, \mp 3, \mp 2, \pm 2), \end{aligned}$$

and from this one obtains the seven-valued function

$$V = (v_\infty - v_0)(v_{\mp 1} - v_{\mp 3})(v_{\pm 1} - v_{\mp 2})(v_{\pm 3} - v_{\pm 2}), \tag{10}$$

where one or the other sign may be taken.

For $p = 11$ one has

$$B = \left(z, -\frac{1}{z}\right), \qquad C = (z, 4z), \qquad U = \left(z, \frac{\mp 1 + z}{2 \mp z}\right),$$

or, represented by cycles:

$$\begin{aligned} B &= (0, \infty)(1, -1)(2, 5)(-2, -5)(3, -4)(-3, 4), \\ C &= (1, 4, 5, -2, 3)(-1, -4, -5, 2, -3), \\ U &= (\infty, \mp 1, \pm 3, \mp 2, \pm 2)(0, \mp 5, \mp 5, \mp 4, \pm 1), \end{aligned}$$

from which arise the two eleven-valued functions

$$\begin{aligned} V = (v_\infty - v_0)&(v_{\mp 1} - v_{\mp 5})(v_{\pm 3} - v_{\mp 5}) \\ &\times (v_{\mp 2} - v_{\mp 4})(v_{\pm 2} - v_{\pm 1})(v_{\pm 4} - v_{\mp 3}). \end{aligned} \tag{11}$$

Here there remains the following remark of interest. The resolvents of degree p, whose roots are the quantities (9), (10), (11), still contain, according to § 78, $\sqrt{\pm p}$ in their coefficients. But the group $\mathfrak{L}_0$ is extended to the group $\mathfrak{L}$ if we adjoin a linear substitution of the form § 78, (13), in which $ad - bc$ is a quadratic non-residue, for instance

$$\begin{aligned} &\text{for } p = 5 && (z, 2z), \\ &\text{for } p = 7, 11 && (z, -z). \end{aligned}$$

By applying this substitution, however, the value (9)

$$V = (v_\infty - v_0)(v_1 - v_4)(v_2 - v_3)$$

passes into its opposite, and from this it follows that, if $p = 5$, V^2 satisfies an equation of the fifth degree with rational numerical coefficients. From this one concludes that in the equation for V the coefficients of the odd powers have the factor $\sqrt{5}$, while the others are rational, or that for the unknown $\sqrt{5}\,V$ one obtains an equation with rational coefficients.

For $p = 7, 11$, the expressions corresponding to the two signs in (10) or (11) pass into one another through the interchange $(z, -z)$. The resolvents of degree 7 or 11 to which one of the two expressions (10), respectively (11), belongs pass into one another by changing the sign of $\sqrt{-7}$, respectively $\sqrt{-11}$.

§ 83. Various resolvents of the fifth degree for the fifth transformation degree.

In the formation of resolvents of degree p we restrict ourselves here to the case $p = 5$.

These resolvents can be given very manifold forms, in that in the function V, (9) of the preceding paragraph, one may not only choose for the quantities v the roots of an arbitrary transformation equation, but may also replace V by many other functions, for example by

$$(v_\infty + v_0)(v_1 + v_4)(v_2 + v_3),$$

or by

$$v_\infty v_0 + v_1 v_4 + v_2 v_3.$$

We shall first apply the transformation equation § 72, (8), in which now, in agreement with the preceding paragraph, v is put in place of x:

$$v^6 + 10v^3 - \gamma_2 v + 5 = 0. \tag{1}$$

Here, if

$$c \equiv 0 \pmod{12}, \qquad z \equiv c \pmod{5},$$

then

$$v_\infty = 5\left(\frac{\eta(5\omega)}{\eta(\omega)}\right)^2, \qquad v_z = \left(\frac{\eta\left(\frac{\omega+c}{5}\right)}{\eta(\omega)}\right)^2. \tag{2}$$

If then we take

$$\begin{aligned} \sqrt{5}\, w_z &= (v_\infty - v_z)(v_{z+1} - v_{z-1})(v_{z+2} - v_{z-2}) \\ &= (v_\infty - v_c)(v_{c+12} - v_{c-12})(v_{c+24} - v_{c-24}), \\ &\quad z = 0, 1, 2, 3, 4, \qquad c = 0, \pm 12, \pm 24, \end{aligned} \tag{3}$$

then, by the concluding remark of the preceding paragraph, w_0, w_1, w_2, w_3, w_4 are the roots of an equation of the fifth degree whose coefficients are formed rationally from γ_2 and rational numbers.

If, however, one observes the relations [§ 54, (14)]

$$\gamma_2(\omega + 1) = e^{-2\pi i/3}\gamma_2(\omega), \qquad \gamma_2\left(-\frac{1}{\omega}\right) = \gamma_2(\omega),$$

then from the form of (1) it follows that under the interchange $(\omega, \omega + 1)$ the roots v, apart from a rearrangement, acquire the factor $e^{2\pi i/3}$, while under $(\omega, -1/\omega)$ they remain unchanged. Thus the w_z only interchange among themselves, and their symmetric functions remain unchanged and therefore depend rationally only on the invariant $j(\omega)$. Moreover the coefficients in the equation for w can become infinite for no finite $j(\omega)$, and consequently are integral functions of $j(\omega)$.

A further indication about these coefficients is obtained from the expansion according to ascending powers of q. By (2), the expansions of the v's have the beginnings

$$v_\infty = 5q^{2/3} + \cdots, \qquad v_z = e^{\frac{c}{12}\frac{2\pi i}{5}} q^{-2/15} + \cdots,$$

and from this, using the known equation

$$4 \sin\frac{2\pi}{5} \sin\frac{4\pi}{5} = \sqrt{5},$$

there follows the beginning of the expansion for w_z:

$$w_z = q^{-2/5} e^{c\pi i/10} + \cdots. \tag{4}$$

It follows from this that the power sums of the w_z, up to and including the fourth, are constants, since they cannot contain even the first power of $j(\omega)$, and that the product of the w_z must be a linear function of $j(\omega)$ in which $j(\omega)$ has coefficient 1.

The equation of the w's therefore has the form

$$w^5 + b_1 w^4 + b_2 w^3 + b_3 w^2 + b_4 w + b_5 = j(\omega), \tag{5}$$

where the b's are rational numbers. These coefficients can be calculated by continuing the expansion (4). We take here another path, which, strictly speaking, belongs to the complex multiplication treated in the next part, but which, because of its simplicity, can nevertheless very well find its place here.

Put $\omega = i = \sqrt{-1}$. Then

$$\omega = -\frac{1}{\omega},$$

and consequently [§ 34, (11), (12), (14)]

$$f_1(i) = f_2(i) = \sqrt[24]{2}, \qquad f(i) = \sqrt[12]{2},$$

hence [§ 54, (5)]

$$\gamma_3(i) = 0, \qquad \gamma_2(i) = 12, \qquad j(i) = 12^3. \tag{6}$$

Furthermore one finds from the transformation formulae for $\eta(\omega)$ [§ 34, (4), (5)]

$$\eta(\omega+1) = e^{\pi i/12}\eta(\omega), \qquad \eta\left(-\frac{1}{\omega}\right) = \sqrt{-i\omega}\,\eta(\omega),$$

if the decomposition

$$5 = (2+i)(2-i)$$

is taken into account:

$$\eta\left(\frac{12+i}{5}\right) = e^{\pi i/6}\eta\left(\frac{2+i}{5}\right) = e^{\pi i/6}\eta\left(\frac{1}{2-i}\right) = \sqrt{1+2i}\,\eta(i),$$

therefore

$$v_2 = 1+2i, \qquad v_3 = 1-2i.$$

Now, since for $\gamma_2 = 12$ two roots of equation (1) are known, for this special value of γ_2 the left hand side of this equation can easily be decomposed into factors:

$$v^6 + 10v^3 - 12v + 5 = (v^2+2v+5)(v^2+v-1)^2,$$

from which one concludes

$$v_\infty = v_0 = \frac{-1+\sqrt{5}}{2}, \qquad v_1 = v_4 = \frac{-1-\sqrt{5}}{2}. \tag{7}$$

(The quantities v_0, v_∞ must be positive, since q is real in them.)

From (3) one then obtains

$$w_0 = 0, \qquad w_1 = w_3 = -5 - 2i\sqrt{5}, \qquad w_2 = w_4 = -5 + 2i\sqrt{5}. \tag{8}$$

These must be the roots of (5) for $j = 12^3$, and from this one obtains the desired resolvent in the form

$$w(w^2+10w+45)^2 = j(\omega) - 12^3 = \gamma_3^2. \tag{9}$$

This equation can also be brought into the form

$$(w+3)^3(w^2+11w+64) = j(\omega) = \gamma_2^3, \tag{10}$$

to which one also comes directly if, instead of putting $\omega = i$, one assumes

$$\omega = \frac{-1 + i\sqrt{3}}{2}.$$

The resolvent found is simplified further if, according to (9), one puts

$$z = \sqrt{w} = \frac{\gamma_3}{w^2 + 10w + 45}.$$

Then

$$z^5 + 10z^3 + 45z = \gamma_3. \tag{11}$$

Or, if according to (10), one puts

$$w^2 + 11w + 64 = y^3, \qquad y = \frac{\gamma_2}{w + 3}, \tag{12}$$

then

$$y^5 - 40y^2 - 5\gamma_2 y - \gamma_2^2 = 0. \tag{13}$$

This equation also results if one makes from the outset the assumption

$$2y = v_\infty v_0 + v_1 v_4 + v_2 v_3.$$

In equation (13), as one sees, the third and fourth powers of the unknown are absent. The same property belongs also, as is easy to prove, to the equation whose roots are

$$x = y(\lambda + \mu z) = \frac{(\lambda + \mu z)\gamma_2}{z^2 + 3}, \tag{14}$$

where λ, μ denote arbitrary parameters which are the same for all five values of x.

If this equation is put in the form

$$x^5 - 5ax^3 - 5b\gamma_2 x - bc\gamma_2^2 = 0, \tag{15}$$

then, after some calculations using the special values of x for $\gamma_3 = 0$ and $\gamma_3 = \infty$, one obtains for a, b, c the following expressions:

$$\begin{aligned} a &= 8\lambda^3 - 72\lambda\mu^2 + \gamma_3(\lambda^2\mu - \mu^3), \\ b &= \lambda^4 + 18\lambda^2\mu^2 - 27\mu^4 + \gamma_3\lambda\mu^3, \\ c &= \lambda^5 + 10\lambda^3\mu^2 + 45\lambda\mu^3 + \gamma_3\mu^5. \end{aligned} \tag{16}$$

One more resolvent of the fifth degree is to be formed by using the function $f(\omega)$.

We put

$$u = f(\omega), \qquad v_\infty = f(5\omega), \qquad v_z = f\left(\frac{\omega + c}{5}\right), \quad c \equiv 0 \pmod{48}, \quad z \equiv c \pmod{5}, \tag{17}$$

and, according to § 73, have between u and v the equation of the sixth degree

$$u^6 + v^6 - u^5v^5 + 4uv = 0. \tag{18}$$

We investigate the influence which the three interchanges

$$(\omega, \omega + 2), \qquad \left(\omega, -\frac{1}{\omega}\right), \qquad \left(\omega, \frac{\omega - 1}{\omega + 1}\right)$$

have on the quantities v_z. Under these interchanges c passes into c_1, c_2, c', which, by § 69, (9), (11) and § 73, (6), are determined by the congruences

$$c_1 \equiv c+2, \qquad c_2 \equiv -\frac{1}{c}, \qquad c' \equiv \frac{c-1}{c+1} \pmod 5. \tag{19}$$

Apart from this change of the index, by § 73, (4) and (9), the v's pass into

$$e^{-5\pi i/12} v, \qquad v, \qquad -\frac{\sqrt{2}}{v}. \tag{20}$$

Thus, if for abbreviation one puts $e^{-5\pi i/12} = \varepsilon$, one has the following corresponding interchanges:

$$\begin{array}{c|ccccccc} & u & v_\infty & v_0 & v_1 & v_2 & v_3 & v_4 \\ \hline \omega+2 & e^{-\pi i/12}u & \varepsilon v_\infty & \varepsilon v_2 & \varepsilon v_3 & \varepsilon v_4 & \varepsilon v_0 & \varepsilon v_1 \\ -1/\omega & u & v_0 & v_\infty & v_4 & v_2 & v_3 & v_1 \\ (\omega-1)/(\omega+1) & \frac{\sqrt{2}}{u} & -\frac{\sqrt{2}}{v_1} & -\frac{\sqrt{2}}{v_4} & -\frac{\sqrt{2}}{v_0} & -\frac{\sqrt{2}}{v_2} & -\frac{\sqrt{2}}{v_3} & -\frac{\sqrt{2}}{v_\infty} \end{array} \tag{21}$$

We now introduce the five quantities w_z by the equation

$$w_z = \frac{(v_\infty - v_z)(v_{z+1} - v_{z-1})(v_{z+2} - v_{z-2})}{\sqrt{5}\, u^3}, \tag{22}$$

that is,

$$\begin{aligned} w_0 &= \frac{(v_\infty - v_0)(v_1 - v_4)(v_2 - v_3)}{\sqrt{5}\, u^3}, \\ w_1 &= \frac{(v_\infty - v_1)(v_2 - v_0)(v_3 - v_4)}{\sqrt{5}\, u^3}, \\ w_2 &= \frac{(v_\infty - v_2)(v_3 - v_1)(v_4 - v_0)}{\sqrt{5}\, u^3}, \\ w_3 &= \frac{(v_\infty - v_3)(v_4 - v_2)(v_0 - v_1)}{\sqrt{5}\, u^3}, \\ w_4 &= \frac{(v_\infty - v_4)(v_0 - v_3)(v_1 - v_2)}{\sqrt{5}\, u^3}. \end{aligned}$$

If in the last row one uses the fact that by (18) the product of the six quantities v has the value u^6, then from (21) there result the following corresponding interchanges of the w's:

$$\begin{array}{c|ccccc} & w_0 & w_1 & w_2 & w_3 & w_4 \\ \hline \omega+2 & -w_2 & -w_3 & -w_4 & -w_0 & -w_1 \\ -1/\omega & w_0 & w_2 & w_1 & w_4 & w_3 \\ (\omega-1)/(\omega+1) & -w_0 & -w_3 & -w_4 & -w_2 & -w_1 \end{array} \tag{23}$$

The functions w can become infinite for no value of u different from 0 and ∞. If therefore we assume the equation whose roots are the five quantities w_z in the form

$$w^5 + A_1 w^4 + A_2 w^3 + A_3 w^2 + A_4 w + A_5 = 0,$$

then, by the principles of § 73, $A_1^2, A_2, A_3^2, A_4, A_5^2$ are integral rational functions of

$$u^{24} + \frac{2^{12}}{u^{24}}. \tag{24}$$

The expansion of (24) according to ascending powers of q begins with q^{-1}, whereas the beginning of the expansion of w_z is

$$q^{-1/10} e^{-c\pi i/60}.$$

The quantities A_1^2, A_2, A_3^2, A_4 therefore cannot contain even the first power of (24), and so are constants, while A_5^2 has the form

$$A_5^2 = u^{24} + \frac{2^{12}}{u^{24}} + C,$$

where C is a constant. The values of the constants can be determined if one knows the values of the w_z for $\omega = i$.

But for $\omega = i$ [compare above (6), (7)] one has

$$u = f(i) = \sqrt[24]{2},$$

$$v_3 = f\left(\frac{i+48}{5}\right) = f\left(\frac{i-2}{5} + 10\right) = e^{-10\pi i/24} f\left(\frac{i-2}{5}\right) = e^{-10\pi i/24} f(i+2) = -i\sqrt[24]{2},$$

$$v_2 = i\sqrt[24]{2},$$

and consequently, by (18),

$$v_\infty = v_0 = \sqrt[24]{2}\frac{1+\sqrt{5}}{2}, \qquad v_1 = v_4 = \sqrt[24]{2}\frac{1-\sqrt{5}}{2}.$$

Consequently

$$w_0 = 0, \qquad w_1 = w_2 = i\sqrt{5}, \qquad w_3 = w_4 = -i\sqrt{5}.$$

First this gives $C = -2^7$, hence

$$A_5^2 = \left(u^{12} - \frac{64}{u^{12}}\right)^2,$$

and from this, by comparing the beginnings of the expansion,

$$A_5 = -u^{12} + \frac{64}{u^{12}},$$

and the equation for w receives the form

$$w(w^2+5)^2 = u^{12} - \frac{64}{u^{12}}. \tag{25}$$

But one can also give it another, very remarkable form.

By the relations existing between the functions $f(\omega), f_1(\omega), f_2(\omega)$ [§ 34, (11), (12)], one has

$$\frac{f(\omega)^{24} - 64}{f(\omega)^{12}} = \frac{[f_1(\omega)^8 - f_2(\omega)^8]^2}{f(\omega)^4}.$$

Thus, if for w we introduce the new unknown

$$y = \sqrt{w} = \frac{f_1(\omega)^8 - f_2(\omega)^8}{f(\omega)^2(w^2+5)}, \tag{26}$$

we obtain the form

$$y^5 + 5y = \frac{f_1(\omega)^8 - f_2(\omega)^8}{f(\omega)^2}. \tag{27}$$

On these formulae rests the solution, created by Hermite and Kronecker, of the equation of the fifth degree by elliptic functions, to which we referred in the fourteenth section of Volume II.

According to Volume I, § 60, the general equation of the fifth degree can be reduced to the form

$$z^5 + 5z = a, \tag{28}$$

and this equation becomes identical with (27) if one puts

$$f_1(\omega)^8 - f_2(\omega)^8 = af(\omega)^2.$$

Adding the equations

$$f_1^8 + f_2^8 = f^8, \qquad f f_1 f_2 = \sqrt{2},$$

one obtains

$$f^{24} - a^2 f^{12} - 64 = 0. \tag{29}$$

By this quadratic equation, f^{12} is determined as a function of a.

Then the roots of equation (28) are represented by the square roots of the expressions (22). The double sign of these square roots is explained by the fact that two opposite values of a lead to the same equation (29). But the roots can also be determined unambiguously by formula (26):

$$z = \frac{a}{w^2 + 5}. \tag{30}$$

In the above-mentioned work of Kiepert, the general equation of the fifth degree is transformed into the form (15), for which only the solution of a quadratic equation is required. If in it a, b, c are regarded as arbitrarily given, then equations (16) serve for the determination of λ, μ, γ_3. This too occurs with the help of a quadratic equation, although this is not apparent at first glance. (Compare F. Klein, *Vorlesungen über das Ikosaeder*, p. 191 f.)

We shall return below to the formation of resolvents for the seventh transformation degree, when we have at our disposal the aids supplied by complex multiplication.

Volume III. Second Book. Quadratic Fields

Ninth Section. Discriminant

§ 84. Definition of discriminants

In the theory of quadratic fields there occur, as discriminants, certain rational integers, positive or negative, which, if they are even, are divisible by 4, and, if they are odd, leave the remainder 1 when divided by 4. Kronecker called such numbers "numbers of discriminant form." We shall here call them simply discriminants, without being disturbed by the fact that the word has many other meanings in algebra. We define:

1. A positive or negative non-zero integer which satisfies one of the two congruences

$$D \equiv 0, \quad D \equiv 1 \pmod 4 \tag{1}$$

is called a discriminant.

Every square number is, by this definition, a discriminant. Since, however, squares occupy a certain exceptional position, we shall call them improper discriminants when a distinction is necessary.

2. The product of two, and consequently of any number of, discriminants is again a discriminant.

3. A discriminant which, apart from 1, contains no quadratic divisor after whose removal a discriminant remains is called a fundamental discriminant. Fundamental discriminants will subsequently be denoted by Δ. If D is not a fundamental discriminant, there is one and only one square Q^2 and one fundamental discriminant Δ such that

$$D = Q^2 \Delta. \tag{2}$$

Then Δ is called the fundamental discriminant, or the stem, of D.

A fundamental discriminant is divisible by no odd square. If, however, Δ is divisible by 4, then necessarily

$$\Delta \equiv 8, 12 \pmod{16}. \tag{3}$$

For if $\Delta \equiv 0, 4 \pmod{16}$, then after removing the factor 4 a discriminant would still remain. Thus, in order to find the stem of a given D, one first removes the greatest odd square divisor and then a sufficiently high power of 4, until the remaining Δ is either odd and $\equiv 1 \pmod 4$, or is $\equiv 8$ or $12 \pmod{16}$.

4. A proper discriminant which is not decomposable into factors which themselves are discriminants is called a prime discriminant. If p is an odd natural prime number, and the sign is so chosen that $\pm p \equiv 1 \pmod 4$, then the prime discriminants are

$$\pm p, \,-4, \,+8, \,-8. \tag{4}$$

5. Every fundamental discriminant can be decomposed in one and only one way into prime discriminants. For if one separates from Δ first all factors $\pm p$, only one of the numbers $1, -4, +8, -8$ remains.

§ 85. The extended Legendre-Jacobi symbol

The Legendre symbol $\left(\frac{m}{p}\right)$, when m is any positive or negative non-zero integer and p is a natural prime number not dividing m, has the value $+1$ if m is a quadratic residue of p, and the value -1 if m is a quadratic non-residue of p. Jacobi extended the meaning of the symbol so that a composite number can stand in place of p. If m is a discriminant, however, a still further extension is often convenient.

If D is a discriminant and n an integer, we define a symbol (D, n) by the following rules:

$$(D, 0) = 0, \tag{1}$$
$$(D, 1) = 1, \tag{2}$$
$$(D, -1) = \begin{cases} 1, & D > 0, \\ -1, & D < 0, \end{cases} \tag{3}$$
$$(D, p) = 0 \quad \text{if } p \mid D, \tag{4}$$
$$(D, 2) = (-1)^{(D^2-1)/8}, \quad D \text{ odd}, \tag{5}$$
$$(D, p) = \left(\frac{D}{p}\right) \quad (p \text{ odd prime}, \ p \nmid D). \tag{6}$$

If $\pm n$ is decomposed into prime factors,

$$\pm n = p_1 p_2 \cdots p_r,$$

then

$$(D, n) = (D, \pm 1)(D, p_1)(D, p_2) \cdots (D, p_r). \tag{7}$$

The symbol is thereby defined without contradiction for all n, and has only the values $-1, 0, +1$. It is zero precisely when D and n are not relatively prime. From (7) follows

$$(D, nn') = (D, n)(D, n'), \tag{8}$$

and, if q^2 is a square prime to n,

$$(q^2 D, n) = (D, n). \tag{9}$$

The formulas belonging to the law of reciprocity take a special form in this notation. If $\pm p \equiv 1 \pmod 4$, then for every n not divisible by p,

$$(\pm p, n) = \left(\frac{\pm p}{n}\right). \tag{10}$$

For odd n one has likewise

$$(-4,n) = (-1)^{(n-1)/2}, \qquad (8,n) = (-1)^{(n^2-1)/8}, \qquad (-8,n) = (-1)^{(n-1)/2+(n^2-1)/8}. \tag{11}$$

Further, if D, D' are two discriminants, then

$$(DD',n) = (D,n)(D',n). \tag{12}$$

The reciprocity law with its supplementary theorems can then be summarized as

$$(D,D') = \pm(D',D), \tag{13}$$

where the upper sign holds when at least one of the two discriminants D, D' is positive, and the lower sign when both are negative.

Since every square is a positive discriminant, one obtains, if m is prime to D,

$$(D,m^2) = 1, \tag{14}$$

and, together with (8),

$$(D,mn^2) = (D,m) \tag{15}$$

whenever mn and D have no common divisor. Moreover

$$(D,n) = (D,n') \quad \text{if } n \equiv n' \pmod{D}. \tag{16}$$

If n, n' in (16) are themselves discriminants, one obtains from (13) and (16)

$$(D_1,D) = \pm(D_2,D), \tag{17}$$

where the lower sign occurs only when the two numbers D and D_1D_2 are both negative. If D_1, D_2 are odd and $D = 4m$, this gives

$$(D_1,m) = \pm(D_2,m), \tag{18}$$

provided

$$D_1 \equiv D_2 \pmod{4m}. \tag{19}$$

The same formula holds in general. The formulas (16) and (18) can be used to compute the symbol (D,n) by the algorithm of the greatest common divisor.

If D is not a square, one can always determine a number β such that

$$(D,\beta) = -1. \tag{21}$$

For if Δ is the stem of D, and β is prime to D, then $(D,\beta) = (\Delta,\beta)$. Decomposing Δ into prime discriminants, one chooses β to make one factor give -1 and the others give $+1$. Therefore, if s runs through a complete system of residues modulo D, and if

$$S = \sum_s (D,s),$$

then multiplication by such a β gives also $S = -S$, and therefore

$$S = 0. \tag{22}$$

Thus in a complete system of incongruent numbers prime to D there are as many numbers α with $(D,\alpha) = +1$ as numbers β with $(D,\beta) = -1$.

§ 86. The Gaussian sums

In the first volume we became acquainted with the Gaussian sums from the theory of cyclotomy. These expressions can be generalized, and, by using the symbol (D, n), take a simple form.

It was shown previously that, if n is an odd prime number, if s is not divisible by n, and if k runs through a complete residue system modulo n, then

$$\sum_k \left(\frac{k}{n}\right) e^{2\pi isk/n} = \left(\frac{s}{n}\right) \sqrt{\pm n}, \tag{1}$$

with the usual choice of sign according as $n \equiv 1$ or $3 \pmod 4$. Put $\Delta = \pm n \equiv 1 \pmod 4$. Then Δ is a prime discriminant, and, by §85,

$$(\Delta, k) = \left(\frac{\Delta}{k}\right) = \left(\frac{k}{n}\right),$$

with the corresponding supplementary signs included. Formula (1) may therefore be written

$$\sum_k (\Delta, k) e^{2\pi isk/\Delta} = (\Delta, s)\sqrt{\Delta}, \tag{2}$$

where $\sqrt{\Delta}$ is positive real or positive imaginary according as Δ is positive or negative. Direct calculation shows that the same formula also holds for

$$\Delta = -4, \quad +8, \quad -8,$$

and hence for every prime discriminant.

Assume the formula true for two relatively prime fundamental discriminants Δ' and Δ''. Since

$$\sqrt{\Delta'}\sqrt{\Delta''} = \pm\sqrt{\Delta'\Delta''}, \tag{3}$$

where the lower sign occurs only when both factors are negative, put

$$k = k'\Delta'' - k''\Delta'.$$

As k', k'' run through complete residue systems modulo Δ', Δ'', the number k runs through a complete residue system modulo $\Delta = \Delta'\Delta''$. Multiplication of the two sums gives

$$\sum_k (\Delta, k) e^{2\pi isk/\Delta}. \tag{5}$$

Using §85, (16) and the reciprocity formula §85, (13), one obtains

$$(\Delta, k) = (\Delta', k')(\Delta'', k''),$$

and consequently (2) is valid for the product $\Delta'\Delta''$. By decomposition into prime discriminants, formula (2) is therefore valid for every fundamental discriminant Δ.

Tenth Section. Algebraic numbers and forms.

§ 87. Ideals and forms in algebraic fields.

The interest which the theory of elliptic functions has for the algebraist arises from its application to the theory of quadratic irrational numbers, to which it stands in an analogous relation as the roots of unity stand to the rational numbers. It offers us the first, and until now the only, example, more exactly known in its laws, of a domain of algebraic numbers which goes beyond the cyclotomic numbers. In order to present the theory of these numbers more closely, we must first place before it some theorems from the theory of forms and algebraic fields, and especially of quadratic forms.

In Vol. II, §§ 163, 169, a connection was obtained between the ideals of an algebraic field Ω of degree n and certain homogeneous functions of degree n in n variables. Before applying this to quadratic fields, let us briefly recall the general theorems.

Let $\mathfrak{a}$ be an ideal of a field Ω, and let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be a basis of this ideal. Further let $\omega_1, \omega_2, \ldots, \omega_n$ be a minimal basis of Ω. Then

$$(\alpha_1, \alpha_2, \ldots, \alpha_n) = (A)(\omega_1, \omega_2, \ldots, \omega_n), \tag{1}$$

where (A) denotes a linear substitution with integral coefficients. To pass to another basis $\alpha'_1, \alpha'_2, \ldots, \alpha'_n$ of $\mathfrak{a}$, make a linear substitution L with determinant ± 1,

$$(\alpha'_1, \alpha'_2, \ldots, \alpha'_n) = (L)(\alpha_1, \alpha_2, \ldots, \alpha_n), \tag{2}$$

whence

$$(\alpha'_1, \alpha'_2, \ldots, \alpha'_n) = (L)(A)(\omega_1, \omega_2, \ldots, \omega_n). \tag{3}$$

By the norm $N(\mathfrak{a})$ of the ideal $\mathfrak{a}$ one understands the absolute value of the determinant A of the substitution (A):

$$A = \sum \pm a_{1,1} a_{2,2} \cdots a_{n,n},$$

and by (3) the norm is independent of the choice of basis.

1. We shall call the basis (α_ν) positive if the determinant A is positive, and therefore equal to the norm of $\mathfrak{a}$. If (α_ν) is positive, then (α'_ν) is positive precisely when the determinant of the substitution is $L = +1$.

From every basis one can derive a positive basis by a substitution with determinant -1, for example by interchanging two of the α_ν, and in what follows we shall mostly use only positive bases.

The basis determined in Vol. II, § 163, (3), is positive. Let $t_1, t_2, \ldots, t_n$ be a system of independent variables and

$$\lambda = \sum_r \alpha_r t_r \tag{4}$$

a basis form with positive basis. Then (Vol. II, § 164)

$$N(\lambda) = N(\mathfrak{a}) T, \tag{5}$$

where

$$T = \sum \pm t_{1,1} t_{2,2} \cdots t_{n,n} \tag{6}$$

is a primitive integral homogeneous form of degree n in the variables t_ν. The $t_{r,s}$ are linear functions of the t_ν.

Applying the substitution (2) to (4), one obtains

$$\lambda = \lambda' = \sum_r \alpha'_r t'_r, \tag{7}$$

where

$$(t_1, t_2, \ldots, t_n) = L'(t'_1, t'_2, \ldots, t'_n), \tag{8}$$

and L' is the transposed substitution of L. If (α'_ν) is likewise a positive basis, then L' has determinant $+1$, and formula (5) gives

$$N(\lambda) = N(\mathfrak{a}) T', \tag{9}$$

where T' is a homogeneous function of degree n in the variables t' which arises from T by the substitution (8).

Two forms which arise from one another by an integral linear substitution with determinant +1 are called equivalent. Sometimes the forms are also called equivalent when the determinant of the substitution is −1, but then with the qualification "improperly." Thus (8) proves:

2. If we call T the form belonging to the basis (α_ν) of $\mathfrak{a}$, then the forms belonging to the different positive bases of the same ideal are equivalent.

Denote the conjugate values of α_s, ω_s by

$$\alpha_{s,1}, \alpha_{s,2}, \ldots, \alpha_{s,n}; \qquad \omega_{s,1}, \omega_{s,2}, \ldots, \omega_{s,n},$$

and put

$$\alpha_{s,r} = \sum_\nu a_{s,\nu}\omega_{\nu,r}. \tag{10}$$

Then

$$\left(\sum \pm\alpha_{1,1}\alpha_{2,2}\cdots\alpha_{n,n}\right)^2 = N(\mathfrak{a})^2\Delta, \tag{11}$$

where Δ is the fundamental number of the field Ω, namely the square of the determinant

$$\Delta = \left(\sum \pm\omega_{1,1}\omega_{2,2}\cdots\omega_{n,n}\right)^2$$

(Vol. II, § 162).

If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the conjugate values of λ, then by (4)

$$\lambda_r = \sum_\nu \alpha_{\nu,r}t_\nu \tag{12}$$

is a linear substitution for the variables t_ν with substitution determinant

$$r = N(\mathfrak{a})\sqrt{\Delta}, \tag{13}$$

and through this, by (5), the form T passes into the product

$$T' = [N(\mathfrak{a})]^{-1}\lambda_1\lambda_2\cdots\lambda_n, \tag{14}$$

because $N(\lambda)$ is the product of the conjugate values of λ (Vol. II, § 151). We can apply invariant theory to this (Vol. I, § 65), and if we form the Hessian determinant

$$H = \det\left(\frac{\partial^2 T}{\partial t_i \partial t_j}\right), \tag{15}$$

then

$$H' = r^{n-2}H. \tag{16}$$

If, however, we form H' from (14) in the variables λ_ν and observe that an n-rowed determinant whose diagonal terms are 0 and all other terms are 1 has the value $(-1)^{n-1}(n-1)$, [24] then

$$H' = (-1)^{n-1}(n-1)N(\mathfrak{a})^{-n}(\lambda_1\lambda_2\cdots\lambda_n)^{n-2} = (-1)^{n-1}(n-1)N(\mathfrak{a})^{-2}T^{n-2},$$

and consequently, by (13) and (16),

$$H = (-1)^{n-1}(n-1)T^{n-2}\Delta. \tag{17}$$

[24] One can easily prove this theorem by adjoining an $(n+1)$st row and an $(n+1)$st column to the determinant, in such a way that in the adjoined row only ones occur and in the column, except for the last element, zeros occur.

§ 88. Ideal classes and form classes.

In Vol. II, § 170, we explained the equivalence of ideals and the ideal classes. Accordingly two ideals $\mathfrak{a}, \mathfrak{b}$ were equivalent if there is an integral or fractional number of the field Ω which is the quotient of $\mathfrak{b}$ and $\mathfrak{a}$, that is,

$$\eta\mathfrak{a} = \mathfrak{b}, \tag{1}$$

and it was shown that the ideals are thereby divided into classes, and that the number of these classes is finite. In some cases it is expedient to take the concept of equivalence somewhat more narrowly and to call $\mathfrak{a}$ and $\mathfrak{b}$ equivalent only when there is a number η satisfying condition (1) with positive norm. This distinction naturally does not arise when the norms of all numbers are positive, as for instance in the imaginary quadratic field. It also does not arise when there are units of norm -1, since, if ε is such a unit, then either $N(\eta)$ or $N(\varepsilon\eta)$ is positive, and ε and $\varepsilon\eta$ simultaneously satisfy condition (1).

But if there are no units whose norm is -1, although there are other numbers in Ω with negative norm, then under the narrower definition of equivalence each ideal class divides once more into two classes. We shall keep here to the narrower concept of equivalence, which for example is in force in some real quadratic fields. [25]

In (1), η may be fractional. But if α is an integral number divisible by $\mathfrak{a}$, then $\eta\alpha$ is an integral number divisible by $\mathfrak{b}$, and from this one obtains the following. If

$$(\alpha_1, \alpha_2, \dots, \alpha_n) \tag{2}$$

is a basis of $\mathfrak{a}$, then

$$(\beta_1, \beta_2, \dots, \beta_n) = (\alpha_1\eta, \alpha_2\eta, \dots, \alpha_n\eta)$$

is a basis of $\mathfrak{b}$; and if the first basis is positive, then, under the narrower concept of equivalence, so is the second.

For the necessary and sufficient condition that (2) be a basis of $\mathfrak{a}$ is that every integral number α divisible by $\mathfrak{a}$, and only such a number, be contained in the form

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n \tag{3}$$

with integral $x_1, x_2, \dots, x_n$. Consequently all numbers

$$\beta = x_1\alpha_1\eta + x_2\alpha_2\eta + \dots + x_n\alpha_n\eta \tag{4}$$

are divisible by $\mathfrak{b}$. Conversely, if β is an integral number divisible by $\mathfrak{b}$, then β/η is divisible by $\mathfrak{a}$ and hence representable in the form (3). Thus β is representable by (4).

Furthermore, using the notation of (9) and (10), § 87, when α is a positive basis:

$$\sum \pm\alpha_{1,1}\alpha_{2,2}\cdots\alpha_{n,n} = N(\mathfrak{a})\sum \pm\omega_{1,1}\omega_{2,2}\cdots\omega_{n,n},$$

$$\sum \pm\beta_{1,1}\beta_{2,2}\cdots\beta_{n,n} = N(\eta)\sum \pm\alpha_{1,1}\alpha_{2,2}\cdots\alpha_{n,n} = \pm N(\mathfrak{b})\sum \pm\omega_{1,1}\omega_{2,2}\cdots\omega_{n,n},$$

and since by (1)

$$N(\mathfrak{b}) = N(\eta)N(\mathfrak{a}), \tag{5}$$

the positive sign must stand at $\pm N(\mathfrak{b})$.

If λ is a basis form of $\mathfrak{a}$, then $\eta\lambda$ is a basis form of $\mathfrak{b}$, and

$$\begin{aligned} N(\lambda) &= N(\mathfrak{a})T, \\ N(\eta\lambda) &= N(\mathfrak{b})T, \end{aligned} \tag{6}$$

[25] Namely, when Pell's equation $t^2 - Du^2 = -4$ has no solution.

and hence the same form T belongs to the two ideals $\mathfrak{a}, \mathfrak{b}$. If equivalent forms T are combined into a class, we can say by 2.:

3. To every ideal class there belongs a determinate form class.

In answering the question whether different ideal classes can belong to one and the same form class, we make the assumption that Ω is a normal field. Suppose, then, that equivalent forms T and T' belong to two ideals $\mathfrak{a}$ and $\mathfrak{a}'$. If λ is a basis form of $\mathfrak{a}$, then

$$N(\lambda) = N(\mathfrak{a})T, \tag{7}$$

and if we carry T' into T by a linear substitution, and understand by λ' a basis form of $\mathfrak{a}'$ (with the same variables t as λ), then

$$N(\lambda') = N(\mathfrak{a}')T. \tag{8}$$

The linear factors of $N(\lambda)$ and $N(\lambda')$ must therefore agree with one another, apart from a constant factor, since an integral function of arbitrary variables can be decomposed into irreducible factors (here linear factors) in only one way (Vol. I, § 20, Vol. II, § 152); and since all these factors belong to the same field Ω, there is among the factors conjugate to λ' one which satisfies the condition

$$\lambda = \eta\lambda',$$

where η is a number in Ω. The functionals λ and λ', and therefore the corresponding ideals, are thus equivalent.

4. If, in a normal field, the same form class belongs to two ideals $\mathfrak{a}$ and $\mathfrak{a}'$, then among the ideals conjugate to $\mathfrak{a}'$ there is one which is equivalent to $\mathfrak{a}$.

§ 89. Composition of forms and multiplication of ideals.

Let $\mathfrak{a}$ and $\mathfrak{b}$ now be two ideals in Ω and let

$$\mathfrak{c} = \mathfrak{a}\mathfrak{b} \tag{1}$$

be the product formed from them. Further let

$$\begin{aligned} \mathfrak{a} &= (\alpha_1, \alpha_2, \dots, \alpha_n), \\ \mathfrak{b} &= (\beta_1, \beta_2, \dots, \beta_n), \\ \mathfrak{c} &= (\gamma_1, \gamma_2, \dots, \gamma_n) \end{aligned} \tag{2}$$

be bases of these ideals. Since each number $\alpha_h\beta_k$ belongs to $\mathfrak{c}$, there is a system of integral rational numbers $c_{h,k}^{(r)}$ such that

$$\alpha_h\beta_k = \sum_r c_{h,k}^{(r)}\gamma_r, \tag{3}$$

and since conversely each number in $\mathfrak{c}$ arises by multiplying one number from $\mathfrak{a}$ and one from $\mathfrak{b}$ and adding these products (Vol. II, § 169), one also has

$$\gamma_s = \sum_{h,k} a_{h,k}^{(s)}\alpha_h\beta_k, \tag{4}$$

where the $a_{h,k}^{(s)}$ are likewise integral rational numbers.

Substituting (3) in (4) gives

$$\gamma_s = \sum_r \gamma_r \sum_{h,k} a_{h,k}^{(s)}c_{h,k}^{(r)} \tag{5}$$

and hence

$$\sum_{h,k} a_{h,k}^{(s)}c_{h,k}^{(r)} = (r, s), \tag{6}$$

where $(r, r) = 1$, $(r, s) = 0$, $r \neq s$.

Thus let x_r denote variables and put

$$\sum_r x_r c_{h,k}^{(r)} = y_{h,k}. \tag{7}$$

It follows that

$$x_s = \sum_{h,k} a_{h,k}^{(s)} y_{h,k}. \tag{8}$$

Now let u_r, v_r, t_r denote three systems of variables and

$$\begin{aligned} \mu &= \sum \alpha_r u_r, \\ \nu &= \sum \beta_r v_r, \\ \lambda &= \sum \gamma_r t_r \end{aligned} \tag{9}$$

basis forms of $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, and let U, V, T be the forms belonging to $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, written in the variables u, v, t. Then

$$\begin{aligned} N(\mu) &= N(\mathfrak{a})U, \\ N(\nu) &= N(\mathfrak{b})V, \\ N(\lambda) &= N(\mathfrak{c})T. \end{aligned} \tag{10}$$

From (3), by multiplication with $u_h v_k$ and summation according to (9), one obtains

$$\mu\nu = \sum_r \gamma_r \sum_{h,k} c_{h,k}^{(r)} u_h v_k, \tag{11}$$

so that, if

$$t_r = \sum_{h,k} c_{h,k}^{(r)} u_h v_k \tag{12}$$

is put,

$$\mu\nu = \lambda, \tag{13}$$

and since $N(\mathfrak{a})N(\mathfrak{b}) = N(\mathfrak{c})$, it follows by (10) that

$$T = UV. \tag{14}$$

Here, however, the t_r are no longer independent variables; rather, they arise from u_h, v_k by the bilinear substitution (12).

5. The form T passes, by the bilinear substitution (12), into the product of the two forms U, V. This substitution, however, has one more essential peculiarity. We call the functions t_r determined by (12) linearly independent modulo a prime number p if from the congruence

$$\sum x_r t_r = 0 \pmod{p}, \tag{15}$$

where the x_r are integral rational numbers, it follows that all these numbers are divisible by p. If this is the case for every arbitrary prime number p, the t_r are simply called linearly independent.

The congruence (15), by (12) and (7), is equivalent to the n^2 congruences

$$y_{h,k} = \sum_r x_r c_{h,k}^{(r)} \equiv 0 \pmod{p},$$

and then (6) gives

$$x_s \equiv 0 \pmod{p}.$$

This says that the substitutions (12) are linearly independent for every modulus p.

According to Gauss (Disq. aritm. art. 235), a form T is called composed or compounded from the two forms U, V if T passes into the product UV by a bilinear substitution for the variables t whose equations are linearly independent for every modulus.

Consequently we have the theorem:

6. The form which belongs to the product of two ideals $\mathfrak{a}, \mathfrak{b}$ is composed from the forms of the ideals $\mathfrak{a}$ and $\mathfrak{b}$. [26]

Eleventh Section. Ideals in quadratic fields.

§ 90. Discriminant of the quadratic field.

A quadratic field arises when one adjoins a square root $\sqrt{d}$ to the field of rational numbers; here d may be assumed to be an integer without a square divisor. The field is real or imaginary according as d is positive or negative. If one denotes the field of rational numbers, the absolute domain of rationality, by $\mathfrak{R}$, and by $\mathfrak{R}(x, x', \ldots)$ the field which arises by adjoining $x, x', \ldots$ to $\mathfrak{R}$, then the quadratic field Ω is to be denoted by

$$\Omega = \mathfrak{R}(\sqrt{d}). \tag{1}$$

Every number of the field Ω can be put in the form

$$\omega = \frac{x + y\sqrt{d}}{2}, \tag{2}$$

where x, y are rational integral or fractional numbers. The number conjugate to ω,

$$\omega' = \frac{x - y\sqrt{d}}{2}, \tag{3}$$

is likewise contained in the field Ω, and consequently Ω is a normal field.

In order that ω be an integer it is necessary that

$$\omega + \omega' = x, \qquad (\omega - \omega')^2 = y^2 d$$

be rational integers; and since d is to have no square divisor, x and y must be integers. But in order that ω really be an integer it is also still necessary that the norm $\frac{1}{4}(x^2 - y^2 d)$ be a rational integer, that is, it must be

$$x^2 - y^2 d \equiv 0 \pmod{4}. \tag{4}$$

If $d \equiv 2$ or $\equiv 3 \pmod{4}$, then this condition can be fulfilled only when x and y are even, and if therefore we replace x, y by $2x, 2y$, there follows:

1. If $d \equiv 2, 3 \pmod{4}$, then all and only those numbers of the field Ω are integral which are contained in the form

$$x + y\sqrt{d}$$

with rational integers x, y.

If, however, $d \equiv 1 \pmod{4}$, then (3) is satisfied when x and y are both even or both odd, hence when x has the form $2x_1 - y$. If one then again replaces x_1 by x, one obtains:

2. If $d \equiv 1 \pmod{4}$, then all and only those numbers in Ω are integral which are contained in the form

$$x + y\frac{-1 + \sqrt{d}}{2}$$

[26] The theorems which Gauss proves at the cited place for binary quadratic forms by a very extensive calculation are here derived in the greatest generality by simple considerations. Cf. Dirichlet-Dedekind, Lectures on Number Theory, § 182. For binary quadratic forms: Dedekind, Crelle's Journal, vol. 129; H. Weber, Göttinger Nachrichten 1907.

with rational integers x, y.

Thus in case 1, $(1, \sqrt{d})$, and in case 2, $\left(1, \frac{-1+\sqrt{d}}{2}\right)$, is a minimal basis of the field Ω. The fundamental number, or discriminant, of the field Ω is therefore, in case 1,

$$\Delta = \begin{vmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{d} \end{vmatrix}^2 = 4d, \tag{5}$$

and in case 2,

$$\Delta = \begin{vmatrix} 1 & \dfrac{-1+\sqrt{d}}{2} \\ 1 & \dfrac{-1-\sqrt{d}}{2} \end{vmatrix}^2 = d. \tag{6}$$

In both cases, therefore, the integral number ω can be put in the form

$$\omega = \frac{x + y\sqrt{\Delta}}{2}, \tag{7}$$

with the condition that

$$4N(\omega) = x^2 - \Delta y^2 \equiv 0 \pmod 4 \tag{8}$$

be satisfied.

The two forms of the minimal basis which we have obtained, namely $(1, \sqrt{d})$ and $\left(1, \frac{-1+\sqrt{d}}{2}\right)$, are therefore

$$(1, \theta) = \left(1, \frac{1}{2}\sqrt{\Delta}\right), \quad \Delta \equiv 0, \qquad = \left(1, \frac{-1+\sqrt{\Delta}}{2}\right), \quad \Delta \equiv 1 \pmod 4). \tag{9}$$

Here the sign of $\sqrt{\Delta}$ may be determined in any arbitrary way, but in the same consideration it is then to be held fixed. We shall assume once for all:

3. If Δ is positive, then $\sqrt{\Delta}$ is likewise to be positive; and if Δ is negative, then $-i\sqrt{\Delta}$ is to be positive, or $\sqrt{\Delta}$ is to be positive imaginary.

From (1), (2) there still follows:

4. The fundamental number of a quadratic field is always $\equiv 0$ or $\equiv 1$ modulo 4 and, apart from 4, has no square divisor. It is therefore a discriminant, indeed a fundamental discriminant.

§ 91. Ideals and forms in quadratic fields.

Let $\mathfrak{a}$ be an ideal of the quadratic field Ω, let α_1, α_2 be a positive basis, and let

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 \tag{1}$$

be a basis form of $\mathfrak{a}$. Then

$$\mathrm{N}(\lambda) = (\alpha_1 t_1 + \alpha_2 t_2)(\alpha_1' t_1 + \alpha_2' t_2) = \mathrm{N}(\mathfrak{a})T, \tag{2}$$

and $\mathrm{N}(\mathfrak{a})$ is the greatest common divisor of the three rational integers

$$\alpha_1\alpha_1', \qquad \alpha_1\alpha_2' + \alpha_2\alpha_1', \qquad \alpha_2\alpha_2'.$$

Thus if we put

$$\begin{aligned} \alpha_1\alpha_1' &= a\,\mathrm{N}(\mathfrak{a}), \\ \alpha_1\alpha_2' + \alpha_2\alpha_1' &= b\,\mathrm{N}(\mathfrak{a}), \\ \alpha_2\alpha_2' &= c\,\mathrm{N}(\mathfrak{a}), \end{aligned} \tag{3}$$

then

$$T = at_1^2 + bt_1t_2 + ct_2^2 \tag{4}$$

is the form belonging to the basis α_1, α_2. We denote it, when the designation of the variables is immaterial, by

$$\varphi = (a, b, c). \tag{5}$$

It is a primitive quadratic form whose discriminant

$$\Delta = b^2 - 4ac \tag{6}$$

is equal to the fundamental number of the field Ω (§ 87). To form the basis α_1, α_2 according to Vol. II, § 163, first seek the least natural number a_{11} divisible by $\mathfrak{a}$, and then the least natural number a_{22} for which a rational number a_{12} satisfying the condition

$$a_{12} + a_{22}\theta \equiv 0 \pmod{\mathfrak{a}} \tag{7}$$

can be determined. Since the congruence (7) can be satisfied for $a_{22} = a_{11}$, namely by $a_{12} = 0$, it follows that a_{11} is divisible by a_{22}, and we put $a_{11} = a_{22}a$. Then we have the basis of $\mathfrak{a}$:

$$\alpha_1 = a_{22}a, \qquad \alpha_2 = a_{12} + a_{22}\theta. \tag{8}$$

Then, according to Vol. II, § 164, (12),

$$\mathrm{N}(\mathfrak{a}) = a_{22}^2 a, \tag{9}$$

and if for $(1, \theta)$ we take the basis § 90, (9), then

$$\theta + \theta' = 0 \text{ or } = -1, \qquad \theta\theta' = -\frac{1}{4}\Delta \text{ or } = \frac{1-\Delta}{4}, \tag{10}$$

according as $\Delta \equiv 0$ or $\equiv 1 \pmod 4$, and

$$(\theta - \theta')^2 = \Delta.$$

In these two cases, if one substitutes (8) in (3) and takes account of (9), one obtains

$$a_{22}b = 2a_{12} \quad \text{or} \quad = 2a_{12} - a_{22},$$

and hence in both cases

$$\alpha_1 = a_{22}a, \qquad \alpha_2 = a_{22}\frac{b + \sqrt{\Delta}}{2}; \tag{11}$$

and the form

$$T = (a, b, c)$$

has here a positive first coefficient.

Put

$$\omega = -\frac{\alpha_2}{\alpha_1} = -\frac{b + \sqrt{\Delta}}{2a}. \tag{12}$$

Then ω is a root of the quadratic equation

$$a\omega^2 + b\omega + c = 0, \tag{13}$$

and

$$\lambda = a_{22}a(x - \omega y) \tag{14}$$

is a functional corresponding to the ideal $\mathfrak{a}$ and at the same time a basis form of $\mathfrak{a}$. The number a is the smallest positive integer for which $a\omega$ becomes an integer.

The necessary and sufficient condition that two integers α_1, α_2 of the field Ω form the basis of an ideal consists in this: first, α_1, α_2 must be a basis of the field Ω; and, if ω_1, ω_2 is a minimal basis of this field, then

$$\alpha_1\omega_1, \quad \alpha_2\omega_1, \quad \alpha_1\omega_2, \quad \alpha_2\omega_2$$

must be representable by the linear form

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2$$

with integral t_1, t_2 (Vol. II, § 163, 164). For then, if α is representable by λ, every product $\omega\alpha$ is also representable by λ. In the present case this condition reduces to the requirement that $\alpha_1\theta$ and $\alpha_2\theta$ be representable by λ. But if (a, b, c) is any quadratic form of discriminant Δ with positive first coefficient a, we can take for θ also $\frac{1}{2}(b + \sqrt{\Delta})$, and since

$$\left(\frac{b+\sqrt{\Delta}}{2}\right)^2 = -ac + b\frac{b+\sqrt{\Delta}}{2},$$

(11) is always the basis of an ideal in Ω. Here the positive number a_{22} may be assumed arbitrarily. If we put $a_{22} = 1$, then

$$\lambda = a(x - \omega y) \tag{15}$$

is the basis form of an ideal $\mathfrak{a}$ completely determined by (a, b, c), whose norm is equal to a; and a basis of this ideal is

$$(a, a\omega). \tag{16}$$

§ 92. Prime ideals in the quadratic field.

A prime number p is, in the quadratic field Ω, either itself still indecomposable, and then is a prime ideal of the second degree, or it decomposes into two prime ideals $\mathfrak{p}, \mathfrak{p}'$ of the first degree. Thus we have to distinguish two cases:

$$\begin{aligned} &\text{a)} \quad p = \mathfrak{p} \text{ indecomposable in } \Omega: \quad \mathrm{N}(\mathfrak{p}) = p^2, \\ &\text{b)} \quad p = \mathfrak{p}\mathfrak{p}', \qquad \mathrm{N}(\mathfrak{p}) = \mathrm{N}(\mathfrak{p}') = p. \end{aligned}$$

In the latter case the two prime ideals $\mathfrak{p}, \mathfrak{p}'$ can be distinct from one another or can also be equal, so that we must distinguish still a third case:

$$\text{c)} \qquad p = \mathfrak{p}^2, \qquad \mathrm{N}(\mathfrak{p}) = p.$$

The matter is now to determine the conditions under which one or another of these cases occurs.

If $\mathrm{N}(\mathfrak{p}) = p$, then p is the number of incongruent numbers modulo $\mathfrak{p}$ in Ω, and consequently, in this case, every number in Ω is congruent to one of the rational numbers

$$0, 1, 2, \ldots, p-1. \tag{1}$$

If, on the other hand, $\mathrm{N}(\mathfrak{p}) = p^2$, then there are also numbers which, modulo $\mathfrak{p}$, are not congruent to a rational number.

But if $(1, \theta)$ is a basis of Ω and, in order to include both cases of § 90, (9),

$$\theta = \frac{-\Delta + \sqrt{\Delta}}{2} \tag{2}$$

is assumed, then in case a) θ is not congruent to a number r, while in case b) or c)

$$\theta \equiv r \pmod{\mathfrak{p}}.$$

If therefore one puts $-x = 2r + \Delta$, it follows:

1. The necessary and sufficient condition for the occurrence of one of the cases b), c) is that there be a rational integer, even or odd simultaneously with Δ, for which

$$\frac{x+\sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}}. \tag{3}$$

If $\mathfrak{p}'$ is the ideal conjugate to $\mathfrak{p}$, then

$$\frac{-x+\sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}'}, \tag{4}$$

and from this follows

$$x^2 \equiv \Delta \pmod{4p}. \tag{5}$$

Conversely, if the congruence (5) is satisfied by $x = r$, then

$$\frac{r+\sqrt{\Delta}}{2} \cdot \frac{-r+\sqrt{\Delta}}{2} \equiv 0 \pmod{\mathfrak{p}},$$

and consequently either $x = r$ or $x = -r$ satisfies condition (3).

The two ideals $\mathfrak{p}, \mathfrak{p}'$ will then be identical when the two numbers (3) and (4), and consequently also their sum $\sqrt{\Delta}$, are divisible by $\mathfrak{p}$, and hence Δ is divisible by p. Then $x \equiv 0$ or $x \equiv p \pmod{2p}$ is the only root of (5). We therefore have the result:

2. The prime factor $\mathfrak{p}$ of the natural prime number p is of the second degree if the congruence (5) has no solution, of the first degree if (5) has a solution, and p is the square of $\mathfrak{p}$ if (5) has only one solution, that is, if p divides Δ.

The latter agrees with the general theorem of Vol. II, § 174.

Using the symbol (Δ, p) which we introduced in § 85, we can also give these theorems the form:

3. If $(\Delta, p) = -1$, then p is indecomposable in Ω.

If $(\Delta, p) = +1$, then p decomposes into two distinct conjugate prime ideals.

If $(\Delta, p) = 0$, then p is the square of a prime ideal.

§ 93. Representation of numbers as ideal norms.

A prime number p is the norm of an ideal only when $(\Delta, p) = +1$ or $= 0$, but not when $(\Delta, p) = -1$. In the latter case only p^2 is a norm, namely that of $\mathfrak{p}$. In general a positive rational integer m can be represented as a norm in several ways. We shall for the moment denote the number of these representations by $\psi(m)$ and determine it more closely.

First take m to be a prime power p^k. If then $(\Delta, p) = +1$ and $p = \mathfrak{p}\mathfrak{p}'$, then p^k can be represented in the following way:

$$p^k = \mathrm{N}(\mathfrak{p}^k), \quad \mathrm{N}(\mathfrak{p}^{k-1}\mathfrak{p}'), \quad \mathrm{N}(\mathfrak{p}^{k-2}\mathfrak{p}'^2), \quad \ldots, \quad \mathrm{N}(\mathfrak{p}'^k), \tag{1}$$

and therefore

$$(\Delta, p) = +1 : \qquad \psi(p^k) = k+1. \tag{2}$$

But if $(\Delta, p) = -1$, then, when k is even, $p^k = \mathrm{N}(\mathfrak{p}^{k/2})$, whereas if k is odd, p^k cannot be represented as a norm. Thus

$$(\Delta, p) = -1 : \qquad \psi(p^k) = 0 \text{ or } = 1, \tag{3}$$

according as k is odd or even. Finally, if $(\Delta, p) = 0$, so that p is contained in Δ, then $p = \mathfrak{p}^2$ and $p^k = \mathrm{N}(\mathfrak{p}^k)$; hence

$$(\Delta, p) = 0 : \qquad \psi(p^k) = 1. \tag{4}$$

These three cases can be combined in one formula. The number p^k has, namely, the following $k+1$ divisors:

$$1, p, p^2, \ldots, p^k. \tag{5}$$

Therefore formulas (2), (3), (4), when $m = p^k$ and n runs through the divisors of m, give

$$\psi(m) = \sum_n (\Delta, n). \tag{6}$$

But this expression holds generally. For if m_1 and m_2 are relatively prime and

$$m_1 = \mathrm{N}(\mathfrak{m}_1), \qquad m_2 = \mathrm{N}(\mathfrak{m}_2),$$

then

$$m = m_1 m_2 = \mathrm{N}(\mathfrak{m}_1 \mathfrak{m}_2).$$

Thus from a representation of m_1, m_2 as norms one can derive a representation of m as a norm.

Conversely, if

$$m_1 m_2 = \mathrm{N}(\mathfrak{m}) = \mathfrak{m}\mathfrak{m}',$$

where $\mathfrak{m}, \mathfrak{m}'$ are conjugate ideals, then, if m_1 and $\mathfrak{m}$ are divisible by a prime ideal $\mathfrak{p}$, m_1 must also be divisible by $\mathfrak{p}\mathfrak{p}'$, and so $m_1 = \mathrm{N}(\mathfrak{m}_1)$ and likewise $m_2 = \mathrm{N}(\mathfrak{m}_2)$.

Accordingly, if m_1 and m_2 are relatively prime, then

$$\psi(m_1)\psi(m_2) = \psi(m_1 m_2). \tag{7}$$

On the other hand, if n_1 and n_2 run through the divisors of m_1 and m_2, then

$$\sum (\Delta, n_1) \sum (\Delta, n_2) = \sum (\Delta, n_1 n_2),$$

and with this formula (6) is proved generally. We state this as the theorem:

4. A natural number m can be represented in the quadratic field Ω with fundamental number Δ in

$$\sum (\Delta, n)$$

different ways as the norm of an ideal, where n runs through all divisors of m.

§ 94. The quadratic reciprocity law.

From the decomposition of prime numbers in the quadratic field, Dedekind derives a new proof of the reciprocity law of quadratic residues.[27]

According to Vol. II, § 167, 3, the necessary and sufficient condition that the prime ideal $\mathfrak{p}$ contained in the prime number p (in any field Ω) be of the first degree is that, for every integer ω of the field Ω,

$$\omega^p \equiv \omega \pmod{\mathfrak{p}}, \tag{1}$$

and in our case of the quadratic field this reduces to

$$\theta^p \equiv \theta \pmod{\mathfrak{p}}. \tag{2}$$

First take $\Delta = -4$. Then $\theta = i = \sqrt{-1}$, and the congruence (2) becomes

$$i^{p-1} \equiv 1 \pmod{\mathfrak{p}}.$$

Since $(1+i)(1-i) = 2$, neither 2 nor $(1+i)$ can be divisible by $\mathfrak{p}$ for odd p; hence $i^{p-1} = 1$, and therefore $p-1$ must be divisible by 4. Accordingly it follows from § 92, 2:

[27] Dirichlet-Dedekind, *Zahlentheorie*, 4th ed. (p. 636, note).

1) The congruence

$$x^2 \equiv -4 \pmod{4p},$$

or, what is the same,

$$x^2 \equiv -1 \pmod{p},$$

has solutions if and only if $p \equiv 1 \pmod{4}$.

Second, take $\Delta = 8$. Then $\theta = \sqrt{2}$, or, if $r = \sqrt{i}$ is an eighth root of unity,

$$\theta = r + r^{-1},$$

and the condition (2) can be written, by the binomial theorem, as

$$r^p + r^{-p} \equiv r + r^{-1} \pmod{\mathfrak{p}}.$$

This requires

$$(r^p - r)(r^p - r^{-1})r^{-p} \equiv 0 \pmod{\mathfrak{p}},$$

therefore

$$r^p \equiv r \quad \text{or} \quad r^p \equiv r^{-1} \pmod{\mathfrak{p}},$$

so that

$$r^{p\pm 1} \equiv 1, \qquad i^{\frac{p\pm 1}{2}} - 1 \equiv 0 \pmod{\mathfrak{p}},$$

and from this, as above, for an odd p,

$$p \equiv \pm 1 \pmod{8}.$$

2) The congruence

$$x^2 \equiv 2 \pmod{p}$$

is therefore solvable if and only if $p \equiv 1$ or $p \equiv -1 \pmod{8}$.

Finally let, if q is an odd prime number,

$$\Delta = \pm q \tag{3}$$

and choose the sign so that $\pm q \equiv 1 \pmod{4}$. In this case

$$\theta = \frac{-1 + \sqrt{\pm q}}{2},$$

and this expression can be represented by qth roots of unity. Let r denote an imaginary qth root of unity, and let a run through the quadratic residues and b through the non-residues of q. We put, as in Vol. I, § 179,

$$A = \sum_a r^a, \qquad B = \sum_b r^b, \tag{4}$$

and have, according to the formulas there, (3), which are derived without use of the reciprocity law,

$$\theta = A, \qquad \theta' = B. \tag{5}$$

The sign of $\sqrt{\pm q}$ depends on the choice of r, but here it does not come into consideration. Now, by the polynomial theorem,

$$A^p \equiv \sum r^{pa}, \qquad B^p \equiv \sum r^{pb},$$

and hence

$$A^p \equiv A, \qquad B^p \equiv B, \quad \text{if } \left(\frac{p}{q}\right) = +1,$$

and

$$A^p \equiv B, \qquad B^p \equiv A, \quad \text{if } \left(\frac{p}{q}\right) = -1.$$

Since A is not congruent to B when p is different from q, it follows from this and from 2 of the preceding paragraph:

3) The congruence

$$x^2 \equiv \pm q \pmod p$$

has solutions if and only if p is a quadratic residue of q.

And this is the reciprocity law with its supplementary theorems.

§ 95. Equivalent forms and ideals in the quadratic field.

If the two ideals [§ 91, (16)]

$$\mathfrak{a} = (a, a\omega), \qquad \mathfrak{a}' = (a', a'\omega') \tag{1}$$

are equivalent, then there is a (integral or non-integral) number η in Ω for which $\eta\mathfrak{a}' = \mathfrak{a}$, and this number η is determined by the ideals $\mathfrak{a}, \mathfrak{a}'$ themselves up to a factor which is a numerical unit. Thus there must be four rational integers $\alpha, \beta, \gamma, \delta$ satisfying the condition

$$\begin{aligned} \eta a' &= a(\alpha - \gamma\omega), \\ \eta a'\omega' &= a(-\beta + \delta\omega). \end{aligned} \tag{2}$$

If, for the moment, we denote by $\eta_1, \omega_1, \omega_1'$ the numbers conjugate to η, ω, ω', then consequently

$$a'^2\eta\eta_1 \begin{vmatrix} 1 & \omega' \\ 1 & \omega_1' \end{vmatrix} = a^2 \begin{vmatrix} \alpha & -\gamma \\ -\beta & \delta \end{vmatrix} \begin{vmatrix} 1 & \omega \\ 1 & \omega_1 \end{vmatrix},$$

that is,

$$a'^2\eta\eta_1(\omega' - \omega_1') = a^2(\alpha\delta - \beta\gamma)(\omega - \omega_1),$$

and since, by § 91, (12),

$$a(\omega - \omega_1) = a'(\omega' - \omega_1') = -\sqrt{\Delta},$$

it follows that, if

$$\varepsilon = \alpha\delta - \beta\gamma$$

is put,

$$a\varepsilon = a'\eta\eta_1; \tag{3}$$

and hence, if, as we have already assumed, a and a' are positive and (according to the sharpened notion of equivalence) $\eta\eta_1$ is positive, then ε is also positive.

But if we interchange $\mathfrak{a}$ with $\mathfrak{a}'$, then η passes into $1 : \eta$, and ε may pass into ε'. Hence also

$$a'\varepsilon' = \frac{a}{\eta\eta_1}, \tag{4}$$

and consequently $\varepsilon\varepsilon' = 1$. But since ε and ε' are positive rational integers, it follows that $\varepsilon = \varepsilon' = 1$. Therefore, by (2),

$$\omega' = \frac{-\beta + \delta\omega}{\alpha - \gamma\omega}, \qquad \omega = \frac{\alpha\omega' + \beta}{\gamma\omega' + \delta}, \qquad \alpha\delta - \beta\gamma = 1. \tag{5}$$

Now ω and ω' are the roots of the two forms

$$(a, b, c), \qquad (a', b', c') \qquad [\S\,91, (13)],$$

and (5) expresses the equivalence of these forms. We therefore have, in agreement with the general theory (§ 88), the theorem:

5. The quadratic forms belonging to equivalent ideals are equivalent.

From this one recognizes the significance of the sharpened notion of equivalence: for if one also wished to allow negative values of $N(\eta)$, then one could also have $\varepsilon = -1$, and the corresponding forms could be improperly equivalent.

Conversely, let $\varphi = (a, b, c)$ and $\varphi' = (a', b', c')$ be two equivalent forms with positive first coefficients a, a', and let

$$\omega = \frac{-b + \sqrt{\Delta}}{2a}, \qquad \omega' = \frac{-b' + \sqrt{\Delta}}{2a'}. \tag{6}$$

Then there is a linear substitution $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ with determinant $+1$ which carries (a, b, c) into (a', b', c'). Then

$$\begin{aligned} a' &= a\alpha^2 + b\alpha\gamma + c\gamma^2, \\ b' &= 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + 2c\gamma\delta, \\ c' &= a\beta^2 + b\beta\delta + c\delta^2, \end{aligned} \tag{7}$$

$$\omega = \frac{\alpha\omega' + \beta}{\gamma\omega' + \delta}, \qquad \omega' = \frac{\delta\omega - \beta}{-\gamma\omega + \alpha}, \tag{8}$$

and a small calculation shows that ω and ω' are corresponding roots, that is, roots with the same sign of $\sqrt{\Delta}$.

Accordingly, for the ideals $\mathfrak{a}, \mathfrak{a}'$ belonging to φ and φ' we obtain two basis forms:

$$\lambda = a(x - \omega y), \qquad \lambda' = a'(x' - \omega' y'),$$

and by (8)

$$\lambda = \frac{a[(\delta x - \beta y) + (\gamma x - \alpha y)\omega']}{\gamma\omega' + \delta}.$$

Thus putting

$$x' = \delta x - \beta y, \qquad y' = -\gamma x + \alpha y,$$

gives

$$\lambda = \frac{a}{a'(\gamma\omega' + \delta)}\lambda'.$$

Since $a : a'(\gamma\omega' + \delta)$ is a number in Ω, it follows that $\mathfrak{a}$ and $\mathfrak{a}'$ are equivalent.

Thus again in agreement with the general theorem § 88, 4., and at the same time in a more precise determination, we have:

6. If (a, b, c) and (a', b', c') are two equivalent forms with positive first coefficients, and if ω, ω' are corresponding roots of these forms, then the ideals $(a, a\omega)$ and $(a', a'\omega')$ are equivalent.

For negative discriminant the first coefficients of equivalent forms always have the same sign, and one considers only the forms in which this sign is positive. For positive discriminant there are forms with positive first coefficients in every form class, and therefore form classes and ideal classes correspond to one another in a one-to-one way.

A basis of the principal ideal η is $(\eta, \eta\theta)$ and

$$\theta = \frac{\sqrt{\Delta}}{2}, \qquad \theta^2 = \frac{\Delta}{4}, \qquad \Delta \equiv 0 \pmod 4,$$

$$\theta = \frac{-1 + \sqrt{\Delta}}{2}, \qquad \theta^2 = \frac{\Delta - 1}{4} - \theta, \qquad \Delta \equiv 1 \pmod 4.$$

If x, y are rational integers and

$$\eta = x + y\theta,$$

then this basis is, in the two cases,

$$(\alpha_1, \alpha_2) = \left(x + y\theta, \frac{\Delta y}{4} + \theta x\right),$$

$$= \left(x + y\theta, \frac{y(\Delta - 1)}{4} + (x - y)\theta\right),$$

and the substitution determinant A [§ 87,(1)] is

$$x^2 - \frac{\Delta}{4}y^2, \qquad x^2 - xy + \frac{1 - \Delta}{4}y^2 = N(\eta).$$

The basis (α_1, α_2) is therefore positive if the norm of η is positive (otherwise α_1 would have to be interchanged with α_2). From the basis form

$$\lambda = \alpha_1 t_1 + \alpha_2 t_2 = \eta(t_1 + \theta t_2)$$

one obtains

$$N(\lambda) = N(\eta)T,$$

and accordingly T becomes the principal form

$$\left(1, 0, \frac{\Delta}{4}\right), \qquad \left(1, -1, \frac{1 - \Delta}{4}\right). \tag{9}$$

If $N(\eta)$ were negative, one would have obtained for T the forms

$$\left(-1, 0, -\frac{\Delta}{4}\right), \qquad \left(-1, 1, -\frac{1 - \Delta}{4}\right),$$

and these are equivalent to (9) only if Pell's equation

$$t^2 - \Delta u^2 = -4$$

is solvable.

§ 96. Discriminants of the orders.

1. *Definition.* If Q is a natural number, then the totality of the integral numbers of the quadratic field Ω which are congruent, modulo Q, to a rational number is called an *order*, and Q is called the conductor of this order. The order with conductor Q is denoted by $[Q]$.

From this definition it follows:

1. All rational integers belong to every order;
2. The sum, difference, and product of two numbers of the order belong to the same order;

that is, within the order addition, subtraction, and multiplication can be performed without restriction, but division cannot in general be so performed.

The totality of all integral numbers of the field Ω likewise forms an order (of conductor 1), the principal order, which would therefore be denoted by [1]. [28]

If Δ is the fundamental number of the field Ω, we put, according as Δ is even or odd,

$$\theta_0 = -\frac{1}{2}\sqrt{\Delta}, \qquad \text{or} \qquad \theta_0 = \frac{-1 + \sqrt{\Delta}}{2}. \tag{1}$$

Then, by § 90, every integral number of the field Ω is representable in the form

$$\omega = x + y\theta_0, \tag{2}$$

[28] Dedekind introduced the expression *order* in the general theory of algebraic numbers because, in the special case of the quadratic field, these orders correspond to Gauss's orders of quadratic forms. Hilbert uses for this the expression "ring" or "number ring". Cf. Dirichlet–Dedekind, *Vorlesungen über Zahlentheorie* (§ 172 of the third, § 170 of the fourth edition). Dedekind: *Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers*, Festschrift zur Säkularfeier von Gauss' Geburtstag, Braunschweig, 1877. Hilbert: *Die Theorie der algebraischen Zahlen*, Bericht der Deutschen Mathematischen Vereinigung, 1894/95, Chap. IX.

where x and y are integers; and this number is congruent, modulo Q, to a rational number a if and only if y is divisible by Q, since then

$$\frac{x-a+y\theta_0}{Q}$$

must be an integer. Accordingly all numbers of the order $[Q]$ are contained in the form

$$\omega = x + yQ\theta_0,$$

or, in the two cases distinguished in (1), if

$$D = Q^2\Delta \tag{3}$$

is put:

$$\begin{aligned} x - y\frac{\sqrt{D}}{2}, &\qquad \Delta \text{ even}, \\ x - \frac{Q}{2}y + y\frac{\sqrt{D}}{2}, &\qquad \Delta \text{ odd},\ D \text{ even}, \\ x - \frac{Q-1}{2}y + y\frac{-1+\sqrt{D}}{2}, &\qquad D \text{ odd}. \end{aligned} \tag{4}$$

Thus, if we put

$$\theta = \frac{\sqrt{D}}{2}, \quad D \equiv 0 \pmod 4, \qquad \theta = \frac{-1+\sqrt{D}}{2}, \quad D \equiv 1 \pmod 4, \tag{5}$$

and replace in the two last formulae (4) the arbitrarily integral numbers

$$x - \frac{Q}{2}y, \qquad x - \frac{Q-1}{2}y$$

by x, we obtain the result:

2. *Theorem.* Every number of the order $[Q]$, and only these, is contained in the form

$$\omega = x + y\theta, \tag{6}$$

where x, y are arbitrary rational integers.

We call the system

$$(1, \theta) \tag{7}$$

a basis of the order $[Q]$.

By substituting (5), the numbers (6) may also be represented in the form

$$\omega = \frac{x + y\sqrt{D}}{2}, \tag{8}$$

where x, y are integers satisfying the condition

$$x^2 - y^2 D \equiv 0 \pmod 4. \tag{9}$$

The number D is a discriminant whose stem is Δ. We call it the *discriminant of the order* $[Q]$.

§ 97. Orders and ideals.

Let $\mathfrak{m}$ be an ideal of the quadratic field Ω. If all whole numbers divisible by $\mathfrak{a}$ are divisible by a rational number m, then the ideal $\mathfrak{a} = m\mathfrak{m}$ is divisible by m; one has

$$N(\mathfrak{a}) = m^2 N(\mathfrak{m}),$$

and the rational number $mN(\mathfrak{m})$, which is smaller than $N(\mathfrak{a})$, is divisible by $\mathfrak{a}$.

If m is the greatest natural number by which $\mathfrak{a}$ is divisible, we shall call m the *divisor* of $\mathfrak{a}$. If the divisor is $= 1$, the ideal is called *primary*; otherwise *derived.* One obtains all ideals derived from a primary ideal $\mathfrak{m}$ by multiplying $\mathfrak{m}$ by arbitrary natural numbers, and all ideals derived from one primary ideal are mutually equivalent.

3. *Theorem.* If $\mathfrak{a}$ is a primary ideal, then the norm of $\mathfrak{a}$ is at the same time the smallest natural number divisible by $\mathfrak{a}$.

For let the smallest natural number divisible by $\mathfrak{a}$ be denoted by a; then the norm of $\mathfrak{a}$ is in any case divisible by a. We put

$$N(\mathfrak{a}) = ma,$$

and, if $\mathfrak{a}'$ is the ideal conjugate to $\mathfrak{a}$, it follows, since by Vol. II, §164 the norm of an ideal is equal to the product of the conjugate ideals, that

$$N(\mathfrak{a}) = \mathfrak{a}\mathfrak{a}' = ma. \tag{1}$$

Since a is to be divisible by $\mathfrak{a}$, we put

$$a = \mathfrak{a}\mathfrak{m}' = \mathfrak{a}'\mathfrak{m}$$

and obtain from (1):

$$\mathfrak{a} = m\mathfrak{m}. \tag{2}$$

Thus m enters the divisor of the ideal $\mathfrak{a}$, and thereby Theorem 3 is proved, since, when $\mathfrak{a}$ is primary, its divisor must be $= 1$.

4. *Theorem.* If $\mathfrak{m}$ is a primary ideal of the field Ω, then every whole number in Ω is congruent modulo $\mathfrak{m}$ to a rational number.

For, by Vol. II, §165, $N(\mathfrak{m})$ is in general the number of numbers incongruent modulo $\mathfrak{m}$ in Ω. If now $N(\mathfrak{m}) = a$ is at the same time the smallest natural number divisible by $\mathfrak{m}$, then the a numbers

$$0, 1, 2, \ldots, a-1$$

are all incongruent, and among them all number classes modulo $\mathfrak{m}$ are represented.

Let now $[Q]$ be an order with conductor Q and discriminant D, and let $(1, \theta)$ be the basis of this order. Further, let $\mathfrak{m}$ be a primary ideal prime to Q. By Theorem 4 there is then a whole rational number t satisfying the condition

$$\theta + t \equiv 0 \pmod{\mathfrak{m}}.$$

We put

$$\begin{aligned} t &= \frac{b}{2}, && \text{if } D \equiv 0 \pmod 4, \\ t &= \frac{1+b}{2}, && \text{if } D \equiv 1 \pmod 4, \end{aligned} \tag{3}$$

and obtain, by §96, (5),

$$\theta + t = \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}, \tag{4}$$

where b is a whole rational number satisfying, by (3), the condition

$$b \equiv D \pmod 2.$$

In addition let

$$N(\mathfrak{m}) = a, \tag{5}$$

so that a is the smallest natural number divisible by $\mathfrak{m}$.

By the congruence (4), the number b is determined only modulo $2a$. For if b, b' are two numbers which satisfy the condition (4), then $\frac{1}{2}(b' - b)$ is divisible by $\mathfrak{m}$.

Further, since

$$N\left(\frac{b+\sqrt{D}}{2}\right) = \frac{b^2 - D}{4}$$

is a whole rational number divisible by $\mathfrak{m}$, it must be divisible by a; if we put it equal to ac, then c is a whole rational number and

$$D = b^2 - 4ac, \tag{6}$$

$$b^2 \equiv D \pmod{4a}, \tag{7}$$

and, if the congruence (7) is fulfilled, then every number b' congruent to b modulo $2a$ satisfies the same congruence.

We infer from this:

5. *Theorem.* Every primary ideal $\mathfrak{m}$ prime to Q, whose norm is $= a$, gives us by (4) one and only one root b of the congruence (7), if by roots one understands not individual numbers but number classes modulo $2a$.

The two numbers

$$a, \qquad \frac{b+\sqrt{D}}{2} \tag{8}$$

if a is relatively prime to Q, cannot both be divisible by one and the same rational prime number; for otherwise

$$\frac{b+\sqrt{D}}{2} - \frac{b-\sqrt{D}}{2} = \sqrt{D}$$

would have to be divisible by p; consequently D would be divisible by p^2, and p would have to be a divisor of Q, contrary to the assumption.

For an odd p this is evident; for $p = 2$, however, it would follow that

$$b \equiv 0 \pmod{2}, \qquad b^2 - D \equiv 0 \pmod{16},$$

and hence

$$D \equiv 0, 4 \pmod{16}.$$

Thus $D/4$ would still be a discriminant and 2 would have to enter Q. This proves:

6. *Theorem.* If a is a natural number prime to Q and b is a root of (7), then one obtains a definite primary ideal $\mathfrak{m}$ as the greatest common divisor of the two numbers (8).

We now prove:

7. *Theorem.* Every number μ of the order $[Q]$, divisible by the primary ideal $\mathfrak{m}$ and prime to Q, is representable once and only once in the form

$$\mu = ax + \frac{b+\sqrt{D}}{2}y, \tag{9}$$

where x, y are whole rational numbers. If x, y are regarded as variables, then μ is called a basis form of the ideal $\mathfrak{m}$ in the order $[Q]$.

For the proof we note that every number of the order $[Q]$, by (4), is contained in the form

$$\omega = z + \frac{b+\sqrt{D}}{2}y, \tag{10}$$

where y and z are whole rational numbers. If now $\mathfrak{m}$ is primary, then

$$N(\mathfrak{m}) = a \tag{11}$$

is the smallest natural number divisible by $\mathfrak{m}$, and therefore, if ω is to be divisible by $\mathfrak{m}$, z must be divisible by a. Thus, putting $z = ax$, we obtain the form (9). Conversely, it is evident that every number of this form is divisible by $\mathfrak{m}$.

That the representation of a number μ in this form is possible in only one way follows from the fact that $\sqrt{D}$ is irrational.

The form μ is therefore called a basis form of the ideal $\mathfrak{m}$ in the order Q. From it one obtains a basis form of the same ideal in the field Ω:

$$\lambda = ax + \frac{b' + \sqrt{\Delta}}{2} y,$$

where b' is determined from the congruence

$$b'Q \equiv b \pmod{2a},$$

which is always solvable, since Q is relatively prime to a, and, for even Q, b too is even.

From (9) one obtains

$$N(\mu) = a(ax^2 + bxy + cy^2). \tag{12}$$

The number μ obtained by assigning definite numbers to x, y in (9) can be decomposed into ideal factors:

$$\mu = \mathfrak{a}\mathfrak{m},$$

if

$$N(\mathfrak{a}) = ax^2 + bxy + cy^2, \tag{13}$$

and the following theorem holds:

8. *Theorem.* If μ is relatively prime to Q, and x, y have no common divisor, then the ideal $\mathfrak{a}$ is primary.

We prove it thus: let p be a rational prime number entering $\mathfrak{a}$; then μ is divisible by p, and from this it follows first that y must be divisible by p; for a number ω of the form (10) can be divisible by a prime number p not entering Q only when both y and $z = ax$ are divisible by p. Therefore, since by assumption x and y are relatively prime, x cannot be divisible by p, and a must be divisible by p.

Put $a = p^k a_1$, where a_1 is no longer divisible by p, and denote by $\mathfrak{p}$ a prime ideal entering p. Then, by (11), $\mathfrak{p}$ must enter $\mathfrak{m}$ or $\mathfrak{m}'$, and, since $\mathfrak{m}$ has been assumed primary, $\mathfrak{p}$ cannot be $= \mathfrak{p}'$. Hence $N(\mathfrak{p}) = p = \mathfrak{p}\mathfrak{p}'$, and if $\mathfrak{m}$ is divisible by $\mathfrak{p}$, it cannot at the same time be divisible by $\mathfrak{p}'$. Consequently $\mathfrak{m}$ is divisible by $\mathfrak{p}^k$. Thus, by the definition of $\mathfrak{m}$ (Theorem 6),

$$\frac{b + \sqrt{D}}{2}$$

is also divisible by $\mathfrak{p}^k$, and consequently

$$\frac{b + \sqrt{D}}{2} y$$

is divisible at least by $\mathfrak{p}^{k+1}$, while ax, and hence μ, is divisible exactly by $\mathfrak{p}^k$. On the other hand $\mathfrak{a}\mathfrak{m} = \mu$ is likewise divisible at least by $\mathfrak{p}^{k+1}$, which is a contradiction. The same holds also when $p = \mathfrak{p}^2$, that is, when p enters Δ but not Q. Consequently no rational prime number can be contained in $\mathfrak{a}$, and $\mathfrak{a}$ is primary.

From what has so far been proved we can formulate the following theorem:

9. *Theorem.* If

$$\mu = ax + \frac{b + \sqrt{D}}{2} y = \mathfrak{m}\mathfrak{a},$$

$$\mu' = ax' + \frac{b+\sqrt{D}}{2}y' = \mathfrak{m}\mathfrak{a}'$$

are two numbers prime to Q, neither x and y nor x' and y' have a common divisor,

$$N(\mathfrak{a}) = N(\mathfrak{a}') = A,$$

and, for one and the same number B,

$$\frac{B+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{a}}, \qquad \equiv 0 \pmod{\mathfrak{a}'},$$

then the two ideals $\mathfrak{a}$ and $\mathfrak{a}'$ are identical, and the numbers μ, μ' differ only by a unit factor.

For both $\mathfrak{a}$ and $\mathfrak{a}'$ are, by 8., primary, and are, by 6., defined as greatest common divisors of

$$A \quad \text{and} \quad \frac{B+\sqrt{D}}{2}.$$

Thirteenth Section. Equivalence according to number groups.

§ 98. Number groups in the orders.

In order to investigate more closely the relations of quadratic irrational numbers to the orders of quadratic forms, we derive from the numbers of the field Ω number groups according to the following points of view.

1. We take an arbitrary natural number Q and exclude first, from the whole numbers in Ω, all those which are not prime to Q. From the remaining ones we also take arbitrary quotients.

The whole and fractional numbers η so obtained we call prime to Q. Let their totality be denoted by O. They form a group in so far as multiplication and division of two of these numbers always again give numbers of the same system.

2. We now proceed in the same way with the numbers of the order $[Q]$, i.e. we exclude all numbers of this order which are not relatively prime to Q, and also form the quotients of arbitrary two of the numbers left over. We denote these numbers by η' and their totality by O', and call them whole and fractional numbers of the order $[Q]$. The numbers O' are all contained in O and likewise form a group with respect to multiplication and division. We call them the group of the order $[Q]$, or briefly an order group.

Every number η in O can be transformed into a whole number ω by multiplication with a whole rational number n prime to Q.

If η' is a number in O', then one can rationalize the denominator and finds that $n\eta'$ is a whole number of the order $[Q]$ and hence is congruent to a rational number modulo Q.

Since n is prime to Q, one can put this number equal to nr, where r is a rational number prime to Q. Accordingly every number η' has the property

$$\eta' \equiv r \pmod{Q}, \tag{1}$$

where r is a whole rational number, and the congruence is to be understood in the sense $n\eta' = nr$.

The congruences (1) can be multiplied and divided, if by r^{-1} one understands the whole number satisfying the condition $rr^{-1} \equiv 1 \pmod{Q}$.

Conversely, a whole or fractional number η' which satisfies a condition (1) belongs to the group O'. For if $n\eta' = nr$ is a number in $[Q]$, then $\eta' = nr : n$ is a quotient of two such numbers.

3. In O' there is again a group O_1, consisting of all numbers η_1 which satisfy the condition

$$\eta_1 \equiv 1 \pmod{Q}, \tag{2}$$

and the numbers of O_1 also give again numbers of O_1 under multiplication and division.

Thus we have three kinds of number groups, each of which is contained in the preceding one:

1. O contains all numbers in Ω prime to Q,
2. O' contains the whole and fractional numbers of the order $[Q]$ prime to Q,
3. O_1 contains the number classes in O' congruent to unity modulo Q.

The groups O, O', O_1 are infinite Abelian groups. But if one takes a complete residue system of rational numbers $r_1, r_2, \ldots, r_\mu$ prime to Q, then every number in O' can be represented as a product of one of these r_i with a number in O_1, and therefore one can put, in the terminology used in Vol. II, § 2,

$$O' = r_1 O_1 + r_2 O_1 + \cdots + r_\mu O_1, \tag{3}$$

and μ is the index of the divisor O_1 of O':

$$\mu = (O', O_1). \tag{4}$$

Thus μ is a finite number, namely

$$\mu = \varphi(Q) = Q \prod_q \left(1 - \frac{1}{q}\right), \tag{5}$$

where $\varphi(Q)$ is the usual symbol in number theory for the number of number classes modulo Q with numbers prime to Q, and where q runs in the product over all rational prime numbers occurring in Q.

If we denote by

$$\omega_1, \omega_2, \ldots, \omega_\nu$$

a complete system of representatives of the numbers in Ω prime to Q, taken modulo Q, then in the same way

$$O = \omega_1 O_1 + \omega_2 O_1 + \cdots + \omega_\nu O_1, \tag{6}$$

and

$$\nu = \psi(Q) = (O, O_1) \tag{7}$$

is the index of the divisor O_1 with respect to O.

The number $\psi(Q)$ was determined generally in Vol. II, § 168, (3), also for the case in which an ideal occurs in place of Q. Since here $N(Q) = Q^2$, we have

$$\nu = \psi(Q) = Q^2 \prod_{\mathfrak{q}} \left(1 - \frac{1}{N(\mathfrak{q})}\right), \tag{8}$$

where $\mathfrak{q}$ runs through the distinct ideal prime divisors of Q.

Now, by § 92,

$$\begin{aligned} &1.\quad N(\mathfrak{q}) = q, \quad \text{if } \left(\frac{\Delta}{q}\right) = +1,\\ &2.\quad N(\mathfrak{q}) = q, \quad \text{if } \left(\frac{\Delta}{q}\right) = 0,\\ &3.\quad N(\mathfrak{q}) = q^2, \quad \text{if } \left(\frac{\Delta}{q}\right) = -1. \end{aligned}$$

We shall denote these three kinds of prime numbers by q_1, q_2, q_3, and remark that every prime number q_1 is the norm of two different prime ideals $\mathfrak{q}$. Accordingly from (8) one obtains

$$\psi(Q) = Q^2 \prod_{q_1} \left(1 - \frac{1}{q_1}\right)^2 \prod_{q_2} \left(1 - \frac{1}{q_2}\right) \prod_{q_3} \left(1 - \frac{1}{q_3^2}\right). \tag{9}$$

Now the index of O' with respect to O is, by Vol. II, § 2, (4),

$$(O, O') = \frac{(O, O_1)}{(O', O_1)} = \frac{\psi(Q)}{\varphi(Q)},$$

and
$$\frac{\psi(Q)}{\varphi(Q)} = Q\prod_{q_1}\left(1-\frac{1}{q_1}\right)\prod_{q_3}\left(1+\frac{1}{q_3}\right).$$
Therefore
$$(O,O') = Q\prod_{q_1}\left(1-\frac{1}{q_1}\right)\prod_{q_3}\left(1+\frac{1}{q_3}\right),$$
for which one can also put, according to the three cases,
$$(O,O') = Q\prod_{q|Q}\left(1-\frac{1}{q}\left(\frac{\Delta}{q}\right)\right). \tag{10}$$
Here the product refers to all prime numbers q occurring in Q.

We can therefore choose a representative system
$$\eta_1, \eta_2, \ldots, \eta_\lambda$$
in O such that
$$O = \eta_1 O' + \eta_2 O' + \cdots + \eta_\lambda O' \tag{11}$$
where $\lambda = (O,O')$.

§ 99. Equivalence in the orders.

10. *Definition.* We shall now call two ideals $\mathfrak{a}, \mathfrak{a}'$ equivalent according to the order $[Q]$ if
$$\mathfrak{a}' = \eta'\mathfrak{a} \tag{1}$$
and η' denotes a number in O'. Here $\mathfrak{a}$ and $\mathfrak{a}'$ are always assumed to be prime to Q.

If one takes a number $\mu = \mathfrak{a}\mathfrak{m}$ divisible by $\mathfrak{a}$ in $[Q]$ in such a way that $\mathfrak{m}$ becomes a primary ideal, and puts $\mathfrak{a}' = \eta'\mathfrak{a}$, then
$$\mu' = \eta'\mu = \mathfrak{a}'\mathfrak{m} \tag{2}$$
and we can also explain the equivalence of $\mathfrak{a}$ and $\mathfrak{a}'$ according to $[Q]$ by saying that $\mathfrak{a}$ and $\mathfrak{a}'$ are transformed into numbers of the order $[Q]$ by multiplication with one and the same ideal $\mathfrak{m}$.

By the ideal $\mathfrak{m}$, according to § 97, a family of parallel quadratic forms (a,b,c) is determined, in which a and b are determined from
$$a = N(\mathfrak{m}), \qquad \frac{b+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}. \tag{3}$$
And this family of forms therefore does not change by (2) if $\mathfrak{a}$ is replaced by an equivalent ideal $\mathfrak{a}'$.

If, however, in (2), one takes in place of the ideal $\mathfrak{m}$ another ideal $\mathfrak{m}_1$, and puts
$$\mu_1 = \mathfrak{a}\mathfrak{m}_1, \qquad \mu_1' = \mathfrak{a}'\mathfrak{m}_1, \tag{4}$$
then
$$\frac{\mu_1'}{\mu_1} = \frac{\mathfrak{a}'}{\mathfrak{a}} = \eta'$$
is a number in O', and therefore $\mathfrak{m}_1$ is equivalent to $\mathfrak{m}$.

If then one puts
$$a_1 = N(\mathfrak{m}_1), \qquad \frac{b_1+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}_1}, \tag{5}$$
one obtains another primitive form (a_1, b_1, c_1), of which we can prove that it is equivalent to (a,b,c).

We put, for abbreviation,

$$A = \frac{b + \sqrt{D}}{2}, \qquad A_1 = \frac{b_1 + \sqrt{D}}{2},$$

and remark that

$$N(A_1)A, \qquad N(A)A_1$$

are numbers in $[Q]$ divisible by $\mathfrak{m}\mathfrak{m}_1$ [by (5)]. Accordingly, by § 97, 7., the whole rational numbers $\alpha, \beta, \gamma, \delta$ can be so determined that

$$A_1 = \alpha A + a(-\beta + \delta\theta), \tag{6}$$

and one proves quite as in § 95 that

$$\alpha\delta - \beta\gamma = 1$$

must hold, that is, the two forms

$$(a, b, c), \qquad (a_1, b_1, c_1)$$

are equivalent. Thus we have:

11. *Theorem.* Every ideal class, represented by an ideal $\mathfrak{a}$ according to $[Q]$, corresponds to a form class A represented by the form (a, b, c), and conversely.

In order to obtain the form class A, take a primary ideal $\mathfrak{m}$ of the reciprocal class A^{-1} and put

$$a = N(\mathfrak{m}), \qquad \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}.$$

Conversely, if the form (a, b, c) is given, then the greatest common divisor $\mathfrak{m}$ of

$$a, \qquad \frac{b + \sqrt{D}}{2}$$

is an ideal of the class A^{-1}.

§ 100. Ideal classes according to the orders.

The concept of equivalence of ideals and the theorems on class numbers can be presented most clearly if, according to Vol. II, § 169, one represents the ideals by means of the functionals discussed in Vol. II, § 153 ff., since in this way one may use the ordinary rules of multiplication and division.[29]

We restrict ourselves here once and for all, in the body Ω, to numbers and ideals which are relatively prime to an arbitrarily assumed whole rational number Q, the conductor of an order.

Under this supposition the whole and fractional numbers of the body Ω form the group previously denoted by O. We enlarge this group by adjoining the functionals, and denote the group so enlarged by $\overline{O}$. To each element of $\overline{O}$ there corresponds then a whole or fractional ideal, and to such elements whose quotient is a functional unit there corresponds the same ideal.

If η is a number in O and ε is any functional unit, then the product $\varepsilon\eta$ corresponds to a principal ideal. Accordingly we denote

$\overline{E}$ the group of functional units,

E the group of numerical units,

and by

$$\overline{E}O$$

[29] Compare also the author's memoir "Über Zahlengruppen in algebraischen Körpern", first communication, *Mathematische Annalen* 48, 438.

the principal class of the functionals.

If we take a system of representatives

$$\varphi_1, \varphi_2, \dots, \varphi_h \tag{1}$$

of non-equivalent whole functionals in Ω, then every whole or fractional functional Φ can be represented in the form

$$\Phi = \varphi_i \omega,$$

where ω is a functional from $\overline{E}O$; and the ideal classes in Ω are the cosets of $\overline{E}O$ in $\overline{O}$. Hence the class number h is the index of the divisor $\overline{E}O$ of $\overline{O}$:

$$h = (\overline{O}, \overline{E}O). \tag{2}$$

Here, in the sharper division into classes, one must in O restrict oneself to the numbers of positive norm.

If now we consider the division of the ideals into classes according to the order $[Q]$, we must regard the functional group $\overline{E}O'$ as principal class, and the class number is

$$h' = (\overline{O}, \overline{E}O'). \tag{3}$$

In order to derive from this the relation between h and h', we make use of the general group theorems of Vol. II, § 2, which we formulate and supplement as follows.

12. *Theorem.* If A is a group and B a divisor of A of finite index (A, B), and furthermore C a divisor of B of finite index (B, C), then, by Vol. II, § 2, (4),

$$(A, C) = (A, B)(B, C). \tag{4}$$

Let A again be a group with the divisor B of index μ. We put

$$A = a_1 B + a_2 B + \cdots + a_\mu B, \tag{5}$$

where $a_1, a_2, \dots, a_\mu$ is a complete system of representatives of A modulo B.

Let now C be a group of elements which can be composed with the elements of A (for example, in our case simply by multiplication). Then from (5) there follows

$$AC = a_1 BC + a_2 BC + \cdots + a_\mu BC. \tag{6}$$

Now one and the same element can occur in the two cosets $a_1 BC$, $a_2 BC$ only if $a_1 a_2^{-1}$ is contained in BC, but not in B.

From this we formulate the theorem:

13. *Theorem.* If A is a group, B a divisor of A of finite index (A, B), and C a group composable with A, of such a kind that A and BC contain no common elements except those of B, then

$$(AC, BC) = (A, B). \tag{7}$$

The supposition of this theorem may also be expressed as follows:

B is the intersection of A and BC;

and it is fulfilled, for example, when C is contained in B, and hence $B = BC$, and also when C has no element in common with A.

14. *Theorem.* If A is a group composed from the two groups A' and B,

$$A = A'B, \tag{8}$$

and if B' is the intersection of A' and B, then

$$(A, B) = (A', B'), \tag{9}$$

provided these indices are finite.

For by (8) there is contained, in every coset aB of B in A, an element a' from A', and one can therefore also represent these cosets by $a'B$, where a' is contained in A'. If then a'_1B, a'_2B are different cosets modulo B in A, then a'_1B', a'_2B' are different cosets modulo B' in A', and conversely; for $a'_1a'^{-1}_2$ is contained in B' if and only if it is at the same time contained in B, and from this formula (9) follows.

We now apply these theorems to the expressions (2) and (3) for h and h'. By (4),

$$(\overline{O}, \overline{E}O') = (\overline{O}, \overline{E}O)(\overline{E}O, \overline{E}O'), \tag{10}$$

and therefore

$$h' = h(\overline{E}O, \overline{E}O'). \tag{11}$$

We now denote, as above, by E the group of numerical units in O. Then $\overline{E}E = \overline{E}$, since the numerical units are contained among the functional units; and every number which is contained in $\overline{E}O'$ is contained in EO'. Hence EO' is the intersection of O and $\overline{E}O'$. If, therefore, we apply (7), putting $O,\ EO',\ \overline{E}$ in place of A, B, C, then B is the intersection of A and BC, and we obtain

$$(\overline{E}O, \overline{E}O') = (\overline{E}O, \overline{E}EO') = (O, EO'). \tag{12}$$

Furthermore, by (4),

$$(O, EO')(EO', O') = (O, O'), \tag{13}$$

and if by E' we denote the group of the numerical units in O', that is, the intersection of E with O', so that $E'O' = O'$, then

$$(EO', O') = (EO', E'O') = (E, E')$$

by (9), since E' is the intersection of $E'O'$ and E. Therefore from (13) we have

$$(O, EO')(E, E') = (O, O'),$$

and from (12)

$$(\overline{E}O, \overline{E}O')(E, E') = (O, O'),$$

and finally, from (11),

$$(E, E')h' = (O, O')h. \tag{14}$$

15. *Theorem.* Formula (14) remains unaltered if O' denotes not precisely the order group, but any number group contained in O, of finite index (O, O'), and if we call two ideals $\mathfrak{a}, \mathfrak{a}'$ equivalent according to O' when their quotient $\mathfrak{a}'/\mathfrak{a} = \eta'$ is a number in O'. If E' is the group of the units contained in O', and if (E, E') has a finite value, then the class number h' is finite and is determined by formula (14).

For example, if O is the group of all numbers in Ω (except 0), and O' the group of numbers of positive norm, then, if there are any numbers of negative norm at all, $(O, O') = 2$. If there are then units of negative norm, then also $(E, E') = 2$. But if all units have positive norm, then $(E, E') = 1$, and we obtain in the first case $h' = h$, in the second case $2h' = h$.

The equivalence according to O is then the general equivalence; that according to O' is the sharpened equivalence.

We return to the orders $[Q]$, and under O understand the numbers relatively prime to Q (if necessary only those of positive norm). Then (O, O') has been determined in the preceding paragraph.

If the discriminant D and its root Δ are negative, then in general both E and E' contain only the two units ± 1. In the two exceptional cases $\Delta = -3, -4$, E contains six and four units, respectively; but E', when $Q > 1$, contains only two; and therefore $(E, E') = 1$ in general, $(E, E') = 3$ in the case $\Delta = -3$, $(E, E') = 2$ in the case $\Delta = -4$. Thus in these cases, by § 98, (10), we have

$$\lambda h' = Q \prod_q \left(1 - \frac{(\Delta, q)}{q}\right) h, \tag{15}$$

where

$$\lambda = 2 \quad \text{for } \Delta = -4, \qquad \lambda = 3 \quad \text{for } \Delta = -3, \qquad \lambda = 1 \quad \text{for } \Delta < -4. \tag{16}$$

This is the relation which was first derived by Gauss and later confirmed by Dirichlet in another way.

If, for example, $\Delta \equiv 1 \pmod 4$ and $Q = 2$, then, according to Gauss's terminology, h is the class number for the forms of the second kind, h' that of the forms of the first kind. It is

$$Q\left(1 - \frac{(\Delta, 2)}{2}\right) = 1, \quad \Delta \equiv 1 \pmod 8,$$

$$= 3, \quad \Delta \equiv 5 \pmod 8.$$

Thus

$$h' = h, \quad \Delta \equiv 1 \pmod 8,$$

$$h' = 3h, \quad \Delta \equiv 5 \pmod 8,$$

and only in the exceptional case $\Delta = -3$ does $h' = h$ occur.

If D is positive, it remains to determine (E, E'). Let

$$\varepsilon = \frac{t + u\sqrt{D}}{2}$$

be the fundamental unit in O, and let λ be the smallest positive exponent for which

$$\varepsilon^\lambda = \frac{t + u\sqrt{D}}{2}$$

is contained in O'. Then E consists of the powers $\pm\varepsilon^\nu$, and E' of the powers $\pm\varepsilon^{\lambda\nu}$, and it is

$$(E, E') = \lambda,$$

and formula (15) gives the class number h'. A further determination of λ cannot in general be given.

If again $\Delta \equiv 1 \pmod 4$ and $Q = 2$, then $\lambda = 1$ or $= 3$; namely $= 1$ when in (17) u is even, and $= 3$ when u is odd. The latter can occur only when $\Delta \equiv 5 \pmod 8$; consequently, for $\Delta \equiv 1 \pmod 8$ one always has $\lambda = 1$, while for $\Delta \equiv 1 \pmod 8$ λ can be both 1 and 3.